Hide and Seek: Remote Sensing and Strategic Stability

by

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Abstract

The competition between the survivability of nuclear arsenals and the counterforce weapons that hold them at risk has been ongoing since the dawn of the nuclear age. A spate of technological and political developments have called the endurance of stability into question once again. This thesis addresses one particular aspect of the competition between technologies of counterforce and strategies of survivability, the use of space-based radar systems to track ground-based mobile missiles carried by TELs (transporter-erector-launcher vehicles).

The work herein describes the development of a set of interlocking models to determine under what conditions space-based radar systems can be used to track TELs, the costs of doing so, and the vulnerability of those systems to countermeasures. This work contributes to an ongoing debate in the security studies literature on the impact of changing technology on strategic stability. I offer policy recommendations on future deployments of SBR systems, evaluate the capabilities of current remote-sensing systems, and provide a framework for evaluating emerging technologies.

In developing these models, the capabilities of radar modalities are reviewed, the tasks that are required to track moving targets are identified, and a scheme for SBR tracking of TELs is developed. A set of satellite constellations are designed and simulated, and their ability to detect TELs is determined by Monte Carlo simulation. A model of the seeker's uncertainty is developed, and geometric tiling is used to model attack planning. Drawing on real-world nuclear forces, model counterforce attacks against a peer and sub-peer adversary are used to determine the number of satellites needed to enable tracking.

SBR systems which are capable of tracking TELs are found to require 50 or more satellites, depending on the conditions, with a total system price tag on the order of a few hundred billion dollars. Countermeasures to SBR tracking are identified and evaluated. SBR systems are found to be vulnerable to a number of countermeasures which are low cost or already at hand, making it unlikely that current or future deployments of SBR systems will meaningfully disrupt strategic stability by undermining the survivability of ground-based mobile missiles.

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"I'm not saying we wouldn't get our hair mussed.
But I do say no more than ten to twenty million killed, tops.
Uh, depending on the breaks."

– *General "Buck" Turgidson*
Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb

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Chapter 1

Introduction

1.1 The Nuclear Revolution

The advent of nuclear weapons fundamentally changed the nature of war. The long human experience with conventional warfare is not applicable to nuclear war [1]. Both Brodie and Schelling argue that the relative strength of military forces is less salient in a world that is defined by the ability to obliterate the valued holdings of adversaries without first subduing their military forces [1,2]. Wars would no longer be contests of strength between two military forces in order to gain control of a battlefield, but an exercise of two states inflicting nuclear harm on one another. Robert Jervis wrote that in the new calculus of war "... what is crucial is each side's ability to cripple the other, and this capability is measured by the match between the country's forces and the targets it seeks to destroy, not between the two sides' forces" [3]. Given how easily nuclear weapons allowed states to cripple each other, Bernard Brodie surmised that "thus far, the chief purpose of our military establishment has been to win wars. From now on its chief purpose must be to avert them. It can have almost no other useful purpose" [4].

The United States was the first country to develop nuclear weapons, but its monopoly did not last for long. The Soviet Union tested its own weapon just four years after the first American weapons were used at Hiroshima and Nagasaki. The competition between the United States and USSR sparked an arms race, which peaked in 1984 with over 68,000 nuclear weapons in existence worldwide. While it was tested a number of times over the course of the Cold War, a tradition of non-use of nuclear weapons was built over time, largely predicated on the fear that any use of nuclear weapons might lead to an apocalyptic nuclear war [3,5]. Such a strategy, though it has been thus far effective, is far from free of risk. General George Lee Butler said of the Cold War that: "we escaped the Cold War without a nuclear holocaust by some combination of skill, luck and divine intervention, and I suspect the latter in greatest proportion." [6]. Avoiding a nuclear nuclear by leveraging threats of societal annihilation, termed deterrence, is a delicate balance that rests on the foundation of mutual vulnerability [2,4]. Accepting mutual vulnerability is recognizing that starting a nuclear war would be suicidal, as any capability that you have to destroy your adversary, they have over you in return. Yet even if mutual vulnerability is accepted, there is still a risk of inadvertant nuclear escalation [7]. If mutual vulnerability were perceived to no longer hold true, then one side might believe it could benefit form a nuclear exchange, and be less deterred from starting one.

With the thawing of the Cold War and the collapse of the USSR, world leaders were able to make drastic cuts in the number of weapons in the world, however we still live in a world that has over 13,000 nuclear warheads, about 4,000 of which are deployed [8] While they are less obvious in today's world, nuclear strategy has not changed significantly since the height of the Cold War. The world has changed, however, including the development and proliferation of many technologies which could undermine the delicate stability of the nuclear balance. Trends in technology are challenging assumptions that underpin nuclear strategy. In particular, the time-worn strategy of hiding nuclear weapons to protect them from attack may be losing effectiveness with the development and proliferation of remote-sensing techniques and computing power. A number of scholars of political science have raised the issue that mutual vulnerability is not to be taken for granted and may inevitably be eroded by ongoing advances in remote-sensing [9,10] Coupled with technical reports outlining advancement and proliferation of new counterforce capabilities, a perception that strategic stability is under attack has been created [11] When considering nuclear stability, it is the perception of mutual vulnerability that matters more than the fact. A state does not have the opportunity to test its belief that it can win a nuclear war until after it has already started one. If a state has decided to set aside the nuclear taboo, then their perception of mutual vulnerability all that they have when deciding whether or not to go to nuclear war. The mere perception that mutual vulnerability is being undermined could increase the risks of nuclear escalation. Thus, it is important to interrogate this issue and determine the extent to which stability is being compromised.

This thesis addresses one particular technological competition between technologies of counterforce and strategies of survivability. Space-based radar systems are a remote-sensing technology that could be used to track ground-based mobile missiles. This would fill a gap in the capabilities of current counterforce technologies. This work analyzes the capacity of space-based radar (SBR) systems to enable or enhance counterforce, and its capability to do so in the face of countermeasures. In doing so, this thesis also develops a framework for understanding what is required for a remote-sensing network to be disruptive to stability. This is used to develop specific policy proscriptions regarding the appropriateness of acquiring radar systems for this purpose in the future. It also weighs in on the ongoing political science debate by putting into relief how far we are today from having a remote-sensing network capable of the task, as well as providing sign-posts of development that can be used to monitor the progression of the competition. Finally, it provides general insights on the inherent challenges faced by the seeker and outlines the requirements of a remote-sensing capability that could meaningfully impact the competition.

This thesis is divided into six chapters. This chapter introduces the concept of deterrence and strategic stability and outlines the current nuclear environment. Chapter 2 introduces the physics underlying radar systems. Chapter 3 presents a model of how nuclear weapons could be applied to a counterforce attack against ground-based mobile missiles. Chapter 4 develops a model of the problem of tracking and how the attacker could manage their uncertainty in the positions of targets by utilizing remote-sensing systems. Chapter 5 and applies the radar technology to it to determine how suitable it would be for the task. Chapter 6 discusses responses that the hider could take to a SBR deployment, and evaluates how decisive those countermeasures would be on the competition, and ties the discussion into the current political science debate.

1.1.1 Is Stability Possible in a Nuclear World?

To address whether SBR systems have the potential to undermine strategic stability, I must first introduce what strategic stability is and how it works. In the first years after the introduction of nuclear weapons to the world, scholars struggled with the question of whether stability was even possible in a world with nuclear weapons. Nuclear weapons were, at the time, considered the weapon of aggression *par excellence*, given their destructiveness and the difficulty in defending against their delivery [4]. If states other than the United States developed nuclear weapons, and there was fierce debate on whether they would, would the proliferation increase or decrease the risk of nuclear war? If nuclear war was inevitable, it might be in the interest of the United States to press its early advantage and fight that war on its own terms. Alternatively, perhaps there was a configuration where it is not in the best interest of any state to employ nuclear weapons which would encourage the non-use of nuclear weapons. This condition is termed strategic stability. Bernard Brodie, one of the seminal thinkers of nuclear strategy, attempted to tackle the problem in *The Absolute Weapon* just six months after the destruction of Hiroshima and Nagasaki [4].

Brodie's conception of nuclear weapons was that they were defined by their immense destructiveness [4]. If a city could be destroyed completely with only a handful of bombs, then it might be possible to erase entire countries with an air campaign much smaller than the one waged in the latter days of the second world war. Both sides of WWII knew the challenge of defending against bombing attacks. At the time, attrition of attacking forces was a few percent of attacking craft per raid [1]. Given the destructiveness of nuclear weapons, an effective defense would need to destroy virtually all of the attacking craft before they were able to release their bombs. Brodie also looked forward to further developments in ballistic missiles, based on the WWII experience of the V-2, that would further make defense nigh-impossible. Based on these assumptions, he concluded that a state with a sufficient number of nuclear weapons would have the capability to destroy an adversary essentially at will.

At the time, the United States was the only country to have successfully developed a nuclear weapon. However, Brodie foresaw the proliferation of weapons to other states [4]. He argued that any perception of scarcity of the raw materials of nuclear weapons, uranium and plutonium, must be weighed against how little material was needed per bomb. Little boy, the bomb dropped on Hiroshima, had only 64 kg of uranium while Fat Man, the weapon dropped on Nagasaki contained only 6 kg of plutonium [12,13]. He also predicted that the technology itself was not uniquely accessible to the United States and would be developed by other states within 5 years. These points may seem quaintly obvious in hindsight, but were fiercely debated at the time. He was proven correct four years later when the Soviet Union tested the RDS-1 device. Faced with the prospect of a world full of nuclear-armed states, each with the power to destroy the cities of the other, Brodie saw the prospect of stability only in a world of assured retaliation.

Brodie posited that nuclear weapons would be weapons of aggression only if the aggressor need not fear retaliation [4]. If they did reasonably expect nuclear retaliation, then the costs of retaliation would outweigh any possible gains from nuclear aggression. Only if each aggression were met with overwhelming retaliation, could a stable system of international security be built. To that end, he advocated a focus not on developing striking power but on survivability: reducing vulnerability of nuclear forces to preemptive attack. The most important conclusion from *The Absolute Weapon* was that stability could be achieved in a world with nuclear weapons, at least in theory. How this might be made to work in practice was developed as the strategy of deterrence.

1.1.2 Deterrence Theory

Deterrence theory is the framework that is largely used to understand nuclear strategy today, as such, it is useful to dwell on the topic for a moment. Deterrence is the art of avoiding war through threat and coercion. Nuclear weapons are particularly well suited to deterrence as they are both massively destructive and difficult to defend against. This means that, if made credibly, the threat of nuclear annihilation can be wielded to influence the actions of your adversaries. If your adversary is also armed with nuclear weapons, however, you must be able to destroy them decisively with a first strike, or you will expect to face retaliation that would result in mutual destruction. Instilling the expectation of nuclear retaliation in your adversaries mind can therefore be used to deter them from taking certain actions. The mechanism of how deterrence should work, and its limitations, requires understanding three main concepts of deterrence: rationality, coercion, and credibility.

Rationality. Rationality, one of the baseline assumptions upon which our conception of deterrence theory is built, is derived from the political science paradigm of realism. This assumption is that states can be treated as rational, unitary actors that exist in an environment of international anarchy. States being the unitary actor in international affairs is the assumption that nation states are the atomic units which interact with each other. Each state will have a set of priorities and goals, typically securing and expanding their physical and economic security, and policies will be set and implemented to pursue those goals at the state level. States have internal structures and pressures that may shape the policies and goals of that state, but the actions of the state are assumed to be controlled by a monolithic entity. A system of international anarchy assumes that there is no global referee which mediates disputes between states, should their goals and policies come into conflict. As such, states must rely on self-help in order to advance their interests. This does not rule out cooperation between states, as groups of states may find it collectively in their best interests to cooperate, but it does assume that each state is ultimately responsible for preserving its own security. Finally rationality implies that states will weigh the costs and benefits to taking actions and select a course of action that best pursues their goals. States may not always correctly perceive the costs or benefits of an action in advance, but those states which excel will see their goals best met, while those who make bad choices will see their fortunes decline. Not every action taken by a state will necessarily be rational, but the system creates a pressure that creates a sort of natural selection pressure that favors rational actions.

Coercion. Coercion is the action of purposefully attempting to alter an adversary's cost-benefit analysis through threats or actions. Since states are assumed to be rational, then the perceived costs and benefits that they weigh when selecting a course of action should be able to be manipulated. Deterrence is the act of preventing an adversary from taking an action. There are two common modes of deterrence, deterrence by denial and deterrence by punishment. Deterrence by denial seeks to reduce an adversary's perceived gains by threatening to destroy their means of achieving their goals. For example, if a state is seeking to capture territory, threatening to destroy their invading forces at the outset of hostilities would constitute deterrence by denial. Deterrence by punishment seeks to increase the perceived costs of an action by threatening destruction of valued holdings in response. For example, threatening the destruction of an adversary's cities if they should invade. Deterrence, however, is the strategy underlying strategic stability. While the concept of deterrence is straight-forward, it is complicated by the fact that it trades in the perceptions of costs and benefits. If you issue a threat in an attempt to deter an adversary from taking an action, and they ignore the threat, subsequent actions to punish or deny your adversary are beyond the scope of deterrence, since deterrence has already failed at that point. The challenge to making deterrence work is selecting threats that are sufficient to deter, and making those threats credible.

If you issue threats in the hopes of deterring an adversary, they must be weighty enough to meaningfully alter their calculus, and credible enough that they believe that you are not bluffing. There is a fundamental tension between these two ends. A large sanction, such as the employment of nuclear weapons, would have more of an impact on an adversary's cost-benefit calculus, it is also inherently less believable than a more conservative response. The employment of nuclear weapons carries great risks, even to the country employing them. This is especially true if both countries are vulnerable to nuclear attack by the other, the condition of mutual vulnerability. Thus, a rational actor would not be reasonably expected to take such an action, unless it was in response to a very serious provocation. Thus, a threat becomes credible when the promised response is proportional to the provocation.

In the case of nuclear weapons, if mutual vulnerability holds, then the only provocation which would be proportional to a nuclear response would be directly threatening the existence of the nuclear-armed state, through invasion or nuclear attack. This leads to what is called the "stability-instability paradox" [14,15]. The stability-instability paradox posits that since nuclear war is so destructive, it will only occur purposefully in response to highly provocative aggressions, any other minor provocation will fall below the threshold at which a nuclear holocaust is an appropriate response. Thus, below this threshold, nuclear weapons do little to deter conflict. This also makes nuclear weapons relatively narrow in their application to statecraft and inter-state coercion.[1]

During the Cold War, one of the fundamental strategic goals of the United States was to keep the USSR from dominating western Europe [16]. The Eisenhower administration, seeking to reduce costs, wanted to draw down the number of United States soldiers stationed in Europe [16]. This would create a significant deficit between the United States and Soviet conventional forces. In order to prevent the Soviets from seeing an easy victory in a European invasion, the White House developed the doctrine of massive retaliation to offset the conventional imbalance. The crux of massive retaliation was that if the Soviets made a provocation in Europe, the United States would respond with overwhelming force, including the employment of nuclear weapons. Massive retaliation was roundly criticized as a strategy because it was not credible. Firstly, the threat of nuclear sanction was not proportional to minor provocations. Additionally, once the Soviets developed their own nuclear weapons, it became questionable if the sovereignty of Western Europe was central enough to United States national interests that it would realistically risk a nuclear war in order to save it [17,16]. The question of whether the United States would trade Los Angeles for Paris highlighted one of the fundamental difficulties in extending deterrence to issues that were not the survival of the state itself.

Credibility. Making retaliation credible is a difficult challenge, especially if assuming that your actors are perfectly rational. If you have been attacked with nuclear weapons, responding with nuclear retaliation gains you little, except perhaps some vindictive satisfaction. In fact, by expending some or all of your weapons, you are expending your leverage. Following through on retaliation requires some departure from rationality, as Brodie noted [1]. One idea for making retaliation credible came from Thomas Schelling's "threat that leaves something to chance" [2]. We have implicitly assumed thus far that nuclear war will only result from a purposeful choice. However, if a general nuclear war could occur not through a purposeful choice but from a loss of control over escalation, then any step up the escalation ladder can be used as a threat to gain leverage. If both actors are aware that they have imperfect control over their own actions, they should be inclined to act with caution, as any small provocation could spiral into an all-out-war. One mechanism that such an event could occur is through an accident involving nuclear weapons. It could also occur through the fog-of-war limiting both actors access to information during a crisis. This could lead to rational decisions, made on faulty premises, leading to an outcome desired by both sides. In this way, any

[1] This assertion has not gone unchallenged. Some strategies for approaching nuclear weapons have hinged on the presumption that escalation following the first use of nuclear weapons may be controllable. This would create scenarios where the limited use of nuclear weapons could be considered. Limited use of tactical nuclear weapons on one's own territory in the face of a conventional invasion (sometimes termed "escalate to de-escalate") is one such example.

confrontation between nuclear-armed states could eventually get out-of-hand and lead to a nuclear crisis. Schelling gave the analogy of two mountain climbers tied together and one approaching a ledge in order to gain leverage over the other climber [2]. If the climber fell, both would die, so it would not be credible for the climber to threaten to do so purposefully. However, simply by approaching the edge, the climber is raising the risk of catastrophe and counting on their partner caving to their demands before disaster struck. In this way, nuclear coercion becomes a contest of wills.

The threat that leaves something to chance allows for a state to extend their nuclear umbrella over non-vital interests but also puts that state in a position where it must signal its resolve. When threatening an adversary with mutual annihilation, the bar for credibility is set high. As such, states have pursued means of making retaliation automatic, either in practice or appearance. United States forces stationed in Berlin and Western Europe more broadly served this function. While the conventional forces were not sufficient to fend off a Soviet invasion, they would act as a trip-wire for United States involvement. Any Soviet invasion of Western Europe would necessitate the spilling of American blood, which would make the United States honor-bound to retaliate. This is the idea of hand-tying, purposefully putting yourself in a situation such that your response to an adversary's provocation is rote. This adds an automaticity that is said to bolster credibility, as the decision to escalate is not made in the heat of the moment, but is presupposed.

While these tacks are designed to decrease the probability of intentional escalation, they run a real risk of being the cause of a nuclear exchange. Harkening back to Albert Wohlstetter's accident problem, purposefully running risks opens the door to catastrophe. Removing decision points from the nuclear escalation process increases its perceived reliability, but also makes it more sensitive to false-alarms.

The theory of deterrence makes the argument that a somewhat stable system of international security can exist in a world of nuclear weapons. In a world where retaliation can be assured, then purposeful nuclear attacks can be deterred. Ensuring peace by threatening nuclear annihilation necessitates running the risk that nuclear war can occur inadvertently, though accident or miscalculation. The stability that can be gained in this way can also be undermined if there is believed to be a significant advantage in striking first in a nuclear exchange. If a nuclear exchange could be perceived to be beneficial to an attacker, then the system of deterrence could break down. Securing retaliatory capability from first strikes has been one of the major challenges in making deterrence work in practice.

1.1.3 Vulnerability and First Strike Incentives

As noted by Brodie, the destructiveness of nuclear weapons and the difficulty in preventing their delivery makes them an attractive weapon for both offense and retaliation. If the fear of retaliation prevents aggression, then attacking an adversary's means of retaliation can provide a means of escaping mutual vulnerability. This is the concept of counterforce, destroying an adversary's nuclear weapons in order to prevent them from employing them. Counterforce requires strategic surprise, as the adversary's weapons must be destroyed before they are launched. This sets up a situation with first strike incentives, which can fundamentally undermine deterrence and strategic stability.

Some of the earliest work on first strike incentives in counterforce pre-date the nuclear age. Giulio Douhet, an Italian air power theorist, developed a conception of air power that captures the importance of striking first in a nuclear war [1]. In Douhet's thinking, air power was fundamentally different from ground and sea power such that it required an entirely new paradigm to understand its strategy. Ground and sea power were focused on‘ gaining control of the battlefield, defeating an adversary's forces through force or guile, and winning control of the space. Air power, on the other hand, moved much faster than the battle lines on the ground. It was also difficult, at the time, to defend against incoming planes with defensive fighters as this was in the era before radar. This meant that area defense in the air would require patrolling oft-empty skies waiting for an incursion. In such a world, where defense was impractical, control of the skies would not be won by fighting in the skies but by bombing an adversary's airfields before they had a chance to respond. Defeating their air power while it was on the ground would provide unfettered control of the skies and the ability to pick apart the country from the air unopposed. Since the primary targets of an air force would be the enemy's grounded air force, there would be an immense onus on striking first. Douhet's premises did not align with the realities of conventional air power, largely due to him massively over estimating the destructive power of conventional munitions, but the logic that followed was a sound description of first strike incentives in nuclear exchanges.

Douhet's assertion was that the only way to defend against an adversary's attack was to preemptively attack their offensive forces while they were still on the ground. This is counterforce, the targeting of an adversary's offensive forces in order to deny the ability to strike at you in the future. In Douhet's conception of air battle, and the current conception of nuclear exchanges, this implies striking first. For counterforce to be successful it must be preemptive. Either as a premeditated attempt to gain advantage over an adversary, or embarked upon when you are certain that the adversary is preparing an attack against you. The latter scenario Brodie noted had been derided as a posture of "I won't strike first unless you do" [1].

If counterforce is both feasible and decisive, then a dyad is not stable as there is a self-reinforcing recursive pressure to attack first as soon as the relationship begins to become fraught. To paraphrase Thomas Schelling, there is pressure on both sides to strike first to prevent "his striking us to keep us from striking him to keep him from striking us" [18]. Even if both sides have parity in their nuclear forces, per Schelling "the situation is symmetrical but not stable when either side, by striking first, can destroy the other's power to strike back" [1]. Escaping this vicious cycle requires developing the ability to weather a first strike with the ability to retaliate intact.

1.1.4 The Competition of Survivability

Albert Wohlstetter opined on the difficulties of achieving a stable second strike capability in his 1958 piece *The Delicate Balance of Terror* [17]. Wohlstetter took exception to a perceived view that deterrence was automatic, created by the mere existence of nuclear weapons. He was concerned with the vulnerability of United States retaliatory capability, having previously raised the alarm over the vulnerability of United States forward airbases to a preemptive Soviet attack [19]. SAC, the strategic air command, operated a number of air bases in close proximity to the Soviet Union. This allowed them, in an era when bombers were the primary delivery system for nuclear weapons, to be able to project their bombing power into the Soviet Union. However, the same proximity that enabled the usefulness of the bases also meant that they would have diminished warning of an incoming attack and would be vulnerable to short and medium range weaponry, allowing the Soviets to reserve long-range striking power to strike at the United States homeland. Ensuring retaliation required clearing a set of hurdles, namely maintaining a stable and cost-effective peacetime posture that was durable to attack, having weapons and delivery systems survive an attack, maintaining the ability to decide to retaliate and communicate that decision, and then reaching and destroying retaliatory targets through enemy active and passive defenses. The hurdles faced by retaliatory forces that have been degraded by a first strike remain today, though were more sharply felt in an era which relied on bombers which are relatively slow and vulnerable to active defenses.

Technological developments, such as the widespread deployment of solid-fueled ICBMs and the development of quiet ballistic missile submarines (SSBNs), would eventually address some of Wohlstetter's concerns, however his main point still stands. A nuclear arsenal without sufficient thought put into its survivability will lead to instability similar to "an old fashioned western gun duel" and that avoiding such a situation takes considerable effort. He also noted that stratagems to reduce the vulnerability of weapons to premeditated aggression, such as alertness, dispersal, and mobility, can often increase the risks of unintentional nuclear escalation, "the accident problem" as he termed it [17]. Still relevant today is his summary that "a protected power to strike back does not come automatically, but it can hardly be stressed too much that it is worth the effort" [17].

Bernard Brodie also took up the question of how to make deterrence work in *Strategy in the Missile Age* [1]. Brodie, like Wohlstetter, worked at the RAND corporation, a United States Air Force affiliated think tank which led the development of much of the analytical thought around nuclear weapons in the 1950s [20]. Thus, many of his concerns were practical in nature, including concern that United States military professionals were not properly adapting to the new realities of nuclear weapons and were instead attempting to fit them into the paradigm of conventional warfare. Much of *Strategy in the Missile Age* builds on his previous work in *the Absolute Weapon* by outlining how nuclear weapons present a fundamental break from conventional conceptions of war, even conventional strategic bombing that dominated the latter half of the second world war. He was also of the belief that a policy of a preventative nuclear war was incompatible with American values, so the United States was forced to find a way to make deterrence work.

He recommended directing resources toward securing the survivability of United States nuclear forces, over the expansion of the arsenal. He argued that it was not the size of the arsenal that was most important, but how confident an enemy could be in reducing its effectiveness to an acceptable level. For example focusing on hardening and protecting SAC bombers rather than building more planes. Living in a resource-constrained world, he recommended focusing the most effort on securing a small core of retaliatory weapons that would be targeted against Soviet cities. Other, redundant weapons that could stand to be less secured, but protecting a dedicated set of weapons designed to threaten what the Soviets were assumed to find most valuable would allow for the most automaticity in retaliatory response. Brodie also struggled with the problem of making retaliatory threats credible. In an envisioned nuclear exchange, where the Soviets struck first, the resultant situation would be the Soviets, with some reserve nuclear forces and the United States with some destroyed cities and assets and a degraded retaliatory capability. In such a situation, the rational response would not be to retaliate and invite more destruction. Thus, making retaliatory threats credible, a central requirement of deterring an attack, required that the enemy expect some amount of irrationality or vindictiveness from the United States if it were attacked. One step which he saw as both promoting the credibility and as a moral imperative was the investment in civil defenses. Civil defenses, such as fallout shelters, would aim to protect civilians in the event of a nuclear war. Brodie argued that possessing effective civil defenses would remove some of the barriers to United States use of nuclear weapons in response to a Soviet attack, therefore increasing the credibility of retaliatory threats, enhancing deterrence.

The discussions surrounding the practicalities of making deterrence work raised a number of points which are still relevant to nuclear weapons today. First, nuclear weapons are both difficult to defend against and hugely destructive. Secondly, a nuclear exchange can only be favorable to an aggressor if they are able to meaningfully degrade their adversary's ability to retaliate. Thirdly, making an arsenal less vulnerable to a preemptive attack can take a considerable amount of work, and is relative to the capabilities of the attack. And finally, deterrence by threatening retaliation also raises a question of credibility that must be addressed in some way.

1.1.5 Strategies for Achieving Survivability

Over the course of the Cold War, both the Soviet Union and the United States took steps to secure their arsenals against preemptive attack and develop a secure second strike. A number of strategies were pursued to ensure the survivability of weapons in their arsenals, namely hardening, alertness, mobility, concealment, and redundancy. These strategies relate to the technical parameters of weapon systems, rather than doctrine. It is here that technical parameters behing to shape strategic stability. Hardening reduces a nuclear weapon delivery system's vulnerability to the effects of nuclear attack while alertness, mobility, and concealment are all means of avoiding being caught in those effects.

Hardening reduces the vulnerability of delivery systems to the direct effects of nuclear weapons by building protective reinforced structures that can withstand weapons effects. Hardening is not limited to delivery systems and the weapons they carry; command and control systems, as well as conventional military forces can be hardened against nuclear attack. The main destructive effects of nuclear weapons that must be hardened against are ionizing radiation, thermal energy, electromagnetic pulses, and blast effects [21].

Ionizing radiation can disable electronic systems and incapacitate personnel. A large, acute dose of radiation, such as from neutrons, can interfere with the proper function of the human neurological system, rendering personnel unconscious. Ionizing radiation can damage semiconductors used in electrical circuits. This can interfere with the proper operation of electronic components of the weapon, such as the fuze or the navigation system. Protecting from the effects of ionizing radiation requires placing shielding materials between the source of the explosion and the personnel and materiel to be protected. Material such as dirt or concrete can be effective shielding material as it is inexpensive and can effectively shield both neutrons and gamma-radiation.

Thermal effects from a nuclear explosion encompass the massive amount of infrared radiation that is emitted by the nuclear fireball. This thermal fluence deposits energy into the materials that it illuminates, which can cause structures to melt or catch fire [21]. Similar to shielding against ionizing radiation, protecting from thermal effects requires placing a barrier between the detonating weapon and the target.

Blast effects are shockwaves created in the air or water by a nuclear explosion. The pressure of the shockwave can destroy structures and overturn vehicles. The degree with which a shockwave is larger than the ambient pressure is referred to as the overpressure. Static overpressure is the maximum pressure experienced by a structure when facing the blast wave. This typically causes fixed structures such as buildings to collapse as it overburdens the structure. Dynamic overpressure is a measure of the flow of air over an object as the shock wave passes. This creates a very strong wind which is able to overturn objects such as vehicles. Hardening against blast involves counteracting these pressures by reinforcing structures, or isolating them from vibrations that would be caused by a ground burst. Vehicles can be hardened against dynamic overpressure, a main mechanism for damaging vehicles, by making them bottom-heavy. During the Cold War, the USSR designed object 279, a boat-shaped tank that was weighted such that it would be very difficult to overturn through the battlefield employment of a nuclear weapon [22].

Hardening is in contention with the accuracy of delivery systems. The blast effects from nuclear weapons decrease with increasing distance from the point of detonation. Thermal and ionizing radiation effects fall off with the square of the distance while blast effects fall off with the cube of the distance. The accuracy of a delivery system describes how likely it is to deliver a weapon to within a certain distance of its aim point. Typically this is quoted as the circular error probable (CEP), the radius of a circle in which 50% of weapons will be expected to fall.[2] If the CEP of a delivery system is large, then the average expected distance between the point of detonation and the aim point will grow, and the average expected intensity of the given blast effects at the target will fall. If a structure is hardened to withstand a specific pressure, then an attacking weapon must achieve some combination of accuracy and yield in order to destroy that target. Missile accuracies have improved over time to the point where hardened structures have been questioned as a reliable means of protecting valuable targets such as nuclear weapons [9,11].

Where hardening still has some utility is in protecting structures that can be buried very deeply. Nuclear weapons, for all of their destructive might, can only move so much dirt. There is also a practical limit on how deeply a ground penetrating nuclear munition can travel, limiting the depth to which a nuclear weapon can reasonably damage a buried structure [23]. Even with larger weapons, the defender can always dig deeper. Buried structures are not invulnerable, however, as they still need links to the surface in order to draw fresh air, through which to communicate, and to allow egress after the attack. There connections can be vulnerable to being damaged, but can also be built to travel significant lateral distances before approaching the surface in order to improve the possibility of surviving.

Mobility can also be used to increase survivability in two ways. First relies on the fact that all nuclear weapon systems have some required delivery time, whether that is the time required to fly a bomber from its base to its target or the flight time of a missile carrying the nuclear weapon. If, during the flight time of a missile, its target moves and the delivery system does not have the capacity to adapt, the weapon can miss its target. This then requires the attacker to either barrage the area encompassing all of the locations into which its targets could move, or have (1) and adaptable delivery system and (2) a coupled remote-sensing system that is able to track the targets, relay that information to the weapons in flight, and have the weapons alter their course. If the attacker fails at any of those tasks, then the attack can fail. The second mode in which mobility can aid survivability also ties in two other methods of providing survivability, concealment and alertness.

Mobility can make targeted weapons difficult to find, making it difficult or impossible for an attacker to successfully target and destroy them. Concealment, denying the adversary the ability to locate your retaliatory weapons does not necessarily rely on mobility, but is most easily achieved by utilizing mobility. A fixed structure, such as a missile silo, could conceivably be hidden, but its fixed location means that as soon as it was detected by the attacker, its location would be forever known, and the defender would have no way of knowing if the attacker knew its location or not. Mobility on the other hand, allows the defender to change the location of weapons continuously, meaning that even if they are detected by the adversary at some point, they have the capability to become lost again. Two vehicles have been developed which rely on mobility and concealment to improve their survivability, transporter-erector-launcher vehicles (TELs) and ballistic missile submarines (SSBNs).[3]

TELs are land based, typically wheeled, vehicles which carry ballistic missiles. Importantly, they are also able to prepare the weapons they carry for launch on their own, meaning that they have free range of a road network to

[2]CEP is less appropriate for some modern fuzing schemes but CEP equivalents can be derived. See [11].

[3]Rail-based launchers have also been investigated in the past but have not seen widespread deployment, so will not be covered in this section.

find a place to hide, from which they can also launch their weapons.[4] This allows TELs, while on patrol, to move between hiding spots, making it difficult for the attacker to keep all of the deployed TELs tracked over time. TELs are typically valued by states with large interior tracts of land in which the TELs can be dispersed, especially those that are mountainous or heavily forested to aid the TELs in evading detection from air- or space-based remote-sensing. The subsequent chapters of this thesis will focus on the relationship between TELs and the remote-sensing systems which attempt to track them.

SSBNs are submarines which carry a number of nuclear-armed missiles (SLBMs). The ability of submarines to evade detection while at sea is largely dependent on how quiet those submarines can be. Sound in some parts of the ocean can travel vast distances, conducted by deep sound channels [24]. A number of submarines processes create sounds that can be detected by sonar receivers. First is broad-spectrum sound that is created by movement of water over the hull. This is typically difficult to detect at range as it is difficult to pick out from the natural background. More easily detected at range are narrow-band signals created by vibrations or rotating machinery within the submarine itself. These tonals can be detected above the background as the receiver can focus on the narrow bandwidth where the signal exists. During the Cold War, the United States and eventually the USSR made their submarines more quiet by damping their tonals in order to make them more difficult to detect with passive systems [25]. The United States pursued this relatively early, but the Soviets did not until near the end of the Cold War. During the 1970s, the United States possessed the capacity to track Soviet submarines throughout their patrols, degrading the survivability of their SSBNs. After submarines tonals were sufficiently quieted, they became much more difficult to track, allowing them to exploit their mobility to gain concealment and make themselves survivable. While not all submarines are capable of achieving an adequate level of quietness, those that do make highly survivable weapons platforms. Those that cannot can instead operate solely in areas of shallow water, such as the Barents Sea, which is too shallow for the deep sound channel to exist and where environmental noise background is very high [25].

A means of gaining survivability, that is somewhat tied to mobility, is alertness. Mobile platforms such as TELs, submarines, or airplanes typically operate out of a base, or set of bases. While not on patrol, they may be in base undergoing resupply, maintenance, crew training, or simply allowing crew members to go off-shift. When mobile platforms are centralized in a known location, they are significantly less survivable. To gain some survivability, they must be dispersed. States must decide how many resources they wish to commit to keeping a baseline level of dispersal at all times. The United States, for example, keeps a third of its SSBN fleet on patrol at any given time [26]. Additionally, war plans can call for mobile platforms to be dispersed in response to anticipated danger. The easier method is dispersal during times of perceived crisis, preemptively making your platforms more survivable at the outset of a confrontation.[5] Alternatively, weapons platforms can be scrambled and dispersed upon warning of an attack. This requires a means of dispersal that is sufficiently fast, as well as an early warning system that gives enough warning. Airplanes have been touted as particularly amenable to dispersal under attack as they can be recalled easily while in flight. This means that the system is relatively insensitive to false alarms. In contrast, silo-based ICBMs can be launched on warning, but cannot be recalled. This method was utilized during the Cold War in order to provide survivability to otherwise vulnerable silos, but is intolerant to false alarms and carries siginificant risk of inadvertant escalation.

[4] Some missiles carried by TELs require special launch pads from which to launch their weapons. Launch pads provide the attacker with a set of fixed targets that can be disabled to effectively disable the TEL-borne weapons, so launch pads need to be made highly redundant to account for this.

[5] Such moves could be seen as escalatory during a crisis, however, so are not free of risk.

The final method for developing survivability is redundancy, expanding the number of targets that an attacker needs to destroy. Such a strategy can provoke a numerical arms race, however. In order to gain security through redundancy, you need more targets than your adversary has weapons. Since weapons themselves are targets, gaining redundancy means building more weapons. The adversary would then see themselves at a numerical disadvantage, and therefore vulnerable to attack, so would be prompted to build more weapons. The difficulty in prevailing in an arms race was recognized by Brodie in 1946, and did not materially change. The added cost of numerical arms race is the exacerbation of the accident problem. With more weapons and weapon systems, there are more chances for low probability events to occur, increasing the overall risk of accidental nuclear escalation [27]. Additionally, the added number of weapons increases the consequences of the failure of deterrence, whether on purpose or by accident.

Each of these strategies have been, or are currently still being applied to nuclear arsenals around the world. As Wohlstetter pointed out in the 1950s, and reiterated by scholars more recently, survivability is context-dependent and takes considerable effort to achieve [17,9]. Survivability is also not necessarily permanent once achieved, the Cold War experience showed that periods of stability and periods of vulnerability could be transient [25,28]. In the following section, I will outline how some select nuclear arsenals around the world are structured and which of these strategies for survivability they utilize.

1.1.6 Deterrence in Practice

In this section, I will summarize and compare the dispositions of the nuclear arsenals of the United States, Russia, and China. I highlight the differences in how each state attempts to make their arsenal survivable and how that makes their arsenals vulnerable, or secure against, different types of technologies. I focus on these three states for the remainder of my analysis for a number of reasons. While the argument in the literature is that counterforce is becoming more feasible, what is meant (and explicitly stated on occasion) is that American counterforce attacks are becoming more feasible [9–11]. As such, it is useful to focus on those countries which may find themselves as American adversaries, such as Russia, China, and North Korea, which have featured in the academic debate [9,29,30]. While there has been discussion surrounding attacks against North Korea, the opacity of the North Korean regime makes it difficult to assess and analyze the disposition of their nuclear forces, or even their doctrine [31]. As such, my focus is placed on the United States-Russia and United States-China dyads.

Figure 1-1 shows the number of weapons each of these states possess. These numbers are based on New START counting rules, so each bomber counts as a single weapon, even if they carry more than one nuclear warhead. The United States and Russia are at numerical parity because of the bilateral New START agreement under which both agree to deploy only 1,550 warheads each. China's nuclear arsenal is approximately one-fifth the size of those of Russia and the United States. If the United States were to attack China, it could launch a relatively inefficient attack, utilizing as many as 3–4 warheads to destroy each Chinese warhead while retaining a few hundred warheads in reserve. The reserve would serve to deter a bedraggled Chinese retaliation as well as an opportunistic attack from the Russians. In contrast, the parity between Russia and the United States require the attacker to destroy at least one target warhead per warhead used, or more than one if the attacker wants to maintain a reserve. Each of these states concentrate multiple warheads on single platforms in some way making this possible even with less than perfectly reliable delivery systems.

All three states base a portion of their arsenal on SSBNs. American and Russian SSBNs are the gold standards

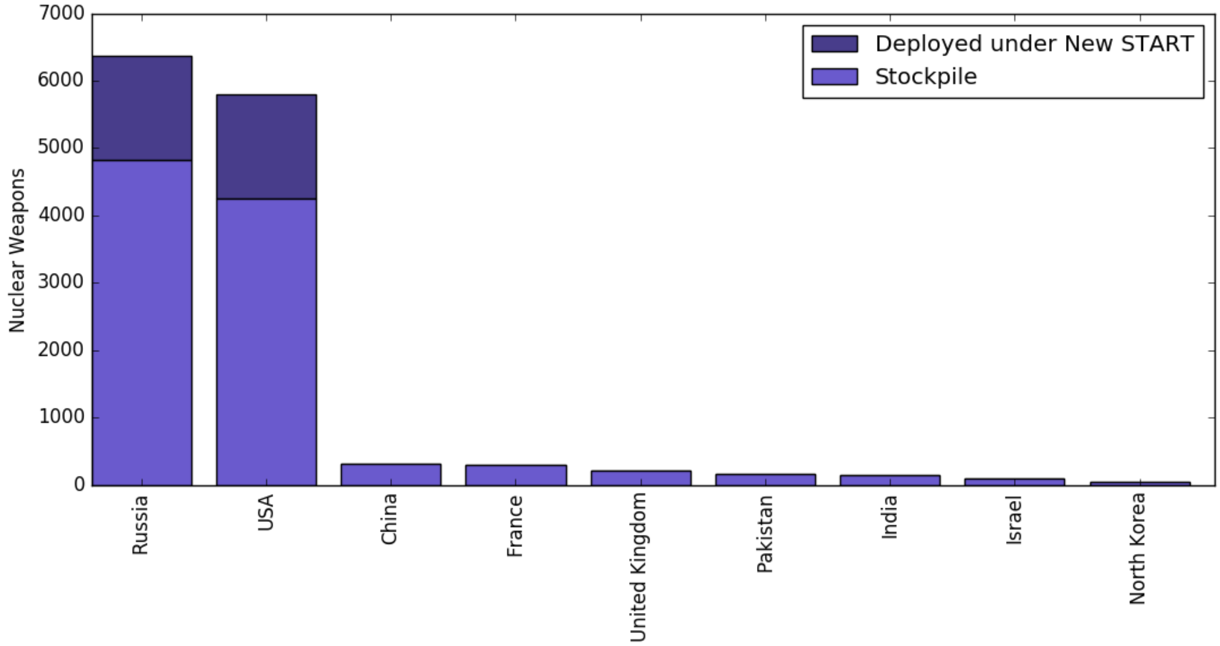

Figure 1-1: The number of nuclear weapons in stockpiles around the world.

for a survivable weapons platform as they have been developed to be very quiet over time [25]. Chinese SSBNs are early in their development cycle, so are noisy and more easily detected [32]. This would require them to operate in shallow waters near mainland China in order to avoid being tracked. The current generation of Chinese SLBM does not have sufficient range to deliver a nuclear weapon to mainland America, limiting their usefulness as retaliatory weapons until further developments are made [33]. An attack against any of these states would require the attacker to successfully locate and destroy all of the deployed SSBNs. If this were to happen, each SSBN constitutes tens to hundreds of warheads. Destroying an SSBN with a handful of nuclear, or even conventional anti-submarine weapons would allow the attacker to break parity. I will note, however, that while concerns that "the seas are becoming more transparent" have been raised in the literature, there is little evidence available supporting this claim [9,10]. In fact, the more recent case for which their information available, long-range passive sonar systems were defeated by advances in making SSBNs quieter [25].

The United States, Russia, and China also all deploy some number of weapons in fixed missile silos. In this realm, the United States possesses an advantage in offensive capability, and has a more favorable disposition of forces. Developments in missile fuzing have made fixed silos vulnerable to counterforce attack. [11]. The deployment of new fuzes on American SLBMs have increaesed their accuracy which is claimed to be sufficient to destroy silos with near-perfect reliability [9,11]. There have not been reports of similar fuzes being incorporated into either Russian or Chinese weapons, although the technology has been in the public sphere since the 1980s [34]. Both China and Russia also deploy multiple warheads on some fraction of their silo-based weapons while all American silo-based weapons have a single warhead [33,35,26]. Coupled with the increased accuracy of American weapons, the United States has an opportunity to gain a numerical edge in a counterforce exchange by destroying Russian or Chinese silo-based warheads on a better than one-for-one basis. This opportunity is not reciprocal. The Chinese arsenal has relatively few MIRVed silo-based weapons, so the advantage gained from attacking MIRVed systems would be small relative to the numerical superiority held by the United States. The Russian arsenal has over roughly 70 silo-based ICBMs which are MIRVed, including 46 launchers that can carry 10 warheads each [35]. This provides a significant opportunity for a U.S. attack to break parity, leaving more warheads to attack other legs of the Russian triad.

All three states deploy nuclear-armed aircraft, however the Chinese deploy only about 20 air-deliverable warheads. The Russian and American aerial components of their nuclear triads are considerably larger. Under the counting rules of the New START treaty, each bomber is counted as a single warhead, despite being able to carry multiple warheads. This counting method incentivizes both parties to emphasize manned aircraft, which derive survivability from alertness and mobility while being poorly suited for offensive counterforce operations due to their slow speed of delivery. This makes air-based weapons more stabilizing than silo-based weapons as they may need to be targeted with a large number of warheads if alerted in advance and would not effectively contribute to carrying out an attack [36]. If the defender were caught unawares and had limited opportunity to load bombers and scramble them away from their air bases, then the attacker could potentially destroy multiple bombers per attacking warhead.

Ground-based mobile missiles on TELs are a basing method that Russia and China employ, but the United States does not. This means that any vulnerability in the basing method would affect only Russia and China without a reciprocal effect on the American arsenal. This is one of the motivating factors for why the competition between remote-sensing systems and TELs is of particular concern. Russia deploys 180 TELs, roughly 100 of which are MIRVed [35]. This fraction will likely increase as Russia retires and replaces Soviet era systems [37]. If advances in U.S. remote-sensing capabilities can deny Russian TELs' concealment, then there is an opportunity for a counterforce attack to destroy TELs on a better than one-for-one basis.

For China, the United States already has numerical superiority, but without remote-sensing the numerical difference must be enough for the United States to barrage every possible TEL position in order to effect a successful counterforce attack. China deploys more than 70 ICBMs on TELs, which are not thought to be currently MIRVed [38]. They also deploy a number of shorter ranged nuclear or dual-capable missiles such as the DF-21 and DF-26 [38]. While these shorter range systems cannot threaten the U.S. homeland, they can hold allied or regional targets at risk.

The arsenals of Russia, China, and the United States all employ multiple basing methods to gain survivability. Successfully conducting a counterforce attack against any of them would require simultaneously defeating multiple independent systems. An attack against China by either Russia or the United States has the lowest remote-sensing requirements, given the numerical imbalance. Even so, this would require the simultaneous destruction of Chinese submarines, TELs, and silo-based weapons. China maintains their warheads in central depots during peace-time, so they would be especially vulnerable to a bolt-from-the-blue attack [33]. Both Russia and the United States have nuclear triads that would need be defeated. The enforced numerical parity between them creates a challenge for a counterforce attack. Weapon delivery systems are not perfectly reliable, and some delivery systems will fail to deliver their warheads. A one-for-one attack, where one attacking warhead is allocated to a single target warhead, will leave the attacker at a numerical disadvantage after the exchange.[6] The concentration of warheads on MIRVed weapons and SSBNs provides an attacker with the opportunity to destroy warheads on a better than one-for-one basis. Whether this is sufficient to create a scenario where a counterforce attack is favorable to the attacker depends

[6] Though, it may be preferable to some attackers to leave a small residual of nuclear weapons in the hands of an adversary in order to ensure that the majority of the nuclear explosions take place in the defender's territory.

on the rate at which warheads can be effectively traded.

For states with diversified basing in their nuclear deterrents, vulnerability in one leg can be exploited to increase the vulnerability of another. For example, if the United States were able to utilize a single warhead to destroy a MIRVed silo-based missile carrying ten warheads, then the United States would have an additional nine warheads to allocate to other targets. These surplus warheads could be allocated to attacking TELs, SSBNs, or bombers which may not be destroyed at parity. Vulnerability in one leg of a deterrent does not necessarily mean that the deterrent is not survivable, but it does raise the survivability requirements of each other leg.

I will assume, for the remainder of this thesis, that the American nuclear deterrent is secure. The United States has long held an advantage in undersea warfare, has a well-financed nuclear arsenal, and the most extensive early-warning system. Thus, if strategic stability is to fail, it will be based on the loss of survivability of either the Russian or Chinese deterrents.

The technological advantages of the United States also make it the best positioned to carry out an attack. It has extensive hard-target kill capabilities on a large number of their weapons, as well as a large number of remote-sensing assets [11]. Russian MIRVed weapons also reduce the number of targets that the United States would have to destroy in an attack. This is not to say that a counterforce attack would be easy, or even feasible. Russian SSBNs would need to be compromised, TELs tracked, and bombers caught on the ground in a highly complex, perfectly timed attack with almost no room for error. While this is a herculean task, I assert that it is the most likely way that strategic stability fails in the near future. For this reason, for the remainder of this thesis, I will focus only on the survivability of TELs. The loss of survivability of TELs is a necessary precondition for a successful attack against either Russia or China. Thus, if it is unlikely to occur, then a loss of strategic stability is similarly unlikely.

1.2 The Impact of Technological Change on Stability

Nuclear-armed states rely on hardening, alertness, mobility, concealment, and redundancy to maintain a survivable second strike capability. None of these strategies are inviolate, hardened structures can be destroyed, concealed targets can be found, and mobile targets can be tracked. This competition between survivability and counterforce has been ongoing since the dawn of the nuclear age [4,17]. As established in Section 1.1.3, finding stability between nuclear-armed adversaries requires that survivability prevail in the competition to avoid destabilizing first-strike incentives. While there have been periods where one side or the other (though usually the United States) held an advantage over their adversary, stability has largely prevailed [25,28]. A spate of technological and political developments have called the endurance of stability into question once again [9,10]. In this section, I will survey a number of the technical and political trends that have changed the nuclear stability landscape since the end of the Cold War. This will provide the backdrop for discussing the implications of space-based radar systems later in this thesis.

One of the primary developments that has occurred is the drastic reduction in the number of deployed weapons since the Cold War. During the mid-1980s there were over 60,000 nuclear warheads worldwide [39]. Today, there are fewer than 4,000 deployed worldwide, with the two primary holders of nuclear weapons, the United States and Russia, each having 1,550 deployed warheads (and another 1,500–2,000 more in storage) [8]. Having fewer deployed nuclear weapons can lead to a reduction in redundancy. This is not always true, as reducing the number of warheads on a MIRVed missile (downloading) reduces the number of warheads without reducing the number of targets. Decreasing redundancy does not inherently improve the feasibility of counterforce if the attacker is similarly limited in the number of weapons they have available to attack with. What it does do is decrease the complexity of the attack, which may lead to it being perceived as more achievable.

The second political trend has been the reduction in alertness levels. As mentioned in Section 1.1.5, increased alertness allows weapons platforms to more easily exploit mobility in response to a perceived attack to increase the odds of surviving the attack. In the case of ICBMs, increased alertness may allow them to be launched after an attack is detected, but before the weapons are destroyed. The cost of increased alertness is a possible increase in the consequences of false alarms. Deciding to launch ICBMs in response to an incoming attack must be done in a matter of minutes [40]. Making such consequential under time pressure with imperfect information creates a potential pathway of inadvertent nuclear escalation [41] Toward the end of the Cold War, many of the steps taken to increase alertness were abandoned in order to increase safety [42]. These were steps such as maintaining bombers loaded with nuclear weapons either in the air or on the runway, which led to a number of near-miss accidents during the Cold War [6]. While bombers have been de-alerted, United States and Russian silo-based ICBMs still maintain launch-on-warning alertness [40,43]. Similarly, United States and Russian SSBN fleets still maintain a number of SSBNs on deterrence patrol at any given time [35,26].

The United States maintains a complex early warning system that utilizes space-based infrared (SBIR) satellites to detect ballistic missile launches. These are integrated with ground- and sea-based radar systems which can then be used to track the trajectories of missiles in flight [44]. Even with these precautions, however, the amount of time available to the president of the United States to make a decision on what to do in response to an incoming attack is on the order of minutes [40]. If a Russian counterforce attack involved an attack on the White House and other command and control nodes utilizing depressed-trajectory SLBMs, the missiles could detonate before the president was even alerted [40]. The Russian early warning system is significantly less capable than the United States system, relying more heavily on ground-based radars and only a handful of early warning satellites in Molniya orbits [45]. Advances and proliferation of space-based monitoring systems may prove useful in increasing the speed and reliability of early warning systems, improving the efficacy of alertness as a means of gaining survivability. However, even with ideal early warning systems, there amount of time available to react to a perceived incoming attack is limited by the flight time of the weapon delivery systems. Launch-under-attack may be feasible for very high readiness weapon systems, but brings a large risk of inadvertant escalation if the early warning system has a false alarm. With mobile missile systems, such as TELs and aircraft, a policy of move-on-warning may provide a means to utilize alertness to improve survivability with less risk of escalation. In order to exploit this, China and Russia would need to continue to invest in their early warning systems.

A number of technological trends have also led to perceptions of a changing landscape of survivability. The improvement and proliferation of highly accurate weapons, particularly among United States SLBMs, has increased hard-target kill capability [11]. A hardened structure can be destroyed by a nuclear weapon though static overpressure from an air burst, or from cratering with a ground burst [21]. In both cases, the probability of destroying the target depends on both the yield of the warhead and the distance between the target and the ground zero of the detonation. During the Cold War, high-yield warheads were developed to hold hardened and deeply buried targets at risk. Aside from a few exceptions, such as the Chinese DF-5, the yields of modern nuclear weapons tend to be considerably smaller, on the order of a few hundred kilotons, than the megaton warheads of the Cold War. In order to threaten hardened targets, smaller warheads require more accuracy.

Significant gains in accuracy have been made in the United States arsenal since the 1990s. One newer development being the widespread deployment of burst-height compensating fuzes [11]. Conventional fuzes trigger the detonation of a warhead at a given altitude set before launch. As a weapon travels from its launch point to its aim point, it can accrue positional errors, caused by unpredictable atmospheric conditions, and any other differences between the predicted and actual conditions. Burst-height compensation fuzes utilize a sensor to determine how far the warhead is from its target trajectory as it reenters the atmosphere and adapt the timing of its detonation to compensate, improving accuracy [34].

Burst-height compensating fuzes were first developed in the 1980s and were deployed on United States silo-based ICBMs [34]. They were later extended to United States SLBMs, which make up a significant portion of the United States deployed nuclear arsenal. By applying the new fuzes to these warheads, the warheads gain sufficient accuracy to be able to threaten hardened targets that they previously could not. This led to a marked increase in the hard-target kill capability of the United States arsenal as a whole, discounting the value of hardening to United States adversaries such as Russia and China [11]. It is unknown if a similar development has occurred in the Russian arsenal. The determination of this development has contributed to speculation that the increase in United States hard-target kill capability is sufficient to enable counterforce attacks against United States adversaries [9,29].

Ballistic missile defenses, while notionally a defensive system, can also be used to enable counterforce. The United States unilaterally withdrew from the anti-ballistic missile defense (ABM) treaty in 2002. The ABM was a bilateral agreement between the United States and Russia, limiting the deployment of strategic missile defenses. The United States pursued development of the GMD (ground-based mid-course defense) system, committing 40 billion USD to its development [46]. The GMD system is comprised of a set of radar systems as well as 44 ballistic missile interceptors based in Alaska and California. Upon detecting and determining the trajectory of incoming ballistic missiles, the system is designed to to physically collide an exo-atmospheric kill vehicle with the warhead reentry vehicle while it is traveling above the atmosphere. This requires sensors to both detect and track the incoming missiles, and also enough interceptors to intercept the warheads of an incoming attack. If each interceptor worked perfectly, the GMD system could be expected to defend against 44 incoming warheads, a minuscule amount compared to what would be expected from an opening volley of a Russia counterforce attack. In practice, the interceptors also have to solve the problem of differentiating between authentic warheads and any decoys that the attacker includes, as well as components of the booster that will be traveling in a cloud with the warheads [47]. It has been questioned if this is a solvable problem, and the problem can be made more difficult by the attacker by including more decoys. In test trials that do not include realistic decoys, the GMD system currently has a track-record of successful intercepts in approximately half of all tests. If each intercept was considered to be independent, then it would require 4 interceptors to achieve a 95% probability of intercept per incoming warhead. Such a system would be able to account for 11 incoming warheads, with a roughly 50% chance of not letting any warheads through.[7] If the attacker were to fire 12 or more warheads, efficacy would fall from there. While such a system would not be an effective defense against a first strike, it could be used in support of an offensive counterforce attack.

The primary concern surrounding effective BMD is that they could be used to "mop up" a retaliatory strike that was reduced in capacity by a counterforce attack [48]. China has expressed concern about this point specifically as

[7] Significantly increasing the number of interceptors, or their reliability, or both, would improve the efficacy of the missile defense system.

their deterrence posture could be characterized as a minimum deterrence posture [49,50]. Increasing the required number of surviving weapons that would be needed to successfully retaliate increases the pressure on their arsenal. They have vociferously protested the expansion of United States missile defense radars on the Korea peninsula over concerns that those radars could be used to support United States ballistic missile defenses against Chinese retaliation and undermine the value of their deterrent [51]. As a consequence, they have been slowly building up the number of nuclear weapons that they operate [38].

United States ballistic missile defenses have also been a point of contention for Russia since the United States withdrew from the ABM treaty during the George W. Bush administration. Since the end of the ABM treaty, Russia has touted the development of advanced nuclear weapons systems, such as hypersonic glide vehicles, nuclear powered cruise missiles, and autonomous submersibles armed with high yield thermonuclear weapons designed to irradiate port cities [52,53]. These weapons would entirely evade the United States GMD system, providing Russia with a more assured retaliatory capability. While these systems may or may not reach deployment, or be effective, they send a clear signal that Russia is invested in preserving the survivability of its retaliatory capability, and is willing to invest resources to do so.

Whether ballistic missile defenses are destabilizing or not depends on the perception of their effectiveness. If the United States perceives that it has an effective missile shield against bedraggled retaliatory attacks, it may be more likely to undertake risky endeavors, confident in its ability to disarm an adversary if push came to shove. If Russia or China feel that United States defenses meaningfully degrade their deterrent, they may be pushed to develop new weapon systems, or simply to increase alertness in order to improve the survivability of their arsenal overall. Increasing alertness can lead to increased risk of accidental escalation.. Importantly, these effects are based on perception of current and future capabilities of the technology, so could have destabilizing effects even if they are not effective.

The final trend that is contributing to the perception of the erosion of strategic stability is the development and proliferation of remote-sensing technologies. remote-sensing technologies are in competition with the survivability strategies of concealment and mobility. One of the fundamental needs when attempting to destroy a target is knowing where it is. Since nuclear weapons are typically deployed in the interior of large states, or underwater, the sensors used to detect and track these weapons must be able to operate at a distance.

In the political science literature, there is a perception that the "seas are becoming more transparent" [54]. The concern is that remote-sensing technologies that can detect SSBNs while they are at sea would make them vulnerable to anti-submarine warfare measures, reducing their survivability. During the Cold War, the United States developed the capability to track Soviet submarines utilizing a combination of long-range passive sensors and attack submarines [25]. The Chinese Jin-class of submarines is sufficiently loud to be tracked at long range using passive sensors, forcing them to operate in shallow water where the background environmental noise is high [32]. By the end of the Cold War, however United States long-range sensors in the Atlantic were rendered obsolete as Soviet subs became sufficiently quiet. In lieu of using long range arrays, newer systems are upward looking sensors that are anchored to the sea-bed and can detect submarines that pass over them [55]. These are limited by being fixed in location, so act only as pickets, and not as tracking sensors. While there is no decisive information that there are any decisive advantages in the competition between SSBNs and remote sensors, a decisive advantage was held in the past so the competition is worth monitoring.

For finding land-based weapons, airplanes and drones can be used as platforms for remote sensors, but are limited by their ability to overfly adversaries' territory. Planes or drones over hostile territory are vulnerable to being downed by air defenses. The downing of an American U-2 spy plane over Soviet territory in 1960 illustrates the risks of relying on planes for reconnaissance deep in hostile territory. It also is illustrative of the actions that an adversary is willing to take if it feels its security is being threatened by spying, and the political toll that such an event can take. More recently, Iran reportedly downed and captured a U.S. drone [56]. Since downing a drone does not directly put at risk the lives of any of the operators servicemen and women, the threshold for escalating to kinetic force or electronic warfare to down a drone may be quite low. The vulnerability of airframes makes them unsuitable for the detection and tracking of an adversaries nuclear weapons that must be done in advance of an attack. An increase in plane and drone overflights could tip-off the defender, causing them to increase alertness, reducing the effectiveness of launching a counterforce attack, and increasing the risk of accidental escalation.

A more suitable platform for such a task are satellite-based remote sensors. While the exact border between airspace and space is muddled, states have been more accepting of satellite overflights than they have historically been of airplane overflights. The ability to conduct observations of adversarial territory with airplanes required the negotiation of the Open Skies treaty, whereas daily satellite overflights have been accepted as the status quo. This provides satellites with unprecedented access to otherwise unreachable areas deep within adversarial states. If those space-based sensors are able to detect targets, then they can defeat concealment, if they can track those targets over time, then they can account for the effects of targets mobility.

There has been an explosion in the number of Earth observation satellites over the past decades [57]. This includes both commercial imaging capacity and state-owned surveillance imaging. The costs of space-launch services have also decreased by an order of magnitude since the turn of the millennium, lowering the cost to expanding space-based remote-sensing capability [58]. The proliferation of remote-sensing has featured heavily in the political science literature surrounding the future of strategic stability as an increase in sensors intuitively leads to a reduction in the effectiveness of concealment and mobility [9,10]. It is this problem in particular which I seek to address in this thesis.

1.3 Thesis Overview

In this chapter, I introduced the notion that the existence of survivable nuclear deterrents support strategic nuclear stability. Over the next five chapters, I introduce and analyze the competition between remote-sensing technologies and nuclear forces based on road-mobile transporter-erector-launcher vehicles (TELs). In Chapter 2, I provide some background on the physics underlying radar systems that is necessary to understand the later chapters. In Chapter 3, I present a method for modeling a counterforce attack against a nuclear armed adversary. The Counterforce Model can be thought of as defining the arena in which hiders and seekers will compete. In Chapter 4, I present a modality-agnostic model of tracking and use that model to survey existing remote-sensing technologies to identify potential counterforce-enabling technologies and develop performance requirements that they would have to meet. In Chapter 5, I introduce how space-based radar (SBR) has the potential to fill the gaps in current capabilities and design a radar-based remote-sensing system for tracking TELs. In Chapter 6 I show that countermeasures to SBR systems already exist and are widely available, making SBR unlikely to significantly undermine strategic stability. In Chapter 7, I argue that the ability for any single technology, such as SBR, to undermine the survivability of TEL must be evaluated in the context of the countermeasures, and offer some concluding thoughts.

To carry out this analysis, I develop a number of interlocking models that together can be used to evaluate whether a given remote-sensing system, including one composed of multiple modalities, could enable a counterforce attack. Figure 1-2 shows a flow-chart of how these models connect. The Counterforce Model defines the performance requirements that a remote-sensing system would need to meet in order to enable a counterforce attack. The Tracking Model determines a given systems ability to meet those requirements. The Radar Model, which in theory could be substituted by a model of a different technology, is used to create a set of design parameters that would create a counterforce enabling remote-sensing system. In practice, this problem must be approached in the opposite direction as there is no unique set of design parameters for a counterforce-enabling system. I search the space of design parameters and each resultant system is evaluated for its ability to enable counterforce. I refer to this configuration as the forward solution. While the model must be solved in the forward direction, the overall problem is better understood in the reverse direction, so I follow the reverse solution over Chapters 3 - 5 as I introduce the component models.

The Counterforce Model compares the number of warheads available to the attacker to the number of targets in order to determine the maximum uncertainty that the attacker can have in the positions of the target TELs. The Tracking Model, outlined in Chapter 4, defines the tasks that are required in order to track TELs. I model how an attacker's uncertainty in the position of a group of TELs will vary over time as a function of the environment and the frequency of the attacker's observations with a remote-sensing system. The combination of the Tracking and Counterforce Attack models are needed in order to define what it means for TELs to be tracked versus untracked. At the end of Chapter 4, utilizing these two models, I survey extant remote-sensing technologies to identify gaps in current capabilities that, if addressed, could enable counterforce by undermining the survivability of TELs. However, before delving into the models, in the next chapter, I introduce some background on radar systems which is crucial for understanding the later chapters.

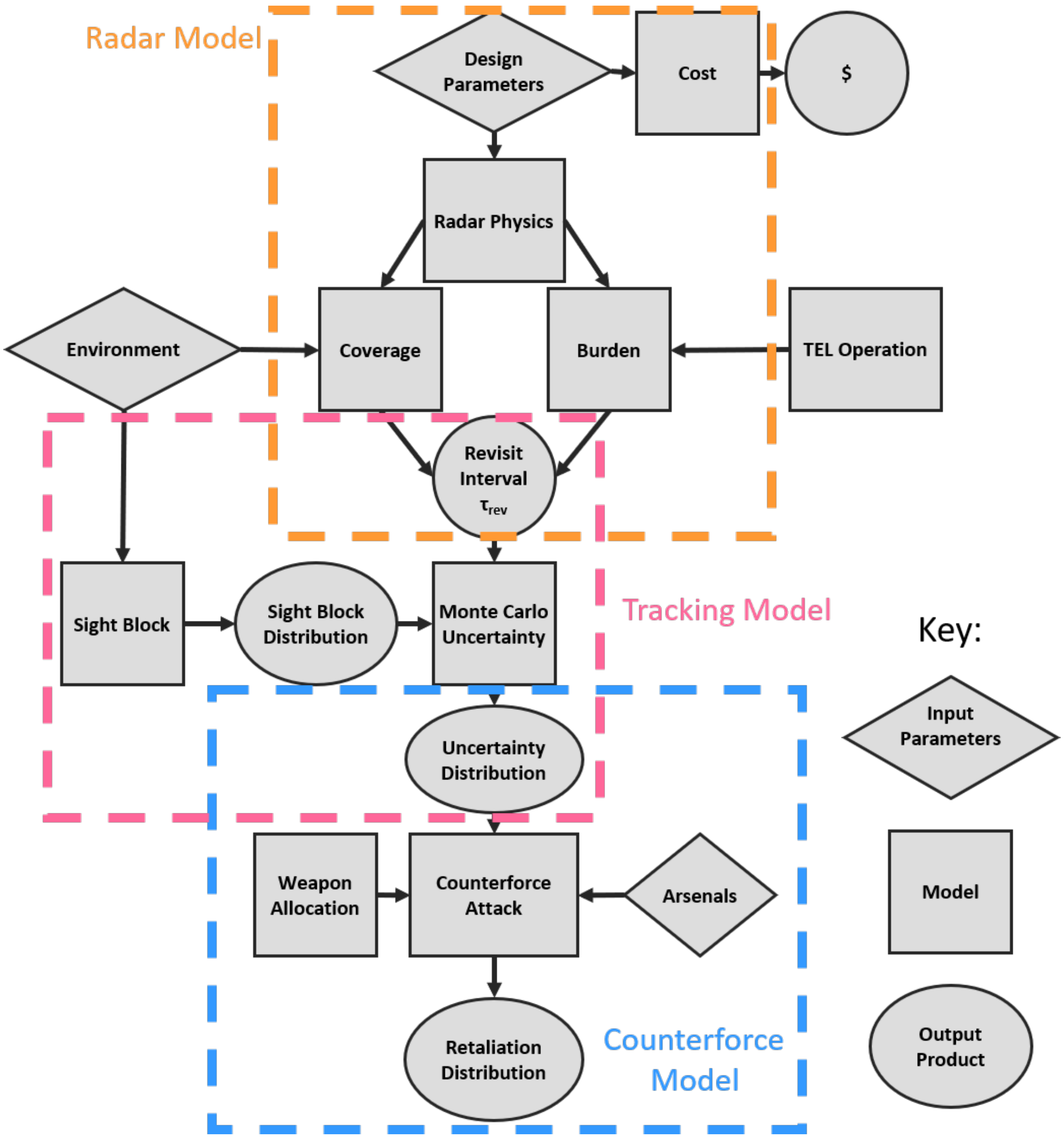

Figure 1-2: Flow chart of the model. The forward solution moves from top-to-bottom in order to determine if a given radar hardware configuration is capable of supporting a counterforce attack. The reverse solution moves from the bottom-up, using the counterforce model to create a set of performance metrics that a radar system is designed to meet.

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Chapter 2

Space Based Radar

2.1 Radar Physics

Radar systems have attracted attention as a technology with the potential to undermine the survivability of TELs. In this section, I outline the physics which underly the operation of radar systems and show how this gives them the capability to operate at night and through cloud cover, potentially closing the gap in current tracking capabilities.

2.1.1 The Fundamentals of Radar Operation

A radar is a remote-sensing system that utilizes electromagnetic radiation to sense objects. Figure 2-2 shows a schematic for the radar detection of a TEL with an SBR system. In this example, the SBR system observes a scene by illuminating it with radio waves. Some fraction of the incident radio power will reflect off of the objects in the scene, such as the TEL. The radar receiver measures the power of the returned signal, which is used decide whether there is a TEL in the scene or not. The processes of emission, travel, and scattering of radio waves used in radar detection together can be used to describe the radar return, the signal that is measured by a radar. When a radar observation is made, the measurement can contain up to four components, target signal, noise, clutter, and interference or jamming. The radar operator is only interested in the target signal, so it must be separated out from the other components.[1] I return to the topics of noise, clutter, and interference in later sections. In this section, I will introduce the factors which impact the magnitude of the measured signal.

The amount of signal power measured in the radar receiver (P_r) can be estimated with the radar range equation

$$P_r = \underbrace{\left[P_t G_D \frac{\epsilon_t}{4\pi R_t^2}\right]}_{Illumination} \cdot \underbrace{\left[\sigma \frac{A_r \epsilon_r}{4\pi R_r^2}\right]}_{Reflection} \cdot \underbrace{\left[\frac{1}{L}\right]}_{SystemLosses} \quad , \tag{2.1}$$

where P_t is the transmitted power, G_D is the directive gain of the transmitting array, ϵ_t is the absorption of radiowaves as they travel from the transmitter to the target, R_t is the distance between the transmitting array and the target, σ is the radar cross-section (RCS) of the target, R_r is the distance between the receiving array and the target, ϵ_r is the attenuation between the target and the receiver, A_r is the receiver aperture area, and L is the loss term. I have broken the range equation into three components. The first term in equation (2.1) describes the amount

[1] In some modalities, the clutter is the signal, see Subsection 2.2.4.

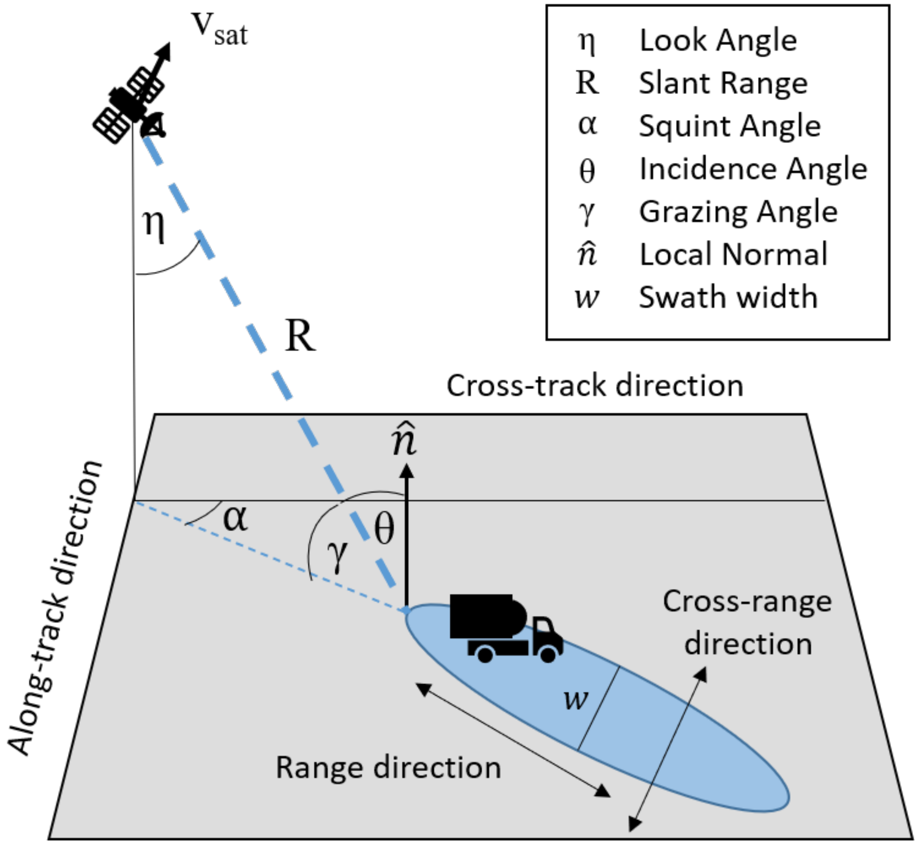

Figure 2-1: Angles and directions in SBR imaging. v_{sat} is, by definition, in the along-track direction. The blue area is the beam footprint, the area between the half-power beam-widths in the range- and cross-range directions. The nearest point on the beam footprint to the satellite is called the heel of the footprint, the furthest point is called the toe. Slant range and look, grazing, and incident angles are shown to the heel of the beam, but they will vary over the footprint.

of power which illuminates the object being observed. The second term describes the fraction of the incident radiation that is scattered from the object into the receiving array. The final term which includes the losses L is used to account for departures from ideal behavior. In the monostatic case where the receiving array is also transmitting array, as is the case in most space-based radar systems, $R = R_t = R_r$, $\epsilon_t = \epsilon_r$, σ is the backscatter cross-section, and $A = A_r$. Assuming the monostatic case and making these simplifying assumptions allow the equation to be restructured as

$$P_r = \underbrace{\left[\frac{P_t G_D A}{L}\right]}_{RadarArray} \cdot \underbrace{[\sigma]}_{Target} \cdot \underbrace{\left[\frac{\epsilon^2}{(4\pi R^2)^2}\right]}_{TransmissionLoss} \quad . \tag{2.2}$$

In this formulation, the factors which influence the measured signal are made explicit. The radar array itself, which is built and designed by the seeker, determines the power, gain, receiver area, and the system losses. The target itself determines how much of the incident signal is reflect toward the receiver through its radar cross section (RCS) σ. Transmission loss is determined by the geometry of the measurement, the distance between the radar and the target and the material present along that path. In the next subsections, covering the radar array, target, and transmission losses, I go over the relevant details for determining the values for each of the variables in equation 2.2.

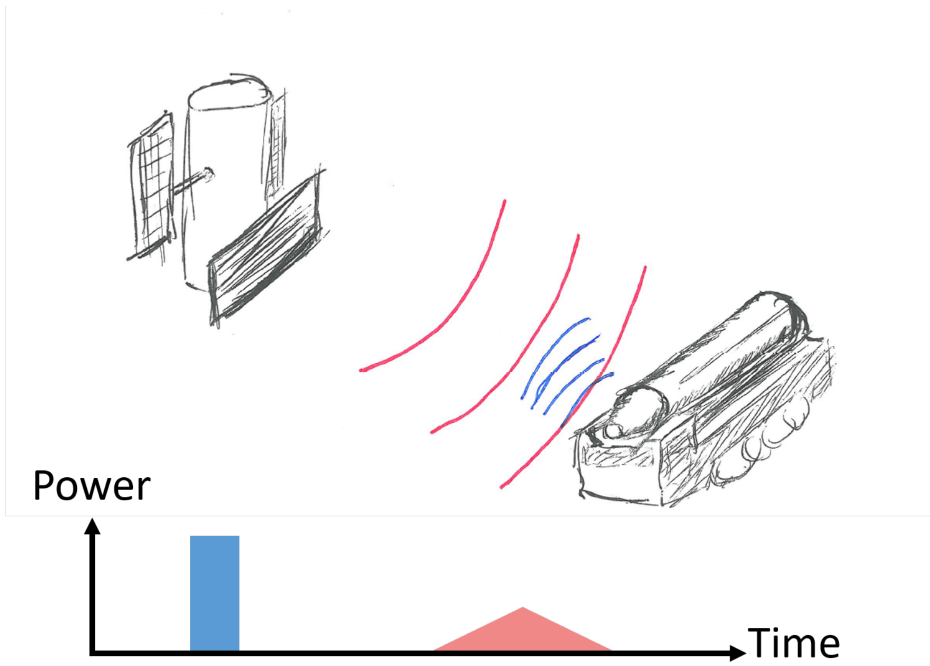

Figure 2-2: Schematic of radar operations. Radio waves are transmitted (red) from a radar system. Objects in the beam reflect radio waves (blue) which can be measured in the radar receiver.

Radar Array

Power. A radar array is composed of a radio frequency (RF) source, an aperture, and a receiver. The RF source produces the radio waves that will ultimately be emitted by the radar.[2] Historically, vacuum tube sources, such as klystrons or traveling wave tubes were used to produce high-power RF signals [59]. In this thesis, I will be considering only active electronically steered arrays (AESAs), as they have beam agility and resistance to interference that is useful for space radars [59,60]. AESA arrays generate their power by combining a large number of low-power solid-state transmit-receive (TR) modules [61]. For the purposes of this thesis, the details of the RF source will simplified to a source that provides power P at some wavelength λ.

Aperture. The shape of a radar beam, the radiation pattern, is defined by the size and shape of the aperture and the wavelength of the radiation. For a line source, the radiation pattern is

$$G(\phi) = \frac{\sin\left[\pi \frac{l}{2\lambda} \sin\phi\right]}{\pi \frac{l}{2\lambda} \sin\phi} \;, \tag{2.3}$$

where $G(\phi)$ is the gain, the relative magnitude of the emitted electric field as a function of angle, l is the total length of the source, λ is the wavelength, ϕ is the aspect angle, the angle between the aperture normal and the direction of emission.

Moving from a one-dimensional source to a two-dimensional aperture, a rectangular aperture defines a plane with two orthogonal principal directions, of length l and width w. Equation 2.3 can be used to describe the radiation

[2] Some radar systems utilize wavelengths of electromagnetic energy that fall into the microwave classification. The boundary between radio and micro waves is somewhat blurry, so throughout this thesis, I will use the term radio wave exclusively for the sake of simplicity.

pattern in each of those principal directions, with the relevant angle measured from the surface normal of the aperture. In two dimensions, the radiation pattern is the product of the two principal patterns

$$G(\phi, \theta) = G(\phi)G(\theta) \ , \tag{2.4}$$

where ϕ is measured from the surface normal $\hat{n}$ of the aperture in the $\hat{n}$-l plane, and θ is measured from the surface normal in the $\hat{n}$-w plane. An array of individual isotropically radiating elements spaced less than λ apart will form a similar radiation pattern as an equivalently sized continuous source. A rectangular array of elements will behave similarly to a rectangular aperture of the same size. This allows me to treat physical apertures and arrays with the same equations.

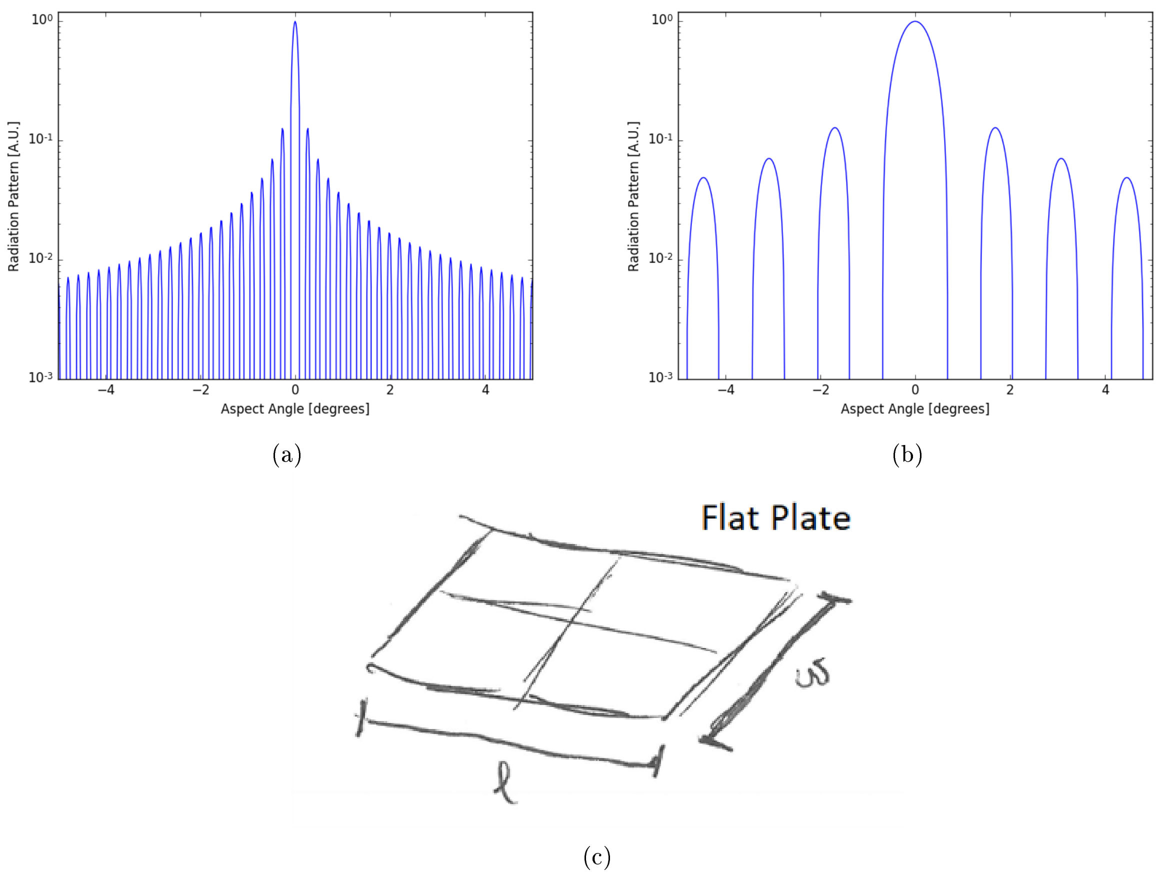

Figure 2-3: Radiation pattern for a rectangular, planar array in planes containing the center of the plate along the (a) length and (b) width directions of the array, respectively for a 40^2 array with $l = 16$m and $w = 2.5$m.

Figure 2-3 shows the radiation patterns in the two principal directions of a 40m^2 planar rectangular array with a length $l = 16$m and width $w = 2.5$m. There are two things to note about this pattern. First, because the radiation pattern is described by a sinusoid, the pattern forms a collection of peaks and nulls. For a transmitting array, no power will be transmitted in the direction of a null. Similarly, for a receiving array, no incoming power from the direction of a null will be registered in the receiver.[3] Between the nulls are peaks which are referred to as lobes. The

[3]For a given aperture, the transmitting and receiving gain patterns are identical, a principle referred to as reciprocity.

main lobe is the largest lobe, into which the most power is radiated. The main lobe defines the beam spot, which is the location where the radar is “looking”. The largest side lobes more than order of magnitude smaller than the main lobe, but can contribute unwanted clutter signal to a measurement.[4] For an AESA array, the size of the size lobes can be reduced by changing the weights of the elements with a taper, which alters the weights and phases of the individual radiating elements to shape the beam [62]. A taper can be useful for reducing the side-lobe magnitudes, but will always come at the expense of broadening and reducing the gain of the main lobe. Throughout this thesis, I assume that no taper is applied when determining the gain of an a radar system, as this is the upper limit of performance for that aperture.

The second item to note is that the lobes are narrower in the longer of the two principal directions. It can be useful to define a beam width. In radar applications, the most common beam width is the half-power (sometimes called the 3dB-) beam width which is the angle between the two points on the main lobe of the radiation pattern where the gain is half of the peak gain. This beam width in radians can be approximated as

$$B \approx \frac{\lambda}{l} \ , \tag{2.5}$$

where B is approximately the half-power beam width, λ is the wavelength of the RF energy, and l is the dimension of the aperture in the plane being considered.[5] The beam is narrower and more focused when the aperture is larger relative to the wavelength of the radiation. Increasing the size of the aperture, or reducing the wavelength of the radiated energy will narrow the beam.

The maximum gain of an aperture can be calculated from the beam widths. The transmitter directive gain G_D is a unitless scaling factor which describes the directionality of a radar array as a ratio of the maximum transmitted power $P_{\max}$ relative to isotropic emission of the total transmitted power P_t

$$G_D = \frac{P_{\max}/\Omega}{P_t/4\pi} \ . \tag{2.6}$$

For a radar array

$$G_D = \frac{4\pi A_e}{\lambda^2} \ , \tag{2.7}$$

where A_e is the effective[6] aperture area of the transmitter and λ is the wavelength of the transmitted radiation. The directive gain G_D is the maximum available gain of the array. The power gain, often just referred to as the gain, G, is the maximum transmitted intensity relative to isotropic emission of the power *accepted* into the antenna. For non-ideal systems, antennas will only transmit some fraction of the accepted power, some amount of the power will be dissipated as heat. This is a subtle, but important distinction, gain G accounts for antenna losses while directivity does not, the two only being equal with an ideal antenna. For simplicity in the following discussion, gain will refer to the idealized case, synonymous with directivity, and antenna losses will be accounted for in the loss term L. For apertures on the order of tens of- to a few hundred square meters, such as those typically considered for space-radar

[4] The side lobes are -13.2 dB ($\approx 1/20$th) of than the main lobe if no taper is applied.

[5] Occasionally, beamwidth will be defined as $B = 0.89\lambda/l$, the factor of 0.89 from measuring the distance between the half-power points from the radiation pattern. The form in equation (2.5) is a simplified approximation of this. The approximate form will be used for the same of simplicity in the following discussion.

[6] The effective aperture is $A_e = \rho_e A_r$ where ρ_e is the efficiency of the aperture, the fraction of incident radiation which it is able to absorb. This efficiency is less than or equal to unity, so the effective area is equal to, or smaller than, the physical aperture [63]. The efficiency is unity for an ideal antenna when no taper is applied, which is what I assume throughout this work.

systems, a typical gain in the X-band will be 5–6 orders of magnitude, or 50–60 dB.

Receiver. Competing with the signal power measured in the receiver, is the phenomenon of noise. Noise is power in a radar receiver is a result of the thermal motion of electrons in the electrical components of the system. For a radar system operating with some frequency bandwidth β, the noise power is

$$N = k_B T \beta \ , \tag{2.8}$$

where k_B is the Boltzmann constant, and T is the receiver temperature. In the radar literature, the receiver temperature is often expressed as the product of a reference temperature (T_0, 290 K) and a scaling factor called the noise figure F_n. For the following, the receiver temperature will be taken as the reference temperature and the noise figure will be subsumed into the loss factor L. Since noise is random, whether a real signal can be detected over the noise depends on the strength of that signal relative to the noise.

Since noise is stochastic, detecting a signal above noise will be as well. There is a relationship between the probability of detecting a signal when it is present P_D, the probability of detecting a signal when it is not present, the probability of false alarm P_{fa}, and the SNR of an observation. The radar operator can define a detectability D_0 by choosing a probability of detection P_d and probability of false alarm (P_{fa}). If a measured signal reaches the D_0, then it will achieve that desired pair of P_D and P_{fa}.

For radar systems, the ratio of the power of a measured signal above the noise background depends on

$$\mathrm{SNR} = \frac{P_r}{k_B T \beta} = \frac{P_t A^2 \sigma}{4\pi\lambda^2 R^4 k_B T_0 \beta L} \ . \tag{2.9}$$

For a pulsed radar system, where radiation is transmitted in pulses of duration τ, there is an optimal bandwidth which, when combined with a matched filter, maximizes the detectability of the signal [59]. This optimal bandwidth $\beta_{\mathrm{opt}} = 1/\tau$, allowing the SNR equation to be rewritten as a function of the pulse length[7]

$$\mathrm{SNR} = \frac{P_t \tau A^2 \sigma}{4\pi\lambda^2 R^4 k_B T_0 L} \ . \tag{2.10}$$

Modulation of the dwell time τ allows the radar operator to reach the detectability with a given measurement. The dwell time cannot be extended indefinitely however as the measured signal can lose coherency as the SBR system and the target move relative to one another. I return to the discussion of coherency later in this chapter when discussing the SMTI modality in Subsection 2.2.3. For now, the relevant information is that the ability of a radar system to detect a target depends on the length of the measurements that it makes.

The characteristics of the radar array are one of the pieces of the puzzle when determining the capability of a radar system. Importantly for the seeker, these design parameters are entirely under the control of the designer of the system. when building a radar system, the designer must choose an aperture, power supply, and measurement time that is tailored to the detection of a specific type of target. I return to this discussion later in this chapter, when discussing the design of a system for tracking TELs.

[7] An additional term, the filter mismatch factor needs to be accounted for in the loss factor L in order to make this substitution, see the discussion in section 2.7 of [63].

Radar Cross-Section

The second component of my formulation of the radar equation, in equation 2.2, is the RCS of the target. The RCS σ is a measure of the radar reflectivity of an object. It is defined as

$$\sigma = \lim_{R \to \infty} 4\pi R^2 \frac{|E_s|^2}{|E_0|^2} \ , \tag{2.11}$$

where R is the range between the radar and the target, E_0 is the electric field strength incident on the target, and E_s is the electric field strength scattered into the detecting array which depends on R [63]. The RCS is equivalent to the cross section of a sphere substituted for the scatterer that would scatter an equivalent power into the radar array [63]. This definition assumes that the scatterer scatters the incident field isotropically, though this is not often the case. Accounting for fluctuations in RCS with aspect angle is done utilizing statistical models (Swerling models discussed below) which have an average RCS value.

If the distance R is sufficiently large that the far-field approximation holds, then

$$E_s = \epsilon \frac{E_0}{4\pi R^2} \tag{2.12}$$

where ϵ is the fraction of the incident power scattered into the solid angle of the radar, which becomes equivalent to the cross-section. This eliminates the apparent dependence of σ on R, seen in Equation 2.11.

The scattering of radio waves by an object depends on the shape and material of the object as well as the frequency (or interchangeably wavelength) of the incident radiation. For the discussion of space-based radar systems, we are primarily concerned with frequencies in and around the X-Band, which has a frequency of roughly ten gigahertz ($\sim$ 3 cm). Additionally, we are primarily concerned with metallic objects with dimensions of meters to tens of meters. Metallic objects are typically strong conductors, meaning that electric fields cannot be sustained inside of them, making them reflect EM radiation such as radio- and microwaves. This allows us to constrain our discussion of RCS to the "optical" domain of scattering, that is when the wavelength of the radiation is small relative to the dimensions of the object. [8]

In the optical domain, the dominant scattering mechanism is specular reflection [64]. Specular reflections are those when the angle between the direction of propagation of the incident wave and the scattering surface normal (θ_0) is equal to the angle between the surface normal and the direction of propagation of the scattered wave (θ_s), as described by Snell's law

$$\sin\theta_0 = \sin\theta_s \ . \tag{2.13}$$

If the scattering object is finite in extent (as all real-world objects are) then the incident field can be described by the surface current set up on the object. The scattered field is equivalent to the field that would be radiated by this current. Analogous to a transmitting radar array, the scattered field will have a radiation pattern dependent on the physical shape of the surface. Even for relatively simple shapes, the radiation pattern can have a complicated structure. For example, the RCS of a flat metal plate in its principal plane (the plane containing the surface normal and the line-of-sight) is

[8] It should be noted that despite the name, the behavior of scattered radio waves in the optical domain can be significantly different than the scattering of visible light. There are different scattering processes that apply in the Rayleigh domain, when the wavelength becomes comparable to the dimensions of the object [63].

$$\sigma = \frac{4\pi A}{\lambda^2} \left| \cos\theta \cdot \frac{\sin(k\ell\sin\theta)}{k\ell\sin\theta} \right|^2 \quad , \tag{2.14}$$

where A is the area of the plate, λ is the wavelength of the incident radiation, θ is the aspect angle, k is the wavenumber of the radiation ($k = 2\pi/\lambda$), and ℓ is the length of the plate in the principal plane.

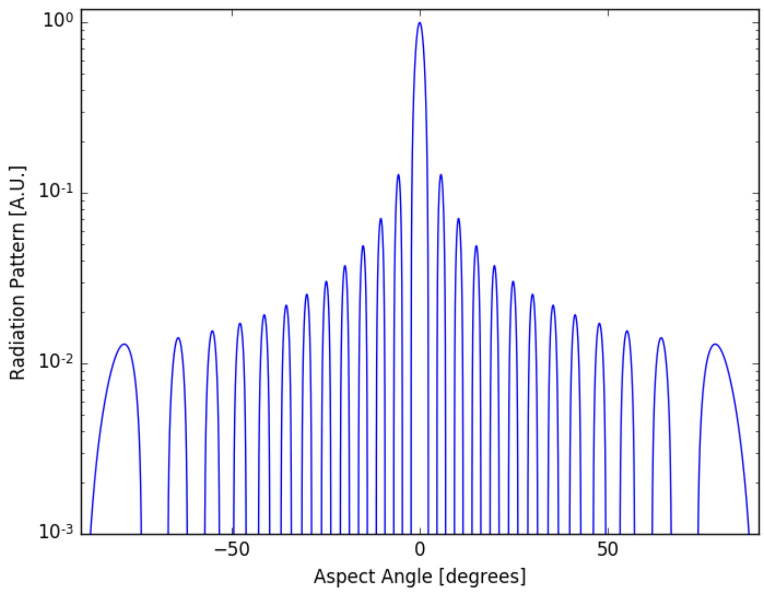

Figure 2-4: Radiation pattern for a line source with $l/\lambda = 25$.

The RCS pattern for a plate of metal, which can be assumed to act as a perfect electrical conductor (PEC) is shown in figure 2-4. It follows very closely the gain patterns shown for a plate in figure 2-3. That is because the RCS is, in a way, the complement of gain: radio waves that reflects off of a PEC, which comprise the RCS, induce an electrical current on the surface of the PEC which then re-radiates that power. The reflected power will be radiated with the radiation pattern of the object and the current induced on its surface.

In figure 2-4, as with the radiation patterns, there are elements of RCS behavior that bear noting. The sinc functions ($\sin(x)/x$) describe a periodic pattern of peaks and nulls in the RCS with changing aspect angle. For objects that are meters in scale, and for radio wavelengths of centimeters, $l/\lambda \gg 1$, so the beam widths of the lobes are very narrow. This means that for even sub-degree changes in viewing angle, the RCS of an object can fluctuate strongly between the peaks and nulls.

The large peak in the RCS at normal incidence is the “specular flash”. This is analogous to the main lobe of a transmitting radar array beam. The off-normal peaks in the radiation pattern are similarly analogous to the side-lobes of a transmitting array. Assuming a monostatic radar array (the transmitter and receiver are co-located), then the largest field scattered by a surface will be measured at the radar receiver when the radar is normal to the surface, and at off-normal aspect angles there will be less reflected energy from the side-lobe contributions. With multiple scattering surfaces, specular reflections from one surface can impinge upon another, undergoing “multiple bounces” off of surfaces in the object. This effect is particularly pronounced at interior corners and cavities, making them large contributors to radar returns from objects.

Complex objects can be thought of as collections of point sources. To first order, the scattered field from a complex object can be approximated as the coherent sum of the scattered fields from the individual scattering centers, creating interference patterns which further add to aspect-angle variations. Physical optics approximates the radiated field by estimating the surface current on the object that would be induced by the incident field. The radar cross section for a surface can be calculated with a surface integral

$$\sqrt{\sigma} = -i\frac{k}{\sqrt{\pi}}\int_S \hat{n}\cdot\hat{e}_r \times \hat{h}_i e^{ik\bar{r}\cdot(\hat{i}-\hat{s})}dS \ , \tag{2.15}$$

where k is the wavenumber, S is the surface, $\hat{n}$ is the surface normal unit vector, $\hat{e}_r$ is a unit vector aligned with the electric polarization of the receiver, $\hat{h}_i$ is a unit vector aligned with the incident magnetic field, $\bar{r}$ is a position vector from a local origin to the surface patch dS, $\hat{i}$ is a unit vector in the direction of propagation of the incident field, and $\hat{s}$ is a unit vector in the direction of propagation of the scattering field. Additional models can be used to calculate returns from edges, surface waves, and creeping waves [65].

The coherent sum of all surfaces and other features can be used to estimate the total RCS of a complex object. For a simple point scatterer or perfect sphere, the RCS is constant and non-fluctuating with aspect angle. For complex objects with multiple scatterers an accurate calculation can become incredibly complicated and onerous. For objects which are much larger than the wavelength of radio waves, the RCS profile will be a quickly varying function of aspect angle. It is therefore useful to turn to a statistical description of RCS.

A set of distributions that can be fit to describe the RCS intensity PDF of real-world targets are Swerling models [66]. Swerling I and II models describe the RCS behavior of complex objects composed of multiple point scatterers added together with random phase, none of which are dominant [67]. They are both described by the same distribution of RCS values

$$dP_\sigma = \frac{1}{\bar{\sigma}}\exp\left(-\frac{\sigma}{\bar{\sigma}}\right)d\sigma \ , \tag{2.16}$$

where dP_σ is the probability that that the RCS from that aspect angle is between σ and $\sigma + d\sigma$, and $\overline{\sigma}$ is the average RCS. To utilize these models, consider a coherent observation (technically a coherent processing interval [CPI], see 2.2.3) made up of a number of individual radar pulses. For a Swerling I object, for each observation, the measured RCS value is drawn from the distribution in equation 2.16 and is constant over all of the pulses in the observation. For a Swerling II object, each pulse is independently drawn from equation 2.16. Frequency hopping, quickly changing the transmitted frequency can be used to cause Swerling I models to behave as Swerling II models, as the slight changes in frequency cause changes in the RCS radiation pattern of the target[67]. Swerling III and IV models describe targets with a single dominant scatterer. Swerling 0 models (sometimes called Swerling V) are those with no fluctuation, where the average RCS is constant from observation to observation. Of course, simply invoking a statistical model does not allow the rapid fluctuations caused by complex, real-world RCS patterns to be ignored. The costs imposed by these fluctuations is accounted for in the calculation of the detectability D_0 of a target as a function of its average RCS (an increase in the required SNR to make a detection).

The Swerling models allow RCS to be treated with an average RCS value $\bar{\sigma}$ when designing a radar system. Unlike the radar array parameters, RCS outside of the control of a radar system designer. Instead, the designer must specify an average RCS value for which they want to be able to detect a given level of signal. In the case of the design of an SBR system for tracking TELs, the U.S. government set a target RCS value of $\bar{\sigma} = 10\text{m}^2$ [68]. This is a conservative

RCS value, simliar to that of a light vehicle.[9] Setting a conservative RCS allows the possibility of detecting support vehicles that can aid in TEL identification. Additionally, it preempts RCS reduction efforts by the hider, which are discussed further in Chapter 6. Through the next sections, I treat the RCS of a TEL as a Swerling II target with an average RCS of 10m^2.

Transmission Loss

Transmission losses are account for the losses of signal strength that occur as radio waves travel between the transmitter, target, and receiver. Losses occur through two modes, path absorption and geometric spreading. Geometric spreading losses occur because power is transmitted into a given solid angle, as described by the radiation pattern of the aperture. The solid angle is the two-dimensional angle that the beam covers. It is defined as

$$d\Omega = \sin\theta d\theta d\phi \ , \tag{2.17}$$

where $d\Omega$ is the differential solid angle, and θ and ϕ are the elevation and azimuthal angles in spherical coordinates. The solid angle eclipsed by an object of area A in the far field, which is at a distance R where $R >> A$, can be approximated by[71]

$$\Omega \approx \frac{A}{R^2} \ . \tag{2.18}$$

For a radar beam, power is transmitted into a solid angle. Equation 2.18 shows that for a fixed solid angle, the area subtended by a fixed solid angle is quadratically dependent on the distance to the area. This means that the power density on the surface of a sphere of radius R is

$$P(R) = \frac{P_t}{4\pi R^2} \ . \tag{2.19}$$

By normalizing for the transmission power, the one-way transmission loss from geometric spreading TL_{geo}

$$TL_{\text{geo}} = \frac{1}{4\pi R^2} \ . \tag{2.20}$$

The one-way transmission loss tells us the ratio of the areal power density at a distance R from the transmitter as a ratio to the power density at R_0, where R_0 is the unit distance from the transmitter. This allows us to account for geometric spreading with distance. In the case of SBR, this becomes important as R can be very large, on the order of a thousand kilometers. Further, since radar provides its own illumination the power is attenuated by TL both on the way to the target, and again as the echo returns from the target to the receiver, giving the two-way transmission loss of

$$TL_{\text{geo,two-way}} = \left(\frac{1}{4\pi R^2}\right)^2 \ . \tag{2.21}$$

The second mode of transmission loss comes from attenuation of the radio waves as they travel through intervening media. Radio waves can be absorbed when the couple with molecular vibro-rotational transitions. This causes a

[9]Estimates of average RCS based on vehicle class range widely in the literature. Per Melvin, automobiles have an RCS of roughly 20 dBsm, trucks roughly 23 dBsm, jet-planes range from 7–20 dBsm, bombers range between 10–30 dBsm, and ships range between 35–50 dBsm [69]. Other references place passenger vehicles between 0–15 dBsm [70].

decrease in the power per solid angle as a function of the path length through a given medium.

The attenuation of electromagnetic radiation can be described by the Beer-Lambert Law. The absorbance $\epsilon(\ell)$ along a path of length ℓ is

$$\epsilon(\ell) = 1 - \exp\left\{ - \sum_{i=1}^{N} \sigma_i(\lambda) \int_0^{\ell} n_i(r) dr \right\} , \tag{2.22}$$

where there are N absorbing species, each with a number density $n_i(r)$ that is a function of the position in the path r, and each of which has an absorption cross-section $\sigma_i(\lambda)$ that will vary with wavelength.

To determine the amount of transmission loss from absorption, we need to know what species exist along the path, and what their interaction cross-sections are. In the case of SBR observing the earths surface, the primary intervening matter will be atmospheric gasses and water vapor in the form of clouds or rain. Atmospheric gas profiles are available for a number of sources, such as the International Telecommunications Union [72]. Tables of attenuation coefficients for these gasses are similarly available. Figure 2-5 shows the zenith path attenuation for a reference atmosphere as a function of carrier wavelength. At frequencies below 10 GHz, attenuation is primarily caused by dry atmospheric gasses. Above 10 GHz, strong water vapor absorption begins to create significant attenuation. As a general rule of thumb, attenuation increases with frequency, though with some fine-structure.

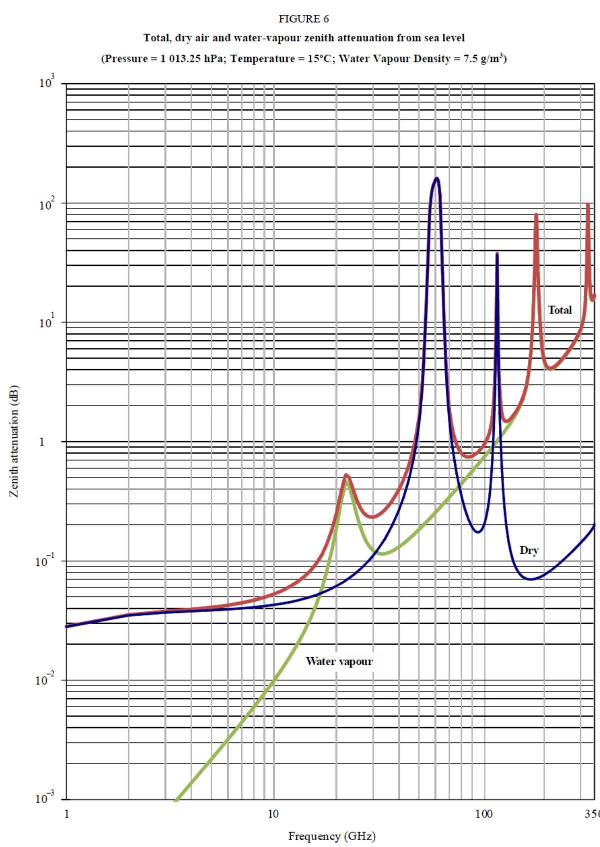

Figure 2-5: Zenith path (shortest path from ground to space) attenuation of electromagnetic radiation as a function of frequency for the dry atmosphere and water vapor with a density of 7.5 g/m^3. Based on a reference atmosphere recommended by ITU-R P.835 [72]. Adapted from International Telecommunication Union Recommendation P.676-11 [73].

Taking path absorption into account the overall two-way transmission loss for the monostatic case is

$$TL_{\text{two-way}} = \left(\frac{\epsilon(\ell = R)^2}{4\pi R^2}\right)^2 . \tag{2.23}$$

Even though the path lengths of the radiation through the atmosphere are large, the low density of the gas species, and the lack of strong absorbances make the absorption ϵ term small relative to the geometric spreading terms. In practice, there is some amount of attenuation that decreases the SNR of a measurement, but that can be overcome by taking longer measurements. This means that if a radar system has sufficient power and gain to contend with geometric spreading (which they do), then attenuation from atmospheric attenuation should be a phenomenon that a radar system can be designed to deal with.

Rainfall will increase the attenuation in the atmosphere, as a function of the severity of the rainfall [74]. Heavy rain ($\sim$ 100 mm/h) can cause one-way attenuation on the order of 10 dB in the X-band [75]. This would lead to two-way attenuation of 20 dB, which may challenge some radar system's ability to operate during periods of strongly inclimate weather (though this weather would also complicate TEL operation).

Atmospheric gasses are relatively transparent to visible wavelengths of light but the water vapor in clouds scatters those wavelengths strongly, making the clouds opaque. Since radiowaves do not interact as strongly with the water vapor in clouds, those clouds remain partially transparent to radar systems. This is centrally why radar systems have been raised in the literature a technology which may have the potential to undermine the survivability of TELs. This point does indeed indicate that radar systems can operate when other systems cannot. It is, however, insufficient to conclude whether SBR systems are capable of tracking TELs independently. As I laid out in the previous chapter, there are a number of tasks that a remote-sensing system must be able to accomplish in order to track a target. In the next section, I introduce the individual radar modalities that could be utilized by an SBR system and outline how they would need to be applied in conjunction in order to complete the tasks of tracking.

2.1.2 Radar Measurables

The radar equation provides us with a means of estimating the ability of a radar system to detect a given target. In this subsection, I will briefly review the information that can be gleaned from a detection. There are physical attributes about that target that can be inferred from a detection, the range and direction to the target, which can be expressed in radial coordinates as R, θ, ϕ, the radial velocity of the target v_r, and its average RCS σ.

Range

The range between the radar array and the target can be determined from time elapsed between the transmission and reception of the radio signal. Consider a pulsed radar system with a short but finite pulse length τ. If a signal is received t seconds after transmission of the pulse, it can be attributed to an object at a distance $d = ct/2$ where c is the speed of light. The finite pulse width imposes a limit on the range resolution δ_r, the minimum distance between two objects at which they can be successfully resolved. For example, if the pulsed radar array were to observe two objects at distances of d_1 and d_2 from the array. The difference in time of arrival between echoes from the two objects will be $\Delta t = 2\Delta d/c$. If this value is smaller than the pulse length τ, then echoes from the two objects will overlap

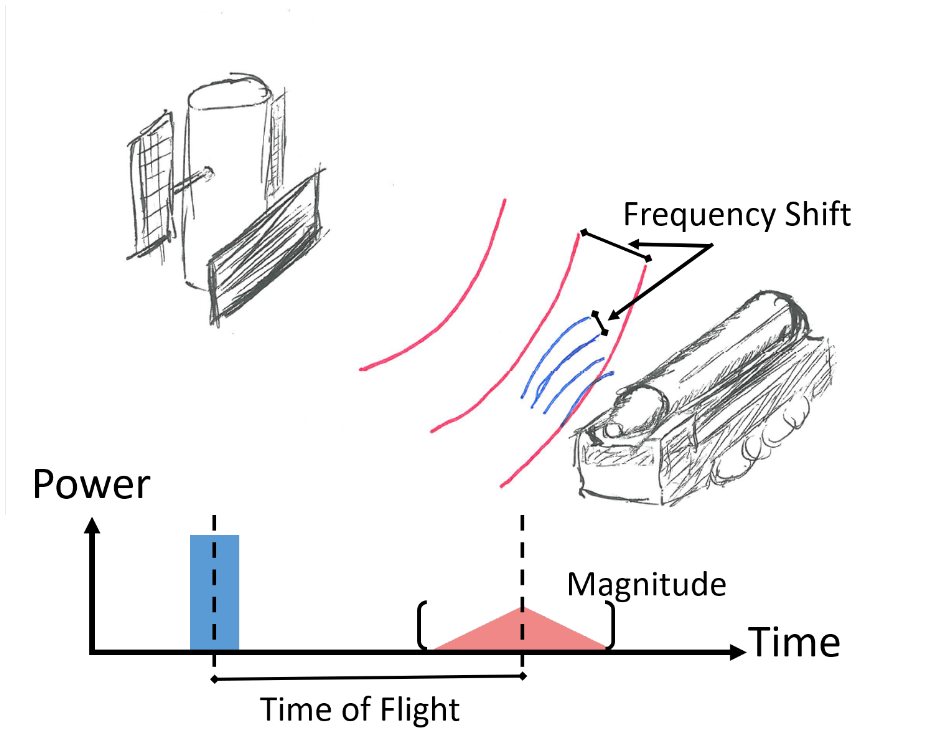

Figure 2-6: Schematic of what information can be determined from a radar observation. Range to the target can be determined by the round-trip time-of-flight of the radio waves. Radial velocity can be determined by the frequency shift between transmitted (red) and received (blue) signal. The time-integrated magnitude of the return can be used to determine the RCS of the target.

in the receiver such that the two objects cannot be resolved.[10] That is, the range resolution of the system will be $\delta_r = c\tau/2$.

A common strategy to improve range resolution is pulse compression. For example, the frequency of the transmitted radio waves (the carrier frequency) can be modulated over the course of the transmission pulse, often a linear increase called a "chirp". This reduces the impact of overlapping returns as radiation transmitted at different points in the pulse will be able to be discerned based on their different frequencies. This improves the range resolution δ_r of the pulsed radar array to

$$\delta_r = \frac{c}{2\beta}\ , \tag{2.24}$$

where β denotes the bandwidth of frequencies covered in the chirp. Chirp bandwidths are typically limited to 10% of the carrier frequency [76].

Direction

The direction in which the main lobe of the radar beam is pointing can be used to determine the direction to a detected target. For example, air traffic control radars continually rotate, sweeping over the full range of azimuthal

[10] The radiation from the beginning of the pulse, scattering off of the far target, arrives at the receiver $2d_2/c$ seconds after the start of the pulse. Radiation from the end of the pulse, scattering off of the near target, arrives at the receiver $\tau + 2d_1/c$ seconds after the beginning of the pulse. The two echoes will arrive simultaneously if their times of flight are equivalent, which occurs when $d_2 - d_1 = \tau c/2$.

angles. If the array is wider than it is tall, it will produce a vertically oriented fan-beam which will subtend some swath of elevations. When a detection of a plane is made, its position in spherical coordinates centered on the radar can be derived within the resolution of the system. When a detection is made, the target can be localized in azimuthal angle to the direction of the beam line-of-sight, within the width of the beam. Since radio waves spread spherically as they travel, the measured range does not specify a point but an arc of constant range subtending the beam. There are methods of improving the azimuthal resolution utilizing multiple sub-apertures of an AESA [77]. For the purposes of tracking TELs, azimuthal resolution is of little relevance as the constant motion of SBR platforms, integrated with Kalman filtering allows for track association with azimuthal resolution well in excess of the sensor resolution.[11]

Radial Velocity

The radial velocity v_r of a moving target can be determined from the Doppler shift of the radar echoes. The radial velocity is the projection of the targets total velocity onto the line-of-sight between the target and the radar. The Doppler shift (f_D) caused by the radial motion of an object relative to the radar is

$$f_D = -f\frac{2v_r}{c} \ , \tag{2.25}$$

where f is the transmitted carrier frequency, v_r is the radial velocity of the object, and c is the speed of light. The radial velocity is the component of the targets velocity which is in the direction of the radar array, which is why it is sometimes referred to as the range-rate, the rate at which the range to the target is changing. In order to estimate the total velocity of a target, the heading of the object must be estimated. Kalman filtering provides a means of integrating contextual information such as the direction of roads on which a TEL is traveling that can be used to estimate the total velocity from the radial velocity.

Average RCS

The average RCS of an object can be estimated by the strength of the signal return. Since the range to the target can be determined from the time of flight of the pulse, and the SNR is measured, the only unknown left in the radar equation is the target cross-section σ. Measuring the strength of radar returns allows σ to be estimated by fitting the measurements to a Swerling model. Frequency hopping, making small adjustments in the carrier frequency between coherent processing intervals, makes each measurement of the RCS independent of the others.[12] This means that any complex target will behave similarly to a Swerling II model. By finding the Swerling II model that best fits the set of RCS measurements, σ can be estimated. Additional information about a target can be gleaned by employing more specialized modalities, which I introduce in the next section.

[11]For example, for a 16m by 2.5m aperture at 1000 km altitude operating in the X-band, the azimuthal resolution of SMTI is on the order of a few kilometers (the range resolution can be sub-meter) which is on the order of a nuclear weapon radius.

[12]Frequency hopping is done to prevent a phenomenon called fading which can cause a target to be lost if the aspect angle enters a low-RCS region of the target.

2.2 The Tracking Capabilities of Radar Modalities

2.2.1 Tasks of Tracking

Tracking a mobile target such as a TEL requires performing a number of discrete tasks: detection, identification, and track association. In this section, I introduce what I identify as the tasks which a remote sensing system must be capable of performing in order to track TELs: detection, identification, and track association. In this section, I introduce each task and then lay out how the radar modalities can be applied to performing them.

Detection. Detection is the act of correctly determining that a TEL is present in an observation. This detection will contain information about the presence of the TEL, and possibly information on its position, depending on the modality used. In any remote-sensing observation of a TEL, there is a signal that is produced by the TEL and some background from which the TEL signal must be distinguished. Depending on the remote-sensing modality, the background will come from noise, clutter, or both. Noise is random fluctuations in a measured signal that is caused by random processes such as thermal motion of electrons in electronics. Clutter is real signal that is produced by objects other than the TEL that exist in the field of observation. Noise can be overcome by collecting signal for a longer time as the random fluctuations tend to average out over time. Clutter, on the other hand, is real signal so will grow as more signal is collected. It is typically overcome by either separating the clutter from the signal by some property of the signal, or by constructing an accurate model of the expected clutter, or both.

Detection of signal from noise is a stochastic process. If a TEL is present and observable, the probability that its signal will be detected is P_D. The probability that a detection is registered without the TEL actually being present is the probability of false alarm (P_{fa}).

Aside from noise and clutter, there are also non-stochastic processes that can interfere with detection. Some of these factors are shaped by the modalities being used. For instance, some remote-sensing modalities cannot detect moving targets while others cannot detect stationary targets. A TEL may be undetectable in some circumstances based on the combination of its state and the modality being used. The environment in which the TEL is deployed can also affect a TELs detectability. Many modalities rely on line-of-sight between the sensor and the TEL to operate; features of the environment, such as terrain, vegetation, or buildings can cause sight-block.

Collectively, these factors determine the probability that a remote-sensing system will successfully detect a TEL when it makes an observation. To avoid ambiguity in terms, I refer to the purely stochastic probability of detecting a given signal above noise as the probability of detection (P_D), and the overall probability that a TEL is successfully detected when an observation is made as the probability of observation (P_O). The probability of observation accounts for all of the processes which impact the task of detection, including non-stochastic processes such as sight-block.

Identification. Identification is the process of discriminating between detected signals that are produced by the TEL being tracked and signals produced by other objects. If something other than the TEL being tracked produces a similar signal to the TEL, if the seeker does not have the ability to discriminate between those signals, they may misinterpret that signal. This could result in the seeker believing that the TEL has moved from its actual position. Alternatively, if the seeker recognized that they were unable to differentiate between the two signals, they would need to track both, which would ultimately increase the burden on remote-sensing system resources and increase the number of targets which would need to be attacked in a counterforce attack.

Identification requires the measurement of a trait or set of traits that are specific to a TEL. These can be physical properties or behavioral patterns. Behavioral patterns can include patterns of movement, such as never exceeding some top speed, traveling exclusively in a certain region, or traveling in a convoy with other vehicles. While these can be useful to the seeker, they can be degraded in utility by an adversarial hider who could vary the manner in which they operate their TELs during periods of heightened tensions. For example, the hider could disperse TELs to new areas, or have a TEL travel without an escorting convoy for short periods of time. As such, while these traits are useful, it is within the power of the hider to render some of them not only ineffective but counterproductive, so they cannot be relied upon exclusively.

Physical properties include traits such as size, shape, and material of the TEL, or emissions given off by the TEL during operation. Some of these traits can be measured by remote-sensing systems. Optical satellites, for example, can image the size and shape of the TEL, allowing for discrimination from smaller or larger vehicles. Alternatively, communications signals given off by the TEL can be detected with electronic intelligence (ELINT) remote sensors. If a trait, or set of traits, used by the seeker to positively identify TELs can be mimicked by a non-TEL, then it creates the possibility of a decoy that could be employed by the hider in order to frustrate the seeker's ability to identify objects. Decoys will be discussed in depth in chapter XXX. The competition between the hider and seeker with respect to the seeker's ability to identify targets is important as this necessary task cannot be carried out once the system becomes overloaded.

Track Association. Track association is the process of correctly associating discrete detections (and identifications) with a given TEL. Detection events provide information that a signal is present at a location (or in an area) at the time of the measurement. Track association is linking these detection events to a TEL, and using those detections to form a belief about both its current position and its position history. Track association is separate from the remote-sensing modalities used. I present a method for track association, Kalman filtering, in more detail in the following subsection.

2.2.2 Radar Tracking of TELs

Radar systems operate by measuring the reflections of transmitted radio waves. There are a number of ways in which this basic premise can be applied, each defining a different radar modality that can operate under different conditions and measure different information. One of the means to distinguish different modalities is based on how they deal with clutter. As I noted in Section 2.1.1, clutter is one of the components of a radar return signal. It is received power reflected off of non-target objects in a radar observation. Clutter is particularly relevant for radar systems which image the Earth's surface, such as SBR systems, as the components of the scene itself: the ground, buildings, vegetation, all produce clutter. It is distinct from noise in that it is a real signal produced by the scene being observed. Unlike noise, it cannot be suppressed simply by taking a longer measurement. Clutter signal scales linearly with the illuminating fluence. Clutter must either be suppressed or utilized, and each modality approaches this problem in a different way.

There are three radar modalities which will be considered, Doppler surface moving target indication (SMTI), synthetic aperture radar (SAR), and inverse SAR (ISAR). Their capabilities are outlined in table 2.1. SMTI is a detection modality capable of monitoring large areas for moving objects, but is incapable of detecting slow-moving or stopped objects, and has limited target identification capability. SMTI exploits the Doppler shift of reflections from moving objects to separate targets from clutter.

	Moving Target Detection	Stationary Target Identification	Moving Target Identification	Wide Area Coverage
SMTI	Yes		Partial	Yes
SAR	Yes	Yes		
ISAR	Yes		Yes	

Table 2.1: Capabilities of SMTI, SAR, and ISAR radar modalities. SMTI has a partial capability to identify moving targets which is not reliable, discussed in Subsection 2.2.3.

Synthetic aperture radar is an imaging modality capable of imaging motionless objects at high resolutions, providing an identification capability, and with a demonstrated potential for moving-target detectio [78]. Rather than try to suppress clutter, as with SMTI, SAR images the clutter itself. SAR observations require more time to collect than SMTI, and can only be collected at certain angles, limiting the area it can monitor.

ISAR is something of a hybrid between SMTI and SAR. ISAR is an imaging modality that can be used to identify TELs, but only when they are in motion and turning. As a moving target turns, it presents a diversity of view angles which can be used to reconstruct an image of the target. However, the ground clutter will also have some relative rotational motion from which the target signal needs to be separated. This clutter must be suppressed. This can either be done with STAP, in the same was as with SMTI, or by forming a SAR image.

A powerful capability of radar systems is that all of these modalities can be performed with the same hardware. Selection of modality is done by selecting an appropriate illuminating pulse shape. This allows a radar system to interleave use of the different modalities as needed.

I begin this section by introducing each modality in turn. In doing so, I present both how the modality works, the limitations on when it can be used, and what unique information can be used to collect. Following the introduction of the three modalities, I present a scheme for how the three modalities can be used in combination in order to complete the tasks of tracking outlined in the previous chapter.

2.2.3 SMTI

Surface moving target indication (SMTI) is the detection of moving objects on the Earth's surface using a remote-sensing system[69].[13] SMTI is a procedure, not a particular modality. It is simply the detection of moving objects on the Earth's surface and can be approached using different measurements and algorithms. Change detection, for instance, can be used to compare sequential images formed through an imaging modality, and look for series of changes in the images that might be consistent with a moving object. SMTI can also be done using raw radar measurements by exploiting the Doppler shift that target motion creates in radar echoes. For simplicity in the following discussion, the term SMTI will used to refer to Doppler-based radar methods for moving target detection.

Doppler-SMTI detects moving objects by converting an observation to Doppler-space to separate signals based on their radial velocities. Radial velocities are measured by illuminating a scene for a period of time that is termed a coherent processing interval (CPI). A CPI is the amount of time over which a signal can be summed coherently with linear integration gain.[14] The signal collected in a CPI has a signal-to-interference-plus noise ratio (SINR) of

[13] Ground moving target indication (GMTI), is occasionally used interchangeably with SMTI, but refers to the more narrow case of mover detection on dry land. SMTI also encompasses detection of moving objects at sea, which is an active area of research for the detection and tracking of sea traffic.

[14] Integration gain is the improvement in SNR or SINR from combining multiple pulses.

$$\text{SINR}_{\text{CPI}} = \tau_{\text{CPI}} \cdot \frac{\overline{P_t} A^2 \sigma}{(4\pi\lambda^2 R^4 k_B T_0 \beta L) + C} \ , \tag{2.26}$$

where SINR_{CPI} is the SINR collected in a single CPI, τ_{CPI} is the duration of the CPI, $\overline{P_t}$ is the time-averaged transmitted power of the radar, and C is the clutter power that overlaps with the signal of the mover. For non-cooperative cases such as tracking TELs, the maximum length of a CPI is limited by target acceleration, which causes a SNR loss[69]. If a single CPI is not sufficient to meet the detectability of the target, non-coherent integration may be needed.

The integration gain from non-coherent integration is lower than that from coherent integration [79]. In practical terms, target acceleration typically limits dwells to fewer than ten CPI which generally do not exceed 50 ms in duration [69].[15] Since making an observation of a mover from an SBR satellite also illuminates the ground, there will be some amount of clutter power C in the return.

For a stationary radar observing a scene of stationary background with movers, then the stationary background clutter can be suppressed by discarding any returns with no Doppler shift relative to the transmitted frequency. For systems based on moving platforms, such as satellites or planes, the process is complicated by the relative motion of the stationary surface clutter and the moving platform.

Consider a radar system based on a platform moving rectilinearly with constant velocity and a stationary surface scene. For stationary objects on the surface, their radial velocity relative to the moving platform depends on the azimuth of the line-of-sight connecting the two, in the frame-of-reference of the platform. The radar beam footprint has a finite width so a single observation will produce returns from a span of different azimuths. Since the radial velocity, and hence Doppler shift, of stationary objects varies with azimuth, the surface itself (and any non-moving objects on it) will span a range of Doppler values in Doppler space, creating background clutter.

If the radial velocity of the target is similar to that of the clutter, its signal needs to be detected above the clutter, effectively increasing the interference portion of the SINR.[16] The edges of the Doppler spread of the stationary clutter can be approximated as

$$D_{\text{clutter}} \approx \frac{2 v_{\text{platform}}}{l} \ , \tag{2.27}$$

where D_{clutter} is the maximum clutter Doppler shift [80]. The corresponding radial velocity associated with this Doppler shift is

$$v_{\text{clutter}} = \frac{\lambda D_{\text{clutter}}}{2} \ . \tag{2.28}$$

This forms the outside bounds of the “clutter notch”, the region in radial-velocity – azimuthal angle space where the clutter has non-zero radial velocity. Objects moving below this radial velocity will be difficult to distinguish from the background clutter. The MDV is the velocity at which the background clutter creates a sufficiently large SINR loss that the operator deems the object “undetectable”, this is typically -5 to -10 dB [69]. The clutter notch forms the upper bound of the MDV, data processing techniques such as space-time adaptive processing (STAP) can be used

[15]Longer coherent intervals are used for ISAR imaging, discussed below, which rely on extensive motion-correction techniques and experience strong blurring of features when acceleration is present [64].

[16]For a satellite moving at 8 km·s^{-1} with an aperture 16 m in extent, operating at $\lambda = 0.03$ m, the stationary clutter will be spread across 15 m·s^{-1}.

to reduce the signal loss caused by the clutter background and lower the MDV. While there are a number of clutter-suppression algorithms, STAP has the best performance [69]. The performance can be difficult to estimate, but a convenient rule-of-thumb is that STAP achieves MDVs a factor of 3–5 smaller than the clutter velocity estimated by equation 2.28 [69].

An important distinction is that this modality does not form an image, it provides range and range-rate, and azimuth at the resolution of the beam footprint. Since the azimuth is measurable only to the beam-width, the position of a detected mover will be known to a constant-range arc across the beam, centered at the detected azimuth. In order to geo-locate the detected movers, this information needs to be combined with other information such as co-registration with a map or image using Kalman filtering.

The coverage rate of a radar modality will be taken as the area for which the radar system can utilize that modality per unit time. This is a function of the time required for the radar system to complete an observation and the size of the beam footprint while it does so. The coverage rate ($\dot{C}$) of a radar system modality is

$$\dot{C} = F/T_{\text{dwell}} \ , \tag{2.29}$$

where F is the beam footprint area, and T_{dwell} is the dwell time required to make an observation.

The footprint area is the segment of the Earth that subtends the beam. Calculating the true footprint area for a spherical Earth requires numerical integration, which I present in detail in the beam appendix. For computational modeling, this is the method that I utilize. However, it can be ungainly, so an approximation is useful. The footprint area can be approximated with the cross-section of the beam at range R,

$$F \approx B_l B_w R^2 \ , \tag{2.30}$$

where B_l and B_w are the beam widths along the two principal planes of a rectangular array, which slightly underestimates the area but is accurate to within 17% of the true value over the range of incidence angles at which SBR satellites can operate.

The SMTI T_{dwell} depends the average RCS of the target and the probabilities of detection and false alarm, P_d and P_{fa} respectively, that are chosen by the radar operator. The SMTI coverage rate for three targets with different RCS values, assuming $P_D = 0.95$ and $P_{fa} = 10^{-6}$ is shown in figure 2-7. A set of design variables, shown in table 2.2 are utilized for this plot. These are representative parameters that serve as a useful benchmark when comparing radar modalities. In Section 5.2, I show how these parameters can be tailored when purpose building a radar system. For the remainder of this section, I utilize these parameters so that different aspects of radar performance can be compared. One of the advantages of SMTI compared to SAR and ISAR, as I will show in the following subsections, is that the dwell time of an SMTI measurement is short compared to the other modalities. This makes SMTI an attractive modality for wide-area monitoring of moving objects.

Parameter	Value	Parameter	Value
Wavelength	0.03 m	Carrier Frequency	10 GHz
Aperture	40 m^2	Transmit Power	10,000 kW
Altitude	1000 km	Duty Cycle	0.15

Table 2.2: Model parameters utilized when modeling a representative radar systems for illustration.

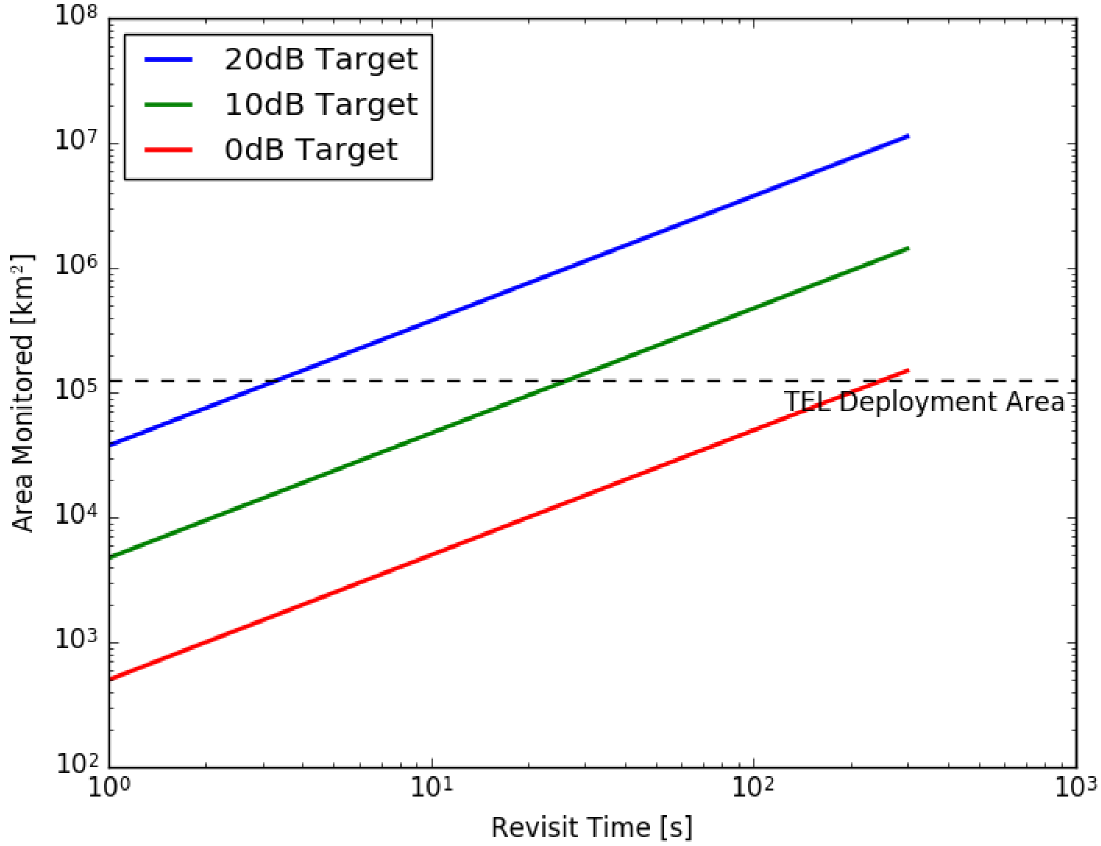

Figure 2-7: Area that can be monitored with SMTI as a function of the interval between observations for three different target RCS values for an idealized constellation with a constant coverage multiplicity $m = 1$.

A slightly modified SMTI waveform can be used to achieve very fine range resolution, referred to as high-range-resolution (HRR) SMTI. This can be used to produce a profile of RCS as a function of range, essentially producing a 1-D image of the target. This method has been investigated as a means of target identification, but the range profiles are highly aspect-angle dependent [81]. Identification with HRR can be done with template-matching, matching the profile to a model of the intended target that must be either gathered from a large number of observations or from a highly accurate RCS model [81]. ISAR, discussed below, improves upon HRR by producing an image with some cross-range resolution in order to make identification more robust toward aspect-angle diversity.

In terms of the tasks of tracking, SMTI provides a means for detecting TELs that are in motion. The ability to estimate average RCS from an SMTI measurement, and the possibility of detecting smaller support vehicles traveling in a convoy with the TEL, provide some capacity for identification, but one that has shortcomings. Any vehicle with similar size and shape to a TEL, such as a large truck, could have a similar average RCS that could not be distinguished with SMTI alone. Behavioral traits such as moving in a convoy and obeying some maximum speed when moving are useful contextual clues but can be broken by the hider, if they chose to, making them unreliable. As such, I consider SMTI as being primarily a detection modality, limited to moving targets, that has the advantage of being able to monitor large areas, as noted in table 2.1.2

2.2.4 SAR

Synthetic aperture radar (SAR) is a modality that can create high-resolution images of motionless objects. SAR is so named because it utilizes data processing to mathematically construct, or synthesize, the signal that a large array would detect from a series of measurements taken over time with a smaller, moving aperture. SAR is a complementary imaging modality to SMTI. SMTI is able to detect moving objects (and only moving objects) while SAR images stationary scenes, any moving objects will be blurred in the resulting image. Rather than suppressing clutter, as SMTI does by exploiting Doppler shifts, SAR utilizes clutter by imaging it.

The signal imaged by SAR is the clutter produced by the terrain and other motionless objects in an observation. In areas with homogeneous terrain, the effective RCS of terrain can be modeled with a "constant gamma" model

$$\sigma_{\text{terrain}} = \gamma \sin(\chi)\, \delta_r \delta_{az} \; = \gamma \cos(\theta)\, \delta_r \delta_{az} \;, \tag{2.31}$$

where χ is the grazing angle, θ is the look angle, γ is a scalar which characterizes the average returns from a type of terrain at a given frequency, δ_r is the range resolution, and δ_{az} is the azimuthal resolution [59]. Values of γ in the X-band for vegetation covered terrain range between -10 – -15 dBsm m^{-2}, while mountainous and urban terrain can reach values of -5 dBsm m^{-2}[82]. Averaged across all terrain types, γ can be approximated as -11 dBsm m^{-2} for the X-band [83]. This model is only valid for intermediate grazing and look angles, breaking down near normal and perpendicular incidence [67]. With this assumed terrain cross section distribution, and by setting the $d_r = d_{az}$ in order to make an image human-readable, the clutter to noise ratio (CNR) for a ground-looking observation is

$$\text{CNR} = \frac{P_t A^2 \gamma \sin(\theta)\, \delta^2}{4\pi \lambda^2 R^4 k_B T_0 L} \;. \tag{2.32}$$

The strength of the signal measured by SAR is therefore dependent on the resolution of the image being formed, with finer resolutions producing less signal per unit dwell time.

In general, the azimuthal resolution (ability to distinguish two closely spaced scatterers of equal intensity) of a radar antenna is proportional to the azimuthal beam width (which is inversely proportional to the array length). The half-power beamwidth B in radians is noted in equation 2.5. While the angular azimuthal resolution of a beam is set by the physical dimensions of an array aperture and the wavelength being used, the spatial azimuthal resolution also depends on the distance between the target and the radar. A rule of thumb for radar systems is that the minimum resolvable distance in the azimuthal direction is the same as the swath-width W, the cross-range extent of the beam,

$$W = \frac{\lambda}{l_{//}} R \;, \tag{2.33}$$

where R is the distance between the radar and the point at which the swath is being measured[84]. For space-based radar systems in particular, the operating distance R is on the order of thousands of kilometers, creating beam-swath widths on the ground on the order of kilometers[17]. For non-imaging modalities such as Doppler-based SMTI, the spatial azimuthal resolution is sufficient when combined with other sources of information. Imaging, however, requires much finer resolution to be useful.

Achieving useful resolution, on the order of a meter or less with a real space-based aperture would require an

[17]Assuming aperture dimensions on the order of tens of meters and wavelength of centimeters, $B \sim 10^{-3}$ m, At ranges of thousands of kilometers, $W = BR \sim 10^3$ m.

aperture on the order of hundred of kilometers in length. SAR improves azimuthal resolution by "synthesizing" an extremely long, virtual array using consecutive measurements taken from a moving platform. Combining measurements in this way constructs a signal equivalent to one captured by an array of the length spanned by the array over the course of the collection period. This allows SAR to construct images with sub-meter resolution, provided the scene has not appreciably changed because of movement or topology.

Since the array is being synthesized by the movement of the platform on which the radar is based, the long dimension of the array is parallel to the platform velocity vector, meaning a side-looking array is synthesized. This defines the range dimension of images created with the system as normal to the velocity vector of the platform, and the cross-range (or along-track) dimension as parallel with the velocity vector. Resolution along the range dimension is, as with ranging, determined by the bandwidth of the compressed pulse, per equation (2.24). Cross-range resolution is determined by the length of the synthetic aperture. For a synthetic aperture of length l_{sa}, the effective angular beamwidth B_{eff} is

$$B_{\mathrm{eff}} = \frac{\lambda}{2l_{\mathrm{sa}}} \ , \tag{2.34}$$

in radians, a factor of two finer than for a physical aperture of the same size. This is the result of the differences in the round-trip phase shifts[63]. Assuming that the synthesized array is much larger than the physical aperture, its length can be approximated as

$$l_{\mathrm{sa}} = vT_{\mathrm{dwell}} \ , \tag{2.35}$$

where v is the platform velocity and T_{dwell} is the dwell time, the amount of time spent imaging the scene. For a radar with its boresight pointed normal to the platform motion, a given point on the ground will be illuminated by a footprint with width $w = BR$, giving a maximum possible synthetic aperture of

$$\max_{\mathrm{fixed}} \{l_{\mathrm{sa}}\} = \frac{\lambda R}{l} \ . \tag{2.36}$$

The maximum synthetic aperture length is inversely proportional to the length of the physical aperture that is parallel to the direction of motion of the satellite. For a fixed-boresight system, a given point on the ground will be "in-view" for a longer period of time, allowing for a longer synthesized array, because of the wider physical beamwidth of the smaller aperture. The finest cross-range spatial resolution δ_{xr} of a fixed-boresight system is then

$$\min_{\mathrm{fixed}} \{\delta_{xr}\} = B_{\mathrm{eff}} R = \frac{l}{2} \ , \tag{2.37}$$

half of the length of the physical aperture. This limit is specific to the side-looking configuration, often called strip-map, or push-broom, imaging where the array boresight is fixed normal to the platform motion. Somewhat counter intuitively, increasing the aperture length in the direction of travel worsens the achievable cross-range spatial resolution. This is because the beam becomes narrower in azimuth, so a given point on the ground is illuminated for a shorter period of time as the satellite flies by.

Allowing for boresight motion allows this resolution limit to be improved upon. Since the limit is set by the amount of time that a point on the ground is illuminated by the array, keeping the boresight pointed at an area of interest over the course of a pass increases the length of the aperture that can be synthesized. This is often referred to as spotlight imaging. The cross-range resolution of for spotlight imaging is

$$\delta_{xr} = \frac{\lambda R}{2vT_{\text{dwell}} \cos\alpha} \ , \tag{2.38}$$

where α is the squint angle, the angle between boresight of the array and the direction normal to the velocity of the platform. The squint angle is a time-dependent variable, but can be approximated as a constant by taking it to be the squint angle at the temporal center of the dwell. The $\cos\alpha$ term which accounts for the projection of the beam onto the along-track direction varies linearly with time, making this a reasonable approximation except in the case of extremely long dwells.

Rearranging equation 2.38, a relationship for estimating the dwell time as a function of the desired imaging resolution can be produced

$$T_{\text{dwell}} = \frac{\lambda R}{2v\delta \cos\alpha} \ , \tag{2.39}$$

where δ is the image resolution, assuming the range and azimuthal resolutions are set equal. Utilizing this relationship, the coverage rate of SAR can be estimated as a function of image resolution. Figure 2-8 shows the amount of area is that can be monitored at a given resolution δ as a function of the time between re-imaging. This plot utilizes the same set of design variables as shown for the SMTI case. A notable point of departure between the two modalities is that, despite using the same hardware, the required dwell times are significantly longer for SAR *versus* SMTI.

SAR utilizes data processing to construct the signal which could be acquired by an array which would not be physically practical. This allows the modality to achieve very high cross-range resolution, but also imposes limitations. SAR relies on objects in the scene being in the same position over the course of the dwell, any objects which are moving will become defocused in the reconstructed image. If an object has moved between the start and end of a SAR scan, the signal collected will not be equivalent to observing that object in the scene from all angles covered at the same time.

The raw images collected by SAR are therefore limited to the detection identification of stationary targets. Since it is an imaging modality, it requires image interpretation by an analyst, or potentially an algorithm. One algorithm that has been explored recently adds a moving target detection capability to SAR images. Detection of moving targets required somehow separating the signal of the moving target from the clutter signal. The signal from movers in a SAR image becomes spread out over a number of pixels. By constructing an estimate of the stationary clutter signal using a subset of the raw SAR data, the background clutter can be estimated and removed in order to isolate the mover signal [85].

SAR imaging therefore has the capability to detect both stationary and moving targets. Further, since it is an imaging modality, with the capability to achieve sub-meter image resolution, it also has the capacity to identify stationary targets, provided that their physical shape is distinguishing. However, the longer dwell times of SAR mean that it is less suited than SMTI to searching wide areas.

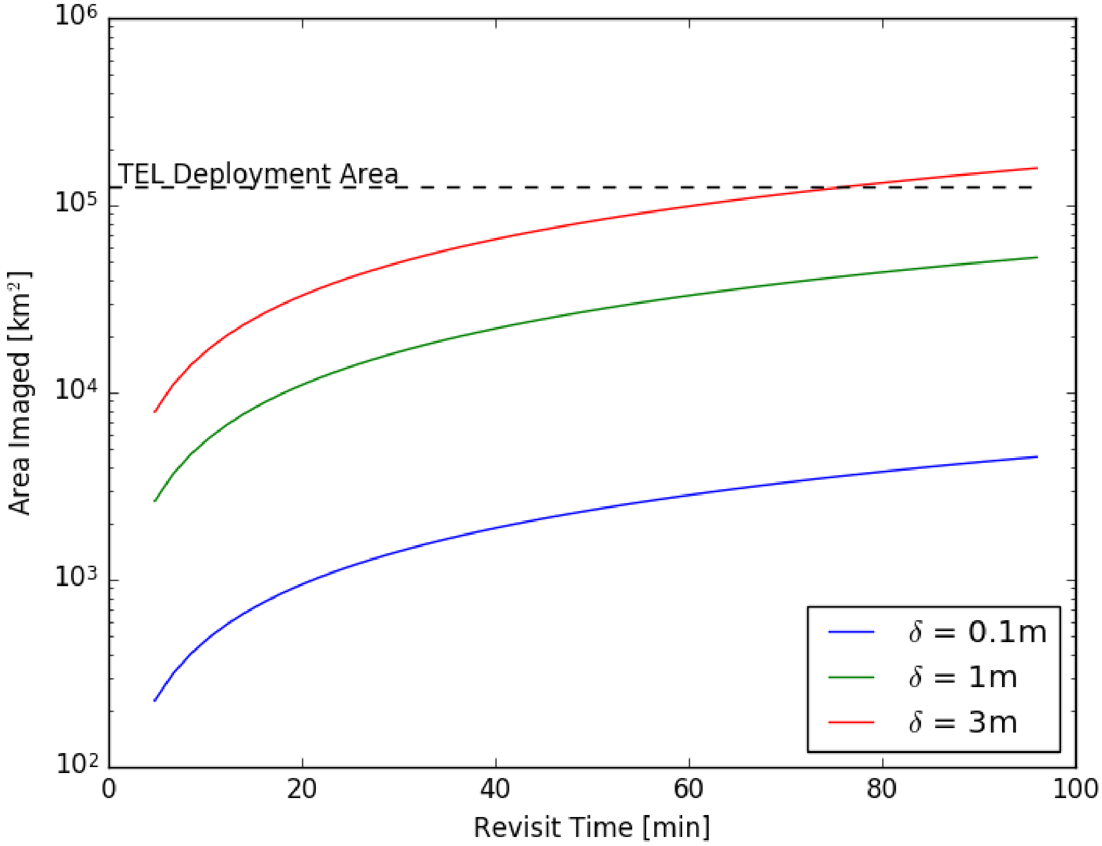

Figure 2-8: Area imaged by SAR at three different resolutions, assuming a single satellite is in view of the area of interest at all times, as a function of the revisit time between re-imaging a location.

2.2.5 ISAR

SAR utilizes motion of the radar array to achieve cross-range resolution finer than the diffraction limit. Inverse SAR (ISAR) similarly exploits relative motion between the radar and a target to achieve fine cross-range resolution, however it is dependent on target motion. ISAR takes advantage of target rotation caused by motion of the object and/or the platform to image the object over a range of aspect angles. ISAR can be used to reconstruct the spatial distribution of the individual scattering centers in a complex object, and with sufficiently fine resolution, this modality could be used to "fingerprint" targets based off of their RCS characteristics[64].

ISAR is not a scene imaging modality like SAR, it is instead used to for target identification or recognition. It can be performed on the same data as SAR, coherent time-series data collected with relative motion between the target and the radar. Conversely, ISAR requires target rotation in order to view it over a large range of aspect angles and the effects of translational motion need to be estimated and removed to develop an accurate image of the target.

The cross-range resolution of an ISAR system depends on the relative rotation rate $\dot{\Omega}$ of the target. When a rigid object rotates about some point, the angular speed (v_{ang}) of an arbitrary point on the surface of that object depends on its displacement from the center of rotation r_{rot}

$$v_{\text{ang}} = \dot{\Omega} r_{\text{rot}} \ . \tag{2.40}$$

If the translational motion of the object can be estimated and compensated for, then the radial velocity caused by the rotation of the object will cause the returns from each scattering center to be Doppler shifted proportionally to their cross-range displacement (r_{xr}) from the point of rotation

$$f_D = \frac{2f\dot{\Omega}r_{xr}}{c} \ . \tag{2.41}$$

The cross-range resolution is derived from the Doppler resolution. If two sinusoids are coherently observed for a period of time T_{dwell}, they can be resolved only if the difference in the frequencies is greater than $1/T_{\text{dwell}}$[64]. This yields a Doppler resolution (δ_{f_D}) of

$$\delta_{f_D} = \frac{1}{T_{\text{dwell}}} \ . \tag{2.42}$$

In order to relate the Doppler resolution to the cross-range resolution, consider two points on a rotating object that have cross-range separation Δr_{xr}. The separation in Doppler space Δf_D of those to objects is

$$\Delta f_D = \frac{2f\dot{\Omega}\Delta r_{xr}}{c} \ . \tag{2.43}$$

Two signals which are separable in Doppler space have a difference in frequency δ_{f_D}, which, when combined with equation (2.42) can be used to express the minimum cross-range separation that produce signals that can be resolved in Doppler space

$$\delta_{xr} = \frac{c}{2f\dot{\Omega}T_{\text{dwell}}} \ , \tag{2.44}$$

which is the cross-range resolution of ISAR.

ISAR provides the capability to identify moving targets, which is a capability that the other two modalities are not capable of. ISAR requires that targets are rotating relative to the observing radar.

The required dwell time of ISAR depends on both the angular rotation rate of the target and the desired cross-range resolution. Since the rotation rate is not controllable by the seeker, they must select a T_D in advance. Since target motion may not be consistent throughout the dwell, and the rotation rate is not predictable, a means of generating usable ISAR images is to dwell for an extended period of time and extract an image from a subset of that data using an automatic time-window selection algorithm [69]. With sufficient target rotation, sub-meter cross-range resolution images can be formed of targets [86]. Further assuming that each ISAR observation is sufficiently spaced in time that the angular rotation rate is independent between observations, then we can characterize ISAR with a probability of identification, analogous to a P_D.

Kapfer and Davis provide an analysis of an example vehicle track for a semi-trailer truck operating on a mixture of highway and surface streets which provides a probability of exceeding a given $\dot{\Omega}$, reproduced in figure 2-9 [87]. Assuming a 0.5 second coherent dwell and a 0.5 m cross-range resolution is required and that there is no appreciable target acceleration over the dwell, the rotation rate will need to meet or exceed 20 $°\text{s}^{-1}$. Utilizing the distribution in figure 2-9, this holds true on roughly 12 percent of the example vehicle's track. Assuming that a TEL moves more slowly than a semi-trailer truck, it would have a smaller turning-rate, allowing the distribution to serve an upper bound of how often ISAR would be expected to be able to produce an image of a TEL with 0.5 m resolution.

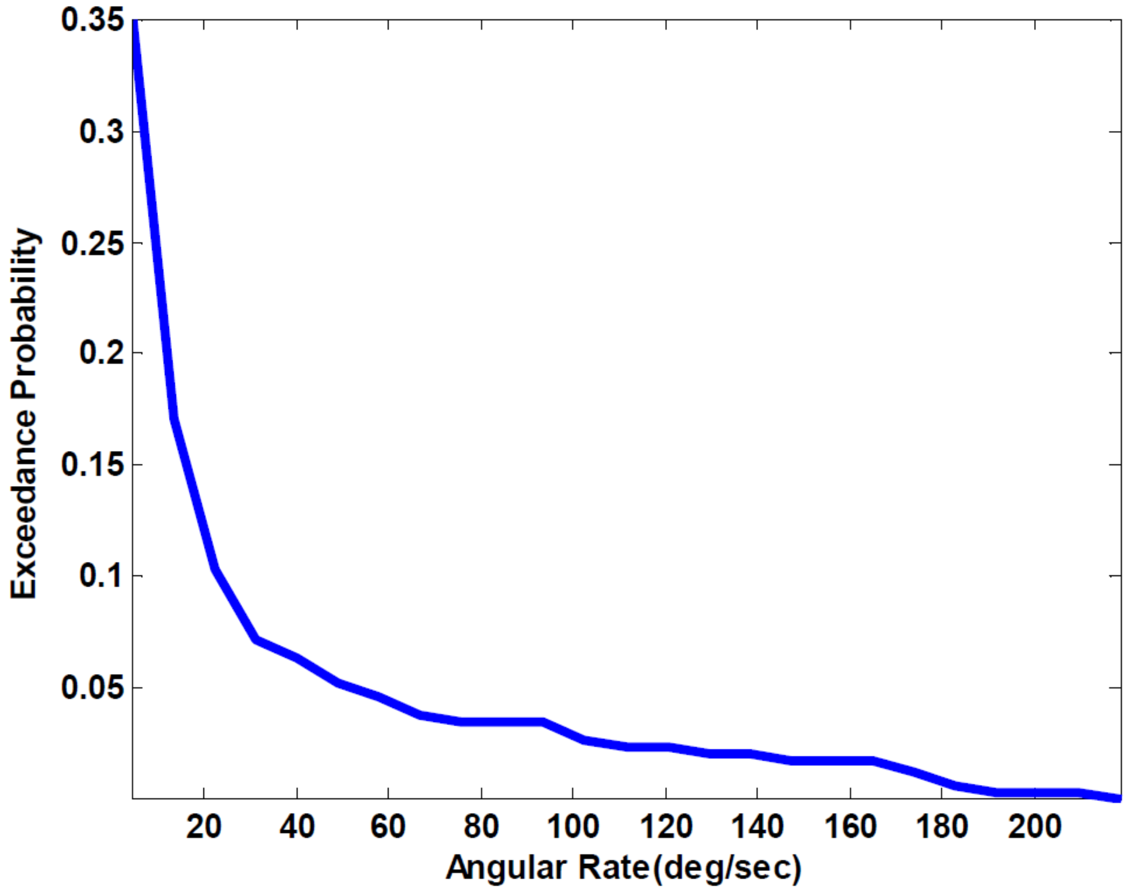

Figure 2-9: Probability of exceeding a given angular rotation rate for a semi-trailer truck traveling on a mixture of highway and surface streets, adapted from [87].

2.2.6 Applications of the Radar Modalities to the Tasks of Tracking

To summarize, the three modalities of SBR: SMTI, SAR, and ISAR provide the seeker with the capability to complete all of the tasks of tracking. Stationary TELs can be both detected and identified with SAR. Moving TELs can be detected with SMTI and identified with ISAR. This can be true even if the hider chooses to move their TELs only at night and/or under cloud cover when other remote-sensing systems are rendered inoperable.

SMTI should be the primary modality for wide-area detection of moving TELs. SMTI is a useful modality for monitoring large areas, such as TEL deployment areas. It can detect large, moving vehicles with relatively short dwell times, allowing large areas to be monitored at a relatively high frequency. Moving TELs are also assumed to be constrained to roads. Off-road travel by TELs is possible, the TELs designed to carry Chinese DF-31A missiles reportedly will have off-road capability. However, off-road travel would be comparatively slow. Additionally, since the SBR beam footprint is large, with dimensions on the order of tens of kilometers across, significant area around roads will be imaged incidentally when the road itself is monitored.

Movement is one of a TEL's principal means of achieving survivability, so being able to detect all large moving vehicles in an area is a powerful capability. It would struggle, however, with identifying whether or not those large vehicles were TELs. Identification with SMTI is limited to measurement of the average RCS, which indicates vehicle class, high-range resolution profiles, and behavioral traits such as moving in a convoy. HRR profiles are unreliable for target identification because of their strong dependence on aspect angle [87]. This creates points of failure as any large non-TEL vehicle could be misidentified as a TEL, and any TEL traveling without the expected convoy would be indistinguishable from a large-RCS, non-TEL vehicle.

SMTI is also incapable of detecting motionless vehicles, and vehicles that are traveling slower than the MDV. If a TEL were to come to a stop while being monitored with SMTI, it would become undetectable. Continuing to monitor the area surrounding its last known position would allow the seeker to rule out the possibility that it had moved, maintaining tracking, were it not for the complication of the MDV. If only SMTI were used to monitor a TEL deployment area, simply operating all TELs slower than the MDV of the system would render them invisible to the seeker, breaking tracking. Thus, SMTI alone is not sufficient for tracking TELs, both SAR and ISAR must be used in conjunction.

SAR should be used to detect and identify stationary objects and has capability to detect moving objects. SAR is able to create an image of stationary objects within an observed scene. This allows SAR to complement SMTI as it can be used to detect stationary targets that are invisible to SMTI. SAR can also be used to identify stationary TELs if the image is formed with sufficiently fine image resolution. Moving target extraction from SAR images further expands the utility of SAR to being able to detect moving objects [85]. This would allow detection of moving objects, such as those detectable with SMTI, as well as slow-moving targets that are moving below the SMTI MDV. It does not however, form images of these objects, so would not be useful for identification of moving targets. The drawback of SAR is that it is time-intensive compared to SMTI, so is not able to monitor as large an area as SMTI per unit time, and is also constrained in its field of regard.

In order to efficiently monitor a large area, such as a TEL deployment area, SAR and SMTI must be interleaved, depending on the state of the TEL being tracked. A moving TEL can be monitored with SMTI, however, if it comes to a stop, the SBR system would need to collect a SAR image of that area in order to continue making detections of the TEL. Since SAR images also contain the information that is available from a SMTI image, nothing except for time is lost when forced to use SAR to make a detection. SAR may not always be available however, as it has a more constrained field of regard than SMTI.

Since tracking is so reliant on SMTI, there is a vulnerability where a TEL could move at or slower than the MDV for an extended period of time in order to move undetected. This therefore necessitates that any stationary or slow moving TEL would need to be periodically re-imaged with SAR to make sure that its location is known.

The final required task is the identification of moving objects. This is required as without it, any large-RCS vehicle that left a shelter and traveled to another without stopping would not be able to be identified, creating the possibility of confusion for the seeker.

ISAR can be used to identify moving objects. SAR and SMTI alone do not have the capability to reliably identify targets. ISAR can add that capability, but under some constraints. ISAR is only capable of identify objects that turn sufficiently over the course of an observation to provide sufficient resolution to identify them. The data for ISAR can be co-produced with a SAR observation, or with a dedicated ISAR observation. The dwell time of ISAR is between that of SMTI and SAR, typically between a few hundred milliseconds to a second [86] The analysis in this chapter will consider the case where the hider is not utilizing decoy countermeasures, so uncertainty growth caused by target ambiguities will be assumed to be negligible. In the next chapter, I remove this assumption, assessing the uncertainty growth. Thus requiring a more capable identification capability. For the time being, I assume that ISAR will be gathered only as a co-product with SAR, or sparingly queued by an SMTI detection, which should be relatively rare given the hider's strategy for avoiding detection.

Sheltered TELs need to be monitored by monitoring the shelter. Shelters are assumed to be any location or structure where a TEL could be undetectable with SBR and other tracking modalities. Example shelters would be structures such as garages or barns built with radar absorptive or reflective materials such as corrugated steel. However, shelters need not be large fixed structures, any structure opaque to radar and optical imaging, such as a combination of wire mesh and camouflage netting would suffice. The seeker can continue to observe the shelter and surrounding area with either SAR or SMTI to attempt to detect the TEL if it were to leave the shelter. This process will be discussed in detail in the following subsection on Kalman filtering, however it is important to note that shelters that are suspected to house TELs must be monitored. If a vehicle is detected leaving a shelter, the seeker must identify it in order to avoid confusions.

Chapter 3

Counterforce Model

With the background established in the previous chapters, I begin introducing my analysis with the Counterforce Model. This model compares the number of warheads available to the attacker to the number of targets in order to determine the maximum uncertainty that the attacker can have in the positions of the target TELs. The Tracking Model, outlined in the next chapter, defines the tasks that are required in order to track TELs. I model how an attacker's uncertainty in the position of a group of TELs will vary over time as a function of the environment and the frequency of the attacker's observations with a remote-sensing system. The combination of the Tracking and Counterforce Attack models are needed in order to define what it means for TELs to be tracked *versus* untracked. At the end of the next chapter, utilizing these two models, I survey extant remote-sensing technologies to identify gaps in current capabilities that, if addressed, could enable counterforce by undermining the survivability of TELs.

3.1 Counterforce Remote-Sensing Requirements

To prevent a counterforce attack, a defender attempts to hide the locations of their TELs through concealment and mobility. In order to carry out a counterforce attack, an attacker must locate and track those TELs using remote-sensing systems. This model determines what is required of a remote-sensing system for it to be capable of tracking. Before delving into the specific tasks of tracking, I must define what I mean by the term "tracking".

To track something is to know where it is over time. A central concept in this work is precision, how well you need to know the location of something depends on the context. If I need to find my laptop computer so that I can write my thesis, I need to know where it is to within a meter or so. If I only know that it is somewhere within my office, it is effectively lost to me and I must spend effort to find it. If I wanted to destroy my laptop with a nuclear weapon, knowing its location within a few kilometers would suffice.[1] If I had a handful of nuclear weapons, perhaps just knowing that it was somewhere in Cambridge would be enough. Whether or not you know where something is depends entirely on what you are trying to do.

In the prelude to a counterforce attack, the attacker searches for the defender's TELs. I cast this competition as a game of hide-and-seek with the attacker as the seeker and the defender as the hider. Whether the attacker can successfully disarm the hider depends on a number of factors: the number of warheads available (n_w), the number of targets (n_{TEL}), the effective kill radius of the attacker's warheads (r_w), the time it takes to deliver those warheads (τ_f),

[1] This would depend on the yield of the warhead and the blast-hardness of my office.

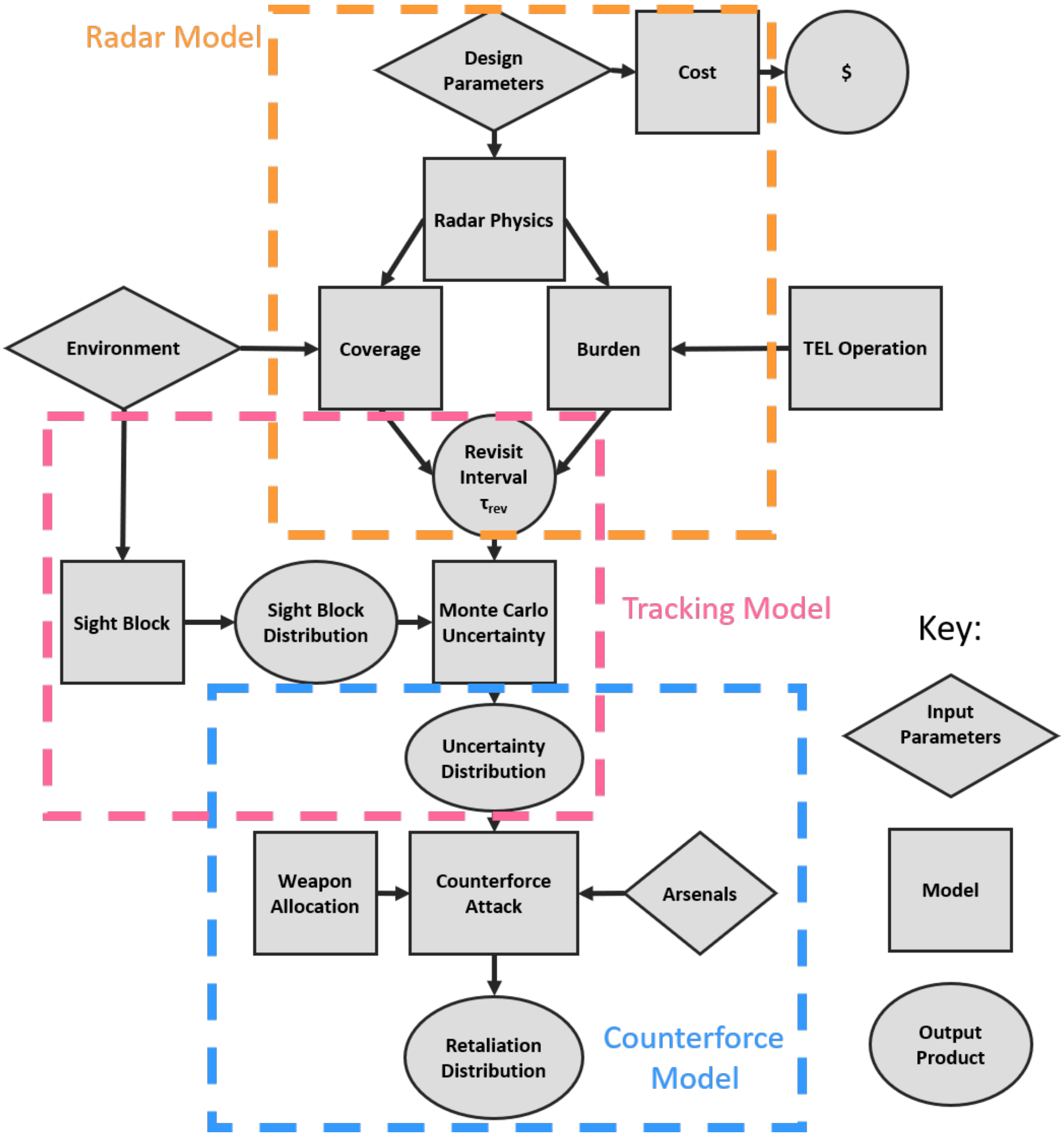

Figure 3-1: Flow chart of the model. The forward solution moves from top-to-bottom in order to determine if a given radar hardware configuration is capable of supporting a counterforce attack. The reverse solution moves from the bottom-up, using the counterforce model to create a set of performance metrics that a radar system is designed to meet.

the speed of the TELs (v_{TEL}), and the attacker's initial uncertainty in TEL position (r_u). Remote-sensing systems, for which I seek to derive performance requirements, are used to manage the positional uncertainty r_u. Finding the largest r_u which still enables a given attack can be used to set a performance metric that the remote-sensing system much meet. The remaining factors are determined by considering a given counterforce attack, and can be patterned on real-world dyads.

The attacker has some number of warheads available to them, each with a weapon radius r_w which defines a circular area within which the warhead can reliably destroy a TEL. The hider operates some number of TELs

(n_{TEL}) which the seeker targets. The seeker has previous knowledge of the TEL locations, which I will capture with an characteristic uncertainty distance r_u. This is the maximum distance the TEL could have traveled undetected since it was last detected by the seeker's remote-sensing system. Later in this section, I will extend the positional uncertainty to capture the seeker's beliefs about where the TEL most likely is by formatting it as a probability distribution, however it is currently useful to treat it as a simple distance in order to introduce the components of the Counterforce Model.

When attacking a given TEL, the initial positional uncertainty bounds the possible locations of the TEL at the time when the attacker launches their weapons. For the i^{th} TEL, the initial positional uncertainty is bounded by $r_{u,i}$. It takes a finite amount of time for a nuclear warhead to be delivered. During that time, the TEL has the opportunity to move, either in response to the incoming attack or through routine repositioning. If a weapon attacking the i^{th} TEL has a flight time of $\tau_{f,i}$, then the TEL could travel a distance $v_{\text{TEL}}\tau_{f,i}$ between the launch and detonation of the warhead. The attacker has no control over or foreknowledge of a TEL's plan to move. To be conservative, the seeker must account for the worst-case TEL movement. At the time when the attacking warheads detonate, the TEL could be anywhere within the area bounded by the total uncertainty $r_{t,i}$, which is

$$r_{t,i} = r_{u,i} + v_{\text{TEL}}\tau_{f,i} \ . \tag{3.1}$$

The Counterforce Model presented in this section determines how the attacker would allocate warheads to targets to account for this uncertainty in the position of the TEL. In the next subsection, I outline two possible modes of attack that the attacker could employ. In Subsection 3.1.3, I present a model which determines the shape of the area bounded by $r_{t,i}$ as a function of the shape of the local road network. In conjunction with the weapon sub-model presented in 3.1.2, the number of warheads that must be allocated to the area to ensure the destruction of the TEL can be calculated. In Subsection 3.1.4, I replace the concept of the radius of uncertainty $r_{u,i}$ and radius of total uncertainty $r_{t,i}$ with probability distribution functions that capture the attacker's beliefs about the potential locations of the TEL. This can be used to evaluate the probability of successful attack, as the attacker would assess it at the time when the attack was launched. Finally, in Subsection 3.1.6, I create profiles for the numbers of weapons and targets for counterforce attacks drawn from real-world dyads.

3.1.1 Modes of Counterforce Attack

When considering how a counterforce attack against TELs may be carried out, I am assuming a concerted attack with nuclear warheads delivered by depressed-trajectory SLBMs. I rule out heavy bombers and ICBMs as delivery systems due to their long delivery times.[2] In their place, I consider depressed-trajectory SLBMs, which as discussed below, have much shorter τ_f so are better suited to attacking mobile targets. I consider only nuclear counterforce and disregard precision conventional weapons due to the difference in scale of their destructive effects.[3] Nuclear weapon effects are on the scale of kilometers, while conventional weapon effects are measured in meters or tens of meters. *i.e.* I assume that conventional counterforce would not offer significant advantages in terms of avoiding nuclear escalation. Any tracking system which is able to track a target sufficiently well to allow a conventional weapon to be delivered

[2]Bombers and other aircraft such as drones also need to penetrate air defenses, so would face a higher level of attrition than ballistic missiles.

[3]I also assume that if a country's nuclear deterrent were directly attacked with conventional weapons, that country would perceive the attack as being part of a nuclear war, so would consider nuclear escalation in response.

to it would also enable a nuclear attack while the opposite is not necessarily true. I also assume that the attack is concerted, meaning that all of the mobile targets are attacked simultaneously (or in quick procession). This is necessary to prevent retaliation (launch-under-attack) or repositioning of TELs with the aid of active countermeasures such as anti-satellite weapons (discussed in Chapter 6).

Under these assumptions, there are two modes of attack that the seeker can utilize: barraging the entire area of uncertainty, or utilizing guided weapons. To barrage a target, the attacker must deliver warheads to the entire area covered by $r_{t,i}$. This takes into account the worst-case TEL movement. By covering the entire area, the attacker eliminates any chance that a TEL will survive the barrage, but at the cost of being inefficient in terms of the number of weapons allocated to each TEL.

Alternatively, the attack could be carried out with guided weapons which could be updated with the targets' location during flight. A maneuverable reentry vehicle (RV) could adjust its aim point while in the atmosphere, or, less flexibly, updating the aim point at the time of the release of the warhead from the bus would allow compensation for target movement up to that point. With a guided weapon, the total uncertainty that must be barraged will fall in a range

$$r_{u,i} \leq r_{\text{guided},i} \leq r_{t,i} \ . \tag{3.2}$$

The minimum is set at $r_{u,i}$ by the scenario where the TEL does not move during the attack. The maximum is set at $r_{t,i}$ by the scenario where the TEL is able to move without being detected. The entire area defined by $r_{\text{guided},i}$ and the local road network must be barraged to ensure the destruction of the target TEL, regardless of what actions it takes.[4] If the TEL attempts to move during the attack and is successfully detected and identified, then the model assumes that it will be destroyed. The attacker must account for the TEL being able to move some distance undetected, or after the ability to change the aim point is lost. The probability that a TEL could move some distance undetected will depend on a number of factors which are developed in the Tracking Model later in Chapter 4. There are differences between evasive maneuvers and a normal TEL movement. However, as the time-scale of an attack is short compared to the length of a TEL patrol and the timing is selected by the seeker, the probability of a TEL having the opportunity to move an appreciable distance without being detected during the delivery of weapons is likely to be low. As such, $r_{\text{guided},i}$ will tend toward

$$r_{\text{guided},i} \approx r_{u,i} + (\tau_{f,i} - t_{\text{release}})v_{\text{TEL}} \ , \tag{3.3}$$

where t_{release} is the last time, measured from the launch of the weapon, when the warhead aim point can be updated.

Since guided weapons would effectively allow the attacker to ignore the uncertainty that is created by TEL movement during the delivery of the attacking warheads, they could increase the warhead efficiency of an attack; the warhead efficiency being the number of warheads that an attacker would need to allocate to each target. Figure 3-2 shows a toy example of barrage and guided attacks. In this example, the seeker has some area of initial positional

[4]If a TEL is detected during an attack, then its instantaneous positional uncertainty can fall below $r_{\text{guided},i}$. However, at the time when the attack is launched, the attacker must account for the TEL either moving or remaining stationary, so must commit enough warheads to barrage the area covered by $r_{\text{guided},i}$ in its entirety.

uncertainty (blue shaded area) centered on the last known location of a TEL, which is operating on an isolated, straight road. The total area of uncertainty takes into account the area into which the TEL could move during the attack (green shaded area). A barrage attack would cover the total area of uncertainty, in this example requiring three warheads to be allocated. Utilizing a guided weapon attack, the entirety of the area of initial uncertainty would be covered with weapons effects, and possible TEL movement accounted for by utilizing a remote-sensing system to detect TEL movements. In this toy example, a guided weapon attack would reduce the number of warheads needing to be allocated to the target to two. When considering a large number of targets, even a modest increase in the warhead efficiency on a per-TEL basis could result in a large difference in the number of warheads required for the attack. This is important particularly for weapons where the flight-time movement uncertainty is large due to a large τ_f.

Guided attacks introduce a new vulnerability for the attacker, however. Any system requiring updating of the target location in order to guide a weapon would require a remote-sensing system to be networked with the weapon. This would introduce two points of failure to the system. Disabling either the remote-sensing system or the communication link would prevent the aim point from being updated which could result in the target not being destroyed. Relying on guided weapons therefore creates a trade-off between warhead efficiency and vulnerability to countermeasures. Specifically, it is in those situations where guided weapons improve warhead efficiency that the vulnerability to countermeasures is introduced. Due to this vulnerability I will consider only barrage attacks for the main body of this text. In Chapter 6, I evaluate the effects of relaxing this assumption and allowing for a guided-weapon attack, representing a scenario where countermeasures are not available or possible.

3.1.2 Weapon Model

The attacker has access to some number of warheads n_w and the seeker will have some number of deployed TELs n_{TEL}. The parameters of each weapon, the weapon radius r_w, the reliability and accuracy, and the flight time τ_f are integral in determining how many warheads need to be allocated to each of the target TELs.

Weapon Radius. The weapon radius is defined as the radius from the ground zero at which the expected number of targets within the radius that have *not* been damaged to a specified level is equal to the expected number outside of the radius that have been damaged to that level, given a uniform field of isotropically oriented targets [88]. TELs that are not damaged to the specified level do not necessarily survive, but receive a lesser amount of damage, which may or may not be sufficient to disable them. For simplicity, for the Weapon Sub-Model, the probability of a TEL being destroyed within the weapon radius is unity, and the probability of it being destroyed outside of the weapon radius is zero.[5] This will overestimate the probability of a kill for targets requiring a single warhead, so is a charitable simplification in favor of the attacker. If multiple warheads are tiled together, the overlapping blast effects will reduce this overestimate.

The weapon radius of a given warhead depends on both the yield of the warhead and the target that it is attempting to destroy. Further, it is important how the effects of the weapon damage or destroy the target, as different weapon effects (peak overpressure, dynamic overpressure, thermal fluence, *etc.*) behave differently as a function of range.

[5]This simplification is made in order to ease computational burden. A distance-dependent kill probability could be implemented but would lengthen simulation time. This assumption favors the attacker, as with most other assumptions made in this analysis. There are additional factors that affect weapon kill probability such as the local terrain that are contextual and not easily captured in a generalized model such as this.

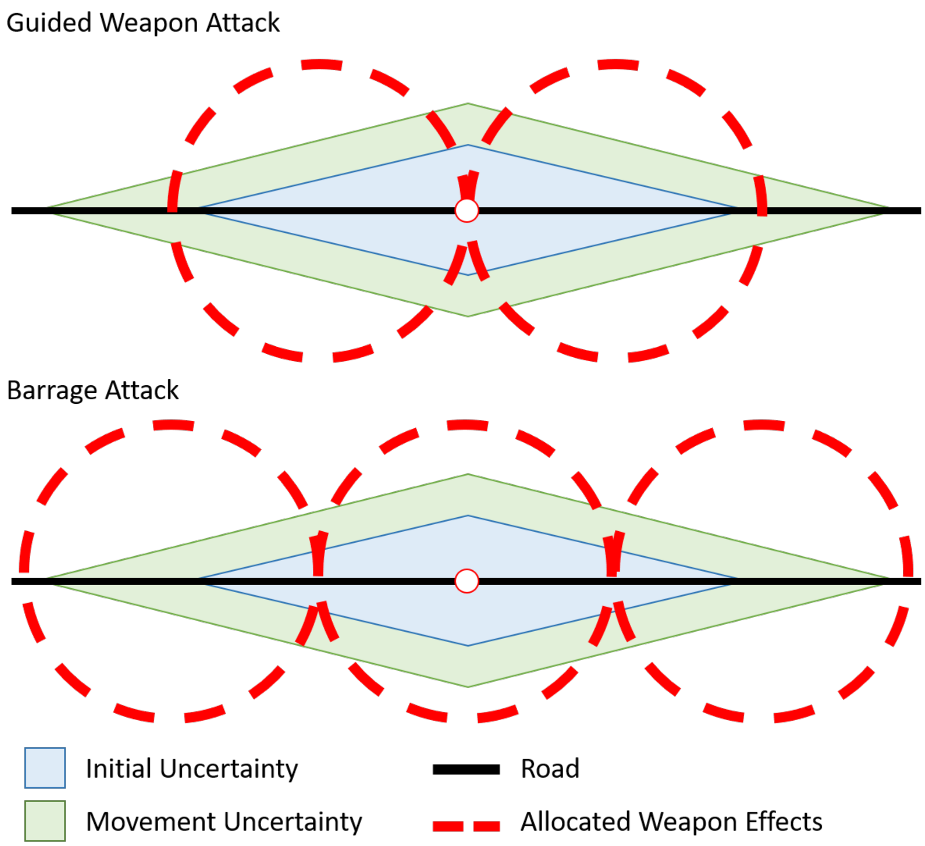

Figure 3-2: A toy example of barrage *vs* guided-warhead attacks. The last detected location of a TEL is designated by the white dot. The seeker's initial uncertainty in the position of the TEL at the time when an attack is launched is shown as the blue shaded area. Uncertainty in the vertical direction is shown for ease of visualization and is not to scale, the true scale would depend on the ratio of on-road to off-road TEL speed. The area into which the TEL could move during the delivery time of attacking weapons is shown as the green shaded area. Red dashed circles show the areas covered by lethal blast effects by the allocated warheads.

Cold War modeling by the RAND corporation envisaged destroying or disabling TELs with dynamic overpressure effects, essentially very strong winds, which can overturn TELs or knock debris onto them in order to damage the missile [89].[6] The kill radii for a number of representative yields, for causing severe damage to TELs, are shown in Table 3.1. For example, a 100-kt warhead, such as the W76 carried by the Trident II, has a weapon radius against TELs of 2.8 km, assuming it is detonated at an optimal height to maximize the weapon radius [89].

Reliability and Accuracy. Reliability is the probability that a delivery system will successfully deliver its payload to its aim point. Accuracy is a measure of how close to the target aim point the warhead can reliably be delivered. Reliability of modern delivery vehicles, such as those deployed by the U.S. is assumed in the literature to be in

[6] The vulnerability numbers system was used by U.S. war planners during the Cold War to classify targets based on how easy they were to damage [89]. Vulnerability numbers are defined for a specified level of damage, and consist of three parts. For example, Russian SS-25 TELs were assigned a vulnerability number of 11Q9. The first number indicates the hardness of the target, relative to some arbitrary benchmark, with larger numbers indicating that the target is more difficult to damage. The second component indicates whether the target is damaged by static overpressure (indicated by a P) or dynamic overpressure (indicated by a Q). The third component indicates the sensitivity of the hardness to the duration of the pressure wave. Increasing numbers indicate increasing sensitivity with 0 indicating damage is independent of blast duration and 9 indicating the largest dependence. Since the duration of the pressure wave depends on the yield of the warhead, larger warheads are more likely to damage a sensitive target than smaller warheads, even at the same overpressure level.

Yield [kt]	Weapon Radius [km]	Example
100	2.8	W-76 (Trident II), SS-27 Mod 2 (assumed)
335	4.0*	W-78 (Minuteman III)
455	4.4*	W-88 (Trident II)
800	5,6*	SS-27 Mod 1 (assumed)

Table 3.1: Weapon radii for representative nuclear weapons for the overturning of TELs. Astrix * indicate values estimated utilizing $w_r \propto Y^{1/3}$ where Y is the weapon yield. Since TELs are vulnerable to dynamic overpressure, which scales super-linearly with yield, these estimates will underestimate the weapon radius,

excess of 80–90% [9,89,90]. When a missile booster suffers a failure, it is typically at launch, so if the attacker is able to quickly identify failures and launch replacement weapons, the issue can quickly be rectified [90]. It is not openly known if rapid retargeting of weapons to account for booster failures is a capability of current nuclear weapons arsenals, but it will be assumed for this model. Thus, reliability will be accounted for by correcting the number of warheads available to the attacker to the product of the total number of warheads and the reliability.

Weapon accuracy can also impact the probability that a given warhead will reach its target. However, it is a negligible effect when considering an attack against TELs, which are relatively soft targets. For example, the weapon radius of a small 100-kt warhead is on the scale of kilometers in radius, while the effects of delivery system accuracy are on the scale of a few hundred meters or less [11].

Flight Time. The flight time of the weapon delivery system is also an important characteristic of an attacking weapon, as it dictates the amount of time between the time of launch and the time of detonation. Nuclear weapons can be delivered on a number of different systems, such as bombers or ballistic missiles. Bombers can be detected leaving their air fields by national technical means, or when approaching the adversaries air space by radar, and so provide a large amount of warning time of an incoming attack that the defender can use to disperse or launch their weapons. They are also vulnerable to being downed by air defenses, and have a relatively slow air speed, making them poorly suited for conducting a counterforce attack. Ballistic missiles are a common basing method for nuclear weapons and have relatively short flight times and can achieve high accuracy. These traits make them attractive for counterforce operations as they can threaten both mobile, and hardened and deeply buried targets.

The flight time of a ballistic missile is determined by the rocket and payload parameters (thrust, mass, *etc.*) the type of trajectory that the missile is flown on, and the range between the target and the launch point. There is a minimum energy trajectory that dictates the maximum range that that missile can deliver its payload to. Missiles can also be flown on depressed trajectories which intentionally sacrifice range in order to decrease flight time [36]. Intercontinental range ballistic missiles (ICBMs) can achieve ranges in excess of 5,500 km (though often can reach in excess of 10,000 km) and are generally based on the home soil of the states that operate them, requiring the missiles to fly near their maximum range. ICBMs traveling the thousands of kilometers between the continental U.S. and Russia, for example, have flight times of approximately 30 minutes [36]. Ballistic missiles can also be launched from submarines (SLBMs) which provide more flexibility in the geographic location of the launch point, allowing for shorter flight ranges. If a missile does not need to be fired on its minimum energy trajectory, the attacker can loft or depress the trajectory, firing the missile at a sharper or shallower angle in order to increase or decrease the flight time to reach a given range. The flight times for a Trident II SLBM, such as those currently deployed on U.S. ballistic missile submarines, are shown in table 3.2 for a number of distances representative of attack scenarios. As an

Scenario	Distance [km]	Time [min]
Bering Sea – Moscow	1,850	7
Bering Sea – Omsk	3,000	10
Montana – Moscow	8,000	30

Table 3.2: Missile flight times.

example, a Trident II launched on a depressed trajectory to minimize flight time can fly 1,850 km in approximately 7 minutes, or 3,000 km in approximately 10 minutes [36].

3.1.3 Warhead-Tiling Model

The Warhead-Tiling Model is used to determine the number of warheads that need to be allocated to a given target in order to ensure its destruction, given the $r_{t,i}$ of that TEL. The total uncertainty in the position of a given TEL, $r_{t,i}$, has so far been developed as a single distance, which must be converted to an area that the seeker must barrage. This requires making an assumption about the local road network shape. I consider two bounding cases and one intermediate case.

Area of Uncertainty. A TEL operating on a single, isolated road only has the choice of moving in one of two directions, assuming off-road travel is difficult or slow. In this case, the area of total uncertainty will grow linearly with $r_{t,i}$. The area of total uncertainty will be along a line with some width w

$$A_{u,i,\text{linear}} = 2r_{t,i}w \ . \tag{3.4}$$

For this lower bounding case, I will assume that off-road travel is slow or restricted by terrain, so the width of the area is negligible and can be set to unit value. Some TELs have been designed with off-road capability. Off-road travel is only relevant if a TEL could travel a distance on the scale of r_w from the road without being detected. If off-road speed is on the order of a few meters per second, to travel a few kilometers off-road would take on the order of thousands of seconds. This is on the scale of the delivery time of an ICBM, but is longer than the delivery time of a SLBM on a depressed trajectory. Additionally, not all off-road terrain would be traversable. While this could be a concern for ICBM-based attacks, off-road travel is not always possible so can be safely ignored for the lower boundary.

In the case of a TEL operating in an area with a dense road network, where the TEL's choice of direction is not constrained, the area of total uncertainty will be

$$A_{u,i,\text{circle}} = \pi r_{t,i}^2 \ . \tag{3.5}$$

This is the upper boundary of the area of uncertainty, which would be difficult to achieve in practice.

I also consider a gridded road system with inter-road spacings of s with shelters placed at each intersection. For the gridded road system, TELs are constrained to moving in a rectilinear Manhattan geometry, so the area of total uncertainty will grow as a square diamond. Assuming a square grid the two largest distances will be in the two principal directions, centered on the initial shelter. The total uncertainty will extend for $r_{t,i}$ in those directions. In the off-principal directions the furthest points where the TEL could reach will fall on a straight line connecting those points. The total area covered by this square diamond is

$$A_{u,i,\mathrm{grid}} = 2r_{t,i}^2 \ . \tag{3.6}$$

Figure 3-3 shows how the area of uncertainty grows for each of these scenarios.

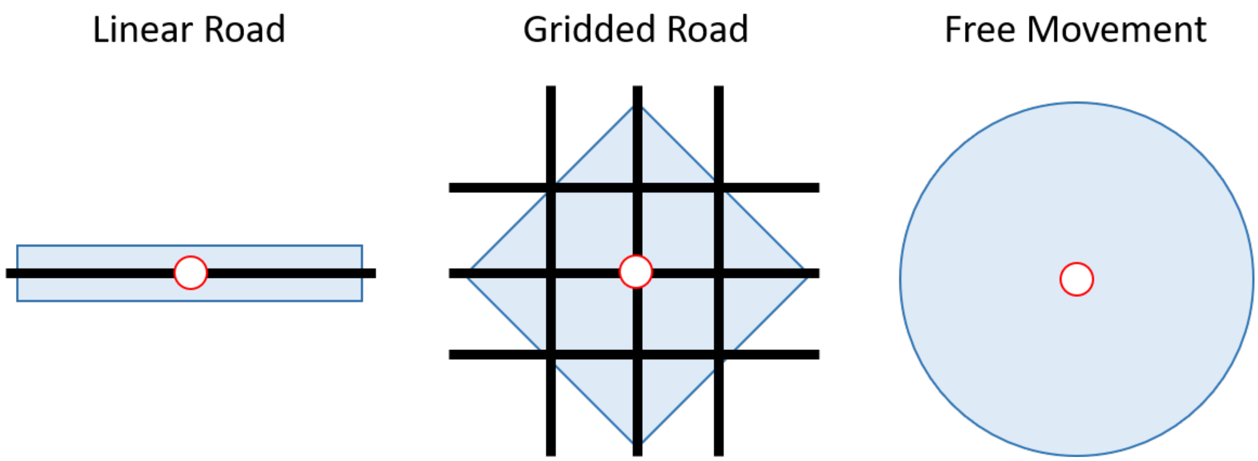

Figure 3-3: Shape of area of uncertainty for for linear road, gridded road, and free movement scenarios.

Warhead Tiling. Determining the number of warheads needed to cover these areas is a geometric tesselation problem of tiling an area with circular blast effects. For a given blast effect, such as overpressure or thermal fluence, the contour lines of constant effect value are circular when projected onto flat ground. Circles are not shapes that can be tiled without either voids or overlap. I assume that the attacker, seeking to minimize the probability of a TEL surviving an attack, will err on the side of overlapping weapons effects rather than having voids.

For the linear road bound, the required number of warheads is

$$n_{w,i,\mathrm{linear}} = \mathrm{ceil}\left\{ \frac{2r_{t,i}}{\sum_{j=1}^{n_{w,i,\mathrm{linear}}} 2r_{w,j}} \right\} \ , \tag{3.7}$$

where i is the index of the i^{th} TEL and j is the index of the jth warhead. If all of the warheads are the same yield, this simplifies to

$$n_{w,i,\mathrm{linear}} = \mathrm{ceil}\left\{ \frac{r_{t,i}}{r_w} \right\} \ . \tag{3.8}$$

If warheads of mixed yield are used, allocating warheads will require an algorithm, as the number of warheads needed depends on the set of warhead yields. To avoid such complicated optimization problems, I will consider only attacks with a single yield of warheads. Regardless of warhead yield, in the linear case, warheads can be tiled without overlap. However, since warheads must be allocated in whole units, targets can be overkilled if $r_{t,i}/r_w$ takes a non-integer value.

The circular area bound and the gridded area require less efficient tiling. The most efficient tessellation of an area with circular blast effects that does not leave voids is tiling with hexagons that are inscribed within the circles. Depending on the number of warheads allocated, and the shape of the underlying area of uncertainty, alignment of the last detected location of the TEL and the tessellated area will have one of three centerings. Centered tiling is where the last known location of the TEL is at the center of a hex, with additional warheads allocated along concentric rings surrounding the center hex. For vertex-centered tiling, the last known location of a TEL is placed between three central hexes. For edge-centric tiling, the last known location is centered along two overlapping edges. These tessellation schemes are shown in figure 3-4. In any scenario where an attacker allocates warheads to cover an area, the most efficient cetnering scheme will be used.

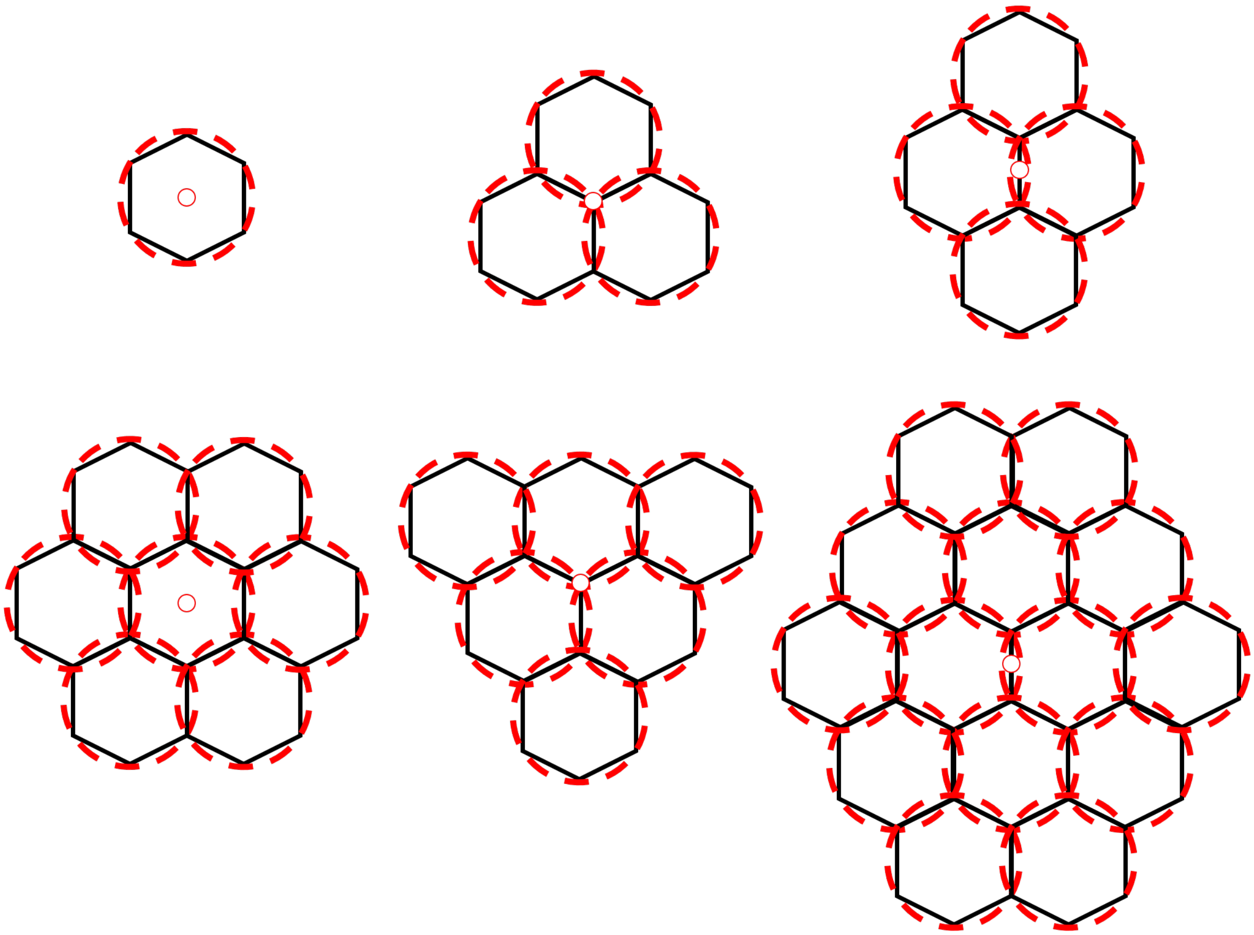

Figure 3-4: Tessellation of an area with circular blast effects without leaving voids, shown for centered, vertex-centered, and edge-centered tiling schemes.

3.1.4 Positional Uncertainty Distribution

To this point, I have been addressing the total area of uncertainty of a TEL by its outer boundary, $r_{t,i}$. With this definition, so long as the attacker is able to barrage the entire area encompassed by $r_{t,i}$, they are able to ensure the destruction of the i^{th} TEL. This is true, but in practice is an unnecessarily stringent condition. As I show in Chapter 4, the process of tracking a TEL has stochastic elements. It is never possible to truly rule out certain scenarios, such as the TEL moving to the boundary $r_{t,i}$ without being detected, even if that scenario has a low probability of occurring. However, if the seeker can estimate the probability of such states (I argue they can in Chapter 4), they can weight these possibilities. As such, the seeker can give more weight to more probable TEL locations. The weights, the seeker's belief in the probability of the TEL being at a given location, can be captured in a probability density function (PDF) $p(r_u, t)$. As I show in the next chapter, the weight of the seeker's belief will be centered on the location where the TEL was last detected and will vary with the distance from that point.

The probability distribution $p(r_u)$ describes the seeker's initial uncertainty in the position of the TELs at the time when the attack is launched, based on the information gathered by their remote-sensing system. The PDF upon which the attacker decides to attack is $p(r_t)$, which also incorporates their beliefs about the distance that the TEL can move during the attack.[7] When treating r_u and r_t as simple distances, they can be related by the simple addition of $v_{\mathrm{TEL}}\tau_f$. For probability density functions, an assumption must be made about how TELs move during the weapon-flight time. For this model, I assume that the seeker plans conservatively and assumes the worst-case TEL movement. This is where the TEL moves away from its last detected position at its maximum speed for the duration of τ_f. This allows the cumulative density functions (CDF) of r_u and r_t, $P(r_u)$ and $P(r_t)$ respectively, to be related by the simple addition of $v_{\mathrm{TEL}}\tau_f$. For example, if r_P defines the radius that encompasses P percent of the probability mass of $p(r_u)$, then $r_P + v_{\mathrm{TEL}}\tau_f$ will capture P percent of $p(r_t)$.[8] How such CDFs of r_t are developed is presented in Chapter 4.

3.1.5 Warhead Allocation and Probability of Kill

Evaluating the success of an attack is an evaluation of the seeker's belief in the success of the attack at the time when the decision is made to attack. That is, the actual outcome of the attack is not relevant from the perspective of deterring an attack, it is the seeker's prediction of the outcome of the attack that matters. The seeker's belief in the success of the attack can be derived using the tiling model and $p(r_t)$ for each TEL. The probability of killing the i^{th} TEL $p_{k,i}$ is determined by the integral of $p(r_t)$ within the boundaries set by the tiling. The probability of destroying all of the TELs (assuming no systemic weaknesses which make the $p_{k,i}$ dependent) is

$$P_k = \prod_{i}^{n_{\mathrm{TEL}}} p_{k,i} \ . \tag{3.9}$$

The attacker, when faced with a set of n_{TEL} TELs, each with some total uncertainty distribution $p(r_{t,i})$, would need to decide on how to allocate the n_w warheads available to bring P_k to an acceptably high value. Since there are a finite number of warheads targeting n_{TEL} TELs, the number of warheads available to be allocated to any individual target depends on how many warheads are allocated to other targets. In a specific scenario with each of these values defined, warheads could be allocated by calculating $p_{k,i}(n_j)$ and $p_{k,i}(n_j + 1)$, the probabilities of killing the i^{th} TEL if j and $j+1$ warheads are allocated to it, respectively, and iteratively allocating warheads to the TEL with the largest $\Delta p_{k,i,j+1}$ until P_k reached an acceptable level or there were no more warheads to allocated. If P_k exceeded the threshold, then the attacker would believe that an acceptably assured attack would be possible and could decide if they wanted to attack or not. If the warheads ran out before the P_k reached the threshold, then an attack would

[7] I assume that the seeker is able to take sufficient measurements over time in order to accurately assess the average road speed of the TELs.

[8] This creates an artifact where there is no probability mass within $v_{\mathrm{TEL}}\tau_f$ of the last detected location of the TEL. As I discuss in Chapter 4, this is inaccurate as keeping a TEL motionless is a valid strategy for the hider. This creates a situation where the model could predict a p_k of zero if $v_{\mathrm{TEL}}\tau_f > r_k$ even if a warhead was allocated to that target. However, under those conditions, the true probability of killing that target though not zero, would be low. The goal of this model is to identify remote-sensing systems that would enable high-probability of success attacks, not differentiate between low-probability attacks, so this artifact is not relevant where the conclusions of this model are applicable.

not be assured and deterrence would hold.

Since the number of warheads available to be allocated to each TEL depend on the number of warheads allocated to each other TEL, such an algorithm requires a specific actualization of each $p(r_{t,i})$ in the set of TELs being targeted. This, in turn, assumes a specific combination of remote-sensing system, environment, attack scenario, and history of when and where the TELs were last detected. This dependence complicates analysis when $p(r_{t,i})$ can take multiple values, which is the case in the Tracking Model presented in the next chapter. I circumvent this dependence by introducing a simplifying assumption.

I define $p_{k,\text{thresh}}$, the attackers minimum acceptable probability of kill for each TEL. For any $p(r_{t,i})$ distribution, the number of warheads needed to meet or exceed $p_{k,\text{thresh}}$, $n_{w,i,\text{thresh}}$ can be determined from the tiling and road models presented above. If $p(r_{t,i})$ can take multiple values, then so will $n_{w,i,\text{thresh}}$. What is important is that $n_{w,i,\text{thresh}}$ of the i^{th} TEL is independent of the other TELs in the target set. I show the application of this assumption in Chapter 5 when I present the output of the Tracking Model. For now, it is useful to understand the costs of this assumption and how to define $p_{k,\text{thresh}}$.

If the $p_{k,\text{thresh}}$ were exactly met for each TEL, then it could simply be defined from $(p_{k,\text{thresh}})^{n_{\text{TEL}}} = P_k$. However, since the threshold can be either met or exceeded, then the probability of destroying all of the TELs is

$$P_k \geq \prod_{i}^{n_{\text{TEL}}} p_{k,\text{thresh},i} \ . \tag{3.10}$$

A conservative hider, who is attempting to determine if their TEL fleet can deter an attack would be unhappy with this inequality. If the hider based their TEL operations off of analysis done with $p_{k,\text{thresh}}$, then it would be possible that a TEL fleets survivability was worse than anticipated, meaning that an unappreciated vulnerability could exist. Defining $p'_{k,\text{thresh}} = P_k$, then

$$P_k \leq \prod_{i}^{n_{\text{TEL}}} p'_{k,\text{thresh},i} \ . \tag{3.11}$$

This definition is can be extremely conservative for the hider, especially if there are a large number of TELs. However, it is conservative with respect to the durability of deterrence, meaning that survivability will always be better than estimated.

What value of P_k constitutes an acceptable risk of retaliation will depend on the temperament of the attacker and how dire they perceive their situation to be when they consider a nuclear attack. Presumably, the attacker would have less tolerance for risk when considering a bolt-from-the-blue attack compared to an attack in a crisis. While P_k cannot be explicitly defined, the treatment above provides some insights that are valuable. Regardless of scenario, any prospective attacker faced with the prospect of being retaliated upon with nuclear weapons will want the probability of destroying all TELs to be high. For an n_{TEL} that is on the order of tens to hundreds of TELs, in order to achieve a P_k close to unity, the individual kill probabilities must be very close to unity. Additionally, the probability of destroying every target is only as good as the probability of killing the most difficult target. As such, I assume that the attacker will be inclined to be conservative, and a charitable assumption for the attacker is to set $p'_{k,\text{thresh},i} = 0.95$.

Weapon	Warheads in U.S. Stockpile	Number Deployed	Yield [kt]
ICBM - W78	600	200	335
ICBM - W87	200	200	300
SLBM - W76-1	1,511	$\sim$ 1,000	90
SLBM - W76-2	25	*	8
SLBM - W88	384	*	455

Table 3.3: The number of ballistic-missile deliverable warheads in the U.S. stockpile, numbers deployed, and yields. There are a total of approximately 1,000 warheads of any variety deployed on SLBMs [91].

3.1.6 Representative Counterforce Attacks

As previously noted, what constitutes a successful counterforce attack depends on the number of attacking warheads, their delivery systems, and the number of targets. To place the analysis of remote-sensing systems into context, two representative counterforce attacks are considered, aligning with potential real-world scenarios. The first scenario will represent the current balance between the United States, as the seeker, and the TEL component of the Russian Strategic Missile Forces. The second scenario will concern a U.S. attack against the current Chinese nuclear forces where the United States holds a roughly five-fold numerical advantage. These two cases, which represent real-world dyads conveniently represent two qualitatively different attacks, one against a peer adversary, and one against a sub-peer.

US Nuclear Forces. The U.S. deploys, under the terms of the New START agreement, 1,550 nuclear warheads on 700 delivery systems with 800 launchers [91,53]. Of those warheads, 400 are deployed on silo-based ICBMs, with yields between 300–335-kt [91]. Against a TEL, the weapon radius, r_w, of a 300-kt warhead is 4.7 km. The flight time of an ICBM from the silos based in the continental U.S. to a distant target in Asia is approximately 30 minutes [36]. SLBM weapons have much shorter flight times of 7–10 minutes when fired on depressed trajectories. The U.S. currently deploys between 900–1000 SLBM warheads at a given time [91]. The total stockpile of SLBM warheads contains 384 W88 warheads with 455-kt yield and 1,511 W76-1 90-kt warheads[91]. The kill radii of the two systems differs significantly, so both are considered. To avoid tiling issues, I assume that attacks will be done using only warheads of a single yield, with each case bounding a mixed attack. Using only the lower yield W76 would represent an attack where the higher yield W88s were reserved for hardened and deeply buried targets. An attack using only the higher yield W88s would represent a scenario where the U.S. loaded attacking submarines with higher yield warheads in preparation for an attack.

U.S.-Russia The Russian strategic nuclear forces, the land-based rocket forces, the strategic naval forces, and the strategic aviation forces, comprise a nuclear triad [37]. The strategic rocket forces deploy 958 warheads on intercontinental-range missiles, 380–450 of which are deployed on TELs [92]. The diversified basing of the Russian nuclear force limits the impact of a vulnerability in any one leg. Russia, being peer to the U.S. in terms of arsenal size, also has a large amount of redundancy in their mobile nuclear force, with between 138–180 TELs, roughly half of which can carry multiple independent reentry vehicles (MIRVs) which allow each missile to deliver multiple warheads to different targets [92].[9]

[9] The SS-25 and SS-27 Topol launchers account for roughly half of the deployed TELs and carry a single warhead with a yield on the order of one megaton. The remaining RS-24 Yars launchers can carry 4 warheads each with a yield on the order of 100-kt [92].

Defender	Latitude	Number of Targets	Flight Times [min]	Number of Warheads Available
Russia	55°N	180	7–10	450
China	28°N,35°N	48	7–10	90–180

Table 3.4: Numbers of targets, weapon delivery system flight times, and the number of available warheads for modeled U.S. attacks against Russia and China.

For the purpose of this analysis, the non-TEL components of the Russian nuclear force will not be considered, and would need to be disabled for an attack to be feasible. Under this scenario, the number of warheads available to the attacker will be kept at parity with the number of warheads carried by the target TEL force. However, because some targets are MIRVed, there are an excess of available warheads to targets. Russia was prepared to operate their mobile nuclear forces in prescribed deployment areas (in peace time) in article V of the START I treaty [89]. These deployment areas were to be no larger than 125,000 km^2. This limitation is not enforced as the SALT I treaty was never ratified, however it provides a useful starting point. For this model, I initially consider the TELs to be deployed only within these areas, though I evaluate the effects of relaxing this assumption in Chapter 6. Roughly half of the Russian missile bases are in the Moscow region, for which a 7 minute flight time will be assumed as they are within approximately 2000 km of potential submarine locations in the Barents and Baltic seas. The remaining half are near the Mongolian border, which is roughly 3000 km from potential attack locations in the Arctic and Pacific oceans, so will be assumed to require a flight time of 10 minutes for SLBMs on depressed trajectories [89]. Both areas are at approximately 55°N latitude. The deployed TELs will be assumed to be evenly split between deployment areas in those two regions.

Chinese Nuclear Forces. The second scenario considers an attack against China. China maintains a nuclear arsenal roughly an order of magnitude smaller than that of the U.S. or Russia. China is not party to nuclear arms control agreements so does not have formal limits on the size of its arsenal and there is concern that they are expanding the size of their arsenal [93,38]. Chinese mobile nuclear forces deploy strategic nuclear weapons on two models of TEL, the DF-31A and DF-31AG. There are assumed to be 24 of each launcher in operation, both of which carry a single 200-300-kt warhead [38].

Chinese operational area appears to be split between the Hunan province, roughly 1000 km from the South and East China seas, and Lanzhou province, roughly 2000 km from the coast [94]. Attacking these areas from the East or South China seas with depressed-trajectory SLBMs would result in missile flight times of roughly 7 minutes or less. Attacking from a safer distance in the Philippine Sea would require flight times of roughly 10 minutes. The Hunan province is at approximately 28°N latitude, and the Lanzhou area is at roughly 35°N latitude. The number of warheads available to the U.S. for an attack against the Chinese arsenal is not bounded by concerns of parity, as it is in the Russian case. Instead, I assume that the U.S. considers a range of warheads to allocate. American Ohio-class SSBNs are assumed to carry roughly 90 warheads each, so I consider attacks which commit all of the weapons from one or two SSBNs.

3.1.7 Summary of Counterforce Model

The counterforce model takes as inputs, the numbers of warheads and targets and the seeker's knowledge of the locations of the target TELs. The numbers of warheads and targets, and the delivery times of the weapons are drawn from representative real-world dyads. The seeker's knowledge of the TEL location, captured as a probability distribution function, is determined by the Tracking Model which is developed in the next chapter. Combining the seeker's uncertainty distribution of TEL position with a Warhead-Tiling Model and an assumption about the shape of the local road network, the probability of destroying a given TEL, as assessed by the seeker, can be estimated. Alternatively, the number of warheads which are required to meet or exceed some threshold probability of destruction of each TEL can be estimated. Given a set of TELs, the total number of required warheads can be compared to the number of warheads available.

Real-world dyads which are interesting in various policy contexts also present qualitatively different attacks. A U.S. attack on Russia models a disarming counterforce attack between peers, where parity in the number of warheads available to each side is more important, and factors which decrease the number of targets per warhead, such as MIRVing may affect stability. A U.S. attack on China models a scenario of a disarming strike where one side holds a strong numerical advantage but the target set is large enough that warhead efficiency is still a concern. The counterforce model can be thought of as a means to define a minimum performance requirement for a remote-sensing system. As the numerical imbalance decreases, the requirements of remote-sensing increases. With a given remote-sensing system, one or more dyads may remain stable while others may be undermined.

In practice, the counterforce model is the last model in the evaluation chain, as the remote-sensing model determines the technical characteristics which the tracking model converts into $p(r_u)$. Conceptually, however, the counterforce model defines the problem that the remote-sensing system is applied to.

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Chapter 4

Tracking Model

4.1 Remote-Sensing for Tracking

The purpose of a remote-sensing system is to limit the seeker's uncertainty in the positions of the target TELs. This occurs in the lead-up to a counterforce attack. The attacker must decide whether to attack or not before knowing the outcome of the attack. They must also make their decision based on their beliefs about the positions of the TELs. They are not able to assess the ground truth. Given the magnitude of the consequences of nuclear retaliation and the recognition of their own uncertainty, the attacker is incentivized to be conservative. Deterrence will hold unless the attacker convinces themselves that the costs of the attack are outweighed by the benefits. While the attacker's appetite for risk cannot be quantified in the abstract, determining the likely perceived risks associated with an attack is useful for qualitatively assessing the durability of deterrence in the face of remote-sensing.

In this chapter, I develop a model for how the attacker utilizes a remote-sensing system to manage their uncertainty. Recalling the tasks of tracking that I identified in Chapter 2, I in introduce Kalman filtering in subsection 4.1.1 as a framework for understanding how the seeker perceives their own uncertainty in the positions of the TELs they are tracking. In subsection 4.1.2, I present a model for how the capabilities of a modality-agnostic remote-sensing system can be used to estimate how the seeker's uncertainty in the TEL locations will develop over time. I then outline what constraints this conception of tracking places on how TELs should be operated by the hider in subsection 4.1.4. In subsection 4.1.5, I present a model for how the estimate of a rate of uncertainty growth from the environment model can be used to estimate the shape of the uncertainty distribution ($p(\mathbf{r}_u)$) over the entire fleet of TELs being tracked. This uncertainty distribution serves as an input to the previously discussed Counterforce Model in order to assess the probability of success of an attack in the forward solution. In the reverse solution of the model, the tracking model translates the requirements set by the counterforce model into requirements for a remote-sensing system.

4.1.1 Tracking with Kalman Filtering

Remote-sensing systems provide detection and identification information that must be synthesized into estimates of TEL locations over time. There are a wide array of tracking algorithms that could be applied to track association, such as joint probabilistic data association filters and particle filters that may have advantages in certain contexts. A relatively simple method for accomplishing the task of track association is with Kalman filtering. In practice, a more specialized filter would likely be required. To avoid having the uncertainty surrounding the algorithm negatively impact my ability to draw conclusions, in the following section, where I model how positional uncertainty grows during the tracking process, I utilize a generalized frequentist model that looks only at average and steady-state behavior. This approach allows for some estimates to be made on the efficacy of tracking without becoming overly dependent on an assumed tracking filter performance. Discussing Kalman filtering here is still useful, however, as it introduces how information gathered by remote-sensing systems is incorporated into a position estimate.

Kalman filtering is the optimal solution to the problem of state space estimation for linear systems with Gaussian noise [95]. In the context of TEL tracking, the state space is a model of the position and velocity of a TEL which can change over time. Both position and velocity are continuous variables but SBR observations are made at discrete intervals. Kalman filtering relies on a model of how TELs can move, coupled with Bayesian estimation in order to produce a position estimate from imperfect measurements.

In the case of tracking TELs, the state vector $\boldsymbol{x}$ includes the position and velocity of the TEL. The Kalman filter estimates the probability density function $p(\mathbf{x}_k|\mathbf{z}_{1:k})$ where $\mathbf{z}_{1:k}$ is total set of measurements of the TELs position up to the kth measurement. This is done through Bayesian estimation in two steps, prediction and update. The prediction step utilizes some assumed equations of motion of the TEL and the PDF from the $(k-1)^{\text{th}}$ observation to create the prior $p(\mathbf{x}_k|\mathbf{z}_{1:k-1})$. This is the prediction of the TELs position at the k^{th} observation. The equations of motion are the seeker's mental model of how a TEL is able to move. The update step computes the positional estimate $p(\mathbf{x}_k|\mathbf{z}_{1:k})$, the posterior, utilizing the likelihood function $p(\mathbf{z}_k|\mathbf{x}_k)$. The likelihood function computes the probability of measuring $\mathbf{z}_k$ for each given possible value of $\mathbf{x}_k$. The posterior generated by this process gives you a new estimate of the position of the TEL, created from a weighted average of the predicted position, given previous measurements of position and velocity, and the new information from the latest measurement. If the measurement $\mathbf{z}_k$ deviates significantly from the predicted value, it will be weighted less. This prevents fluctuations created by noise from drastically altering the estimate of $\mathbf{x}_k$. Through Kalman filtering, the seeker can estimate the position of a TEL with greater accuracy than a single sensor measurement can provide.

Consider a simple example of a TEL moving in a straight line on a flat tarmac. The seeker possesses a satellite-based sensor that can measure only the range to a target and its radial velocity. The seeker makes a measurement of the TEL z_0 at t_0 at a range r_0 with radial velocity v_{r0}, allowing them to construct the pdf of the state x_0. The uncertainty in the position of the TEL will be relatively large, as it could be anywhere in the arc of range r_0 from the satellite at t_0. Measuring only radial velocity, the estimate of the speed of the TEL v_0 will be bounded at the minimum by v_{r0} and at the maximum by the TEL top speed v_{TEL}. These constitute the estimate of the TEL state at t_0, which is a PDF $p(x_0|z_0)$ which contains j possible states each with probability $p(x_0 = x_j)$.

At t_1, the seeker makes another observation of the TEL. The satellite and the TEL will have moved during this time. Using the Kalman filtering framework, the seeker will make a prediction of where the TEL could be at t_1 based off of the observation made at t_0, $p(x_1|z_0)$. If the seeker assumes that the TEL is moving at a constant speed and constant direction, then the model of motion for the TEL is that it will have traveled a distance $d = (t_1 - t_0)v_0$ in the direction that it was previously traveling. The convolution of this model of motion and $p(x_0|z_0)$ creates the prior $p(x_1|z_0)$. After making the observation z_1, for each actualization of the state x_j in $p(x_1|z_0)$, the seeker applies the likelihood function $p(z_1|x_1 = x_j)$ to determine the probability that the measurement z_1 could have been created by the state x_j. This can allow the seeker to achieve better localization of the TEL than the resolution of the sensor, as there can be states in x_1 that could not be reached from x_0, based on the model of motion. The convolution of the likelihood function with the prior gives the posterior distribution $p(x_1|z_0, z_1)$. As the seeker makes more measurements of the TEL over time, the PDF will narrow to those locations consistent with the total set of measurements $\{z_{1:k}\}$, provided that the assumed model of motion is correct.[1]

A constant-speed and direction model is useful for illustration, but a more nuanced model of motion would be needed in practice. The seeker could apply knowledge of the road network to better localize TEL detections by assuming that TELs traveling above a certain speed must be on a road. This would also allow better estimation of heading, which would allow better estimate of speed from the radial velocity. The seeker would also have to account for the actions of the TEL driver however, which would complicate the evolution of states between measurements. A TEL driver traveling on a road can speed up, slow down, come to a stop and reverse direction, turn onto a different road, *etc.* The seeker must account for each of these possibilities, but can bound and weight the possibilities. If the seeker knows, or estimates, the maximum speed of the TEL, and its maximum acceleration, it can bound the maximum distance that it could have moved since the last observation. If the seeker can estimate the maximum braking and turning speeds of the TEL, it can also determine the maximum distance in the opposite direction that it could be after executing a U-turn. Based on observations of how the hider's TELs have operated in the past, the seeker can refine their estimates of maximum speed, acceleration, *etc.* Accounting for these processes will add complexity to the prediction step, but the process remains the same.

Further complexity is required to account for the possibility of missed detections. This can occur in two ways, from statistical noise in the sensor, or from non-stochastic processes such as sight-block. For example, the seeker may make an observation of the area where the TEL is positioned, but not detect it because of intervening terrain sight-blocking the measurement. In this case, information is still gained on the possible states of the TEL, as there are areas of the scene which are not occluded, allowing the seeker to determine in which locations the TEL is not. This can be captured in the likelihood function of the Kalman filter. If a measurement returns that the TEL is not detected in the scene, but the model of motion limit that the TEL could not have left the scene, then the TEL must be in an occluded state within the scene. While there may be many occluded areas in the scene, possible TEL locations in the posterior will be limited to those occlusions which could have been reached by the TEL from a position with non-zero probability in the prior. This also requires that the seeker build an accurate model of the terrain, vegetation, and buildings in the area in order to be able to know which areas would be occluded from a given view angle.

As an example, consider a TEL operating on a road in the forest. The trees of the forest occlude segments of the road from view from some angles. The seeker knows the location of the TEL well at the beginning of the example, so posterior for the TEL state from t_{-1} is relatively narrow. Assume that the seeker assumes a uniform distribution for possible TEL actions when creating the prior, and that the seeker's sensor has a P_D of unity. The TEL is not detected in the measurement z_0. The posterior at t_0 will be confined to areas that are occluded in z_0. If the TEL is not detected in the measurement z_1, then the posterior at t_1 will be only those locations occluded in z_1 that could be reached, according to the model of movement, from the areas occluded in z_0. Even though the seeker has failed to detect the TEL, they can be relatively confident in its location by observing where it is not provided that their underlying models and assumptions are correct.

The certainty with which the seeker can rule out a TEL from a position is limited by statistical noise, however.

[1] This example corresponds to the operation of a Doppler moving target indication radar, which I introduce in the following chapter.

Statistical noise causes random missed detections with a probability of $1 - P_D$. If a detection is missed, then every possible location in the prior will have non-zero probability in the posterior. If multiple detections are missed in a row, the distance bounding the area which has non-zero probability grows as $d_n = (t_j - t_{j-n}) \cdot v_{\text{TEL}}$. The probability of missing n detections from statistical noise alone is $(1-P_D)^n$, which can be accounted for in the likelihood function. For P_D close to unity, or with a short interval between observations, $(1 - P_D)^n$ will be small.

Using the Kalman filter for tracking, if the seeker is making successful sequential detections of a TEL, they can localize it to a finer resolution than sensor resolution alone can offer. This may be of limited use because nuclear weapons effects are large relative to the uncertainty that the seeker would have in the position of a continuously detected TEL. In order for the seeker's uncertainty to grow large enough to be comparable to nuclear weapons effects, and therefore increase the required number of warheads tasked to destroy it, detections must be missed. This creates situations where traditional Kalman filters (and other filters, for that matter) struggle to maintain an accurate track association. Through the remainder of this section, I introduce a probabilistic model, taking a frequentist approach, that allows for the expectation of positional uncertainty to be estimated under a steady-state.

4.1.2 Sight-Block Model

The TEL deployment environment is a major factor affecting the seeker's uncertainty in TEL position. The objects that comprise the environment can occlude parts of the seeker's observations, causing missed detections. This also limits the amount of information that in the update step of Kalman filtering. I will differentiate between two states of occlusion, occluded areas and shelters. Occluded areas are locations, such as road segments, that are transiently occluded by environmental features (*e.g.*, hills, trees, buildings, *etc.*) for a set of view angles. Shelters are locations which occlude for all observation angles. While the name shelter implies a fixed, purpose-built structure, the definition here is based on function. A natural ravine, a tunnel, an underpass, or even an area where a TEL could quickly deploy camouflage could serve as a shelter. Purpose-built structures, such as garages or barns are also shelters.

If a TEL is outside of a shelter, it may or may not be sight-blocked by occlusions. The seeker can make observations using a variety of look angles, and on a random schedule, in order to reduce the probability that a single location remains occluded for an extended period of time. Mounting remote sensors on moving platforms such as satellites or airframes can provide diversity in observation angle. If observation times are random and angles are varied, the hider will have difficulty in avoiding detection over an extended period of time outside of a shelter.[2] If the probability that a given location is sight-blocked by occluded areas in an observation is P_B, and the observations are varied in time and observation angle such that each observation can be considered independent, then the number of observations that a TEL would spend unobserved while out of a shelter can be modeled with a geometric distribution. The expected number of observations until the first detection is $1/(1 - P_B)$. So, if a TEL is traveling down a road for which half of the road segments are occluded in a given measurement, it will, on average, be observed once every two measurements. Depending on how closely spaced measurements are in time, this can create significant amounts of transient uncertainty in the TEL's location. However, since the uncertainty collapses at every successful detection, this makes for a poor strategy for remaining undetected for an extended period of time.

If the seeker was making observations at some frequency, the probability of successfully detecting the TEL in k

[2] A truly random schedule is nigh impossible for a space-based system to actually achieve given the constraints on the location of the sensor, and limitations on the agility of the sensor aperture. In this context, the seeker wants to prevent the hider from being able to time movements between observations, so true randomness is not required, only unpredictability.

measurements can be determined from the geometric distribution CDF

$$P(X \leq k) = 1 - P_B^k \ , \tag{4.1}$$

where $P(X \leq k)$ is the probability of successfully detecting the TEL in k or fewer measurements. For example, consider a very highly occluded area, where 90% of road segments are sight-blocked in any given measurement. Over a single day, with an observation once every twenty minutes, $k = 72$, and the probability of a TEL in the open remaining undetected is less than 0.05%.

Shelters are important as they provide locations for TELs to park and evade detection for extended periods of time. This gives TELs safe locations to move between where they can exist with no risk of detection, which allows them to preserve any accumulated positional uncertainty. To the seeker, who must account for all possible movements of the TEL, as per section 4.1.1, shelters present locations where missing TELs can reside. Occluded areas provide transient channels between shelters. The purpose of the uncertainty model is to estimate the rate at which uncertainty grows to include additional shelters.

Tree Sight-Block Model

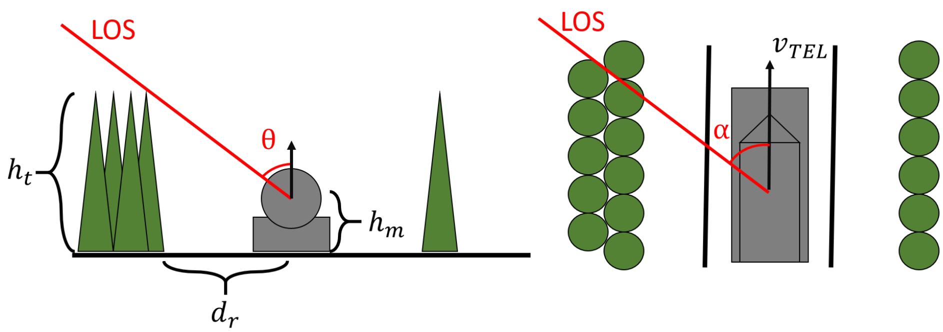

Figure 4-1: Schematic of sight-block by trees lining a road. The observable point of the TEL, the lowest point on the TEL which must be in view, is at a height h_m. Tree height is denoted as h_t, and the lateral displacement between the TEL and the tree line is d_r. TEL velocity is into the page. The aspect angle α is the angle between the TEL velocity and the line-of-sight (LOS) vector, as projected onto the local surface of the Earth. The maximum incidence angle at which a TEL can be viewed, $\theta_{\max}(\alpha)$ is the maximum angle, measured between the LOS and the local surface normal, at which the observable point is not intersected by the trees.

Figure 4-1 shows a schematic for how trees in a forest can cause transient occlusion of sight lines. The aspect angle α is the angle between the projection of the line-of-sight onto the Earth's (locally flat) surface and the heading of the TEL. The incidence angle θ is the angle between the line-of-sight vector and the local surface normal. The lowest point on the TEL which must be in view for the TEL to be detected by a SBR satellite is h_m. The tree-line runs parallel to the road with a height of h_t and a lateral distance d_r from the center of the road. The distance of closest approach between the center of the TEL and the edge of the tree-line is

$$d(\alpha) = \min\left\{\frac{d_r}{\sin\alpha}\ , d_h\right\}, \tag{4.2}$$

on $\alpha \sim (0, \pi/2]$, where d_h is the distance to the horizon. The minimum value of the two is taken to avoid the discontinuity when $\sin\alpha$ approaches zero.

Considering the geometry in figure 4-1, the maximum incidence angle $\theta_{\max}(\alpha)$ at which a TEL is visible to a SBR satellite is

$$\theta_{\max}(\alpha) = \frac{\pi}{2} - \tan^{-1}\left[\frac{h_t - h_m}{d(\alpha)}\right] . \tag{4.3}$$

If both the incidence and aspect angles of an observation are known, tree-sight block can be represented by a boolean variable W where

$$W(\alpha, \theta) = \begin{cases} 0, & \text{when } \theta > \theta_{\max}(\alpha) \\ 1, & \text{otherwise} \end{cases} , \tag{4.4}$$

where $W = 0$ indicates that the observation is sight-blocked by trees.

To illustrate the effects of sight-block from forest cover, the fraction of road segments that would be visible, as a function of θ can be estimated using equation 4.3. Inserting equation 4.2 into equation 4.3 and solving for α gives an expression for the maximum α at which a road segment would be visible at a given θ,

$$\alpha_{\max} = \sin^{-1}\left[\frac{d_r}{h_t - h_m}\tan\left(\frac{\pi}{2} - \theta\right)\right] . \tag{4.5}$$

The geometry is symmetric in terms of both direction of rotation about α, and direction of travel on the road, so α only needs to be evaluated on $(0, \pi/2]$. If the satellite orbits and road directions are, on average, randomly oriented relative to one another, then the fraction of road segments that would be visible (F_{Forest})at a given θ is the fraction of possible aspect angles from which a TEL would be visible,

$$F_{\text{Forest}} = \frac{\alpha_{\max}}{\pi/2} = \frac{2}{\pi}\sin^{-1}\left[\frac{d_r}{h_t - h_m}\tan\left(\frac{\pi}{2} - \theta\right)\right] . \tag{4.6}$$

Figure 4-2 shows the fraction of road segments on which a TEL would be visible to a SBR satellite given assuming isotropically random aspect angles and a 4 m tall TEL, roughly the height of a Russian Topol-M. It is assumed that at least one meter of the top of the TEL must be visible to the satellite for it to be able to detect the TEL, making $h_m = 3$m. The road is assumed to have two 4m-wide lanes with 1m shoulders.[3]

Terrain Sight-Block Model

Terrain features such as mountains, hills, and valleys can block line-of-sight in the same way that trees do, except the scales are larger. While there is no strict definition of a hill or mountain, they are common features that can rise hundreds, in the case of hills, or thousands of meters above the surrounding land. *Lieber and Press* utilized the available GIS data of North Korea to estimate the fraction of roads which were sight-blocked as a function of the incidence angle of the observation [9]. Figure 4-3 shows an approximation of their line of best fit. While terrain features are significantly taller than trees, sight-block is dominated by objects that are closer to the TEL.

Terrain sight-block in a given observation can be modeled as a random variable H where

[3] The tree-height is assumed to form a continuous wall with a 20m height. Russian forests in the areas surrounding their TEL bases largely consist of various species of larch, pine, spruce, fir, and birch, and are mostly middle-aged to mature trees [96]. Mature specimens *Larix sibirica*, Siberian larch, the most common species of tree in Siberian forests range between 20–50m in height. Other common species include Siberian Spruce, which ranges from 15–35m, Siberian Pine (30–40m), Siberian fir (30–35m), and Scots pine which grows up to 35 m. Since tree-tops may not fully block line-of-sight, I utilize 20m height as an estimate for the height to which sight will be fully blocked.

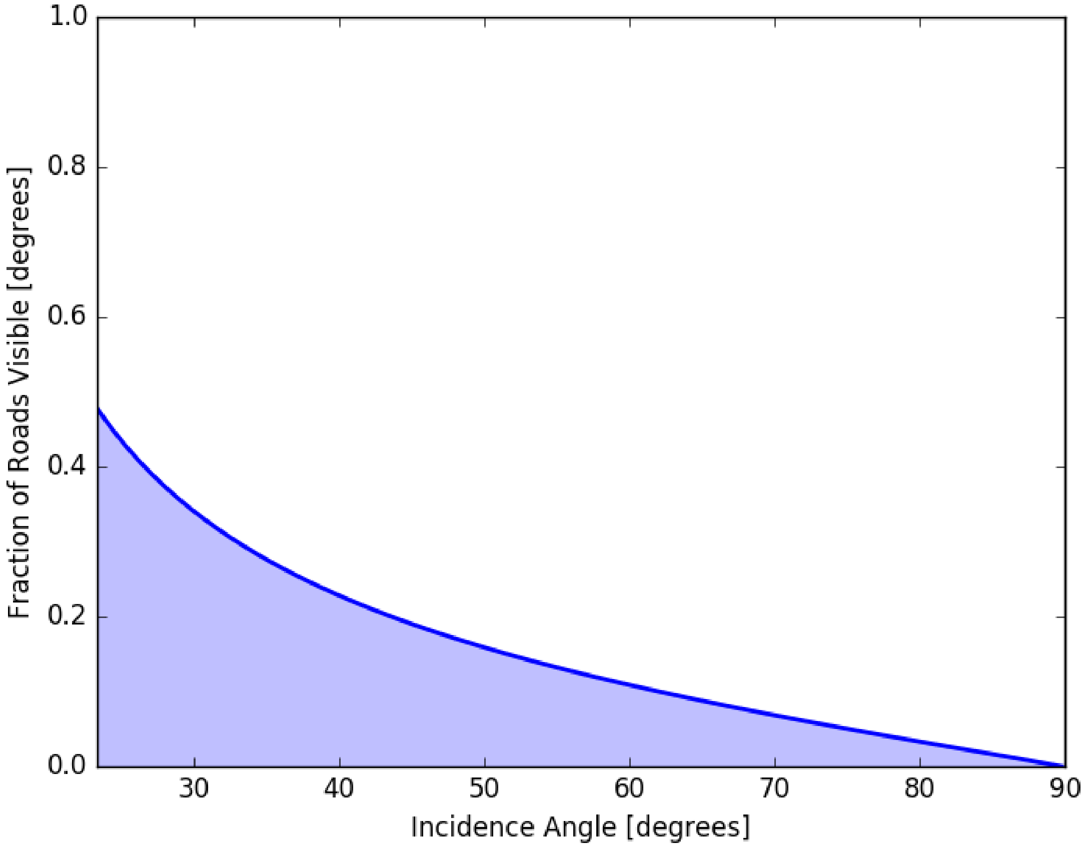

Figure 4-2: The fraction of road segments on which a TEL would visible to a SBR satellite for a 4m tall TEL driving on a 10 m wide road lined by 20 m tall trees. The shaded area indicates the visible fraction. I assume the aspect angle between the line-of-sight and the TEL heading is random and isotropic. The lower limit of the plot, 22 degrees, is the incidence angle is set by the nadir hole.

$$H = \begin{cases} 0, & \text{if } x \leq P_B(\theta) \text{ where x} \sim U(0,1) \\ 1, & \text{otherwise} \end{cases}, \tag{4.7}$$

where $H = 0$ indicates that the observation is sight-blocked, the parameter $P_B(\theta)$ is the probability that a road segment is visible at a given θ (the function shown in figure 4-3), θ is the incidence angle, and $U(0, 1)$ is a standard uniform distribution. The lack of visibility when $0 \leq \theta \leq 20$ is an effect of the nadir hole which effects radar satellites. Very strong specular returns off of the Earth's surface cause saturation of the receiver, making radar satellites effectively unable to "look straight down". This phenomemon would not effect optical imaging systems.

4.1.3 SBR Modalities and Information Balance

As an active sensing modality, SBR operates by illuminating the beam footprint with energy. This affects the expected balance of information between the hider and the seeker. If a TEL is equipped with a radar antenna, and is monitoring the frequencies used by SBR systems, it could detect if it were observed in real time. Due to the great distances

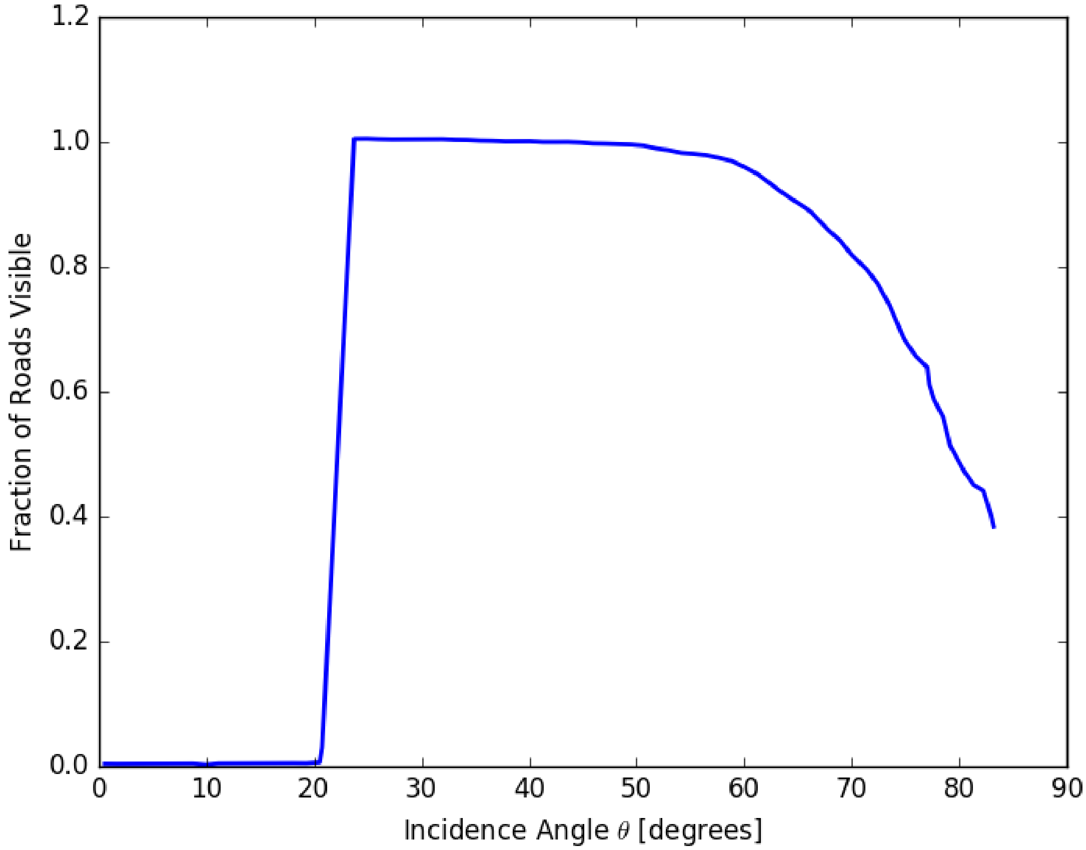

Figure 4-3: Fraction of North Korean roads visible to an observation from a satellite with an altitude of 700 km when only considering terrain sight-block. Estimated from [9].

involved, detecting observations requires significantly less sophisticated hardware than does making an observation. SBR systems measure radar echoes, so the illuminating field is a factor of $4\pi R^2\sigma^{-1}$ stronger than the echo at the SBR receiver. Since R in this case is on the order of 10^6 m, and σ is on the order of 10 m^2, the illuminating radio energy is 12 orders of magnitude higher at the TEL than the echo at the radar receiver. This provides considerable leeway for the TEL operator to use antennas with small apertures and inexpensive, noisy receivers and still be able to detect illumination. Resultantly, the hider can determine when they have been observed. The primary consequence is that while the actual locations of the TELs are known only to the hider, the seeker's knowledge of possible TEL detections is shared information.

Beyond detecting when a TEL has been observed, the hider can also learn information about how the seeker operates their radar system. By inferring values for the variables in the radar equation 4.8,

$$S = \frac{P_t G_t \sigma A}{4\pi R^2 \cdot 4\pi R^2} \ , \tag{4.8}$$

the hider can estimate the performance of the system. Optical or radar measurements of the spacecraft could reveal the size of the radar aperture A, as well as the size of the solar panels which power the craft.[4] Carrier

[4]For example, Russia has recently deployed a set of "inspector" satellites whose stated purpose is to image Russian satellites

wavelength λ can be measured directly from the incident field. Knowing the size of the aperture and the carrier wavelength, the gain of the aperture G_t can be estimated. Measuring the strength of the incident field of the beam on the ground, the transmit power P_t can be inferred as the one-way transmission loss $1/4\pi R^2$ can be calculated with knowledge of the SBR satellite orbit. Together, these values would allow the hider to determine S, the signal measured by the seeker when observing a TEL with cross-section σ. Conservative estimates of system losses and receiver noise would further allow the hider to estimate the SNR of any given measurement, which indicates the probability of detection of that measurement. Placing receivers in TEL deployment zones (such as on shelters) would allow the hider to determine τ_{rev} of each modality. By estimating the technical capability, and manner of operation of the SBR system, the hider would be able to estimate the seeker's uncertainty in TEL positions and assess whether TELs were vulnerable or not.

One consideration is that, due to this imbalance, the hider can adjust their TEL $p_{\min}$ in response to being detected or their assessment of the state of the seeker's uncertainty. In the normal run of operations, the seeker already hedges against this possibility by assuming the worst-case uncertainty distribution of TEL position at any given time. Additionally, since the seeker constructs their uncertainty as a CMF, determining the probability that the TEL has made *fewer* than m moves, given that no detections have been made, the possibility that the TEL does not move when presented the opportunity is also accounted for. So, while the hider does have the opportunity to adjust operations on-the-fly, doing so should not affect the seeker's uncertainty. I do explore one method of exploiting the imbalance in information to temporarily increase the seeker's uncertainty during a crisis by reducing the rate of detection in Chapter 6.

4.1.4 TEL Operations to Defeat Tracking

In addition to considering the physical effects of the TEL environment on remote-sensing, the seeker must also account for how the hider may alter their behavior to exploit environmental effects. The hider is not prescient so can only make use of information that is available to them about the seeker's remote-sensing system. The information available will depend on the modality being used.

For satellite-based sensors, the hider can determine sensor view angles from satellites orbits. Satellites are capable of maneuvering to change their orbits, but at the cost of fuel which is necessary for station keeping, corrections performed to keep the satellite in its orbit. After a maneuver, new measurements of the position and velocity of the satellite would still allow predictions of its new trajectory. This allows the hider to determine the current and probable future positions of all satellites in the seeker's constellation. Information for hiding from satellites could be disseminated to the TELs by radio broadcast.

The goal of the hider is to instill uncertainty in the seeker. Importantly, the seeker's uncertainty will grow if a TEL is undetected whether the TEL moves or not; it is the possibility of movement that creates uncertainty. Actually moving a TEL invites risk, as the seeker's uncertainty in that TELs position will begin to collapse down to the true location of the TEL if it is detected while moving. Taken to the extreme, the hider can choose never to move a TEL from a shelter, shielding that TEL from ever being detected. During the time that the TEL is in the shelter, the

for damage for maintenance purposes [97]. In practice, a sensor suite capable of that task can be utilized to inspect the satellites of others. These satellites have been detected "stalking" U.S. reconnaissance satellites with orbits designed to obtain a diverse set of aspect angles to observe the target satellite while on the sunward side of the Earth. Additionally, these satellites have been observed conducted a suspected non-destructive anti-satellite weapon test(anti-satellite weapons are discussed in Chapter 6). China also has an extensive space surveillance network of its own [98].

seeker's uncertainty about its position will grow as opportunities for the TEL to move have occurred.

The limit of this strategy is that the seeker can learn the hider's behaviors over time. I assume that a TEL, at least during peacetime, will have a finite amount of time that it can spend away from its base. I refer to this as the patrol lifetime which has some length L. When a TEL begins a patrol, it leaves its base of operations and moves to a shelter where it will begin to evade detection and grow the seeker's uncertainty in its position. At the end of the patrol lifetime, the TEL will need to return to its base in order to resupply, change crew, *etc.* If the seeker can monitor the TEL base to detect when a TEL begins its patrol, they can determine what shelter the TEL begins is deployment from. When the TEL ends its patrol, the seeker has the opportunity to detect which shelter it ended its patrol in.[5] This gives the seeker a means to evaluate which TELs, if any, the hider chose to move. If the seeker detects a TEL leaving the same shelter that it deployed to, the seeker can assume that the TEL never moved. The seeker can use this information to construct their model of TEL movement that they incorporate into their tracking Kalman filter.

If the seeker becomes convinced that the hider is not moving TELs, the seeker could assume that if they saw a TEL enter a shelter, it remains in that shelter. If the seeker holds this belief, their perceived uncertainty in the position of an undetected TEL will grow slowly or not at all. If the belief is mistaken, then the seeker may put themselves in a situation where they launch an attack that they perceive will be effective, but could end up being unsuccessful. This would be detrimental to the goals of both the hider and the seeker. The seeker therefore has an incentive to be conservative and respect the ability of the hider's TELs to move undetected. The hider has an incentive force the seeker to adopt this position by moving at least some of their TELs some of the time, despite the risk of detection.

If the hider must move some TELs, they will want to do so in such a way as to maximize the seeker's uncertainty growth. This requires minimizing the number of successful detections that the seeker is able to make. To maximize the chance that a TEL can move from one shelter to the next undetected, a TEL driver should move at maximum speed between the two, preferably at a time when the satellites which are in view of the route are at a low angle in the sky, thus are operating at a large incidence angle (see figure 4-1). Moving when satellites are operating at a large incidence angle increases the probability of sight-block, and moving at top speed reduces the number of possible observations.

The hider's incentives are to be as conservative as possible, minimizing the risk of detection of their TELs, without becoming predictable. Since satellite orbits are predictable, the hider is able to estimate the probability that they are able to move from one shelter to the next without being detect, when leaving at a given time. The model I present in the following section outlines how the probability of observation of a given TEL movement can be estimated. The hider can use this information to balance the frequency with opportunities to move arise and the probability of success of those movements to maximize the seeker's uncertainty growth.

[5] A TEL could attempt to deploy to its initial shelter, or return to its base at the end of a patrol without being detected. As I will show later in this section, the safest way to operate a TEL is to move it some minimum necessary distance in the least risky manner which is also the slowest. If a TEL has some finite number of moves that it can make during its deployment lifetime, it could choose to center that movement on its initial base. If it instead centered that movement on some distant shelter, it could grow its uncertainty to the same amount, the only loss being the time taken to travel to and from the shelter. If multiple TELs attempt to center their evasive maneuvers at their base of origin, their areas of uncertainty may begin to overlap, which will improve an attacker's warhead efficiency. It is therefore better to disperse patrolling TELs throughout a deployment area.

4.1.5 A Model of Seeker Uncertainty

In this subsection, I present a simulation for estimating the rate at which the seeker's uncertainty grows over time, as a function of remote-sensing capabilities, the environment, and how the hider operates their TELs. An important note is that this model describes the seekers beliefs in the possible locations of the tracked TEL, given the information available from remote-sensing. Unless the seeker is repeatedly detecting the TEL, the seeker does not have enough information to determine what the TEL is actually doing. Instead, they must form their beliefs on what the TEL could be doing. As such, the seeker's uncertainty is the set of possible TEL locations that are consistent with the information available from remote-sensing. The actual location of the TEL will be some location within that larger set of possibilities.

Since shelters are the only locations where TELs can remain undetected for an extended period of time, the seeker's long-term uncertainty in the position of a TEL will only grow if the TEL has a non-zero chance of reaching the next shelter without being detected. Uncertainty grows continuously in the short term, but since Kalman filtering can make use of information in negative detections to rule out areas where TELs are not, the seeker can eventually rule out all non-shelter locations over time. Thus, the long-term positional uncertainty grows only as it encompasses new shelters, in increments of the inter-shelter spacing s.

The hider needs to move some of their TELs but should choose to do so at the times when they have the highest chance of moving undetected. The simulation presented below estimates the probability that a TEL can make it to the next shelter without being detected if it leaves its shelter in some window of time. These probabilities can be estimated by the hider and the seeker in advance since both can have knowledge of projected satellite positions and the shape of the local road network and sight-blocking features. The model developed here is a Monte Carlo simulation that estimates the rate $R(p_{\min})$ that windows with probability $p_{\min}$ of moving undetected to the next shelter are expected to arise.

Simulating the Sky

The simulation includes a component to determine the view-angles from which satellites could observe a moving TEL. The simulation is initialized by populating a Walker Constellation of remote-sensing satellites.[6] Initial shelter locations for TELs are placed at a set of latitudes, as the coverage density varies as a function of latitude.[7] Starting at $t = 0$, satellite and the Earth's motion is evolved with some time resolution, in this case $t_{\text{res}} = 1\text{s}$. At each time step, satellites move and the Earth rotates. Satellites are advanced in their orbits through an arc of length $v_{\text{sat}}t_{\text{res}}$. The Earth rotates $2\pi t_{\text{res}}/\tau_{\text{day}}$. The rate of motion of the TELs is very small compared to those of the Earth and the satellites, so their motion relative to the Earth is neglected. The simulation is evolved for a length of time, $t_{\text{sim}} = 1$ day. The position history of each point of origin and every satellite is calculated in this way.

For each point of origin, for each time step, the position of every satellite in the overhead sky is modeled utilizing these position histories. For each satellite in range of the point of origin, the range, incidence angle, squint angle, and aspect angle are calculated. This information can be used to create a time-history of the sky from the point of view of a TEL which includes the elevation, range, and aspect angle of each line-of-sight between the TEL and a

[6] Walker constellations are discussed in more detail in Chapter 5. In short, they are constellations of circular orbits designed to maximize coverage for a given number of satellites.

[7] The chosen latitudes are 10 degree increments spanning $[30, 60]$. These latitudes span the south of China to northern Russia, which encompass the areas where TELs are deployed in the real-world dyads considered.

satellite. The range and squint angles are used to determine if the TEL is within the field-of-regard of each satellite at each time point. If the area being simulated is forested, the aspect and incidence angles are used with equation 4.5 to determine if the observation is sight-blocked by trees. Tree sight-block depends only on the geometry of the observation and the road heading, so is not stochastic (though the road heading is chosen stochastically). If the area being simulated is hilly, the incidence angle is used with the terrain sight-block model to determine the probability of terrain sight-block. For systems effected by weather, such as optical imaging, the simulation may be applied only to those situations where the weather is permissive of tracking.

Calculating $R(p)$

Traveling from one shelter to the next requires a window of time $t_w = s/v_{\mathrm{TEL}}$. If a TEL leaves a shelter at time t, the probability that it could travel in the open for t_w without being detected ($p_w(t)$) can be estimated utilizing the Sight Block model, modality-specific probability-of-detection models,[8] and the frequency of observation.

For each run, a starting time point is randomly selected from the first satellite orbit period of the simulation. The heading to the next shelter at random from a uniform distribution as the direction of any particular road cannot be presumed by the seeker. The road is assumed to be straight for the entire length s. For each time step, the number of satellites in view is determined from the simulated position history. Since the TEL motion is small relative to the rotation of the Earth and the orbiting of the constellation, the TEL location can be approximated by only considering the rotation of the Earth. With this assumption, the sky seen by a traveling TEL is equivalent to a time window in the simulation of length t_w. The total number of potential observations (n_{obs}), where a satellite has a clear line-of-sight to the road segment is summed over the time window at the resolution of t_{res}.

Since the seeker randomizes the observation schedule, as long as $t_{\mathrm{res}} << 1/\tau_{\mathrm{rev}}$, the probability that any given satellite in view makes an observation at any given time step can be modeled as

$$p_{\mathrm{obs},i} = \frac{t_{\mathrm{res}}}{\tau_{\mathrm{rev,i}}} \ , \tag{4.9}$$

where the index i denotes different modalities, and $\tau_{\mathrm{rev,i}}$ is the average time between observations with the i^{th} modality, defined on a per-satellite basis. For example, if m satellites were in view, each operating a single remote-sensing modality with average revisit intervals of τ_{rev}, the TEL would experience an effective revisit interval of observation of $\tau_{\mathrm{rev,i}}/m$. The probability that the TEL is able to traverse the road without being viewed by a given modality, if it left at time t is

$$p(t) = \prod_{j=1}^{n_{\mathrm{obs}}} \left(1 - P_{O,j} p_{\mathrm{obs},j}\right) \ , \tag{4.10}$$

where $P_{O,j}$ is the probability that a TEL would be detected in the jth observation if an observation were made, and $p_{\mathrm{obs},j}$ is the probability that the TEL makes an observation at the jth opportunity.

Repeating this process for every window of t_w in a run produces a vector of the probability of success if the TEL left its shelter in a period of time of width t_{res}, which is small, in this case, one second. To represent a period of time more manageable by a TEL crew to time their movements down to, a moving average of the probability is calculated over a window of length $\tau_w = 30$s. For each run, the number of windows where $p_w(t) \geq p_{\min}$, the minimum

[8] Models for space-based radar systems are presented in Chapter 5.

acceptable probability at which the hider is willing to move a TEL, is tallied.[9] The total tallied number of successful windows is averaged over the total number of runs to estimate the average rate at which a window exists for $p_{\min}$. Evaluating a range of $p_{\min}$ allows the construction of an $R(p)$ curve, the rate of occurrence of opportunities to move as a function of $p_{\min}$. By selecting a $p_{\min}$, the hider able to estimate how often opportunities to move will arrive. This curve describes the interaction between a given remote-sensing system and an environment and provides the relevant information for modeling how the seeker's uncertainty will behave over time.

Figure 4-4 shows an example $R(p_{\min})$ surface, as a function of both τ_{rev} and $p_{\min}$ for an example constellation of SBR satellites.[10] Figure 4-5 shows projections of that surface onto the τ_{rev} and $p_{\min}$ axes, respectively. For all $p_{\min}$ the $R(p_{\min})$ increases with increasing revisit interval. Similarly, $R(p_{\min})$ decreases with increasing $p_{\min}$, though the slope is less steep for smaller τ_{rev}. This is because at very short τ_{rev}, TELs spending any appreciable time in a location where they can be detected will be detected. Under these circumstances, windows occur due to gaps in coverage or where sight is reliably blocked for the entire duration that it takes to move a TEL.

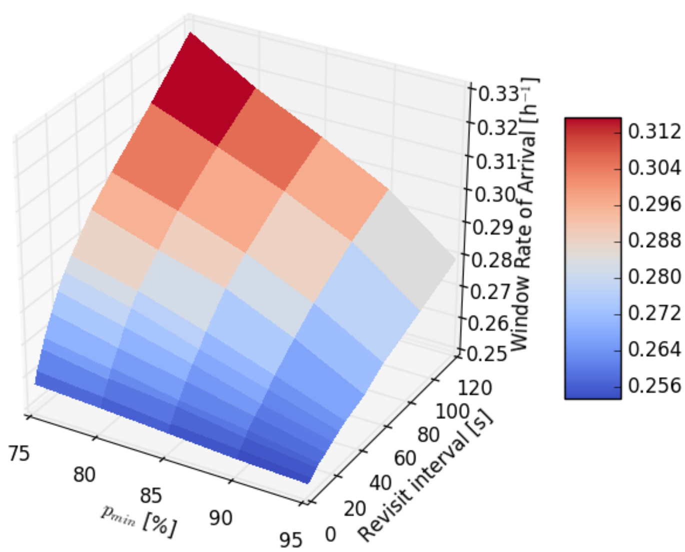

Figure 4-4: An example $R(p_{\min})$ surface, as a function of τ_{rev} and $p_{\min}$ for an SBR constellation with configuration 55:40/10/5 and a target at 55° N. Surface shows the mean value of three Monte Carlo simulations, each with 1000 runs each simulating 1/5 of a day for a straight road with a randomly selected heading. (Figure 4-5 shows projections with error bars.)

4.1.6 Evolution of Uncertainty Over Time

Hider Strategy and Moves Since Last Detection The hider must select a strategy for how they will move their TELs, defined by the parameter $p_{\min}$, the minimum probability of moving undetected at which they will move their TELs.

[9] If a window of length $n\tau_w$ exists, it is tallied floor(n) times where floor(n) is an integer.

[10] The constellation is a Walker Constellation with configuration 55:40/10/5, a 40m^2 aperture with 10 kW peak power at 1000 km altitude with a target TEL at 55° N latitude. Further details on how these parameters affect $R(p_{\min})$ can be found in Chapter 5.

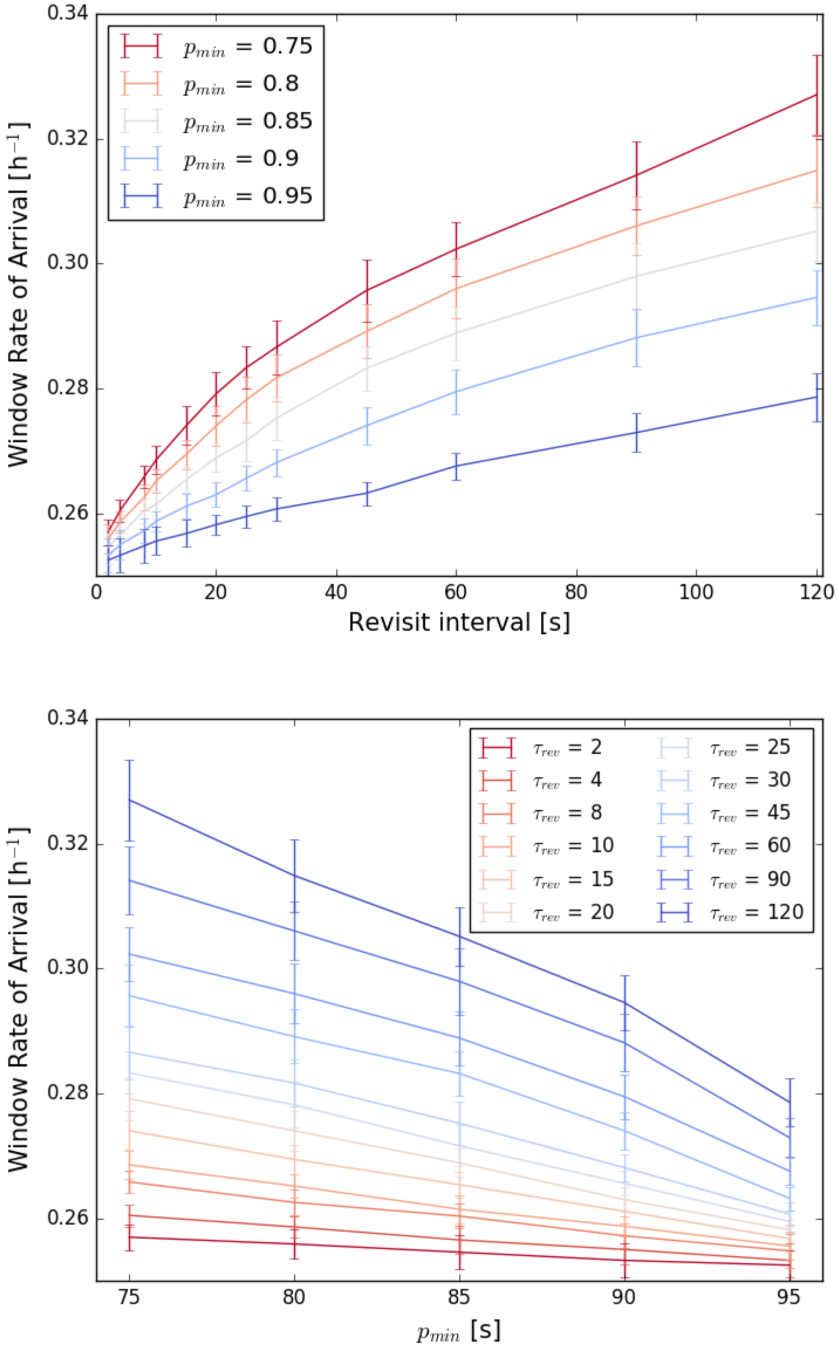

Figure 4-5: Curves of $R(\tau_{\text{rev}})$ and $R(p_{\text{min}})$, repectively, projections of the surface in figure 4-4. Error bars show the standard deviation of the means from three separate Monte Carlo simulations, each with 1000 runs each simulating 1/5 of a day for a straight road with a randomly selected heading at 55° N.

The hider's goal is to maximize the seeker's uncertainty in the position of the TEL. The hider can do this by signalling to the seeker that they are able to move their TELs without being detected, which creates an incentive for the hider to move their TELs. When a TEL moves, it will have some probability p_w of reaching the next shelter without being detection, and a complementary probability $1-p_w$ of being detected while moving. Each time a TEL is detected, the seeker's uncertainty collapses down to the true location of the TEL, decreasing the seeker's uncertainty, presenting a cost associated with moving a TEL. Thus, the hider must select a $p_{\min}$ which presents enough opportunities to move in order to be able to signal their capability to move undetected while minimizing the probability that their TELs are detected while doing so.

The rate-of-arrival rate simulated in the previous section, $R(p_{\min})$, provides the rate at which opporunities to move arise. In selecting a minimum acceptable probability of moving $p_{\min}$, the hider can expect an opportunity to move once every $1/R(p_{\min})$ on average. For simplicity, I assume that the p_w of each of these windows is exactly $p_{\min}$, even though the calculation of $R(p)$ calculates the rate for which $p \geq p_{\min}$.[11] The i^{th} TEL must begin its patrol at some time $t_{\text{deploy},i}$, and end its patrol at the time $t_L = t_{\text{deploy},i} + L$. Between those two times, the seeker's uncertainty in its location will evolve as opportunities to move arise.

If a TEL moves at every opportunity where $p_w \geq p_{\min}$, then the sequence of moves can be modeled with a Bernoulli process. A Bernoulli process is a sequence of discrete random variables which take a binary value with some probability. In this case, a trial will be represented by the random variable X which takes the values

$$X = \begin{cases} 1, & \text{if } x \leq p_{\min} \\ 0 & \text{otherwise} \end{cases} , \tag{4.11}$$

where $x = U[0,1)$ is a standard uniform random variable. The outcome $X = 1$ represents the TEL moving to the next shelter without being detected while the outcome $X = 0$ represents the TEL being detected.

For the i^{th} TEL, the number of possibles moves which could have been made at an arbitrary point in time t is, on average,

$$k = \text{floor}\left\{R(p_{(min)})(t - t_{\text{deploy},i})\right\} . \tag{4.12}$$

At time t, k random variables will comprise the Bernoulli process for a given TEL. The sequence of Bernoulli trials can be used to determine the number of moves since last detection for the TEL. The moves since last detection, can be determined by counting the number of successes (moves without detection) starting at the end of the sequence and terminating at the first failure. This is equivalent to the moves until next failure in the forward direction. Since the probabilities are assumed to be fixed, a probability mass function for the number of moves since last detection can be formulated.

I define a random variable $M_i(p_{\min}, k)$ which is the number of moves that the i^{th} TEL has made since it was last detected, assuming it is moving at every opportunity. If there have been k opportunities to move since it began its patrol, the PMF for M_i is

$$P(m, p_{\min}, k) = \begin{cases} p_{\min}^m & \text{for } m = k \\ (1 - p_{\min})p_{\min}^m & \text{for } m \in [0, k) \\ 0 & \text{otherwise} \end{cases} . \tag{4.13}$$

[11] This simplification will underestimate the spread of the seeker's uncertainty. However, the error will be largest for strategies where $p_{\min}$ is small, *i.e.* the relative error in treating a pair of moves with p_w of 0.95, and 0.99 as 0.95^2 is smaller than the relative error in treating windows with p_w of 0.5 and 0.99 as 0.5^2. As I show below, strategies with a small $p_{\min}$ quickly become irrelevant. As such, the error is smallest in the cases which matter most.

Figure 4-6 shows such a distribution for the case where $p_{\min} = 0.95$ and $k = 35$.

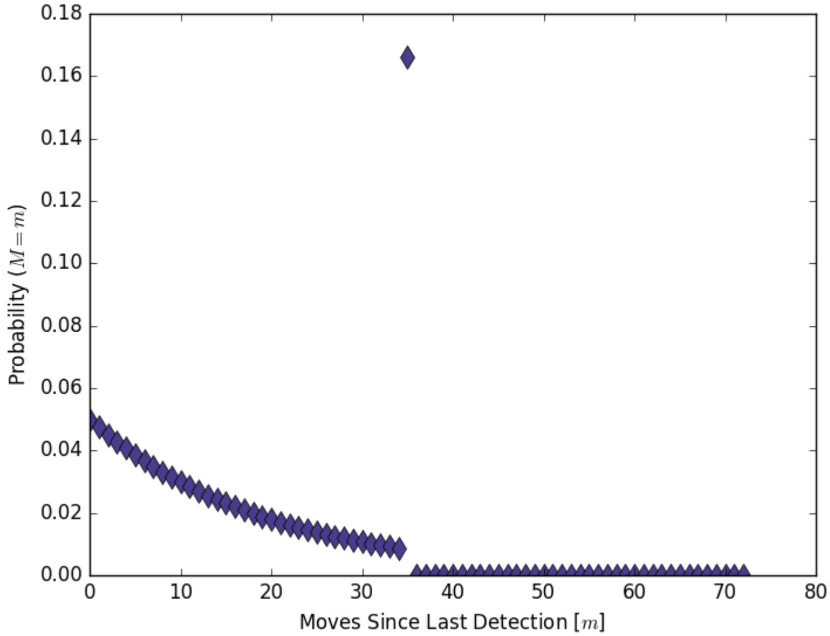

Figure 4-6: Distribution of $M_i = j$ for $p_{\min} = 0.95$ and $k = 35$.

This distribution is necessary in order to account for the possibility that the seeker detects a TEL while it is moving. The value of k will be determined for each TEL, based on how long it has been deployed. A distribution for k is presented below in section 4.1.6. The hider seeks to maximize the expected value of $M_i(p_{\min}, k)$ for each of their TELs by selecting a $p_{\min}$.

In a given scenario, when the seeker decides whether to attack, there will have been a specific history for each of the target TELs, meaning each $M_i(p_{\min}, k)$ will have be actualized. Therefore, the seeker's uncertainty will be based on knowing that $M_i = m_i$ for each TEL. Thus, the PMF above describes the likelihood of a number of different, discrete scenarios, each of which the hider must expect with some probability. This is distinct from the seeker's uncertainty in the position of the TEL which is described below (though the two have similarities in how they are derived).

Determining the Positional Uncertainty Distribution. After making the last detection of a TEL, the seeker loses access to some information on what the target TEL is doing. The TEL could stop, move at the $p_{\min}$ with which it had previously been moving, or select a new $p_{\min}$. The seeker does not know what $p_{\min}$ a TEL is using at a given time.[12] The only information that the seeker has is the location and time of the last detection, and the knowledge that the TEL has not been detected since. This second point, which may seem trivial on the surface, allows the seeker to weight their beliefs on the location of the TEL. In other words, the seeker is able to rule out locations where they are confident that the TEL would have been detected if it had been there.

The seeker must consider every $p_{\min}$, though, as I shortly show, the smaller $p_{\min}$ strategies quickly become

[12]Over time, if the a TEL, or set of TELs, only moved with a single $p_{\min}$, the seeker could estimate the $p_{\min}$ that the hider was using. This would be limited by the hider's ability to change $p_{\min}$ over time.

irrelevant. When evaluating a strategy, the seeker must consider how many moves between shelters j that the TEL could have made with that strategy since the time it was last detected as

$$j(p_{\min}) = \text{floor}\left\{R(p_{\min})(t - t_{\text{detected},i})\right\} \ , \tag{4.14}$$

where $t_{\text{detected},i}$ is the time when the i^{th} TEL was last detected by the seeker. This time point is always equal to, or more recent than, the $t_{\text{deploy},i}$ for that TEL. The number of moves j allows the seeker to weight the possible positions of the TEL at time t given that the TEL has not been detected with the cumulative probability mass function

$$P(x < m) = \begin{cases} 1 & \text{for } m > j \\ 1 - p_{\min}^{m} & \text{for } m \in [1, j] \text{ and } j \geq 1 \end{cases} \ , \tag{4.15}$$

where $P(x < m)$ is probability that the TEL has made fewer than m moves. The cumulative mass function (CMF) represents the probability that a TEL has made fewer than m moves, given that the seeker has not detected it. The CMF must be formulated as an inequality as the hider has the option of stopping at any point. The function is undefined for $m = 0$ as it is impossible to make fewer than zero moves. In the special case where $j = 0$, the probability that the TEL is in the 0th shelter is unity.

The cumulative probability mass function in equation 4.15 can be defined for each strategy $p_{\min}$. Since the seeker does not know which strategy the hider is utilizing, the seeker must account for all possibilities, and, in order to be conservative, accept the worst-case distribution at the time of attack. In order to compare distributions between each other, the seeker can utilize the counterforce model. For example, with $p_{\min} = 0.5$ and $j = 3$ and a TEL on a one-directional road, the probability that the TEL could move to the third shelter is $p^3 = 0.125$. Given that the seeker has made no detections, the probability that the TEL has made fewer than 3 moves is $1 - p^3 = 0.875$. The seeker can interpret this to mean that if they allocate enough warheads to destroy the 0th, first, and second shelters, they will have an 87.5% chance of destroying the TEL (assuming unity for p_k of the weapons). Evaluating a different distribution with a different $p_{\min}$ and therefore j would produce a different CMF.

The uncertainty CMF in equation 4.15 links to the counterforce model introduced earlier in this chapter. Since the distribution is one-dimensional, it must be overlaid with a road network in order to determine the area of uncertainty and the function that relates the number of allocated warheads to the probability of kill. Comparisons can be made either through the number of warheads required to achieve some desired kill probability, or through the kill probability given some number of available warheads.

In order to keep notation clean, I define the conservative uncertainty distribution U as

$$U = \min_{0 \leq p_{\min} \leq 1} \left\{CF[u(p_{\min}, t), n_w]\right\} \ , \tag{4.16}$$

where $CF(u, n_w)$ is the Counterforce Model that returns the probability of kill for a target TEL with n_w warheads allocated to it and a CMF represented by $u(p_{\min}, t)$. The counterforce model relates the number of warheads to the probability of kill for the target, given some assumed road network. By choosing the uncertainty distribution which produces the lowest p_k for a given n_w, U represents the worst-case CMF for the seeker *vis-a-vis* conducting a counterforce attack at time t.

From the CMF in equation 4.15, I can draw some conclusions about differing strategies and how they contribute to U. Figure 4-7 shows an example of how the CMF evolves over time for a number of $p_{\min}$, assuming that the seeker makes no detections. The j for low $p_{\min}$ strategies will grow quickly as $R(p)$ tends to be large when p is small. However, the CMF will very quickly approach unity at relatively small m because p is small. Conversely, high $p_{\min}$ strategies will have broad PDFs with thick tails that will grow slowly as $R(p)$ will be smaller. This means when $(t - t_{\text{detected},i})$ is small, the lower $p_{\min}$ will dominate but will remain largely centered on the point of last detection. As time passes, the larger $p_{\min}$ strategies will begin to dominate. If $s << r_w$, the shelter spacing is small relative to the warhead radii, then in some situations, the lower $p_{\min}$ strategies can be disregarded altogether as their contribution may be limited to an area that can be accounted for with a relatively small number of warheads.

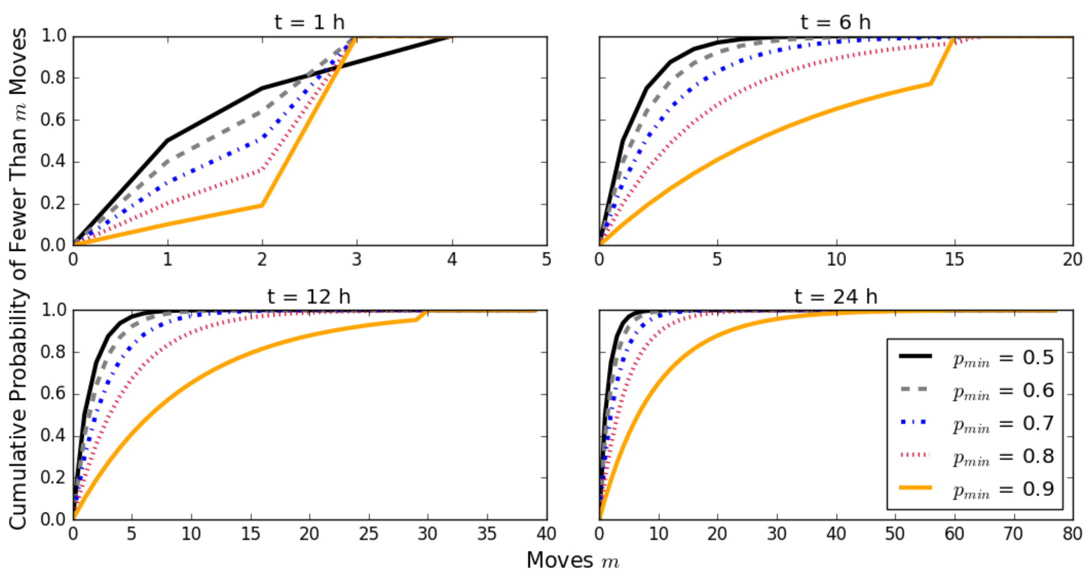

Figure 4-7: An example of CMF evolution over time for a number of different $p_{\min}$. The four subplots show the CMFs for each $p_{\min}$ at different time points. The j for each $p_{\min}$ will depend on the $R(p_{\min})$ and the elapsed time. When $m > j(p_{\min})$, the CMF goes to unity, causing a step in the CMF. The $R(p_{\min})$ curves utilized in this plot were created for a SBR constellation with $n_{\text{sat}} = 20$, ranging from 3.2 windows per hour for $p_{\min} = 0.5$ to 2.4 windows per hour for $p_{\min} = 0.9$. These values are used only as an example, evaluation of the $R(p_{\min})$ curves for SBR constellations is discussed in more detail in the next chapter.

Steady State Uncertainty

The positional uncertainty distribution U, when combined with the counterforce model, provides a conservative estimate for the likelihood of destroying a given TEL as a function of the time since it was last detected. The hider strategy $p_{\min}$ provides a means for estimating the expected time-since-last-detection for a TEL as a function of how long it has been deployed. The seeker, when evaluating whether they can successfully conduct a counterforce attack needs to evaluate their ability to destroy all of the TELs in the target fleet. By making an assumption about how TEL deployments are distributed in time, I can estimate what the steady-state distribution of uncertainties will look like for a fleet of TELs.

For a TEL deployment area there will be some number of TELs, n_{TEL}, that are on patrol at a given time. The only constraint that I place on n_{TEL} is that it is less than or equal to the number of TELs which are based in that deployment area. The fraction of TELs which are deployed will likely vary with international tensions, with more TELs being placed on patrol during periods of heightened tension. The steady state uncertainty distribution depends only on n_{TEL} and not the fraction of TELs which are on patrol, so there will be no difference between attacking a base with three TELs, all of which are on patrol *versus* attacking a base with 30 TELs, 3 of which are on patrol.

Since the time-since-last detection distribution for a given TEL depends on how long ago that TEL was deployed, an assumption needs to be made about how TEL patrols are distributed in time. I assume that TEL patrols last an average lifetime of L. This lifetime is set by the operational capabilities of the TELs, and I examine how varying L affects uncertainty in Chapter 6. The simplest TEL deployment scheme is to evenly space the TEL deployments uniformly in time. In order to avoid predictability, and possible resonances with satellite coverage, I assume that this holds on average but that there is some variation in practice. In this case, the average time between TEL deployments will be L/n_{TEL}. At an arbitrary point in time t, the distributions of times-since-deployment $(t - t_{\mathrm{deploy},i})$ will be, on average,

$$\mathbf{t}_{\mathrm{deploy}} = \left\{(1-0.5)\frac{L}{n_{\mathrm{TEL}}}, ..., (i-0.5)\frac{L}{n_{\mathrm{TEL}}}, ..., (n_{\mathrm{TEL}}-0.5)\frac{L}{n_{\mathrm{TEL}}}\right\} \;, \tag{4.17}$$

where $\mathbf{t}_{\mathrm{deploy}}$ is a vector of the times-since-deployment and i is the index for the i^{th} TEL.

Connection with Counterforce Model

For each TEL there is an associated distribution of possible values of the random variable $M_i(p_{\min}, k)$. When considering an attack on a group of TELs, each TEL will have an actualized value of $M_i = j$ when the attacker evaluates the attack. For each $M_i = j$, there will be a positional uncertainty distribution U. Warheads can be allocated against that specific combination of actualizations and a P_k determined. However, since there is dependence between TELs when allocating warheads, this process needs to be done for every permutation of the set of $\{M_1 = j_1, ..., M_{n_{\mathrm{TEL}}} = j_{n_{\mathrm{TEL}}}\}$. The number of possible permutations of the set of js can be intractably large, going on the order of $\sim k^{n_{\mathrm{TEL}}}$. This necessitates the introduction of the threshold $p_{k,\mathrm{thresh}}$ introduced in the previous chapter.

Utilizing the $p_{k,\mathrm{thresh}}$, for each actualization of j, the number of required warheads to meet or exceed that threshold $(n_{w,j})$ can be calculated independently of the other target TELs. This allows the distribution in j for each TEL that arises from the PMF 4.13 and equation 4.14 to be converted into a distribution of required warheads $n_{w,j}$. Figure 4-8 shows the distribution of possible values of j on the left, and the associated PMF of the required number of warheads to meet or exceed $p_{k,\mathrm{thresh}}$ on the right.

The convolution of the distributions of the required number of warheads for each TEL $n_{w,j,i}$, and the distribution of deployment times in equation 4.17 provides a distribution of the total number of warheads required to destroy n_{TEL} with the desired $p_{k,\mathrm{thresh}}$. Figure 4-9 shows the result of convolving the required warhead distributions of $n_{\mathrm{TEL}} = 18$ TELs, deployed with $L = 3$ days according to the deployment scheme in equation 4.17. The fraction of the distribution which falls below the total number of available warheads n_w is the fraction of patrol lifetimes which are expected to be vulnerable to attack.

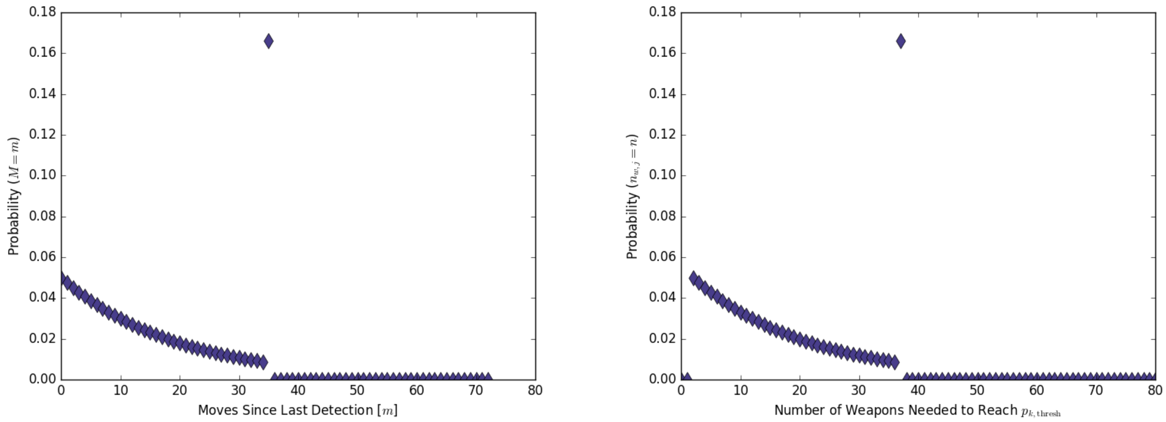

Figure 4-8: **Top:** Probability distribution of number of moves since last detection for a TEL when $p_{\min} = 0.95$ and $k = 35$. **Bottom:** Probability distribution of number of warheads needed to meet or exceed $p_{k,\text{thresh}} = 0.95$ for a that TEL with a linear road model, $r_w/s = 1$, and $\tau_f = 2r_w$.

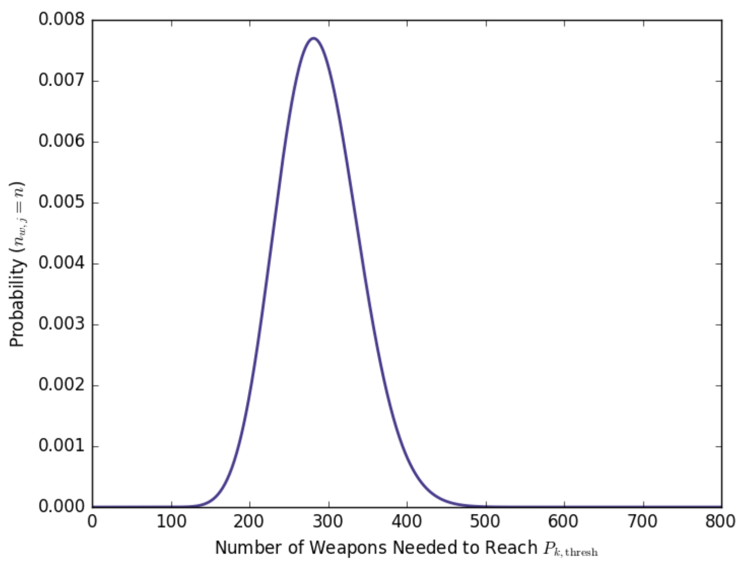

Figure 4-9: Probability distribution of number of warheads needed to meet or exceed $p_{k,\text{thresh}} = 0.95$ for a set of $n_{\text{TEL}} = 18$ TELs, deployed evenly over $L = 3$ days using a linear road model, $r_w/s = 1$, and $\tau_f = 2r_w$.

4.2 Conclusions from Tracking for Counterforce

The tracking and counterforce models are directly interlinked such that it makes sense to take them as a whole when deriving conclusions from them. In this section, I discuss what these two models can tell us about what is required of a remote-sensing modality to be useful for tracking TELs. I then apply these lessons when surveying existing technologies to identify gaps in current capabilities.

Tracking must be defined in the context of a counterforce attack. How well one needs to know where something is in order for it to be considered tracked can only be defined in a given context. In the case of tracking in support of a counterforce attack, the context is provided by the number and yields of warheads available to the attacker, the number of targets of the defender, and the speed of the delivery systems. The number of warheads and their yields determine the area within which the attacker can reliably destroy a TEL. The number of targets and the delivery time of the weapons, as well as the local shape of the road network on which each TEL is operating, determines the area that the attacker needs to cover. If the attacker has a sufficient number of warheads to cover all of the targets and the locations into which they can move, then they will additionally be able to tolerate some initial uncertainty in the positions of the targets. Remote-sensing systems are utilized to manage and minimize this uncertainty. There will be a threshold of uncertainty below which the attacker can be confident in limiting retaliatory damage below some maximum acceptable level. Under those conditions, the hider's TELs can be considered fully tracked.

One important point is that the distinction between tracked and not tracked is evaluated at the level of the entire fleet of the defender's TELs. It is nonsensical to attempt to define whether a single TEL is sufficiently tracked or not, as the number of warheads that are able to be allocated to its destruction depend on how well the locations of the other target TELs are known. Conversely, this also means that the detection of any subset of TELs is not necessarily deleterious to the survivability of those TELs. In fact, there is a direct parallel between a deployed TEL which has been detected and those TELs which are not currently deployed but are in their base of origin. In both cases, the seeker knows their locations and could potentially destroy them with a modest number of warheads.[13] While the individual TEL may be vulnerable to destruction, it is the deterrent value of the entire fleet which shapes the attacker's perceived chances of success. In the literature, strategies such as deploying TELs in mountainous terrain which are highly sight-blocked but increase the chances of mechanical failure of TELs, risking their detection, have been criticized as inviting disaster [99]. However, so long as the defender maintains sufficient uncertainty over the remainder of their fleet, such a failure could be tolerated while maintaining a survivable deterrent.

Delivery systems with long flight times, such as ICBMs, are unsuited to attacking mobile targets such as TELs. I discuss countermeasures to remote-sensing in Chapter 6, however here it is relevant to point out that countermeasures can constrain how an attacker carries out a counterforce attack. In the most conservative case, the attacker barrages each possible location into which the target could move. If the attacker has the capability to update the aim point of warheads while they are in flight, they can reduce how much uncertainty is generated by TEL mobility during the flight time of the weapons. If a TEL were to attempt to move during the attack and was detected, the aim point of the allocated warheads could be updated to take into account the new detection. This could allow for more efficient attacks that utilize fewer warheads per target, changing the delineation between tracked and untracked fleets. If the attacker utilizes fewer warheads than a barrage would require and countermeasures can interfere with the attacker's remote-sensing system during the attack, then the attack may fall below the desired probability of success. In these circumstances, where retargeting warheads provides a benefit to the attacker, countermeasures introduce a vulnerability. If countermeasure exists to a remote-sensing modality, a conservative attacker should revert to a barrage attack.

If retargeting is not reliable, ICBMs are not well suited to counterforce attacks against TELs. The long flight times, roughly 30 minutes, provide mobile TELs with the opportunity to move tens of kilometers during the flight of the weapon. Without retargeting, this can decrease the efficiency of an attack significantly.[14] The weapons that

[13]The attacker would only need to account for TEL mobility during the delivery time of the weapons. Since delivery times are on the order of hundreds of seconds and TEL speeds are on the order of tens of meters per second, the flight time uncertainty is on the same scale as the destructive effects of nuclear weapons, so one can assume that no more than a few warheads would be needed to destroy a tracked TEL.

[14]For example, for a 30 minute flight time, a TEL moving at 10 m/s can travel 18 km. Assuming a straight road and a

are best suited for counterforce attacks against TELS are SLBMs launched on depressed trajectories. Depressing a trajectory decreases the flight time of the weapon, reducing the distance that a TEL could move. SLBMs, while often heralded as the most survivable basing method for nuclear weapons and thus stabilizing, is also one of the best suited for conducting counterforce attacks against TELs and therefore destabilizing.

The hider must have shelters in order to compete. Shelters are a necessity to compete with tracking. Any TEL which remains outside of a shelter can eventually be detected if the seeker is observing the area. Shelters provide locations for a TEL to exist where it cannot be detected with remote-sensing. The distance between shelters can strongly affect the shape of the $R(p)$ curve. If shelters are close together, shorter windows with high probability of moving undetected can be exploited by the TELs. For any given scenario, there will be some minimum s that the hider must possess in order for their TELs to remain survivable.

If the minimum s falls below what the hider possesses, they will need to build additional shelters or risk their deterrent. This can create an arms-race dynamic between the hider and seeker with a competition between remote-sensing assets and purpose-built shelter structures. Which party is advantaged in the arms race depends on the cost of the remote-sensing systems and shelters as well as the relative means of the two competitors. Broadly speaking, sight-blocking shelters can be of simple construction and relatively inexpensive, while remote-sensing systems, especially those based on spacecraft, can be extremely expensive. However, if shelters need to be constructed on the order of one per kilometer over thousands of kilometers of road, the cost may become appreciable to some hider's, depending on their means.[15] In Chapter 6, I evaluate the costs of one such competition, constructing shelters as a countermeasure to space-based radar tracking. While remote-sensing systems remain sufficiently sparse that there are significant gaps in coverage, hider's may be able to avoid the construction of dedicated shelters and instead rely on existing buildings, terrain, *etc.* to shelter their TELs. If gaps in coverage are reduced, the costs of operating a TEL-based deterrent will be increased. In this way, the cost of constructing and maintaining shelters should be considered as part of the cost of operating TELs in a remote-sensing laden environment.

The qualitative effect of multimodality on tracking depends on whether modailities are complementary or redundant. One highly discussed aspect of tracking is the potential for fusing information from multiple different sensor modalities. While all modalities will have some limitations, the combination of different sensors can allow one sensor to cover the vulnerabilities of another. However, while fusing the information gathered from multiple modalities can be advantageous, it is not enough simply to utilize multiple modalities. The argument has been made in the literature that the employment of different modalities necessarily reduces the concealment of TELs[9]. However, in order to enable tracking, there are specific tasks which need to be accomplished. Whether the addition of some remote-sensing modality will qualitatively change the feasibility of tracking depends not just on whether or not it can detect or identify a TEL, but if it can do so under conditions where existing systems cannot do well enough.

The hider must signal their ability to defeat tracking in order to deter attack. Signaling between the hider and the seeker plays a major role in deterrence, including in the competition between TELs and remote-sensing systems. For the purposes of deterrence, the survivability of one's TELs does not matter in-and-of itself. The goal of a nuclear deterrent is not to be able to retaliate if attacked, it is to prevent an attack in the first place by relying on the threat of retaliation. Therefore the ability of a TEL to survive an attack is only valuable if this fact can be communicated to

warhead with $r_w = 3$ km, this would require 6 ICBM-based warheads just to account for the movement during the flight time.

[15]For a U.S. attack against Russian TELs at numerical parity, the U.S. would have roughly 800 nuclear warheads at their disposal. If each has a weapon radius of 3 km, roughly 5000 km of linear road could be attacked. This is the minimum amount of road that would need to have shelters for a TEL-based deterrent to be survivable.

the potential attacker. Communicating this must also be done in a way that does not compromise the survivability of the overall TEL fleet.

The hider must balance the risks of moving their TELs with the risks of becoming predictable. If the hider chooses to never move their TELs, the seeker will be able to determine this retrospectively. This will signal to the seeker that the hider is unwilling to move their TELs. Even though the seeker may never detect those TELs because the are not moving, they may be confident that they know where the TELs are. This creates a risk that the seeker may feel confident in an attack. In order to mitigate this risk, the hider must signal their willingness to move their TELs and their ability to do so without being detected.

Signaling willingness to move entails moving TELs. This requires taking on some risk of detection. If the hider takes too many risks, the seeker is likely to detect their moving TELs. In this case, the seeker is able to utilize the information that they gather from their remote-sensing system detections to estimate how many TELs are moving. If the seeker can confidently know how many TELs are moving and detect those TELs, they could believe that they know the locations of all of the hider's TELs. Managing the seeker's beliefs therefore sets limits on both how conservative the hider is able to be and how much risk they should be willing to take.

The most advantageous strategies to the hider are those which minimize the risk of detection while also allowing them to make the minimum required movements in order to signal their capability. If a TEL needs to make some number of moves over its lifetime, minimizing its probability of detection implies maximizing p, the probability of making a move undetected, for each movement. $R(p)$ curves monotonically decrease with increasing p, meaning that there are more opportunities to move at lower p but higher risk of being detected. If a TEL is unable to know if it has been detected, it is therefore advantageous to make the minimum number of moves over the longest period of time.

The corollary to this is that, for a seeker who is designing a remote-sensing system, the goal should be to minimize the frequency of opportunities to move with a high p. The seeker does not need to make it impossible to move without being detected in order to track TELs, they must simply reduce the opportunities to reliably move without detection. Predictable gaps in can be particularly deleterious to the capability to track TELs.

Gaps in coverage can cause tracking to fail. One of the primary means through which a remote-sensing system can fail is through gaps in coverage. A gap in coverage is a period of time where there are no sensors capable of observing an area. If a TEL is in that area, it would have the opportunity to move with 100% probability of not being detected. In the context of the Tracking Model, during a gap in coverage, the probability of detection is zero, and the TEL has to opportunity to move freely during the gap. Regardless of whether the TEL moves during the gap, the seeker's uncertainty will grow by $v_{\text{TEL}}\tau_{\text{gap}}$ where τ_{gap} is the duration of the gap.

If the gap in coverage is small, a TEL exploiting the gap will not be able to move far. Once the gap becomes long enough that the overall probability of success of moving to the next shelter without being detected approaches or reaches unity, then the hider has access to a low- to no-risk means of achieving their minimum movement required to signal their willingness to move to the seeker. For shelter spacings on the order of a few kilometers, and TEL speeds on the order of 10 m s^{-1}, gaps of length from a few minutes to tens of minutes are sufficiently long that a TEL can move between shelters without risk of detection. Shorter gaps in coverage also contribute to uncertainty growth as they will raise the $R(p)$ curve.

The simplest conception of a gap in coverage is caused by not having enough remote sensor platforms. If there is an appreciable time between overflights of a satellite, for example, the target will have some opportunity to move undetected. Gaps in coverage can also be caused by the paucity of sensors which are capable of completing one of the tasks of tracking. For example, if a remote sensor network did not have a modality capable of identifying targets, then there would effectively be a gap in identification coverage. Gaps can also be caused by environmental conditions. The most common example is remote sensors that are not capable of operating at night. This would create a predictable, hours-long gap in coverage on a daily basis which are well in excess of the gap size which would allow inter-shelter movement. In the following section, I survey current remote-sensing technologies to identify gaps in coverage that may prevent extant remote-sensing networks from tracking TELs.

4.2.1 Gaps in Current Capabilities

This conception of tracking can be used to evaluate the current remote-sensing capabilities in order to build context for the current balance of the competition between remote-sensing and TEL survivability. There are a number of remote-sensing modalities that currently exist which could be applied to tracking TELs. Three which have attracted attention in the literature for tracking TELs are optical imaging, electronic intelligence, and unattended ground sensors[9,10]. To close this chapter, I survey these technologies and identify gaps where these technologies are unable to perform the tasks of tracking.

Optical Imaging. Optical sensors utilize visible, infrared, or near-ultraviolet light to produce images or video surveillance. Images and video can reveal the size and shape of a vehicle once interpreted. The images formed with optical imagery are traditionally interpreted by human analysts, however machine learning techniques are well suited to image feature extraction, so interpretation has the potential to be automated [100]. With proper interpretation, these images can be used detect TELs and potentially to identify them. The ability to identify TELs from other vehicles of similar size depends on the resolution of the sensor and the size of identifying features. One of the primary identifying features of a TEL is the missile canister which has dimensions of tens of meters by a few meters which should be resolvable with optical imaging sensors. The hider could disguise TELs or non-TELs, I discuss the effects of this on identification in Chapter 6.

While optical imaging is capable of both detecting and identifying TELs, it is constrained in its availability. Optical imaging is done with passive sensors that collect light scattered into them. This allows them to operate discretely, without giving any outward indication of what they are imaging but requires an independent illumination source, such as the sun, or for the target to be radiating light (such as thermal infrared radiation). Situations which interfere with the operation of optical sensors are night time, when there is no illumination from the sun, and during periods of bad weather where suspended water droplets in the atmosphere attenuate light.

At night time, there is no passive source of illumination for illuminated scenes. Imaging with visible light is therefore not possible as there is no signal to measure. Imaging with infrared at night is possible in theory, as any hot object such as a TEL engine block will radiate blackbody radiation, however it would not be useful in practice. It is unlikely that current space-based infrared sensors are well suited for the detection of objects such as operating ground vehicles. Overhead infrared sensors, such as early warning satellites, are designed to detect very large, very hot objects such as rocket exhaust plumes in the upper atmosphere. Comparatively, a vehicle on the ground would radiate less infrared energy, there would be a larger background signal from the ground itself, and there would be more atmospheric attenuation.[16] If a specialized infrared sensor did exist that was capable of detecting a vehicles operating engine, it would be difficult to properly identify a TEL based on its external temperature alone. Night therefore creates a gap in coverage for optical sensors. Visible-light sensors have a complete gap in capability, while infrared sensors have a gap in identification capability.

A second major limitation of optical sensors is the attenuation and scattering of optical wavelengths by cloud cover. Visible and infrared light are strongly scattered by water vapor droplets, making clouds opaque. Cloudy weather therefore creates gaps in coverage for all optical sensors. Taken together, while optical sensors are capable of completing the required tasks of tracking, there are significant gaps in coverage that undermine their ability to track TELs. Night time causes predictable gaps in coverage that are hours in duration. Clouds, while less predictable in advance, can create gaps in coverage that last for days. Some areas are regularly overcast, for example, a U.S. Navy study of the environment of the Korean peninsula found that any given location in Korea would experience, on average, 150–200 cloudy days per year [101]. The combined effects of weather and night render optical sensors unable to track TELs the majority of the time. If the seeker were relying entirely on optical sensors for tracking, a dedicated hider could exploit these predictable gaps in coverage to move their TELs entirely undetected by optical sensors. A gap in coverage of a single hour would allow a TEL moving at 10 ms^{-1} to travel 36 km in any direction. An 8 hour overnight gap in coverage would allow hundreds of kilometers of travel, putting each TEL outside of the capability of any attacker to barrage effectively.[17]

Electronic Intelligence. Electronic intelligence (ELINT) is the detection and interception of communications or other electromagnetic signals by a target. Any electromagnetic signal which is transmitted by the TEL can be detected by radio antennae which are monitoring the relevant frequencies. For example, radio communications between a TEL and its security convoy, or between the convoy and the main base can be intercepted with ELINT sensors. ELINT sensors can be based on the ground, on airframes, or on satellites. ELINT monitoring has the capability to both detect TELs which are transmitting radio signals, and potentially identify them by the frequencies or types of signals which they are transmitting.

While ELINT techniques are useful sources of intelligence which can both detect and identify TELs, they are not well suited to form the backbone of a TEL tracking system. This is because ELINT methods rely emissions from the TEL itself, as such, its availability and utility is dependent the behavior of the hider. Effectively, ELINT suffers from gaps in coverage which are entirely under the control of the hider. By observing radio silence, a TEL can create a gap in ELINT coverage for as long as they require to move to a new shelter. This does place operational restrictions on the TEL, as they are not able to freely communicate with the hider's broader command and control systems. This can be circumvented in a number of ways, such as installing hard line communications in TEL shelters. Alternatively, a TELs orders can be given as it deploys which it can then execute without communicating with its base. The hider can still communicate with their TELs by broadcasting encrypted signals to the TELs which receive but do not transmit signals. This is a similar principle under that U.S. SSBNs operate under. Operating in this manner also does not have to compromise the physical security of deployed TELs as, in an emergency, a TEL can

[16] The total power radiated by a blackbody $P \sim T^4$. Rocket exhaust plumes are at temperatures on the order of a few thousand Kelvin, compared to a few hundred Kelvin for a vehicle engine. Overhead infrared sensors detecting which detect rocket plumes also operate in wavelength bands which suppress radiated infrared background from the Earth itself, which is not possible for a see-to-ground sensor which needs to detect objects on the Earths surface.

[17] Assuming a straight road, 8 hours of travel and a r_w of 3 km, a barrage would required 96 warheads per target TEL per night.

transmit signals back to the hider's command and control systems. As noted before, unmasking the location of a single TEL does not necessarily undermine the survivability of the TEL fleet as a whole, meaning that this modus operandi does not require such a strict adherence as to be a detriment to security.

Another weakness of ELINT methods is the susceptibility to spoofing countermeasures. If a seeker relied entirely upon ELINT to track TELs, a hider could feint the movement of TELs make TEL-like transmissions from locations where TELs are not actually deployed. Such a decoy would require only equipping a non-TEL vehicle with the same radio system that actual TELs were equipped with. If the hider did so, the seeker would not be able to distinguish between actual signals and spoofs, undermining the utility of ELINT as a whole.

Unattended Ground Sensors. Unattended ground sensors (UGS) are static remote-sensing systems that can be deployed along a route that detect and identify passing mobile targets. UGS can use a variety of sensing modalities, such as acoustic, magnetic, laser, or optical. The UGS collects these measurements and periodically transmits relevant data, via some communications uplink, to the seeker. In order to be useful over an extended period, UGS must be disguised in some way, and be able to transmit their data in a timely and difficult-to-detect manner. UGS must also have the ability to supply their instruments with sufficient power over the lifetime of the system. One of the main challenges with UGS is emplacing them, as they must be either dropped or physically installed along a route of interest.

UGS were utilized by the U.S. during the Vietnam war in order to monitor traffic in areas that were under tree cover and not amenable to aerial photography[10]. The challenge facing UGS in the context of tracking TELs is that TEL deployment areas can be deep within the interior of an adversary's territory. These sensors would need to be emplaced well in advance of a considered counterforce attack. They would additionally need to be able to survive any route clearance efforts made by the hider. Sensors able to meet these conditions might provide the seeker with a detection and identification capacity for passing TELs.

The limitation of UGS is that they are statically emplaced. This means that they do not necessarily have temporal gaps in coverage, but instead have spatial gaps in coverage. Assume that UGS are emplaced along a road at a spacing d. UGS would be able to detect if a TEL were between two UGS sensors. If $d/r_w \leq \epsilon$ where ϵ is the average number of warheads available to be allocated to each TEL during an attack, then a network of UGS would be capable of tracking TELs. Assuming a parity attack against Russian TELs which each carry 4 warheads, this would require a UGS sensor to be placed once every 24 kilometers along every road in a TEL deployment area. This could require covertly emplacing hundreds of UGS within Russian territory.

In this way, UGS are modality most capable of acting as the central technology for a remote-sensing system. The only gaps in coverage are spatial, which can be overcome simply by emplacing more sensors. The challenge of basing a tracking system on UGS is that it would be extremely fragile. UGS need to be covertly emplaced along every route through which TELs would be expected to travel. Building the system therefore risks its detection, and the more capable a UGS based system would be, the more likely it would be to be detected. If it were detected or even suspected by the hider, it could be quickly rendered inoperable by the hider. Routes along which TELs traveled could be cleared by the hider utilizing metal detectors and visual inspection. Since the routes are within the hider's own territory, the seeker would not be able to interfere, short of starting an armed conflict. Once a route was cleared, there would be a persistent gap in coverage along that entire route.

UGS are useful for supporting other remote-sensing technologies however. Providing an opportunity to detect movers, UGS would reduce the expected uncertainty distribution of each deployed mover . If UGS were placed, on average, once every d along all of the routes in a deployment area, then there would be a probability $p_u = s/d$ that a TEL would be detected each time it moved between shelters.[18] This would effectively reduce p of the mover, increasing the detection rate and lowering the expected uncertainty. Additionally, UGS could prevent the spread of uncertainty of undetected TELs along the routes on which they are placed.

Gaps in current tracking capability. Optical imaging, ELINT, and UGS, in concert have the capacity to detect, identify, and track TELs under some conditions. Each of them have potential gaps in their coverage which limit their ability to act as the linchpin of a tracking system. Multimodal tracking utilizing all three modalities also has limitations on its coverage.

Optical imaging and ELINT both suffer from gaps in coverage that cannot be overcome by simply deploying more sensors. Optical sensors experience gaps in coverage at night and when it is overcast, which can constitute significant and predictable periods of time in some parts of the world. ELINT experiences gaps in coverage at the whim of the hider. These two modalities can contribute to tracking, however for a system to be capable of tracking TELs, a modality or set of modalities which are capable of detecting and identifying TELs at night and during periods of overcast weather is required. These technologies would need to be capable of this entirely independently of optical and ELINT systems.

As noted above, UGS in theory would have the capacity to do this alone, though the system would be fragile. As I argue in Chapter 6, the capability of a technology to close a gap in coverage must be evaluated in light of the actions and countermeasures available to the hider. A UGS sensor network would need to be covertly built and maintained, and any detection of the effort or suspicion from the hider could prompt route clearance and disassembly the system. If the hider respected the possibility of UGS, regular route clearance could prevent a UGS-based system from ever reaching sufficient density to be useful. Because of this, UGS alone are unlikely to enable tracking.

In the next chapter, I introduce space-based radar (SBR), a modality which has the capacity to operate at night and through clouds. Since it would be required to operate independently during the gaps in coverage for optical and ELINT systems, and given the fragility of UGS, I evaluate its capacity to perform the tasks of tracking in isolation of other systems, though I note where UGS would be useful to support the system. I then design a SBR system that would be capable of closing the current gaps in coverage in order to estimate its cost before evaluating its durability to countermeasures in Chapter 6.

[18]This probability holds for $s \ll d$.

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Chapter 5

Tracking With SBR

5.1 A Summary of SBR-enabled TEL Tracking

Recall from Chapter 2 that the three radar modalities have the capability to perform the tasks of tracking. I present a scheme for how I envision SBR should be applied to tracking TELs and some of the attendant assumptions required:

1. The seeker utilizes Kalman filtering to estimate the location of individual TELs from SBR measurements.
2. Observations are made of the area surrounding each TEL from when it leaves its base to when it ends its patrol.
3. Observations are randomly scheduled to prevent the hider from being able to time when to move.
4. SMTI observations are made with an average revisit interval $\tau_{\mathrm{rev,SMTI}}$ which detects moving targets and cues identification with ISAR.
5. SAR observations are made with an average revisit interval $\tau_{\mathrm{rev},SAR}$ and provide detection and identification of stationary targets. In addition, co-production of ISAR data allows for moving target detection and identification from the same observations.
6. The seeker's uncertainty grows when TELs have the opportunity to move between shelters without being detected by either SMTI or SAR. Since moving target detection can occur through either modality, the effective revisit interval of the two modalities in combination must be sufficiently small to prevent a TEL from having the opportunity to move between shelters without detection.
7. If a TEL enters a shelter, the seeker can be confident that the TEL remains in the shelter, even though it cannot be imaged directly, by monitoring the exit(s) of the shelter. The revisit interval of $\tau_{\mathrm{rev},SAR}$ must be sufficiently high to prevent TELs from being able to move from one shelter to the next at the MDV of the SMTI modality.
8. Identification is assumed to be sufficiently accomplished with SAR and ISAR imaging with contextual clues from target behavior. In the next chapter, I investigate how uncertainty can grow from ambiguity in target identity caused by the possibility of decoys.

The ability to complete all of the tasks of tracking alone means that SBR could enable a counterforce attack if sufficient coverage is achieved. With a sufficient number of SBR sensors, the seeker would be able to track the hiders

Constellation Label	Altitude (km)	Aperture (m^2)	Power (W)
Large Aperture	1000	100	10 000
Baseline	1000	40	10 000
Higher Power	1000	40	15 000
Medium Altitude	700	40	10 000
Low Altitude	300	20	5 000

Table 5.1: System design parameters evaluated in this section. For each set of parameters, the minimum number of satellites needed to enable TEL tracking is calculated.

TELs no matter what state they are in. The hider cannot defeat a sufficiently dense SBR constellation by simply changing the way in which they operate their TELs. Direct countermeasures are available to radar tracking, but discussion of those will be reserved until Chapter 6. For now, I consider an adversary who, despite operating their TELs in such a fashion to minimize their vulnerability to SBR tracking, has not yet developed or is unable to deploy technical countermeasures.

What is left in this chapter, is to determine how many satellites an SBR constellation would need in order to actualize this potential for tracking. In the following section, I will develop a model of how TEL uncertainty can be expected to grow over time, and how much coverage is needed to keep this to a manageable level. Since this needs to be achieved at night and during bad weather, I design a radar system that is self-sufficient in performing tracking.

5.2 SBR System Design

Designing an SBR system for tracking TELs requires making a large number of decisions. Some of these decisions are easily constrained and have a "best" answer. Many of these decisions do not, or are intimately linked to other decisions. The work that I presented in the previous chapter provides a performance goal and a means to evaluate an SBR constellation for tracking TELs. With the large number of design variables that need to be fixed, there are an infinite number of possible constellations that reach meet the requirements. One additional constraint on any defense project that can be used to compare amongst these possibilities is cost. However, even within the smaller set of "reasonable" constellation possibilities, one could design a constellation of highly capable satellites in high orbits (within LEO), or a larger number of less capable satellites in lower orbits which may be able to achieve the same tracking performance. In this section, I do not seek to determine the global minimum cost SBR system that is capable of tracking, only to identify a handful of possible constellations. Where there is a clear "best" option, I will take it. In those scenarios where there are a range of possibilities, I will evaluate cases spanning as much of that range as is feasible, using cost as a basis of comparison between the resulting cases.

I divide the design of an SBR system into two major design projects, the design of the radar array, power source, and spacecraft which comprise each individual SBR satellite, and the design of the constellation of those satellites. Through this section, I introduce each design choice and then narrow down the possibilities to a best choice where possible, or a range of values to be considered. Table 5.1 summarizes the set of designs which are considered in this section. Any design parameter not included in this table is held constant across designs, *e.g.* carrier frequency. For each set of parameters, the smallest constellation which is capable of enabling tracking for counterforce attacks against China or Russia is calculated utilizing the Tracking Model from Chapter 4.

5.2.1 Radar Hardware Selection

Source Power and Wavelength

The transmitter power is a key component in determining the amount of signal collected during an observation, as per the radar equation. From the perspective of a maximizing the SNR of an observation, more power is better. The limits on what is practical for a power supply are set by the power budget of the system, and the acceptable cost of the system.

SBR satellites must be self-sufficient, thus any power used in the operation of the radar array must be generated and stored on the spacecraft. For spacecraft, power can be generated from an on-board generator, such as a radioisotope thermoelectric generator (RTG), or through solar panels. Generated power is stored in batteries which can then be discharged to power the radar system. The required power of the system then depends on the desired transmit power of the system, and its expected duty cycle. The power budget of a spacecraft balances the power consumption of the spacecraft against the power generated and stored. Assuming a constant duty cycle, the power balance for an SBR spacecraft can be expressed as

$$P_{\text{supply}} - CF \cdot P_t DC - P_{\text{spacecraft}} = 0 \ , \tag{5.1}$$

where P_{supply} is the average supplied power, DC is the duty cycle of the radar transmitter, P_t is the peak transmit power of the radar system, CF is the capacity factor, and $P_{\text{spacecraft}}$ is the average power consumed by the non-radar systems of the spacecraft. The duty cycle is the fraction of time where the radar is transmitting power while in operation. Inefficiencies in converting input power to radiated radio waves are already accounted for in the loss term of the radar equation, so only P_t will be considered here. The capacity factor is the fraction of the satellite's orbit where it needs to be able to operate. A SBR satellite designed to monitor TELs would only need to operate when in range of a TEL-deployment area, for example. When designing an SBR constellation, P_t, DC, and CF are determined by the mission parameters and a power supply must be selected that can support that operation.

RTGs. RTGs produce power by producing heat with radioisotope decay, and extracting power from that heat using the thermoelectric effect [102]. They have been used to power spacecraft in the past, including for deep-space probes [103,104]. RTGs have a roughly 6% efficiency in converting heat to electricity, requiring relatively large masses to achieve even modest powers [105]. High performance, flight tested RTGs, such as the Mutli-Hundred Watt RTG, have electrical power outputs on the order of 100 W with a mass of roughly 40 kg [103]. For radar system with a transmit power of 10 kW and a duty cycle of 15%, a 150 We RTG would allow for a capacity factor of only 10%, without even considering spacecraft power consumption and inefficiencies in storing and discharging the power in batteries. Utilizing RTGs in Earth orbiting satellites also runs political risks, as evidenced by the crash of an RTG powered Soviet satellite in the Canadian arctic in 1978 [106]. Given their relatively low power output and other drawbacks, RTGs are not well suited to being the power supply for an SBR satellite.

Solar Panels. Solar panels produce power during a portion of a satellites orbit, unless that orbit is sun-synchronous. Power generated when in sunlight must be stored in batteries for the portion of the orbit when the satellite is in darkness. Assuming the duty cycle is constant, regardless of whether the satellite is in darkness or not, the amount of power generated by the solar panels must be

$$f_{\text{illuminated}} P_{\text{solar}} - CF \cdot P_t DC - P_{\text{spacecraft}} = 0 \ , \tag{5.2}$$

where P_{solar} is the average power generated by the solar panels when illuminated, DC is the duty cycle, P_t is the average power used when the radar system is in operation, CF is the capacity factor, and $f_{\text{illuminated}}$ is the fraction of the orbit spent in sunlight, and $P_{\text{spacecraft}}$ is the average power consumed by the non-radar systems of the spacecraft. The amount of power generated by the solar panels when illuminated is

$$P_{\text{solar}} = \Phi A \cos\theta \epsilon_{\text{cell}} \epsilon_{\text{battery}} EOL \ , \tag{5.3}$$

where Φ is the solar flux, A is the solar panel area, ϵ_{cell} efficiency which the solar panel can convert light into electrical power, $\epsilon_{\text{battery}}$ is the ratio between the output and input of power stored in the spacecrafts batteries, θ is the angle between the panel normal and the sunlight incidence, and EOL is the end-of-life factor which accounts for degradation of the solar panels over time.

The area of solar panels that would be needed to power an SBR satellite can be estimated by assuming some model values for the variables in equations 5.2 and 5.3. The model values are summarized in Table 5.2. For LEO orbits, satellites are eclipsed roughly one third of the time, so $f_{\text{illuminated}} \approx 2/3$ [107]. Solar power density in low Earth orbit (LEO) is approximately $\Phi = 1400$ W/m^2 at normal incidence [107]. Solar cell efficiency has almost doubled since the 1990s, with modern solar cells having ϵ_{cell} approaching 30% [108,107]. I will assume that panel pointing misalignments will be minimal, so the total solar panel area will be taken as the effective area. When power is stored in and then discharged from batteries a fraction is lost as heat, $\epsilon_{\text{battery}} = 0.85$ is a typical efficiency [108]. Solar cells in the hostile radiation environment of space degrade over time, losing roughly 5% output per year [108]. For a satellite with a 10 year lifetime, the power supply at end of life would be degraded by a factor of $EOL = 0.95^{10} = 0.6$, which must be compensated for in the initial design.

The payload parameters are determined by the mission. As an example, an SBR system analyzed by *Post et al.* utilizied a transmit power of $P_t = 10$ kW with a duty cycle of $DC = 0.15$, giving an average power consumption of $\bar{P}_t = 1500$ W[68]. I will assume a capacity factor of unity in order to determine the upper bound of power requirement.[1] In practice, it would likely be less, but setting it to the upper limit allows me to determine a solar panel size that could support any possible mission. The spacecraft will also draw power to regulate temperature, communicate, and monitor and regulate all of the subsystems. *Fortescue* shows a simplified power budget for a communications satellite where the payload accounted for roughly 80% of the total power budget, with the remainder drawn by the spacecraft systems [107]. I will use that fraction and assume that $P_{\text{spacecraft}} = 0.25\bar{P}_t$.

To support a 10-kW radar system with a capacity factor of unity until its 10 year end-of-life, the minimum area of solar cells that would needed is $A_{\text{min}} = 13\text{m}^2$. This is an area significantly smaller than the array apertures being considered. Given the amount of power which they can generate, solar panels would be well suited to powering an SBR satellite. Solar panels do add weight and therefore cost to a satellite, which I discuss in the cost model in subsection 5.2.4.

From previous analyses, a power supply of 10 kW is suitable to power a 40m^2 aperture at a 1000 km altitude

[1] In practice, the capacity factor, the fraction of time when a satellite is imaging, will be significantly less than unity as much of the Earth's surface is ocean. However, radar imaging may also be utilized for detection of ships to monitor maritime traffic, so there are scenarios where a duty factor approaching unity is not inconceivable.

Parameter	Value	Parameter	Value
$f_{\text{illuminated}}$	0.67	EOL	0.6
Φ	1400 W/m^2	P_t	10,000 kW
ϵ_{cell}	0.3	DC	0.15
$\epsilon_{\text{battery}}$	0.85	$P_{\text{spacecraft}}$	$0.25 \cdot P_t \cdot DC \cdot CF$

Table 5.2: Model power budget for estimated required area of solar cells.

[68]. Since 1000 km, the upper edge of LEO is the upper bound of of the altitude range that I am considering, the majority of my cases use a 10-kW power supply. To investigate a potential for cost savings, I consider smaller power supplies when considering lower-altitude constellations. I also consider a handful of cases with higher-power supplies to determine if there are meaningful performance gains which can be realized by increasing P_t.

Transmitted Frequency

The transmitted frequency (also known as the carrier frequency) of the radar is important in determining the gain and resolution of the system. Higher frequencies allow for narrower beams which increase the gain of the system, improving the signal-to-noise ratio. Higher gains also yield smaller beam footprints, however, meaning that a smaller area is illuminated in each observation. The gain can be purposely reduced in order to increase the beam footprint by changing the weighting of the elements in the array, it cannot, however, be improved beyond the limit set by the physical aperture. Additionally, if the desired gain is set (based on some desired performance metric), then the size of the physical aperture can be reduced which can result in cost savings and ease engineering challenges. This means that short wavelengths are attractive because they give the system operator the flexibility to operate with a high- or low-gain beam, depending on application. For high-resolution imaging with modalities such as SAR, the X-band of radio waves provides a good balance between resolution and transmission through the atmosphere [60].

5.2.2 Aperture

Radar antenna aperture size and shape affect both the transmission gain of the illuminating beam and the solid angle of the receiver. In this thesis, I will only be considering AESA, large panels embedded with a large number of TR modules. The size and shape of the panels are drivers of both the performance and cost of an SBR system. As seen in the radar equation 2.10 the signal-to-noise ratio of an observation has a quadratic dependence on the size of the aperture. A larger array allows for detection at longer ranges and decreases the impact of surface clutter in Doppler images. The downside of large apertures is the difficulty in deploying them, and the increased cost that comes with utilizing more TR modules. This makes aperture an important design choice when designing an SBR system.

For satellite-based radars, it is advantageous to have an array that is longer than it is tall. This gives a relatively narrow beam azimuthally, but a wide beam in elevation. For SMTI, where the width of the clutter notch depends on the azimuthal extent of the beam, a narrower beam means a smaller MDV, as noted in section 2.2.3.

Given that the beam is wide in elevation, the beam will span a range of distances. While a wide beam in azimuth can lead to poor cross-range resolution, range is resolved by timing the time-of-flight of echoes in the pulse. This is determined by the pulse compression and the speed of the receiver digitizer. As such, having a wide swath in elevation allows for a large area to be simultaneously illuminated while maintaining fine range resolution. The height of the aperture cannot be made arbitrarily small, however, or else range ambiguities will occur.

Pulsed radar systems operate in a transmit-receive cycle where radiation is emitted during a transmission pulse of length τ before the system switches into a receive mode and listens for returning echoes. The frequency at which this cycle occurs is called the pulse repetition frequency (PRF) of the system, which is inversely proportional to the period of the system T,

$$T = \frac{1}{\mathrm{PRF}} \ . \tag{5.4}$$

The duty cycle of the system is the ratio of the transmit pulse duration to the duration of the pulse-listen cycle

$$DC = \frac{\tau}{T} \ . \tag{5.5}$$

The SNR of an measurement is dependent on the time-integrated power (Φ) of the object's radar echoes which is proportional to pulse length and number of pulses Increasing either the duty cycle or the PRF increases the fluence on target, therefore the signal collected. The transmission of radar waves to and from the target is not instantaneous, but requires time

$$\tau_{\mathrm{roundtrip}} = 2\tau_{\mathrm{transmission}} = \frac{2R}{c} \ , \tag{5.6}$$

where R is the range to the target and c is the speed of light. If the radar does not wait $\tau_{\mathrm{roundtrip}}$ between pulses, then echoes will arrive either during illumination, where they will not be detected, or during the listen portion of the next pulse-listen cycle. These are called "nth time around" signals. They create range ambiguities, as the echoes "wrapped around" into the next cycle cannot be distinguished from returns that would happen from an object at a nearer range. This places a limit on the maximum range at which the radar can detect targets without range ambiguities, which is a function of the PRF

$$d_{\max} = \frac{c}{2\mathrm{PRF}} \ . \tag{5.7}$$

If the beam range-swath (the distance between the heel and toe of the beam footprint) exceeds $d_{\max}$, then the system cannot operate without in-beam range ambiguities. This places a lower limit on the w dimension of the aperture. This becomes more stringent as the altitude of the platform increase, as the beam swath on the ground grows with increasing R.

For systems which operate at high altitudes, such as satellites, the transmission time $\tau_{\mathrm{transmission}}$ is large relative to the differences between the echoes at the heel and toe of the beam. Under such circumstances, multiple pulses may be sent in a row before the satellite begins to listen for the echoes. The transmitted pulses must still be separated by T however, in order to allow separation of the echo pulses.

Aside from limitations set by the physics of radar systems, there are also practical and cost-based limits. For electronically steerable arrays, a large number of transmit-receive modules are placed on panels which make up the array. The TR modules need to be embedded in a grid with an inter-module spacing of no more than $\lambda/2$, meaning that for a system operating at 10 GHz, roughly 4,500 TR modules are needed per square meter of aperture. These can constitute a major component of cost of the system, both in the cost of the modules themselves and in the additional weight added to the spacecraft. Additional components add additional power requirements, and additional costs to launch services.

In addition, the means of transporting SBR satellites into orbit places limits on their physical size. The array must fit within the fairing of the launch vehicle that carries it into orbit. The aperture panels must be kept nearly perfectly planar in order for the array to operate at peak efficiency. If the array is larger than the fairing, then it must unfold after attaining orbit. These practical considerations place limits on the size of the aperture that can reasonably be utilized.

When considering performance of a radar, the mapping rate of a modality is independent of the gain of the system. The gain depends on the size of the aperture. As the gain increases, the transmitted power is incident on a smaller footprint, decreasing the dwell time needed to achieve some desired SNR. When considering coverage rate, while the dwell time decreases, so does the area of the footprint. When selecting an aperture size, the motivation for increasing the size of the aperture is either to allow the system to operate at longer ranges, or to decrease the MDV of SMTI as the decreased swath width decreases the clutter Doppler spread. If minimizing costs, the smallest aperture should be chosen that allows the satellite to achieve a given field of regard with an acceptable SMTI MDV.

I consider a range of aperture sizes, between 20–100 m^2. Smaller apertures lack the gain to achieve manageable beam-widths on high-altitude satellites, so will only be considered for lower-altitude constellations. Larger apertures are effective at increasing gain and improving SMTI performance at any altitude. As I show in the next section, lower altitude constellations require more satellites to achieve a given level of coverage, which can exacerbate the increased cost of larger aperture. As such, the consideration of larger-aperture cases will be skewed toward higher-altitude constellations.

5.2.3 Constellation Design

Constellation design refers to determining the number of satellites that will be deployed, and the orbits in which they will be placed. The orbit can be classified by altitude and the eccentricity. The number of satellites needed is determined by the coverage and the access of satellites in the orbit, as well as what modalities those satellites will utilize. As with the selection of radar hardware parameters, there are meaningful design choices to be made when designing a constellation. Constellation design choices are slightly more constrained once hardware parameters have been selected. After selecting an altitude and constellation shape, the number of satellites required to meet performance requirements set out by the Tracking Model can be calculated.

Orbit. There are a number of orbits into which satellites can be placed. The most important characteristic of the orbit is the altitude at which the SBR satellite is operating because it impacts the range to targets within the field of regard of the satellite. The measured signal-to-noise of the system has a R^{-4} dependence, while the size of the beam footprint has a roughly R^2 dependence. The range also factors into the resolution of some modalities. The R^{-4} makes medium Earth (MEO) and geostationary (GEO) orbits impractical. Highly eccentric orbits (HEO), such as Molniya orbits are attractive as they can spend significant fractions of their time over a given hemisphere of interest. However, the altitudes of the apogees of such orbit can rival or even exceed the altitude of GEO, making them impractical for radar imaging.

Low Earth orbits (LEO) span altitudes from 300–1000 km above the surface of the Earth. The trade-off in selecting the altitude of the orbit is that higher orbits require fewer satellites, as the field of regard and the beam footprints are larger, but those satellites must be powerful in order to compensate for the R^{-4} dependence of measured signal. Conversely, at lower altitudes satellites have a smaller field of regard, and spend less time over a given target. This allows for less powerful satellites, but requires more of them to achieve the same coverage. Altitudes near the bottom of LEO also can experience some drag from the Earth's atmosphere, decreasing the lifetime of the satellite. In the system designs that I consider, altitudes spanning LEO will be evaluated with a lower bound of 300 km, and upper bound of 1000 km, and a midpoint altitude of 650 km.

This analysis will also assume the constellation of satellites entails circular orbits in a Walker configuration. Walker configurations, specifically the delta patterns, create a spherical lattice using a number of co-rotating orbital planes in order to maximize coverage with the fewest satellites [109]. Walker constellations can be described with a notation $i : n/p/f$. Such a constellation would contain n satellites at some inclination i evenly divided amongst p equally spaced planes, with a phase offset between the satellites in each plane, described by the number f.

Number of Satellites The number of satellites needed for a given constellation depend on the hardware of the radar systems themselves, the orbit in which the satellites will operate, and the task which the system is designed to complete. For the purposes of tracking TELs, the SBR system will need to utilize some combination of SMTI, SAR, and ISAR. The number of satellites needed to enable a given counterforce attack can be determined utilizing the Counterforce and Tracking Models outlined in Chapters 3 and Chapter 4, respectively. For a given n_{sat}, the Walker configuration with the largest coverage (covered in the next section) is selected. The revisit intervals for each radar modality that a satellite in that constellation can maintain is determined (see section 5.2.3), and the rate of growth of uncertainty U is determined utilizing the Tracking Model simulation outlined in Chapter 4. The calculated U serves as an input for the Counterforce Model, which can be used to determine if the number of warheads available to the attacker in a given scenario is expected to be sufficient to attack the target set of TELs. The enabling constellation for a given counterforce attack (*e.g.* The U.S. attacking Russia or the U.S. attacking China) is the one with the smallest number of satellites where the attacker is expected to have a sufficient number of warheads to carry out the attack.

Coverage

Coverage is the fraction of some area of interest, *e.g.*, Russia, China, North Korea, the world, which is within the operable range of at least m SBR satellites at a time. The variable m will be taken as the multiplicity of coverage, the average number of satellites within range of an arbitrary point in the area of interest over time. When m is unity, there is on average one satellite overhead of a point at a given time for all of the points within the area of interest, which will be termed "continuous coverage". Coverage is set by a combination of the number (and orbits) of satellites in the constellation, the definition of the area of interest, and the size and shape of the field of regard (FOR) of each of those satellites.

The FOR of an SBR modality is the set of ranges and angles at which it can operate. Each imaging modality will have its own field of regard which will define some area on the ground that the satellite is able to monitor with that modality. Figure 5-1 shows the fields of regard for SMTI and SAR/ISAR. The FOR for SMTI is limited in azimuth only by the physical limitation of how far the aperture itself can be turned. The ranges over which it can operate are set at the maximum by the power of the system and the radar cross-section of the object it is attempting to detect. Fields of regard are shown for SMTI detection of Swerling II targets with mean cross-sections of 10 m^2 and 1 m^2. The 10 m^2 target represents a vehicle such as a heavy truck or tank, it is the target that the U.S. government set

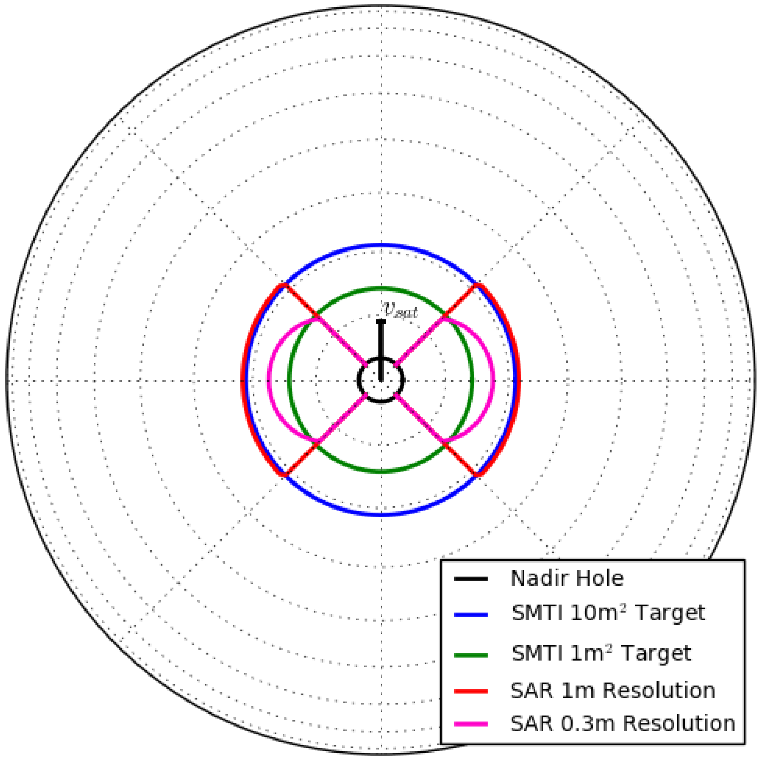


Figure 5-1: The fields of regard for SMTI and SAR imaging for an SBR system with the parameters of system B in Table 5.1, relative to the direction of travel $\hat{v}_{\mathrm{sat}}$. The plot area represents a projected view of a spherical Earth. Assuming a 10-degree minimum grazing angle. SMTI fields of regard for targets with average radar cross-sections of 1 m^2 and 10 m^2 (cool colours). SAR field of regard as a function of desired image resolution with in-range and cross range resolution set equal to 0.1 m, 0.3 m, and 1.0 m (warm colors). Nadir hole of 20° in look angle (black).

as a requirement to be able to track when planning a potential SBR system in the 2000s [68]. Security vehicles in a TEL convoy, for example, would likely have an average RCS of 10 m^2. The maximum distance of this FOR is limited not by power, but by geometry, so vehicles with larger RCS values (such as TELs themselves) would have the same FOR.[2] The 1 m^2 corresponds to a target such as a passenger vehicle, and is shown to illstrate the effects of RCS reduction (RCSR), which are discussed in the next chapter. This minimum range defines the nadir hole, a hole in the field of regard directly below the SBR satellite where clutter returns saturate the radar receiver. These features create the donut shaped SMTI FOR.

The field of regard for SAR, also shown in Figure 5-1, has some commonalities with the SMTI FOR. The limits in look angle are set by the system power and the nadir hole, however SAR is restricted in the azimuths at which it can operate. The virtual array synthesized by SAR is oriented along the satellite's direction of motion, as such, what is important is the projection of the physical aperture onto the along-track direction. SAR cannot operate

[2]The larger RCS of a TEL compared to the reference target will decrease the required dwell time for SMTI observations, but the dwell time for single observations with SMTI is already sufficiently short that the limiting factor for an SBR system is sight-block rather than the frequency of SMTI observations.

while looking ahead or behind, the physical aperture must be pointed perpendicular to the direction of motion. The angle between line-of-sight to the scene being imaged and perpendicular is referred to as the squint angle. As shown in equation 2.38, the spotlight SAR cross-range resolution is inversely proportional to the cosine of the squint angle. The further the beam is directed further away from broadside, the more coarse the SAR image becomes, or the more dwell time is required to achieve a given resolution. In the case of AESA which are electronically steering, there is a further $1/\cos\alpha$ dependence from gain loss [68].

Collectively, these place a practical limit on the squint angle, beyond which it is inefficient to utilize SAR. This creates a bow-tie like shape of the FOR of SAR, shown in figure 5-1. For this analysis the squint will be restricted to $\pm$ 45°, which corresponds to a point where the dwell time for a given resolution is doubled.

Coverage is determined by the total area covered by FORs of the satellites within the constellation. As satellites are added to a constellation, more areas fall within the FOR of a satellite, and coverage increases. For small numbers of satellite, coverage grows linearly, but becomes sub-linear as satellite FORs start to overlap[68]. Given the geometric inefficiencies of tiling the surface of a sphere, such as the Earth, with bowtie or donut FOR, overlap will begin to occur before continuous coverage is achieved. It also requires a large number of satellites to achieve complete coverage due to the nadir hole in each FOR. A designer may specify a desired ground coverage between some latitudes of interest as a target for an SBR constellation, or the maximum or average time between overflights of an SBR at some location of interest. Figure 5-2 displays the number of satellites required to achieve near-continuous single satellite coverage (95% of the surface of the Earth covered at a given time, ignoring nadir holes) as a function of the minimum grazing angle (maximum incidence angle) at which the system is capable of operating, assuming the constellation is formatted as a Walker constellation.

When designing a constellation for tracking TELs, coverage provides a measure for the presence of gaps in coverage. Different constellations may have similar levels of coverage with different $i : p/n/f$ configurations. Ultimately, it is the rate of uncertainty growth that is the most important measure of a constellation's capability, but a good rule of thumb when on a constellation shape is one that minimizes the length of gaps in coverage. Another important caveat is that simply having a satellite in view of a TEL is not necessarily sufficient to prevent its uncertainty from growing. In order to integrate the capabilities of the individual satellites into constellation design, it is useful to define a metric for the availability of remote-sensing resources, which I call burden.

Burden and Revisit Interval

Tracking TELs requires utilizing multiple modalities in tandem. Each of these modalities will be required to make repeated observations of the TEL deployment area at some revisit interval $\tau_{\text{rev},i}$. The revisit interval is defined for each satellite for each modality, $\tau_{\text{rev,SMTI}}$, $\tau_{\text{rev,ISAR}}$, and $\tau_{\text{rev,SAR}}$. Each modality requires a different amount of time to make an observation, so the proper balance between modalities must be found in order to maximize the amount of area that a given satellite can monitor with all three modalities in conjunction. This requires defining a quantity that I call the burden. Burden describes the fraction of a remote-sensing system's resources that a given task would consume.

In order to make some observation i, a remote-sensing system will need to collect signal for some dwell time T_{Di}, which is determined by the modality employed. If that observation is to be repeated, the maximum frequency at which an observation can be repeated is $f_{\max} = 1/T_{Di}$. Repeating that observation at $f_{\max}$ would consume all of the

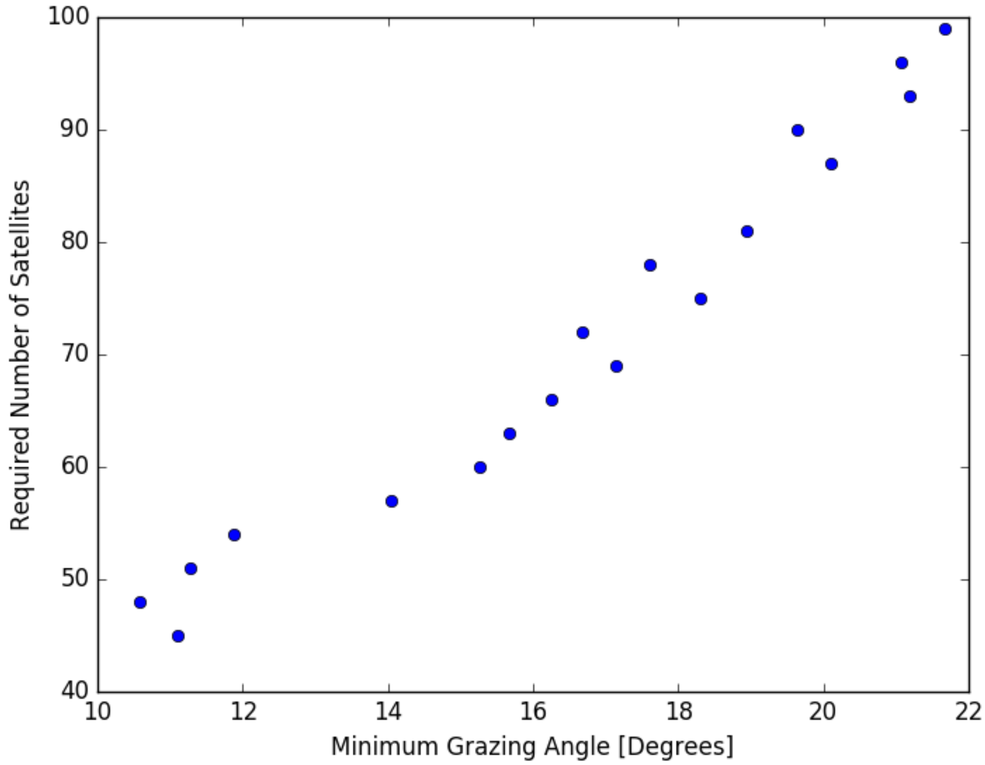


Figure 5-2: The number of satellites required to attain continuous coverage with SMTI (90% of Earths area) utilizing a Walker constellation as a function of the maximum look-angle of the system determined by Monte Carlo simulation. For each n_{sat}, the positions of the satellites in each possible Walker constellation were generated. Over 10,000 samples, a random point on the Earth's surface is sampled and the smallest grazing angle to the nearest satellite is determined. For each n_{sat}, the smallest angle which 90% of samples are below is determined.

available system resources. If the sensor performs the observation at a lower frequency, then sensor time could be utilized for other tasks. The burden of a task (ζ_i) is the fraction of a sensor's total capacity that that task consumes, and can be simply calculated as $\zeta_i = f_i T_{Di}$. For a remote-sensing system to perform n different tasks, each with its own individual f_i and T_{Di}, the total burden placed on the system (ζ) would be

$$\zeta = \Sigma_{i=1}^{n} \zeta_i = \Sigma_{i=1}^{n} f_i T_{Di} \ . \tag{5.8}$$

For a remote-sensing system to be capable of performing all n tasks, the total burden of those tasks must be less than the number of available sensors, *i.e.* For a satellite with single sensor, which I assume SBR systems to operate as, the maximum burden is 1.[3] For a constellation of multiple satellites, I assume the average burden of each modality to be constant across satellites. Across an entire constellation, the maximum burden at any location must be less than the multiplicity m.

Each modality, such as SMTI or SAR, will have some characteristic dwell time for a single observation $T_{D0,i}$.

[3] An AESA array can be operated as multiple sub-apertures. Doing so splits the power between each sub-aperture and decreases the gain of each sub-aperture. While the decreased gain and increased number of sensors can be used to observe multiple locations at once, the decreased power density will result in the mapping rate staying equal. In addition, the decreased gain will decrease the performance of SMTI as the MDV will increase, as such, I assume all operate is as a single sensor.

In a tracking situation, an SBR satellite will be tracking some number of satellites, or monitoring some area, each of which may require some number of individual measurements. Since the burdens sum linearly, if q observations are needed to monitor an area, or q target TELs, then the total dwell for that modality $T_{Di} = q \cdot T_{D0,i}$. For area monitoring

$$T_{Di,\text{area}} = \frac{A_{\text{area}}}{\dot{A}_i} T_{D0,i} \ , \tag{5.9}$$

where A_{area} is the area being monitored and $\dot{A}_i$ is the mapping rate of the i^{th} modality. If the seeker knows the number of TELs in the deployment area, then the seeker can focus on observing only the areas surrounding those TELs and

$$T_{Di,\text{focused}} = (n_{\text{TEL}} f_d + 1) T_{D0,i} \ , \tag{5.10}$$

where f_d is the fraction of TELs at the base which are deployed at a given time. The f_d is assumed to be as close to unity as possible, as the hider has a vested interest in keeping their TELs deployed, but TELs need to be resupplied and crews exchanged periodically. The additional target that needs to be tracked is the TEL base itself. Since the deployment area is small relative to the field of regard of the satellite, I assume that all TELs within the deployment area are in view simultaneously.[4]

A SBR beam footprint, depending on the altitude of the satellite and the gain of the aperture, is approximately a few kilometers in width, and a few tens of kilometers in length. Characteristic uncertainty distances are on the order of ones to tens of shelter spacings s, in this analysis $s = 3$ km, making them of a similar scale to a beam width.[5] As such, the focused-tracking dwell time $T_{Di,\text{focused}}$ to scan the entire uncertainty area of a TEL will be proportional to the size of its uncertainty area. The rate at which the uncertainty area grows is dependent on the revisit interval, which depends on the focused-tracking dwell time $T_{Di,\text{focused}}$. Thus, the linear uncertainty (which must be projected onto a road network) is exponential, with the rate of uncertainty growth $R(p_{\min}, T_{Di,\text{focused}})$ dependent on the uncertainty. As a TEL becomes more lost, it becomes more difficult to track.

For modeling purposes, this non-linearity makes it difficult to estimate the dwell times, and therefore rates of uncertainty growth, over time for a general case. When a constant dwell time is assumed, the general case can be assessed. There are two bounding cases, assuming a single observation with dwell $T_{Di,0}$ is sufficient to track each TEL, and assuming that the entire deployment area needs to be monitored. When a TEL first deploys only a single measurement is needed, so $T_{Di,0}$ is accurate. However, if $R(p_{\min}, T_{Di,0})$ is non-zero, the required number of observations per dwell will grow and $T_{Di,0}$ underestimates $T_{Di,\text{focused}}$. Conversely, if the seeker monitors the entire deployment area, areas where TELs are not suspected to be will be included in each dwell, overestimating $T_{Di,\text{focused}}$ which is uncharitable to the seeker.[6]

The two boundaries can produce significantly different results. If a single observation is assumed, 17 observations are needed per revisit to monitor 16 deployed TELs and one base. For a 125,000 km^2 deployment area, assuming a beam footprint 3 km in width and 25 km in length, more than 1,500 observations would need to be made per revisit.

[4]SBR operable ranges are on the order of 2 $\times 10^3$ km, resulting in a FOR area of 1.2×10^7 km^2. A 125,000 km^2 deployment area represents less than 1% of that area.

[5]Alternative shelter spacings are explored in Chapter 6.

[6]A seeker may want to monitor wide areas to maintain situational awareness, but likely would not do so at the expense of giving up the ability to track TELs.

To better approximate true $T_{Di,\text{focused}}$, using a constant value that is computationally tractable, while remaining charitable to the seeker, I assume the following. When a TEL initially deploys, its uncertainty will grow as $R(p_{\min}, T_{Di,0})$. I calculate the expected number of moves that TEL's uncertainty would span if uncertainty grew constantly at this rate, without detection

$$n_m = L/R(p_{\min}, T_{Di,0}) \ , \tag{5.11}$$

where L is the patrol lifetime. If uncertainty grew linearly, the uncertainty of the TEL would span $\bar{n}_m = n_m/2$ shelters on average. I base a constant dwell time $\bar{T}_{Di,\text{focused}}$ on requiring $\bar{n}_m$ observations per dwell. This will overestimate the rate of uncertainty growth at the beginning of a TEL's patrol, and underestimate it in the later parts. Since the average $\bar{n}_m$ is calculated based off of $R(p_{\min}, T_{Di,0})$, the smallest rate of uncertainty growth, this assumption is charitable to the seeker and closer to the true $R(p_{\min}, T_{Di,\text{focused}})$ than the lower bound.

Balance of Modalities. Burden, when used in conjunction with the Tracking Model, is a useful tool for determining how much area a single satellite can effectively cover. A SBR operator must balance the use of each modality to minimize the rate of growth of uncertainty from every mode of uncertainty growth. The simplest mode of uncertainty growth is the TEL having the opportunity to move from one shelter to the next without being detected by any modality. A TEL can also move at the MDV of the SBR system and attempt to sneak to the next shelter, which is only detectable by SAR. A TEL can also move to another shelter, but avoid being identified, making it ambiguous if it was indeed a TEL or confusable object, which could happen if the TEL is not observed with ISAR. A different rate of growth can be calculated for each of these modes as a function of the revisit interval.

Burden can be used to select the set of modality revisit intervals that minimizes overall uncertainty growth. The burdens of all modalities obey

$$1 \geq \zeta_{\text{SMTI}} + \zeta_{\text{SAR}} + \zeta_{\text{ISAR}} = \frac{T_{D,\text{SMTI}}}{\tau_{\text{rev,SMTI}}} + \frac{T_{D,\text{SAR}}}{\tau_{\text{rev,SAR}}} + \frac{T_{D,\text{ISAR}}}{\tau_{\text{rev,ISAR}}} \ , \tag{5.12}$$

If there are j different modes of uncertainty growth, the proper mix of modalities to be used by the seeker is the one which sets each of the mode-specific rates of growth equal to each other,

$$U_1 = U_2 = \ldots = U_j \tag{5.13}$$

where U_i are the mode specific rates. Since reducing any one rate decreases the resources available to the other two, increasing them. In situations where the U_j cannot be set equal, the minimum $U = \min\{U_i\}$ is used.[7]

The specific revisit intervals which satisfy these relationships can be derived from the Tracking Model where

$$\tau_{\text{rev},i} = f^{-1}(U_i) \ , \tag{5.14}$$

where $f^{-1}(U_i)$ is the inverted Tracking Model which provides the $\tau_{\text{rev},i}$ that produces a given U_i.

Balancing the revisit intervals in this way minimizes the overall rate of uncertainty growth, making the best use of the capabilities of each modality. From a modeling perspective, this is the means by which the Tracking Model can be applied to set the performance goals of a radar system. Since the Tracking Model considers the whole constellation,

[7]The sum of rates is not used as the modes of uncertainty growth are not indepedent, *i.e.* a gap in coverage can create an opportunity to move in multiple ways, but the TEL must select only one.

each satellite can be tasked with the same balance of modalities to achieve the desired level of coverage.

5.2.4 SBR Constellation Cost

While there are a large number variables in the design of an SBR system, they will all impact the cost of the system. Cost is ultimately the basis of comparison between different constellations. In this subsection I present a model for roughly estimating the costs of an SBR system as a whole.

In 2007 the Congressional Budget Office conducted a cost estimate study for a set of notional SBR constellations that were then under consideration for procurement[68]. Their estimated life-cycle costs ranged from $32 billion dollars for the low-cost estimate for a 5 satellite constellation to $118 billion dollars for the high-estimate for a 21 satellite system (adjusted for inflation). Given the capacity of the CBO, I treat this model as an authoritative cost estimate, recognizing that any estimate will, by its very nature have significant uncertainty. Unfortunately, the CBO report was not perfectly explicit about their methodology. The report did, however, present line items for different components of the system, which I use to reverse engineer the relevant portions of the model. All estimates are inflation-adjusted from 2007 to 2020 dollars, a factor of 1.26. All of the following costs are given in 2020 dollars. In addition, the cost estimates for the individual components are updated with additional information for trends in costs where available, *e.g.* changes in the costs of launch services over time. Their cost estimates, by category, for a 21-satellite constellation are presented in Table 5.3.

The CBO cost estimate can be roughly broken into payload and spacecraft, launch vehicles, and research and operations. A number of constellations were examined, with a planned 20-year operational lifetime, assuming that each individual satellite had a lifetime of 10 years[68]. As an example, breaking down the cost of a 21 satellite constellation, physical hardware accounted for $38–46 billion in cost and launch vehicles accounted for $3.4 billion, assuming each satellite had to be replaced once. Up-front research and development costs, followed by 20 years of operations, accounted for the remaining $41–69 billion[68]. I will evaluate each of these categories in turn in order to develop a model for estimating the costs of the constellations considered in this section.

Item	Cost [$B]	Percentage of Total
Launch Services	**3**	**4–3**
Hardware (Total)	**38–46**	**45–39**
- Ground Hardware	6–9	8–7
- Spacecraft and Integration	8–10	10–8
- Payload	24–28	28–23
Operations	**24–41**	**28–34**
Research and Development	**14–23**	**17–19**
Total	**83–119**	

Table 5.3: Summary of costs, in 2020 USD, and as a fraction of the total project cost, for the 21 satellite constellation analyzed in [68]. Ranges show the differences between their low- and high-cost estimates. Displayed sub-costs and percentages are rounded, so may not sum to the total.

Launch Services.

The decreased cost of space-launches has been lauded as making future space-based systems more affordable [58]. At the time of the CBO estimate, launches would have been carried out utilizing Atlas or Delta launch vehicles. Both had launch costs of approximately $20,000 per kilogram of payload [68,58]. Currently, launch services aboard a Falcon heavy rocket are advertised at approximately $1,700 per kilogram of payload, representing an order of magnitude reduction in cost [58]. In my model, launch services are therefore discounted, relative to the CBO estimate, by a factor of ten (after inflation adjustment). However, since launch vehicles only accounted for 3–4% of the total cost, this development does not represent a significant savings.

My cost model for launch services is

$$L(n_{\mathrm{sat}}) = 2n_{\mathrm{sat}}0.1L_r \ , \tag{5.15}$$

where $L(n_{\mathrm{sat}})$ is the estimated cost of launch services for a constellation of n_{sat} satellites, if it were planned today, and L_r is the per-satellite launch-services cost estimated by the CBO, with both costs in same-year dollars.[8]

Hardware

Hardware consists of the satellite itself, including the spacecraft and payload (the radar), as well as ground equipment needed to operate the satellites. For the 21-satellite system, hardware accounted for 38%–45% of the total cost of the program, but this fraction will scale with the number of satellites required. Ground hardware accounted for 16–19%, spacecraft and integration for 20–21%, and payload for 60–62% of the hardware costs.

Ground Hardware. I assume ground hardware costs to scale linearly with the number of satellites, representing communication links and computational and data handling capacity.

Spacecraft and Integration. The spacecraft refers to the satellite exclusive of the radar payload. Integration refers to the assembly of the different modules (radar payload, power systems, *etc.*) into a functioning SBR satellite. I assume that the spacecraft and integration costs scale linearly with both the number of satellites and with the power requirements. Per the approximation from *Fortescue* introduced in section 5.2.1, the radar payload accounts for the majority of the power needs of the system, so I assume that cost will scale one-to-one with payload power [107].[9]

Payload. The radar panels, transmit-receive (TR) modules, electronics, and communications systems comprise the payload. For example, the transmit-receive modules accounted for roughly a quarter of the total payload costs. However, the CBO estimate assumed that the price per module would decrease significantly from 2007 costs, based on presumed technological improvements, allowing them to predict cost of $3100 per module, of which 2600 would be needed for each square meter of aperture. I take this value as the current cost of a TR module. This corresponds to a cost of $8.2 million per square meter of aperture, or $330 million for a 40 m^2 aperture and $820 million for a 100m^2 aperture satellite. Per the CBO, the radar panels, TR modules, and electronics scale with the area of the aperture while the communications systems have a fixed cost. As such, I assume that the payload costs will scale linearly with both the number of satellites and with the area of the radar aperture used.

Hardware Cost Model. My cost model for the hardware of a prospective SBR system is

[8] This equation does not discount future expenditures. It assumes the treasury rate is equal to inflation.

[9] Other systems such as the communications and heat management systems will also face increased power needs as payload power increases.

$$H(n_{\text{sat}}, A, P_t) = n_{\text{sat}} \left[\frac{A}{A_0} PL_r + \frac{P_t}{P_0} SC_r + \frac{P_t}{P_0} GH_r \right] \ , \tag{5.16}$$

where A is the radar aperture area of the prospective system, A_0 is the aperture of a reference radar with a cost estimate from the CBO, in this case 40 m^2; PL_r is the CBO per-satellite payload cost estimate for the reference system; P_t is the prospective system peak radar power; P_0 is the power of the reference system; SC_r is the reference per-satellite spacecraft cost; and GH_r is the per-satellite reference ground hardware cost. I assume that the ground hardware, which I assume to include data-processing hardware will scale with the mapping rate of the system, which scales with the transmit power.

Research and Operations.

Research and development covers investments in designing and testing components of the SBR system before full-scale production. The CBO estimates for R&D ranged between $14–24B. I assume these values as a fixed cost, which do not scale with the eventual size of a prospective constellation.

Operations, over a 20-year lifetime include satellite operations and tasking. These costs account for building, maintaining, and operating the facilities which control, process, and analyze the information gathered by the satellites. This also covers program management costs. Since the amount of information that needs to be handled will scale with the number of satellites in operation, I assume that the operation costs will grow linearly with the number of satellites.

Advances in artificial intelligence could reduce the costs of interpreting radar observations into usable intelligence products. The line-item for data interpretation accounted for 10–15% of the total cost of the system, as estimated by the CBO, which would need to be spent on hardware, infrastructure, and personnel to interpret collected images. Image interpretation algorithms could reduce the need for personnel, but would still require hardware and support infrastructure. It would also require a considerable research and development effort, such as project MAVEN, an ongoing research effort by the U.S. Department of Defense to develop image interpretation algorithms. Real-time tracking of multiple targets utilizing both SMTI and SAR would necessitate a similar or larger effort.

My research and operations cost model is

$$RO(n_{\text{sat}}) = n_{\text{sat}} O_r + RD_r, \tag{5.17}$$

where O_r is the per satellite operations cost, and RD_r is the reference R&D costs.

Constellation Cost Model

My overall cost model is

$$C(n_{\text{sat}}, A, P_t) = n_{\text{sat}} \left[\frac{2}{10} L_r + \frac{2A}{A_0} PL_r + \frac{2P_t}{P_0} SC_r + \frac{P_t}{P_0} GH_r + O_r \right] + RD_r \ , \tag{5.18}$$

where the factor of 2 accounts for a single hardware replacement during the lifetime of the system.

5.2.5 SBR Systems for Tracking TELs

Based on the design constraints presented in this section, I have identified a number of SBR systems that are capable of tracking TELs. Since tracking must be defined in the context of a given counterforce attack, I present the constellations which are capable of tracking in each of these contexts.

China

Base Region	Latitude	Number of Bases	TELs/Base	τ_f [min]
Southern China	28°N	2	12	7
Central China	35°N	2	12	10
Total	35°N	4	48	

Table 5.4: Parameters for each TEL base considered for the modeled U.S. counterforce attack against China.

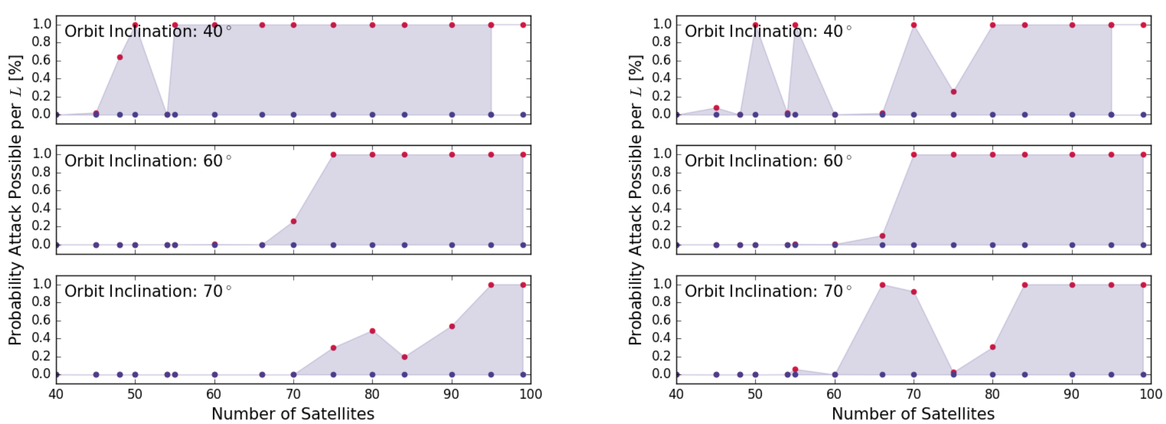
Figure 5-3: Approximate locations of Chinese TEL bases which operate TELs carrying ICBMs. Number denotes the PLARF Base number. Figure derived from a map under the GNU Free Documentation License 1.2 [110].

Some relevant parameters of a U.S. attack against the Chinese nuclear arsenal, used in the Counterforce model, are presented in Table 5.4. There are four bases in the People's Liberation Army Rocket Force (PLARF) which operate TELs carrying DF-31 class ICBMs [111]. Each base is roughly 1000 km from any other, and, based on latitude, I sort them into two groups, bases 62 and 63 in Southern China at roughly 28° N, bases 64 and 66 at roughly 35° N. Figure 5-4 shows the ability of the Baseline System (see Table 5.1) to track TELs as a function of the number of satellites and inclination of the constellation against each of these regions utilizing warheads with $r_w = 3$ km. Table 5.5 summarizes the different attacks analyzed in this section and presents the minimum number of satellites at which there is a non-zero probability of a successful attack existing against all targets on a linear or gridded road system, respectively, as laid out in Chapter 3. Since the probabilities presented in the associated figures are the probability of an attack existing for a randomly selected deployment lifetime L, with enough time, a bolt-from-the-blue attack will eventually be possible for any non-zero result (though a window existing during a shorter crisis is less likely).

Figure	Number of Warheads	Warhead r_w	Flight Time (τ_f) [min] (Southern China, Central China)	Required n_{sat} (linear, grid)
5-4	180	3.0 km	7, 10	50, -
5-5	180	5.0 km	7, 10	45, 50^a
5-6	180	3.0 km	0, 0	50, 55

Table 5.5: Parameters and enabling constellations for attacks presented in the figures indicated. In all attacks, warheads are allocated against a deployment area in each Southern China and Central China regions. Of the 24 TELs at each base, 22 are assumed to be on patrol at any given time, including monitoring of the base, 23 targets are assumed to be monitored simultaneously. The required number of satellites is determined for both the linear and gridded road scenarios. The - symbol indicates that no constellation tested enabled an attack against any deployment area, under the road system assumed. The ^{a} symbol indicates partial capability against the Southern China deployment area.

In the first scenario, with results presented in Figure 5-4, none of the gridded road model attacks are possible. In fact, even with no positional uncertainty at the time of attack, these attacks would not be assured due to the distance that TELs could move during the flight of the warheads. For flight times of 7–10 minutes, a TEL moving at $v_{\text{TEL}} = 10$ m·s^{-1}, the TEL could move 4.2–6.0 km. On a gridded road system with roads spaced evenly every 3 km, 7 warheads with a r_w of 3 km are required to tile the area into which the TEL could move which implies that some TELs would survive if the number of available warheads and targets are similar. Tracking of a TEL on a linear road system, which serves as the lower, more-forgiving bound becomes possible with 48–50 satellites. In particular, the 50:48/12/7 constellation has capability against both deployment areas simultaneously.



(a) Southern China

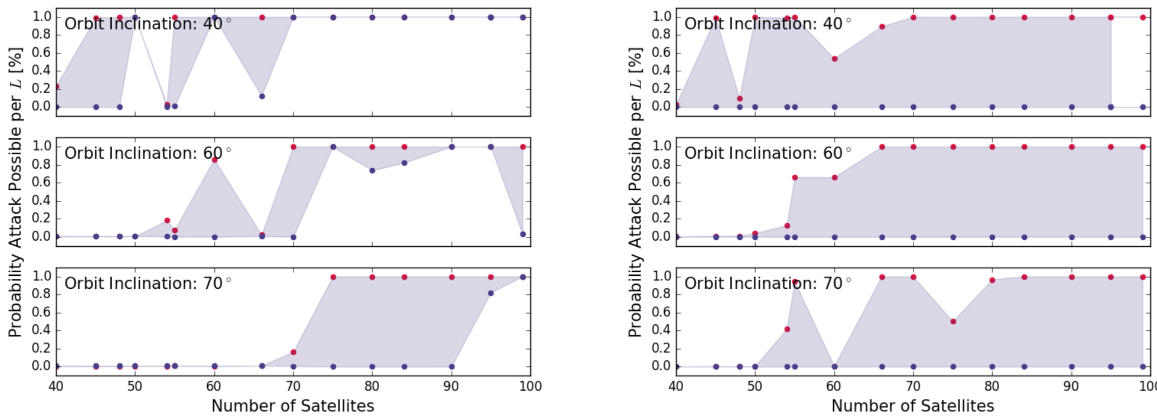
(b) Central China

Figure 5-4: Probability of successful attack existing during a deployment lifetime $L = 3$ days against Chinese mobile missiles in Southern China (a) and Central China (b) as a function of n_{sat} for a number of different constellation inclinations. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

A second scenario is one in which the U.S., in preparation of an attack, uploads two submarines with higher yield W88 warheads which have a r_w of 5 km. The U.S. maintains approximately 400 W88 SLBMs in its inventory so would have the capacity to do so.[10] Figure 5-5 shows the probability of successful attacks existing under these conditions.

[10]This includes W88 warheads in storage which would need to be deployed, requiring significant premeditation. Such an attack would likely be illegal, violating the law of armed conflict (LOAC). However, such a scenario could be a concern from

With the higher yield warheads, attacks against a gridded road system become possible with a few constellations against targets in Southern China, the smallest having 50 satellites, though no such attacks are possible against Central China.



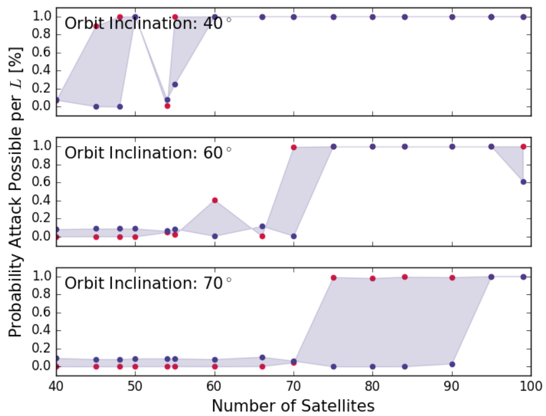
(a) Southern China

(b) Central China

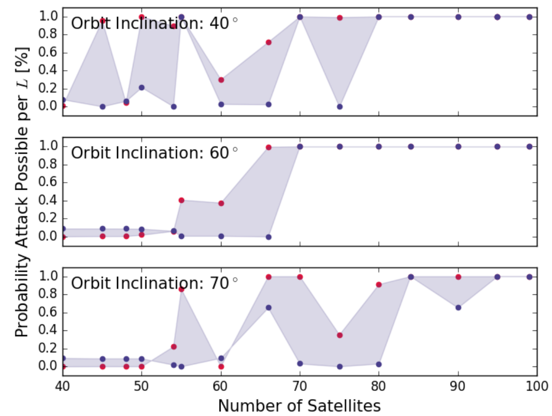
Figure 5-5: Probability of successful attack existing against Chinese mobile missiles at Southern China (a) and Central China (b) if, utilizing W88 SLBMs with $r_w = 5$ km. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

A third scenario to consider is one where, given China's lack of an early-warning system, Chinese TELs, unaware of impending attack, do not move during missile flight. Given that the attacker has control over the exact time of launch, it could be planned for a window where opportunities to move at the estimated $p_{\min}$ are not available in order to minimize the chance of the target TELs moving of their own volition during the attack. Figure 5-6 shows the probabilities of successful attacks existing for this scenario. Under these conditions, attacks against TELs deployed on a gridded road become possible at 60–70-satellite constellations, as the only driver of warhead requirements is the initial positional uncertainty.

the perspective of China.



(a) Southern China



(b) Central China

Figure 5-6: Probability of successful attack existing against Chinese mobile missiles in Southern China (a) and Central China (b) if, given China's lack of early warning system, the target TELs do not move during missile flight. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

Russia

Figure 5-7: Approximate locations of Russian TEL bases which operate TELs carrying ICBMs. Figure derived from a map under the GNU Free Documentation License 1.2 [112].

Some relevant parameters of a U.S. attack against the Russian nuclear arsenal, used in the Counterforce model, are presented in Table 5.6 and approximate locations of Russian TEL bases are shown in Figure 5-7. The Russian missile bases are clustered into view areas based on the FOR of a satellite. There are three bases within a few hundred kilometers of Moscow. The two bases in Central Russia at Novosibirk and Barnaul are within 250 km of one another Other bases, such as Irkutsk, are isolated. Due to the proximity of bases in the Moscow and Central Russia regions,

Region	Latitude	Number of Bases	Bases in View	TELs/Bases	τ_f [min]
Moscow	55°N	4	2	18	7
Central Russia	55°N	4	2	18	10
Irkutsk	55°N	2	1	18	10

Table 5.6: Parameters for the modeled U.S. counterforce attack against Russia for each qualitative type of deployment area. Number of bases indicates the number of deployment areas of that type, bases in view indicates the average number of deployment areas that are simultaneously in view of a satellite for an average overflight.

Figure	Number of Warheads	Warhead r_w	Flight Time (τ_f) [min] (Moscow, Central Russia, Irkutsk)	Effective Constellation (linear, grid)
5-8	450	3.0 km	7, 10, 10	66, -
5-9	450	5.0 km	7, 10, 10	50, 66[a]
5-10	450	3.0 km	4, 7, 7	66, 66[a]

Table 5.7: Parameters and enabling constellations for attacks presented in the figured indicated. In all attacks, warheads are allocated against four deployment zones in each Moscow and Central Russia deployment areas, and two in the Irkutsk region. At each base, 16 of 18 are assumed to be on patrol at a given time. Including the base, there are 17 targets that need to be tracked at a time, per base. Two deployment areas are assumed to be in the satellite field of regard simultaneously in the Moscow and Central Russia regions. The - symbol indicates that no constellation tested enabled an attack against any deployment zones. The [a] symbol indicates partial capability against only the Moscow region.

I assume that in any given pass, an average of two bases are in the FOR at a given time.[11] For the isolated bases such as Irkutsk, I assume a single area is in view at a given time, decreasing the number of targets to be tracked from 34 to 17.[12] Table 5.7 summarizes the different attacks analyzed in this section and presents the minimum number of satellites at which there is a non-zero probability of a successful attack existing against a target on a linear or gridded road system, respectively.

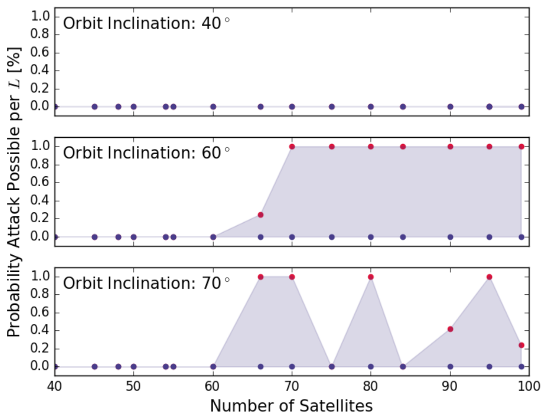
The first attack scenario considered is an at-parity attack against Russian TELs from a U.S. SLBM utilizing W76 warheads with a $r_w = 3$ km. The probability of a successful attack existing in a randomly selected patrol lifetime of length $L = 3$ days is presented for each classification of deployment area in figure 5-8. Under these conditions, there are 450 U.S. warheads available to target the 180 TELs.[13] Given that the ratio of warheads to targets is 2.5, no attack against targets on a gridded road system is assured since a minimum of 7 warheads is needed to account for TEL movement under these conditions. Since all deployment areas have similar latitudes, they behave similarly with respect to the inclination of the SBR constellation. High-inclination constellations begin to have tracking capability with 50 satellites for the favorable linear-road boundary. While there are some differences between regions in terms of flight times of weapons and number of targets to be tracked, none of these effects are strong enough to effect the outcome of attacks between areas.

The second scenario, similar to the Chinese case, involves utilizing larger warheads with $r_w = 5$ km. The U.S. maintains only 400 W88 SLBM warheads in its arsenal, however I assume that the larger warheads can be

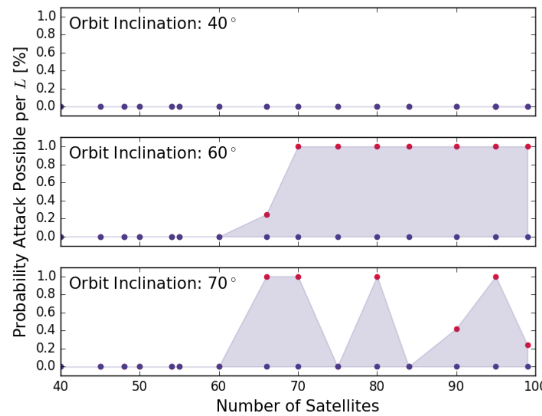
[11]The scenarios where only one of a cluster of bases is in view at a given time are largely those where the satellite is very far from the cluster, so would be observing from a large incidence angle and would therefore be of limited effectiveness. At intermediate incidence angles where satellites are more likely to have unobstructed views of TELs, it is much more likely that areas separated by only a few hundred kilometers will be within view simultaneously.

[12]Of the 18 TELs allocated to a base, I assume 16 are on patrol at a given time. In addition, the base needs to be monitored to detect the beginning of patrols. I examine the effect of varying the fraction of TELs deployed in Chapter 6.

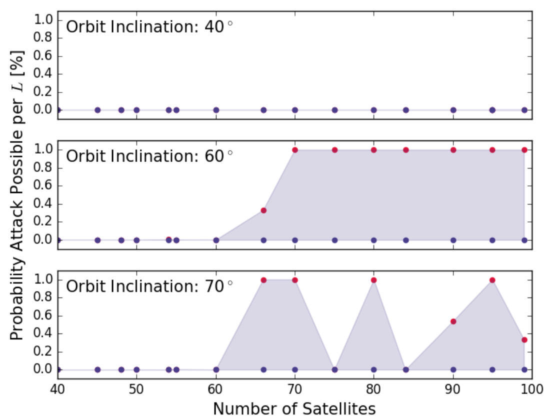
[13]The RS-24 Yars missiles are MIRVed with 4 warheads, reducing the ratio of targets to warheads.



(a) Moscow



(b) Central Russia



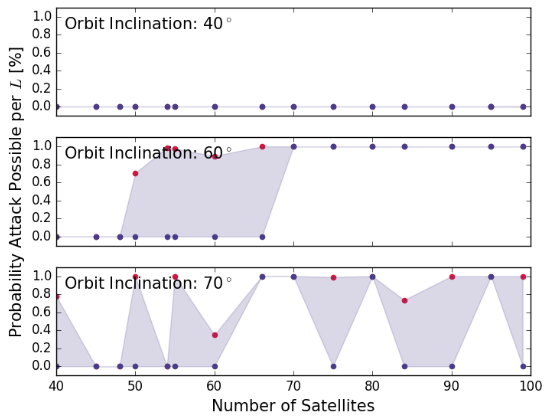
(c) Irkutsk

Figure 5-8: Probability of successful attack existing during a deployment lifetime $L = 3$ days against Russian mobile missiles in the (a) Moscow, (b) Central Russia, and (c) Irkutsk regions, respectively, as a function of n_{sat} for a number of different constellation inclinations. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

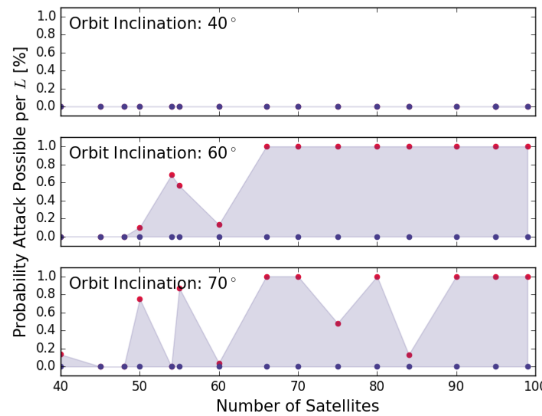
supplemented with smaller W76 in order to make up the balance without losing coverage.[14] Figure 5-9 shows the results of such an attack against each deployment area. Under these conditions, the 60° inclination constellations continue to perform the best. The shorter flight time to the Moscow region bases increases the probability of a successful attack existing for the linear-road cases, and begins to show some capability against the gridded road system. Due to the longer flight times to the Central Russia and Irkutsk bases, attacks against gridded road systems are not possible, but there is an increase in the number of constellations which have capability against linear road systems. The difference between the Central Russia and Irkutsk regions shows the effect of the increased tracking burden that comes from clustered deployment areas.

The third attack scenario, similar to the Chinese case, involves a decreased missile flight time representing TELs moving only when warned by the Russian Early Warning System (EWS). The Russian EWS consists of a set of three

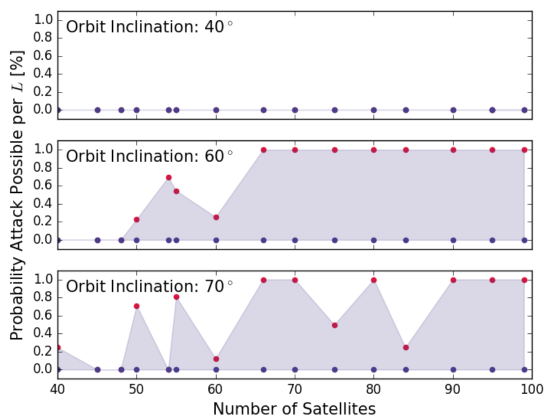
[14]Weapon effects of different sized warheads do not tessellate without significant overlap. However, I assume that there will be situations where only a fraction of the larger warhead's effects will be utilized where a smaller warhead could be substituted without loss in efficacy. I am assuming that 1 in 9 warheads can be substituted thusly for this analysis.



(a) Moscow



(b) Central Russia



(c) Irkutsk

Figure 5-9: Probability of successful attack existing against Russian mobile missiles at (a) Moscow, (b) Central Russia, and (c) Irkutsk, if the attacker were utilizing W88 SLBMs with $r_w = 5$ km. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

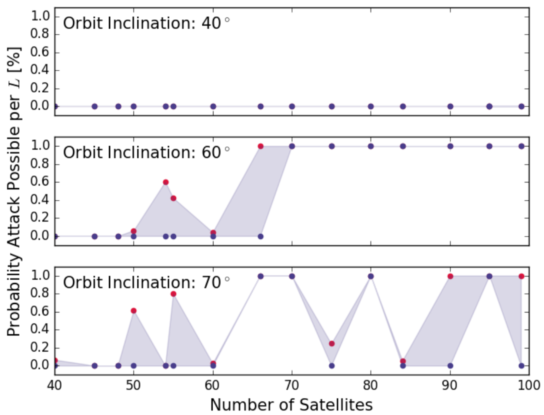
satellites and a network of ground-based radar [45]. I assume that the Russian EWS will operate on a similar timeline to the U.S. EWS.[15] The U.S. system is capable of detecting Russian launches within 1 minute and 2 minutes of launch, for space sensors and ground-based radars respectively, it then takes some amount of time to confirm and communicate the detections. In the case of incoming attack, TELs can begin to move when the attack is first communicated to them, a "move-on-warning" posture, since, unlike launch on warning, moving TELs carries considerably less risk of escalation.[16] As such, I assume that it requires 3 minutes from the time of launch to make initial detections, and broadcast a move-on-warning signal to TEL deployment areas.[17] This has the effect of shortening the effective flight time of incoming weapons to 4–7 minutes, depending on the region.

[15]Though less capable than the U.S. system, especially in terms of space assets, the launch of SLBMs from near the Russian coast will mean that the Russian ground-based radars will be operable earlier in the flight of the incoming missiles.

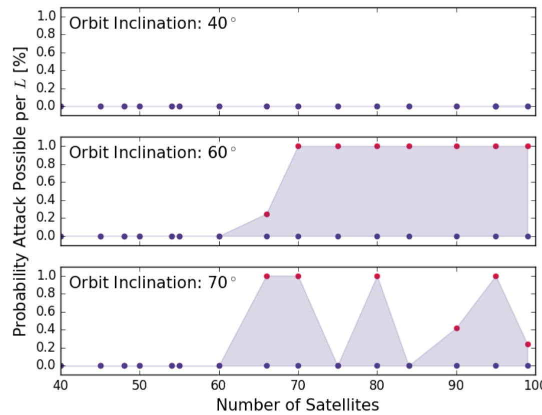
[16]Launch-on-warning involves launching at risk weapons between when the incoming attack is detected and before the warheads arrive.

[17]TELs in hiding may have reduced ability to move on short notice, for example, if they are deployed under camouflage netting. I assume that, if TEL operators chose to commit to a move-on-warning posture, they would amend operations to allow for quick movement.

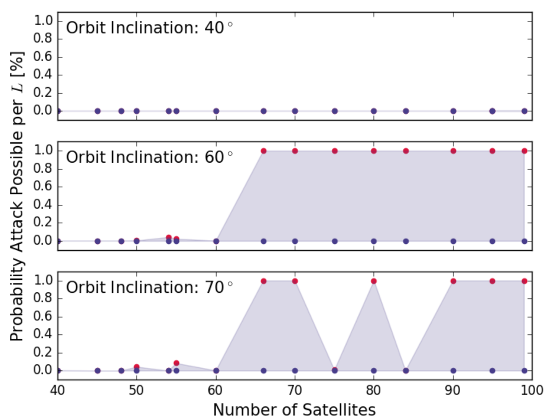
Figure 5-10 shows the results of such an attack. The decreased warning time of the attack causes attacks against the Moscow area to become more feasible, including on a gridded road system with 54 satellites. Central Russia and Irkutsk are sufficiently in the center of the Asian continent that the decrease in warning time does not effect the probability of an attack against these deployment zones existing, compared to the attack in figure 5-8.



(a) Moscow



(b) Central Russia



(c) Irkutsk

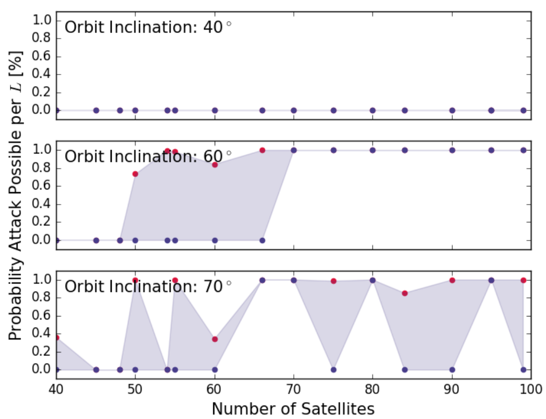
Figure 5-10: Probability of successful attack existing against Russian mobile missiles at (a) Moscow, (b) Central Russia, and (c) Irkutsk, if the attacker assumes the target TELs will not move except after the attack is detected. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

5.2.6 Alternative Hardware and Constellation Designs

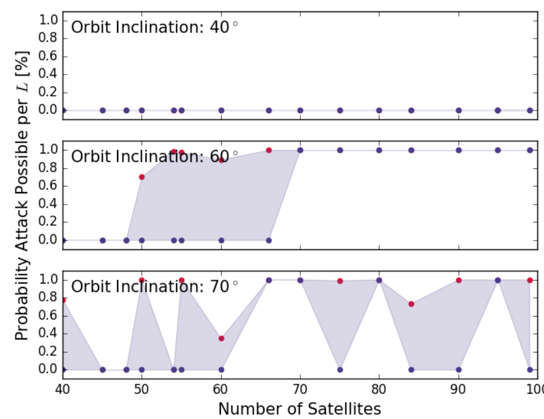
The previous section looked specifically at the Baseline System, which utilized a 1000 km altitude, 40 m^2 apertures, and 10 kW peak power. There are alternative choices that can be made for each of these variables, noted in Table 5.8 which are explored here.

Figure 5-11 shows attack plots against targets in the Moscow region, utilizing weapon with $r_w = 5$ km, for constellations at 1000 km altitude varying the size of the aperture and the power supply. There are effectively no differences in the curves. This is for a number of reasons. Increasing the size of the aperture increases the gain, which can allow a system to collect images at longer distances, however the Baseline system has sufficient power to image the entirety of its field of regard with SMTI for 10 m^2 targets. Increasing the gain decreases the dwell time of SMTI measurements, but also decreases the size of the beam footprint, so does not improve the revisit interval. A larger aperture will have a lower MDV as the beam swath-width (and therefore clutter Doppler-spread) is smaller. In a highly sight-blocked environment such as a forest, MDV effects are small however, as those view-angles where the radial velocity is likely to fall below the MDV (*i.e.*, approaching orthogonal to the velocity vector) are also the aspect angles that are likely to be sight-blocked by objects lining the road (such as trees).

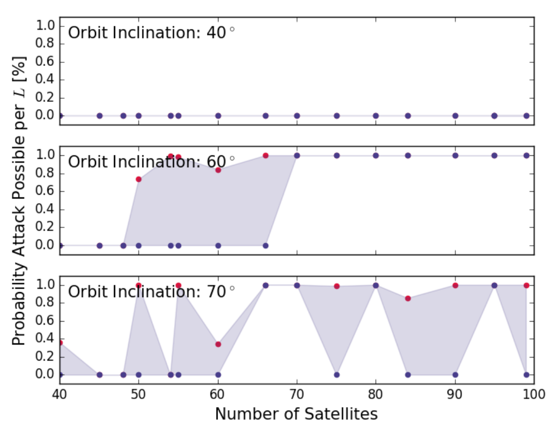
Increasing the power does improve the revisit interval by decreasing the SMTI dwell time. It also increases the range at which SAR images can be collected (as does improving the gain). However, in focused-tracking mode, the revisit interval with the baseline system is already sufficiently short that TELs moving in the open–even for a short period–are likely to be detected. Growth in uncertainty is largely driven by sight-block, which is not affected by the revisit interval.



(a) Large Aperture



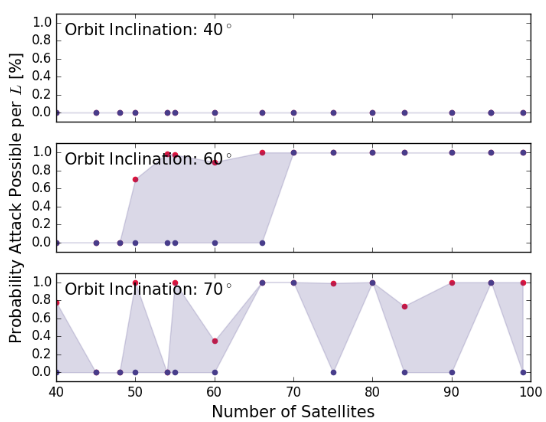
(b) Baseline



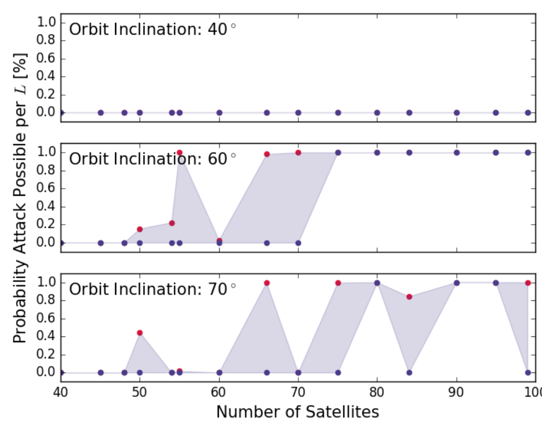
(c) Higher Power

Figure 5-11: Probability of successful attack existing against TELs in the Moscow Region during a deployment lifetime $L = 3$ days using weapons with $r_w = 5$ km as a function of n_{sat} for a number of different constellation inclinations for the Large Aperture (a), Baseline (b), and Higher Power (c) systems. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

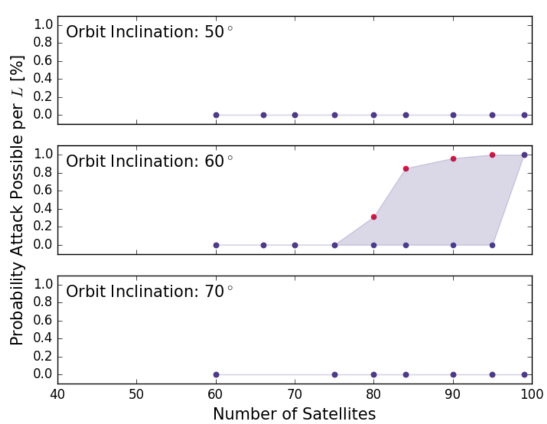
The effect of varying the altitude of the constellation is shown in Figure 5-12. Comparing the Baseline to the Medium Altitude system ,decreasing the altitude from 1000 km to 700 km has two effects. First, it decreases the average slant-range, increasing SNR, decreasing beam footprint size, and reducing the revisit interval for both SMTI and SAR observations. Secondly, it decreases each satellite's field of regard, effectively reducing coverage. This increases the number of satellites necessary to track TELs. For example, 70 satellites in the Baseline constellation at 60°inclination can track TELs on gridded roads under these conditions, but 75 Medium Altitude satellites are required. The Low Altitude system has even less coverage per satellite, greatly increasing the number of satellites needed to track TELs. The first Low Altitude system which shows tracking capability under the linear road assumption has 80 satellites, compared to 40 satellites in the Baseline system. The trade-off is that the individual satellites in the Low Altitude system are cheaper than in the Baseline system as they need less power and aperture.



(a) Baseline



(b) Medium Altitude



(c) Low Altitude

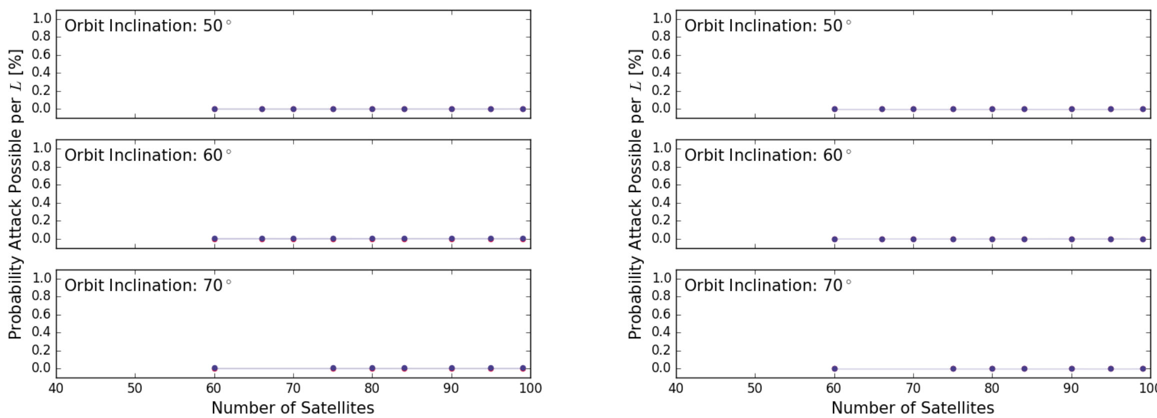
Figure 5-12: Probability of successful attack existing against TELs in the Moscow Region during a deployment lifetime $L = 3$ days using weapons with $r_w = 5$ km as a function of n_{sat} for a number of different constellation inclinations for the Baseline (a), Medium Altitude (b), and Low Altitude (c) systems. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

Coverage is more difficult to achieve near the equator. The decreased field of regard of satellites in the Low Altitude system are particularly detrimental to their effectiveness at tracking TELs at lower latitudes, such as those

Label	n_{sat} China	n_{sat} Russia	n_{sat} Both	Cost Per Satellite [$B]	Total Cost [$B]
Large Aperture	50	45	70	1.6 – 1.9	280 – 370
Baseline	50	45	70	0.76 – 0.90	220 – 300
Higher Power	50	45	70	0.85 – 1.0	230 – 310
Medium Altitude	55	55	80	0.76 – 0.90	254 – 340
Low Altitude	> 99	85	> 99	0.38 – 0.45	270 – 370

Table 5.8: Constellation design parameters evaluated in this section. For each set of parameters, the minimum number of satellites needed to enable TEL tracking is calculated for the Chinese and Russian attacks, utilizing $r_w =$ 5 km warheads. The minimum number of satellites needed to track both Chinese and Russian TELs is calculated. The cost per satellite is the marginal cost of the hardware of a single SBR satellite, excluding R&D and operational costs, launch services, operational cost *etc.*. The total lifetime cost of a system capable of tracking both Chinese and Russian TELs is presented in the final column.

of the Chinese bases. Figure 5-13 shows the lack of efficacy of the Low Altitude system at tracking targets in China.



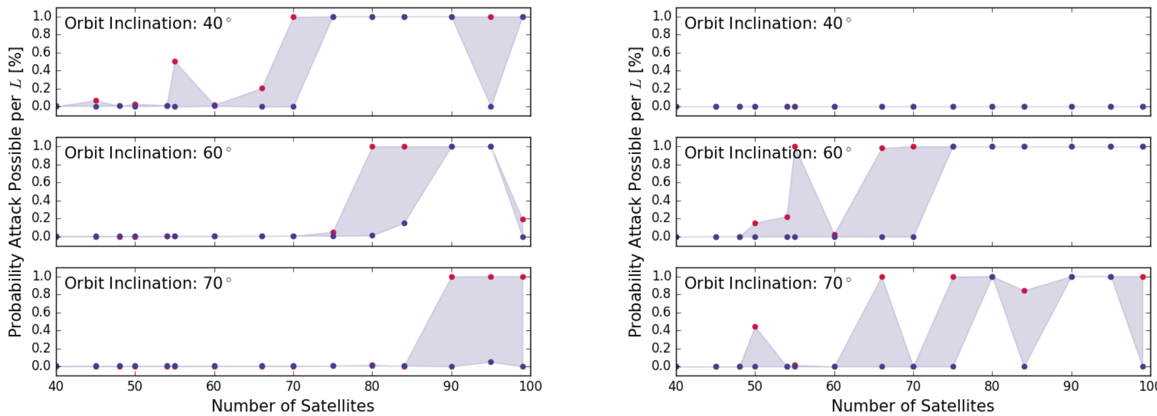
(a) Low Altitude Southern China

(b) Low Altitude Central China

Figure 5-13: Low Altitude tracking of Chinese TELs. Probability of successful attack existing during a deployment lifetime $L = 3$ days using weapons with $r_w = 5$ km as a function of n_{sat} for a number of different constellation inclinations against TELs in the (a) Southern China and (b) Central China regions. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

Lowering altitude also makes it more difficult to design a constellation which is capable of tracking TELs across multiple latitudes. Figure 5-14 shows the capability of the Medium Altitude system against the Moscow and Southern China regions. While a constellation of 55 satellites has some tracking capability (under a linear road assumption) against either region, 80 satellites are needed to have tracking capability against both.

Table 5.8 shows the minimum number of satellites of each constellation design which are capable of enabling counterforce against each China and Russia, assuming the attack structures presented in figures 5-4 and 5-8, respectively.



(a) Medium Altitude Southern China

(b) Medium Altitude Moscow

Figure 5-14: Probability of successful attack existing during a deployment lifetime $L = 3$ days using weapons with $r_w = 5$ km as a function of n_{sat} for a number of different constellation inclinations against TELs in the (a) Southern China and (b) Central China regions. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

5.3 SBR Tracking Conclusions

In this chapter, I presented how SBR modalities can be used in tandem to complete the tasks of tracking outlined in Chapter 2. SBR therefore has the capability the current gap in tracking capability because of its ability to operate at night and through cloud cover. I then presented how the SBR modalities would need to be balanced in order to minimize the seeker's uncertainty in TEL positions. Utilizing this information, I showed how an SBR constellation could be designed for the purposes of tracking TELs and identified constellations which have the potential to track TELs in the contexts of the different counterforce attacks presented in the previous chapter. From the process of designing and evaluating these SBR constellations, there are some conclusions.

Road networks greatly shapes counterforce probability of success. The local shape of the road network greatly affects the feasibility of counterforce attacks against mobile targets. If the incoming attack can be detected (or the attacker chooses to respect the possibility that the target TELs may move), then the target TELs will have the ability to move some distance before the warheads detonate. On a road system with relative freedom of movement, such as a grid, the additional area into which the TEL can move can be prohibitive for a barrage attack given limits on warhead numbers. In the case of a 7-min flight time and a gridded road system with roads spaced 3 km apart, a minimum of 7 warheads are needed to barrage a TEL's potential movement area, even if its location was known exactly at the time when the attack was launched.

While management of positional uncertainty with a remote-sensing system is clearly a requirement for enabling counterforce, it is not sufficient in-and-of itself. A sufficient number of warheads need to be made available, or the time available for the TELs to move during the attack needs to be reduced. Against a state which possessed an EWS, such as Russia, warning time can only be reduced to the point where the EWS can detect a launch, communicate it to the TELs, and the TELs are able to act upon it.[18]

[18] China has recently expressed an interest in developing an EWS as well. While this could make their TELs more survivable to attack, it also introduces risk of accidental escalation[7].

If the attacker could successfully leverage guided weapons such as hypersonics or maneuverable RVs, then this additional warhead cost per target could be circumvented. Under those circumstances, the attacker could barrage the area of positional uncertainty, and if the target TEL moved during the attack, could adjust the aim points accordingly. The modeled attack against Chinese forces with no warning time, presented in figure 5-6 can also serve as the model for this scenario. In Chapter 3, I discuss guided weapons and their vulnerabilities to countermeasures which make guided-weapon attacks unreliable.

Diversity in latitude improves survivability. The Chinese deployment areas are separated in latitude by roughly 7°. This improves the overall survivability of their mobile forces. The attack plots in section 5.2.5 show the efficacy of SBR constellations for a number of constellation inclinations. In many cases, constellations which are effective at tracking TELs in one deployment area are less competent at tracking in the other. Since an SBR constellation designer must commit to a single inclination when building the constellation, it becomes challenging to find a single constellation design which can track TELs at diverse latitudes.[19]

This becomes especially true when considering the separation in latitude between Russian and Chinese deployment areas. Russian TELs are deployed at significantly high latitudes, so require constellations with larger inclinations (which yield better coverage at higher altitudes) to track. Conversely, Chinese TELs are best tracked by constellations with lower inclinations (which provide better coverage closer to the equator). Designing a constellation with capability to track both would require more satellites than one designed to track the TELs of one in particular.

The enabling constellations identified by this analysis all require more than 50 SBR constellations. Per the Cost Model, the minimum cost of a TEL tracking constellation is in excess of $164B. Such a constellation would need to be designed to target either Russia or China. As such, a general purpose SBR constellation would be more complex and costly than the ones identified here.

Countries could conspire by deploying TELs at different latitudes. While a single country may be limited in diversity of latitudes available in which TELs can be deployed, multiple countries could conspire to deploy their TELs across different latitudes in order to make a comprehensive SBR system more expensive.[20]

The required number of satellites increases with decreasing satellite altitude. A major design trade off is the selection of a satellite orbital altitude. Altitude drives a number of hardware selection choices, such as power, and aperture size. Loosely, the choice can be reduced to choosing between a small number of powerful satellites in orbits near the upper edge of LEO versus having a larger number of smaller satellites closer to Earth.

There is some nuance in the design decision in balancing the altitude with desired SMTI MDV. For a fixed aperture size, higher altitudes result in larger beam footprints, which will widen the Doppler signal of ground clutter. Satellites at lower altitudes will have a narrower beam footprint, and more power on target (assuming a fixed power). However, not decreasing the aperture or power of the satellite while reducing altitude will increase the number of satellites required without reducing the per satellite cost.

As noted in the cost model, the largest driver of cost is in the number of satellites. The hardware - power and aperture - only affect the hardware procurement costs, which is only a fraction of the overall cost. As such, the most cost effective constellations will be those in higher orbits (to the limit where the required power becomes infeasible due to transmission loss).

[19] Satellites in orbit can change the inclination of their orbital planes, but it is a very costly maneuver in terms of fuel [113].

[20] This need not be a premeditated conspiracy *per se*, but simply accounting for where TELs are deployed in other countries in order to make one's own more difficult to track.

The hider can take countermeasures. In the design of these constellations, I have implicitly assumed that the hider is aware enough of the need to defeat tracking that they are willing to shape the way the operate their TELs in order to make tracking more difficult. I have stopped short of considering ways in which the hider could actively interfere with tracking. In that way, designing a constellation by the process outlined here would be the seeker embracing a fallacy of the last move.

As I show in the next chapter, there are a number of options available to the hider through which they can contest the seeker's ability to track with SBR. I will reserve the detailed discussion of those for then, but it is relevant now to realize the limitations of the process outlined here. If the seeker were to design the minimum system capable of tracking, any countermeasures taken by the hider which made tracking more difficult could defeat that tracking capability. The question then becomes one of the cost of the countermeasures to the hider versus the cost of the SBR system to the seeker. As I have shown, the price point on a tracking-capable SBR system is in the hundreds of billions of dollars. In the next chapter, I explore the countermeasures available to the hider and how their costs may compare to that of an SBR system.

The design process presented here is not invalidated by the presence of countermeasures, however. The countermeasures simply add a new constraint or alter an input parameter, such as the probability of detection. The modular nature of the forward model allows this design process to be adapted to the changing conditions.

Chapter 6

Countermeasures to SBR Tracking

6.1 Countermeasures to SBR Tracking

In Chapter 5, I developed a model for determining the number of SBR satellites needed to enable a counterforce attack against a fleet of TELs which was behaving under a set of constraints. The TELs were operating with knowledge of the seeker's SBR system and were attempting to operate in way to maximize the growth of the seeker's uncertainty, but were not taking active measures to interfere with the operation of the SBR system. Under these conditions, SBR systems have the capacity to track TELs if deployed in sufficient enough numbers to overcome the effects of sight block and reduce the opportunities for TELs to move between shelters with a high probability of being undetected. In this chapter, I introduce a number of methods that the hider could use to interfere with the operation of SBR tracking. These countermeasures are roughly ordered in terms of effort and provocativeness. The first three countermeasures, discussed in subsections 6.1.1, 6.1.1, and 6.1.1, involve ways to increase the tracking burden placed on the seeker by altering the way in which TELs are operated. These alterations would not necessarily be sustainable in the long run, but could be enacted in a crisis to improve survivability in a unprovocative manner. Subsections 6.1.2, 6.1.3, 6.1.5, and 6.1.4 cover countermeasures which improve survivability in an unprovocative manner, but require an investment in time, money, or technological development. In subsections 6.1.6, and 6.1.7, I discuss means of actively interfering with the operation of SBR systems with electronic warfare (jamming) and anti-satellite (ASAT) weapons, active interference being the most provocative countermeasures.

This is followed, in section 7.1, by a discussion of the dynamics between remote-sensing technologies and countermeasures to them. In subsection 7.1.1, I define the term critical technologies as those technologies which are essential to tracking, and so provide targets for countermeasures. In subsection 7.1.2 , I provide an estimate for the size of an SBR constellation that would be capable of enabling tracking of TELs. In subsection 7.1.3, I discuss how different countermeasures could be used to increase the costs of tracking, or defeat tracking outright. This is followed by a discussion of how the attainability of countermeasures can shape the trajectories of arms races in subsection 7.1.4. I provide a historical example to compare the case of SBR to that of sonar tracking of SSBNs in order to identify hallmarks of technologies which are likely to be destabilizing. Finally, in section 7.1.5, I conclude this thesis with policy recommendations for how to avoid or defray the risks to strategic stability from the deployment of SBR systems in the future.

6.1.1 Location and Length of Patrols

A driver of the warhead efficiency of counterforce attacks, and the burden of tracking TELs is the dispersal of TELs across their deployment areas. The hider must spread its TEL patrols in order to prevent TEL positional uncertainties from overlapping such that a single warhead could threaten multiple TELs. They must also decide where within their territory to deploy their TELs. Choices such as the range of latitudes across which deployment areas are spread, and the number of TELs operated in a deployment area, impact the survivability of TELs. The length of patrol lifetimes impacts the average and maximum positional uncertainty of TELs, and provides a means for the hider to increase the seeker's uncertainty, subject to logistical limitations of keeping those TELs supplied and operating.

Location of Patrols

Patrol areas, more specifically the areas that the seeker's uncertainty is expected to encompass, need to not overlap one another if the hider wishes to get the most deterrent value out of each TEL. The way in which mobility and concealment provide survivability is by making an attack require a prohibitive number of warheads per target. If the hider allows uncertainty areas overlap by having TELs patrol near one another, a single warhead may have effect against multiple target, increasing the warhead efficiency of the attack. The hider must therefore maintain proper spacing of TELs within a deployment area.

Beyond maintaining the minimum distance to prevent overlap of uncertainties, significant advantage is not gained by dispersing TEL patrols further in space, unless the seeker is using wide-area monitoring. As discussed in Chapter 4, I assume that the seeker is utilizing focused tracking - monitoring only the area surrounding where the TEL was last seen.[1] Increasing the dispersion of the patrol areas does not, in most cases, increase the burden of tracking.[2] The same number of areas need to be observed each revisit interval, even if they are slightly more spread out. Spreading out deployment areas may actually decrease tracking burden as it counters some of the benefits of clustering.

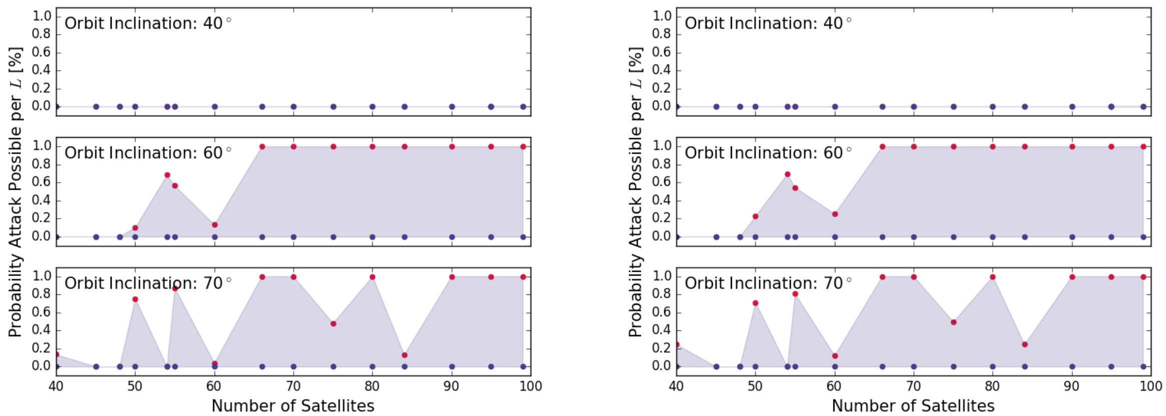
Clustering of deployment areas, as happens with some Russian TEL bases, was identified in the previous chapter as a factor which can make tracking more difficult. Clustering causes more TELs will be in view of each overflying satellite at a given time, increasing the number of looks that the satellite needs to make during each revisit interval. This same effect can be generated by increasing the number of TELs deployed at a single base. Figure 6-1 shows the difference in the attack plots between clusetered bases in the Central Russia region versus an isolated base in Irktusk, assuming the same weapon flight time to both.

Similar to clustering of bases, placing deployment areas at a variety of different latitudes can also create difficulties for a seeker. Satellite coverage varies by latitude. Orbits can be designed to optimize coverage at one latitude or another, with higher inclination orbits having better coverage at high latitudes and lower inclination orbits having better coverage at low latitudes. Figure 6-2 shows how constellations which are capable of tracking TELs at one latitude may have poor performance at tracking another. By having deployment areas span a number of latitudes, the hider can prevent the seeker from being able to utilize an efficient, specialized constellation.

Where TEL bases are located, and the number of TELs at each base are not parameters that can be changed on

[1] More specifically, monitoring the area encompassed by the positional uncertainty and the routes to the surrounding shelters.

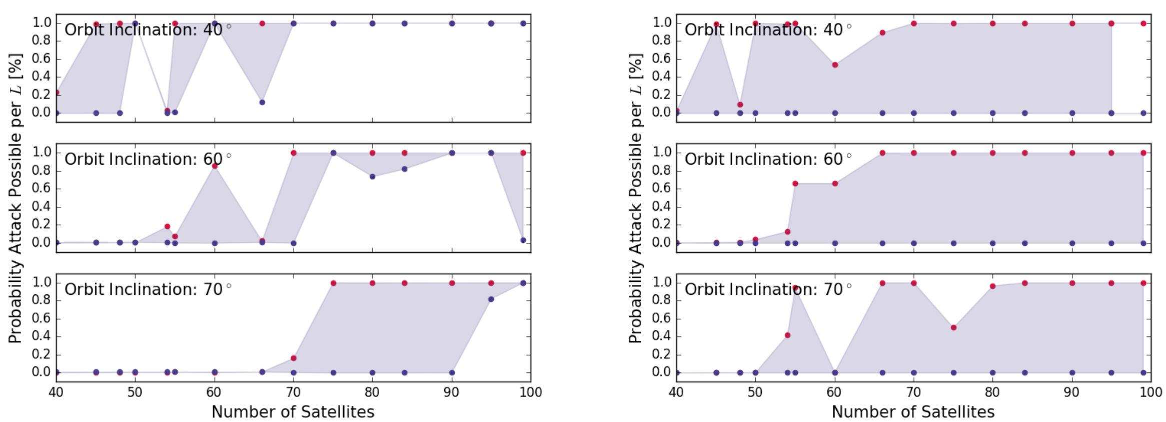
[2] Deployment areas are small relative to the field of regard of satellites. Dispersal could cause issues for tracking if the deployment area were large enough that it was consistently not wholly within view, for instance by spanning the nadir hole so that the array needed to by physically slewed in order to get the whole area in view. The nadir hole is 200–700 km in diameter, depending on satellite altitude, so dispersing patrol areas out would only place patrols on opposite sides of the satellite if they were dispersed that far, and then only for direct satellite overflights.



(a) Central Russia

(b) Irkutsk

Figure 6-1: Difference in the probability of successful attack existing for clustered versus isolated bases. For a deployment lifetime $L = 3$ days against Russian mobile missiles in Central Russia (a) and Irktusk (b) regions, respectively, utilizing $r_w = 5$ km warheads. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.



(a) Southern China

(b) Central China

Figure 6-2: Probability of successful attack existing against Chinese mobile missiles at Southern China (a) and Central China (b) if, utilizing W88 SLBMs with $r_w = 5$ km. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

short notice, but are decided at the force planning stage. While I would not consider clustering bases and having a diversity of latitudes a countermeasure *per se*, a well-planned deterrent force can exploit its benefits. Where possible, bases should be placed at significantly different latitudes. Bases should either be clustered together, or the number of TELs assigned to each base should be as large as is manageable, in order to increase the tracking burden on overflying TELs.[3] Clustering should be done on such a scale that multiple bases appear within an overflying satellites field of view while maintaining enough surrounding area for TELs to patrol without having their uncertainty areas

[3]The advantage of having a cluster of bases over one large base is that, presumably each base in the cluster would need to be attacked, increasing the number of targets.

overlap.[4] Clustered bases separated on the order of a few hundred kilometers, such as the Russian bases at Barnaul and Novosibirsk, leaves sufficient area to keep patrols from overlapping while keeping both within satellite fields of regard for a majority of overflights.

Increased Patrol Length

Increasing patrol length increases the average TEL uncertainty across the fleet. From Chapter 4, the maximum number of moves that the i^{th} TEL could have moved since it was deployed is

$$k_i = \text{floor}\left\{R(p_{(min)})(t - t_{\text{deploy},i})\right\} \ . \tag{6.1}$$

Increasing the patrol lifetime L means that t can become larger, increasing k_i on average. Increasing k_i increases the number of shelters that the positional uncertainty of a TEL can be expected to encompass, provided that a non-zero $R(p_{\min})$ exists. Figure 6-3 shows the Irktusk attack with $r_w = 5$ km, with the deployment lifetime L varying from two days to two weeks. The ability of the attacker to threaten TELs on the linear road model decreases as L increases. Selection of L provides the hider with a degree of control over the seeker's uncertainty.

The cost of increasing the lifetime is that TELs need to return to their bases periodically to exchange crew, resupply, and undergo maintenance. The upper boundary on the feasible L depends on the hider's ability to keep their TELs supplied with fuel, fresh crew, *etc.* This can be approached by increasing the amount of supplies that a TEL convoy carries with it, or by resupplying TEL convoys in the field, or a combination thereof. Convoy size cannot be increased indefinitely and will be limited by practical concerns, such as how many vehicles can simultaneously fit in a shelter.

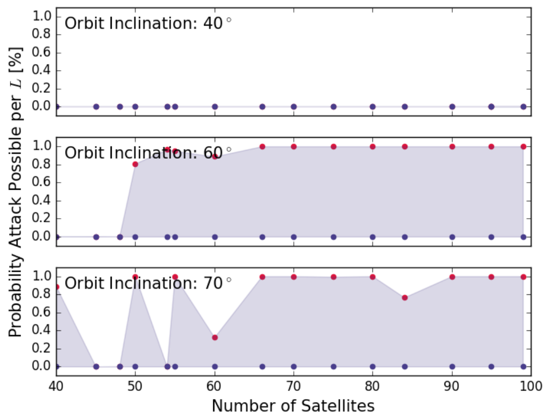
The alternative is to resupply TELs while they remain on patrol. This could be done by sending vehicles containing fuel, food, fresh crew, *etc.* to rendezvous with the TEL convoy. If the seeker were able to track the resupply convoy, they could track it to the TEL convoy location, so the hider would need to obfuscate the locations of the rendezvous. The most direct means of doing so would be to operate resupply convoys like TEL convoys, moving only during gaps in observability. Alternatively, TELs could leave supplies at a shelter in advance of a TEL visiting, essentially creating caches of supplies throughout the deployment area. Resupply convoys could also take circuitous routes through the deployment area, visiting many shelters along the route, one of which contained the TEL convoy.[5]

Resupply vehicles could also be useful to the hider as a low-risk means of bluffing. Sending a conspicuous resupply convoy to an area could bluff the presence of a TEL in that area. In conjunction with not moving a TEL (the value of which is discussed in 6.1.1), the hider could feign TEL movement by sending a resupply convoy to the edge of the TEL's area uncertainty. While such feints are not cost free, there would only be so much resupply capacity, the resupply could create a supply cache at some point along the route, or intersect its route with an actual TEL along the way.

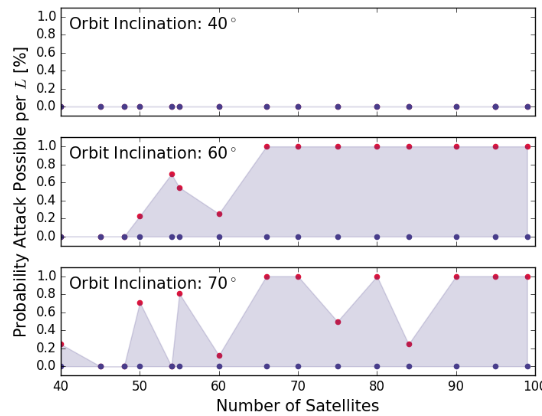
Increasing deployment lifetime is attractive as an option at or near the outset of a crisis. The added uncertainty takes time to accrue, but the action is not initially detectable by the seeker, so would not be provocative and would

[4]Fields of regard typically span a few thousand kilometers, while individual patrol areas are likely on the scale of a few tens of kilometers in radius. Uncertainty areas can become larger when the quality of tracking is poor, under those conditions, some amount of overlap is tolerable as an attack is likely to be infeasible regardless.

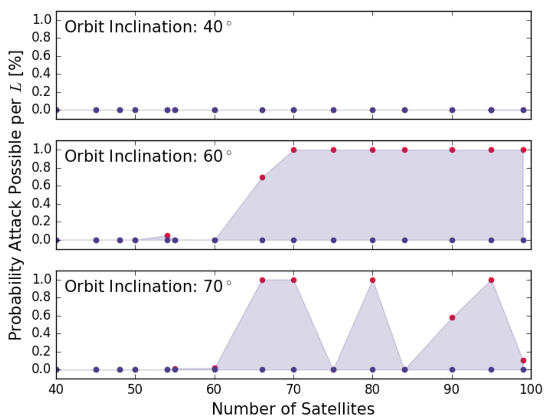
[5]At each stop, the resupply convoy would need to stop for a roughly equivalent amount of time, sufficient to actually perform a resupply. The convoy would also need to take care to appear outwardly the same before and after resupplying to avoid indicating where supplies were offloaded.



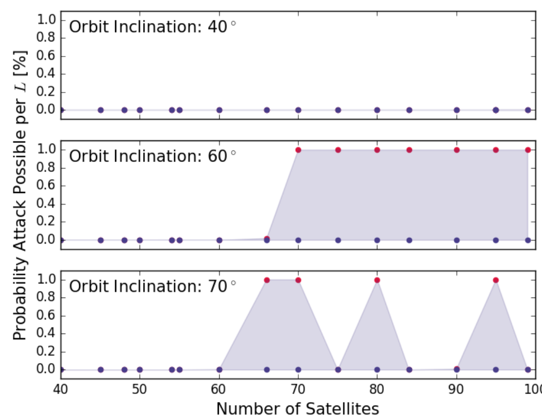
(a) $L = 2$



(b) $L = 3$



(c) $L = 7$



(d) $L = 14$

Figure 6-3: Difference in the probability of successful attack existing for an attack against Irktusk utilizing $r_w = 5$ km warheads, for $L = 2$, 3, 4, and 7 days, respectively. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

not risk inviting escalation. This means that it can be an option that is readily taken whenever tensions begin to develop in order to preserve the value of a deterrent.

Increasing patrol lifetimes has a cost in terms of personnel and fuel. The increased deployment lengths would increase the demands placed on missile crews. My baseline assumed lifetime is three days. This is a short patrol length, an assumption in favor of the seeker, but one that can account for possibilities such as missile dispersal in a rapidly escalating crisis. Russian TELs have demonstrated the capability to conduct patrols with lengths in excess of three weeks [114]. meaning that extending patrol lifetimes is an accessible countermeasure, though requires advanced planning to enact.

The seeker is unable to adapt to increased deployment lifetime outside of increasing constellation tracking capacity. Patrol lifetimes can also be extended after a constellation has been placed into orbit in order to degrade its tracking capability. As such, correctly predicting the hider's ability to sustain patrol lifetimes is a source of uncertainty to a SBR system designer.

Fraction of TELs deployed

The fraction of TELs assigned to a base that are on patrol at a given time determines the number of targets which need to be simultaneously tracked in a deployment zone. In the analysis in this and previous chapters, I have assumed that 2 TELs are in base undergoing resupply or maintenance at a given time, out of the 18 or 24 TELs assigned to that area. This assumption implies that the hider is cognizant of the need to make their TELs survivable, so is keeping as many on patrol as they sustainably can. The hider may vary the fraction of TELs that they keep on patrol over time, as a function of their concern over the survivability of their TELs.

During periods of low tensions, TEL operators may keep a smaller complement of TELs on patrol at any given time in order to save on costs and preserve crew and TEL readiness. During such periods, there will be fewer targets to track, reducing tracking burden, and also more targets concentrated at the TEL base, increasing warhead efficiency of an attack. Figure 6-4 shows the attack plots for an attack against Irktusk with $r_w = 5$ km and $L = 3$ days, for $n_{\text{TEL}} = \{4, 8, 16, 18\}$. With fewer TELs on patrol, more constellations have tracking capability, with attacks against a gridded road system possible with 4 TELs deployed.

If TEL operators keep a smaller fraction of their TELs on patrol during periods of low tensions, then they may become vulnerable to a bolt-from-the-blue attack. The hider therefore needs to balance between cost savings and their fear of a bolt-from-the-blue. If the hider perceives a worsening of the political situation, they could choose to increase the fraction of TELs on patrol. Doing so would be detectable by the seeker. If, with the TELs concentrated in base, the seeker believes that they are capable of a disarming attack, this would place a pressure on the seeker to attack before TEL dispersal removed the option.[6] Such an issue exists with the Chinese nuclear deterrent currently, as it is believed that the nuclear weapons themselves are kept in central storage and meant to be dispersed if tensions rise.

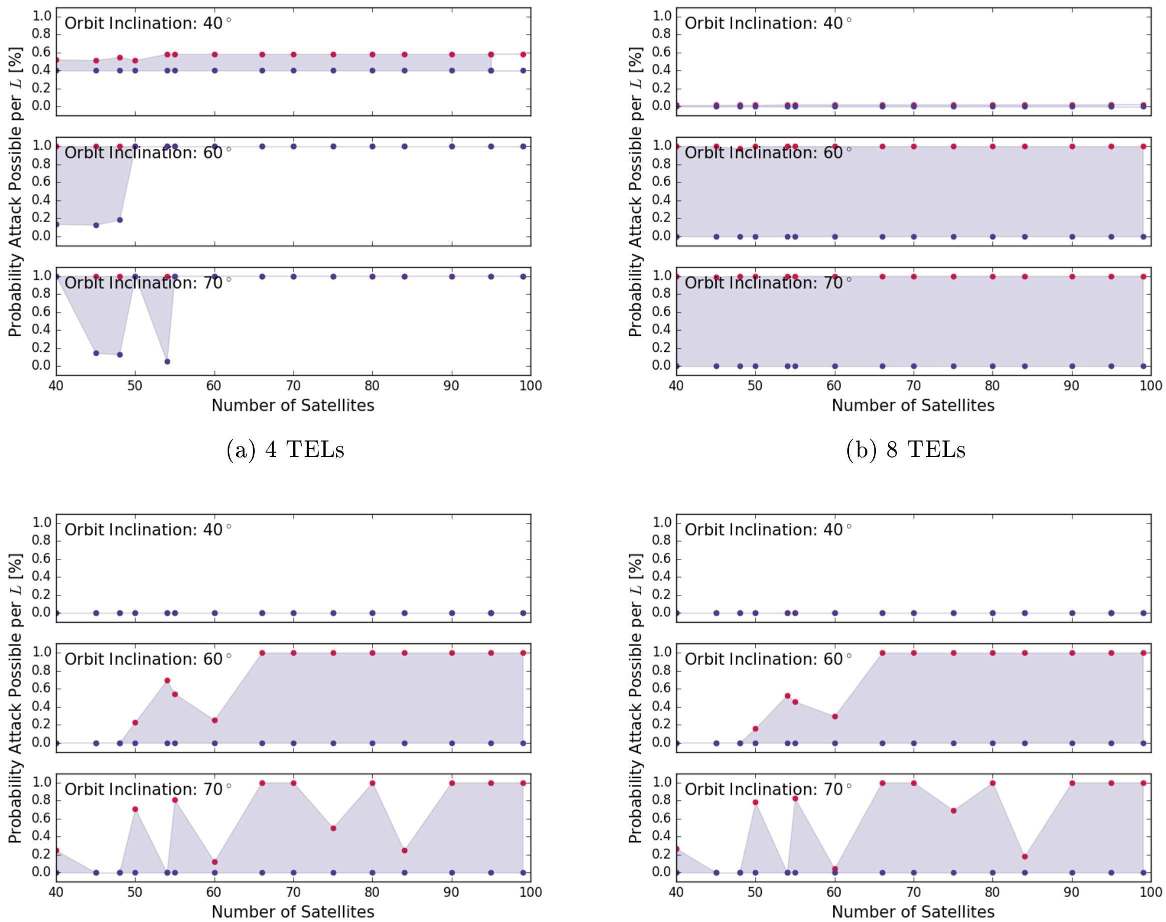
States relying on such a posture need to be cognizant of the risk of escalation that comes from dispersing TELs. Ideally, a state would set a precedent for modulating the fraction of TELs on patrol during peacetime, to reduce the perceived novelty of dispersal. It may also be valuable to ramp up the fraction deployed slowly. States should also disperse TELs onto patrol early in a crisis while tensions are relatively low when the other pressures to conduct a preemptive counterforce attack are low.

Not Moving TELs in a Crisis

One advantage that the hider possesses is that they determine when to move their TELs. The complement to this observation is that the hider can also choose when not to move their TELs. As noted in Chapter 4, moving TELs usually entails risk of detection, which can be construed as a cost to moving. The benefit to moving is in signaling, the hider uses movement to demonstrate their ability to defeat tracking, forcing the seeker to respect that ability and not attempt an attack. Since the seeker's ability to determine if a TEL moved during its patrol is retrospective, during the patrol of a given TEL, the seeker is unable to distinguish between the TEL moving undetected, and the TEL not moving (and therefore not having the opportunity to be detected if it is in a shelterg).

Since moving entails a risk of detection, not moving a TEL allows the hider to keep a TEL from being detected, while allowing the seeker's uncertainty in its position to grow. If done regularly, the seeker can retrospectively learn

[6] The pressure is placed on the seeker regardless of its magnitude or whether the seeker is swayed by it. The existence of a pressure toward nuclear use does not imply that nuclear use must follow, but it does increase the chance that it will.



(a) 4 TELs

(b) 8 TELs

(c) 16 TELs

(d) 18 TELs

Figure 6-4: Difference in the probability of successful attack existing for an attack against Irktusk utilizing $r_w = 5$ km warheads, for $L = 3$ days, with 4, 8, 16, and 18 TELs deployed respectively. Red dots show a linear road model, blue dots show gridded road model, shaded area shows where the two differ.

that the hider is not moving TELs, and adjust their beliefs about how likely TELs are to have moved accordingly.[7] If done sparingly, such as during a crisis, the seeker will not learn of it until the TEL ends its patrol and, presumably, the crisis has passed.

In choosing to not move a TEL, the hider is increasing the probability that the TEL will be destroyed in an attack in exchange for increasing the seeker's uncertainty in the TELs position, in the hopes of deterring the attack in the first place. The advantages can be marginal, it is most effective when $p_{\min}$ is close to unity, which is also the case where the probability of detection is already small. Under these conditions, the probability distribution of moves since last detection, introduced in Chapter 4 becomes a delta function at its maximum value k.[8] This method provides the hider a means of the risk of TEL detection.

[7]Assuming that TELs are moving less than they are capable of narrows the shape of the uncertainty PMF towards the initial shelter, which will always increase (or leave constant) the perceived probability of destroying the TEL in an attack.

[8]Except in the case where a TEL started its patrol by moving and then later stopped, in which case uncertainty will grow from the time of that detection, and no further detections will be risked.

6.1.2 Shelter Spacing

Shelters are an integral part of the hider's ability to keep TELs undetected for an extended period of time. As noted in Chapter 4, shelters are any location in which a TEL cannot be detected by remote-sensing systems. This could be a natural area, or a permanent or temporary structure.[9] It is under the hider's control how many, and how closely spaced these shelters are.

Fixed Shelters

Fixed shelters are permanent or semi-permanent locations for TELs to hide. Electromagnetic radiation, such as micro- and radio-waves are reflected by conductive surfaces, as discussed in Chapter 2. Dielectric substances, such as wood can also reflect or absorb radio waves in sufficient thickness. The hider could use a number of materials to build shelters.

A simple material for reflecting radiowaves is a conductive metal mesh. The size of holes in a mesh determines if it is radar-reflective. Radio waves are reflected by the metal of the mesh, but the holes in the mesh act as waveguides which can allow radiation to pass through if it has a high enough frequency. The cutoff frequency of a rectangular waveguide, the lowest frequency that can travel through the waveguide, is

$$\omega_c = c\sqrt{\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2}\ , \tag{6.2}$$

where c is the speed of light, n and m are mode numbers associated with the waveguide dimensions a and b, respectively, where $a \geq b$ The lowest frequency mode that is allowed is the TE_{10} mode where $n = 1$ and $m = 0$. The cutoff frequency of a waveguide is therefore

$$f_c = \frac{c}{2a}\ . \tag{6.3}$$

Assuming a square mesh (*i.e.* $a = b$), the smallest a that can support a X-band radio waves where $f = 10 \times 10^9$ Hz, is 1.5 cm, or half the wavelength of the radiation. For a conductive mesh to reflect X-band radiowaves, the mesh must have holes less than 0.015 cm, roughly the spacing of fine chicken wire.

If woven into camouflage netting, chicken wire could be used to create lightweight radar reflectors that could be used to create shelters using natural formations or pre-existing permanent structures unrelated to TEL deployment. A steep ravine or alleyway between structures can be transformed into a shelter by using conductive netting to block line of sight in any direction not sight blocked by the existing structure. Such structures could be created on a semi-permanent or temporary basis, allowing the hider to quickly increase local shelter density in an area in response to, or in preparation of, an increase in tracking capability.

A more permanent shelter could be built using lightweight construction materials such as sheet metal. As of the writing of this thesis, bulk sheet aluminum can be bought on Ali Baba for 2.05 USD per kilogram. For a 1mm thick sheet, the raw material price is 5.50 USD per m^2. A cuboid structure that is 10m wide, 30m long, standing 5m tall with covered ends (fashioned into doors) would require 700 m^2 of sheeting. This would cost, without including the costs of labor and upkeep, roughly 4000 USD in sheet metal. Assuming that labor costs account for roughly half of the total installed cost, I will assume that each shelter will cost 10,000 USD.

[9]Temporary structures cannot be so temporary as to only exist when the TEL is in it.

Achieving shelter spacings on the order of a few kilometers would cost tens to hundreds of millions of dollars per deployment area, assuming a gridded road system and complete and uniform coverage of the deployment area. This is the cost required to be able to compete with a SBR constellation. For the Russian case, with ten deployment areas, which is the largest case considered, the upper bound of the cost would be on the order of hundreds of millions of dollars to build shelters, an order of magnitude less than the cost of a single SBR satellite.

Covered Roads

The hider can also create road segments that permanently cannot be observed with radar systems by creating covered roads. This effectively reduces the inter-shelter spacing to zero. If a TEL enters a covered road section, after a time C/v_{TEL} where C is the length of the sheltered road segment, the seeker's uncertainty will span the covered segment. The hider can deter an attack from an attacker with a risk tolerance of P_k and n_w available by building a covered roads totaling at most

$$\sum_i C_i \leq P_k n_w 2 r_w \ , \tag{6.4}$$

of length, where C_i is the length of the $i$the covered segment and r_w is the weapon effect radius. The equivalence holds if all n_{TEL} are deployed within the covered segments, and the covered segments are built in multiples of r_w. I will take this as the upper bound. In practice the hider could keep TELs deployed outside of the covered roads in addition to the covered segments, and force the seeker to allocate inefficient numbers of warheads by building covered segments of non-integer multiple lengths of r_w. Covered roads could also be utilized as tunnels to connect naturally well-sight-blocked areas in order facilitate spread of uncertainty.

The cost of building covered roads is one of material, labor, and maintenance. For the case of a counterforce attack against Russian TELs, there are approximately 180 TELs, each with up to 4 warheads, constituting 720 target warheads. In order to force an adversary to trade at parity, utilizing 100 kt warheads with $r_w = 2.8$km, would require roughly 4000 km $/P_k$ of covered road segments in the worst case. Finding an inexpensive and durable building material would be required for a construction effort of that magnitude.

Chain link fencing could be utilized as a covering material. The upper estimated commercial price of an installed 4m tall fence is 120 USD per linear meter. Assuming that the density of the weave needs to be double the commercial density, fences on both sides of a road, a covering arch that costs the same as the sides, then the price of a fence covering would be roughly 1000 USD per linear meter. For 4000 km of road, this again would cost roughly 4 billion USD to have installed at the commercial market rate.

Such a large construction project would have overhead costs which would increase the total cost, as well as be able to secure bulk pricing which would decrease the cost. As a first-order cost estimate, a sufficient amount covered roads to defeat any SBR constellation would cost approximately the same as a single SBR satellite. Given that a competent constellation requires tens of satellites, this countermeasure is more cost efficient by an order of magnitude than the tracking system, and could be built in response to the development of the SBR capacity. The covering could also be built up piecemeal as extensions of shelters that are expanded over time in order to cope with an expanding SBR constellation.

6.1.3 Sight-Blocking Structures

The hider can build sight-blocking structures alongside roads if they want to decrease the probability of detection of satellites which are at large incidence angles. Sight-blocking structures, as discussed in Chapter 4, can impinge on lines-of-sight to a road segment as a function of their height and lateral displacement from the road. The maximum incidence angle ($\theta_{\max}$) at which a TEL on a road segment is visible is

$$\theta_{\max}(\alpha) = \frac{\pi}{2} - \tan^{-1}\left[\frac{h_{sb} - h_m}{d_r(\alpha)}\right] , \tag{6.5}$$

where h_{sb} is the height of the sight blocking structure, h_m is the highest point on the TEL which the seeker needs line-of-sight to, and $d_r(\alpha)$ is the distance to the sight blocking object from the center point of the TEL as a function of the azimuthal angle from the heading of the TEL α. Sight-blocking structures are most effective when they are tall, close to the road, or both.

Sight-blocking structures need to extend along the length of a road in order to be useful to the hider. In this way, they can be a resource intensive construction project, similar to that of a covered road. Unlike with a covered road, TELs are not immune to detection when traveling along uncovered roads, even with sight-blocking structures, especially against dense constellations which can achieve a small average incidence angle. Being similar in scope to covered roads, but less effective, building sight-blocking structures is only an efficient countermeasure if the sight-blocking materials are inexpensive, if they can later be built upon to cover a road, or if the hider needs to extend sight-blocking objects onto small segments of road.

I consider sight-blocking structures built as part of the process of creating a covered road, for example by building fences abutting roads that can later be covered with radar reflective netting, as part of the strategy of covering roads that can be approached in a piecemeal fashion. If the seeker builds a small constellation with a large average look angle, the hider could build road-abutting structures which would increase the rate of uncertainty growth as the overall probability of detection at any given point in time would decrease. This could cause the seeker to choose whether to expand the constellation to overcome these structures. If the seeker did, the hider could response by covering the roads.

While man-made structures would be effective at inhibiting tracking, they can also require a significant investment. An inexpensive alternative would be to plant tall vegetation abutting the road, or alternatively to cut narrow roads through vegetation. Roads would need to be wide enough for TELs to be able to turn when necessary, but would not need to be as wide as a civilian road. The hider is incentivized to keep their deployed TELs well separated within the deployment area, so these roads, so long as they were blocked to civilian traffic, would not need two lanes. In the case of extant forests, the average height of the mature trees would define the capacity to sight-block. If the hider wished to add abutting vegetation, trees would be a poor choice due to their slow rate of growth.

Bamboo is a family of tall grasses, which, depending on the species, can grow to heights exceeding 30 m [115]. Bamboo also grows extremely fast, some species elongating at more than a meter per day, and can reach full height in a single growing season [115]. The range of bamboo is limited climatically, but can be grown in both China and on the Korean peninsula, however it would depend on the suitability of local soil. Planting bamboo would be an inexpensive option for either of these two countries to, within a period of months, bolster sight-blocking along their TEL-deployment-area roads. Maintenance of bamboo-abutted roads would be necessary to clear fallen shoots to keep roads traversable. This could be done in advance of TEL movement by a smaller vehicle in the TEL convoy, or could be cleared regularly in advance of TEL deployment.

6.1.4 Radar Cross-Section Reduction

Radar cross-section reduction (RCSR) is the process of controlling the shape and the magnitude of an objects radar cross section in order to make it more difficult to detect with radar systems. The RCS of a target determines the magnitude of its signal, and how easily that target is distinguished from other targets. The magnitude of a signal determines how the required dwell times of observations. The relative RCS between a TEL and other vehicles determines how easy it is to identify a TEL, and how large decoys need to be. RCSR can be approached through changing the surface material of the vehicle, or through purposefully shaping the surface of the vehicle to minimize the occurrence of highly radar-reflecting features.

Radar-Absorbing Material

There are two primary approaches for modulation of the surface reflection characteristics, radar-absorbing materials (RAM) and surface-shape modulation. TELs are primarily made from metal, which is a near-perfect reflector of radio waves. Metal can be coated with RAM which absorbs a portion of the incident field, decreasing the magnitude of the reflected electric field.

RAM operate by diminishing the magnitude of radar reflections by coupling the incident field into a partially conductive, lossy medium [116]. At a boundary between two media (such as air and a TEL), some fraction of the incident energy will reflect off of the boundary and some will transmit into the second medium. Figure 6-5 shows a schematic of a wave approaching a boundary. At normal incidence, for a wave traveling from medium 1 to medium 2,

$$R = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad , \quad T = \frac{2\eta_2}{\eta_2 + \eta_1} \ , \tag{6.6}$$

where R is the reflection coefficient, η_i is the wave impedance of the i^{th} medium.[10] The reflectivity (r) and transmissivity t of the boundary are

$$r = |R|^2 \quad , \quad t = |T|^2 \ . \tag{6.7}$$

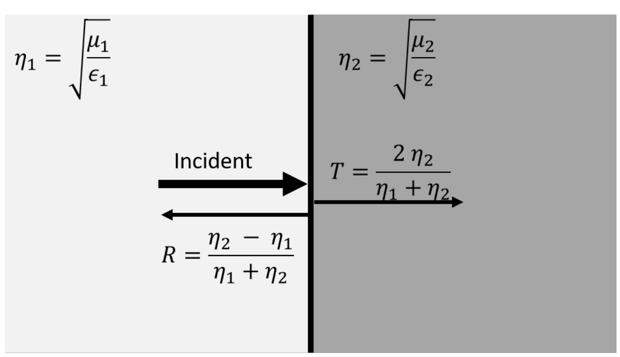


Figure 6-5: A schematic of an electromagnetic wave incident on a boundary of two materials.

[10]A more complicated treatment is necessary for oblique incidence, but the general principles are the same.

If the second material can be modeled as a perfect electrical conductor, *e.g.* it is made of metal, the wave impedance η_2 is zero so, with $r = 1$ and $t = 0$, the wave is full reflected. This is approximately the case for a TEL that is not covered with RAM.

RAM operates by placing a medium between the air and the TEL which is able to absorb some of the incident field. This presents a challenge. The incident field must be coupled into the RAM, and the RAM must be sufficiently conductive that a significant portion of the field is attenuated as it transits the RAM. Transmission is greatest and reflection the smallest when the wave impedances are closely matched. Coupling a large fraction of the incident field into the RAM necessitates that the wave impedance of the RAM be similar to that of air (which is often approximated as free space). Absorption of energy requires an impedance closer to that of a conductor. An ideal RAM will therefore have an impedance similar to free space, and similar to a perfect conductor simultaneously, which is not possible.

This problem can be approached by creating a layered RAM [65]. The air-facing layers have an impedance similar to that of air, while the inner layers are more conductive and therefore lossy. Utilizing multiple layers, the impedances can be gradually matched, allowing for efficient coupling of the incident field into the RAM. Figure 6-6 shows the spectral absorbency of an example multilayer RAM coating. One of the challenges with RAM is that impedance matching depends on both the frequency of the incident field, and the thickness of the layers. This creates a wavelength dependency in the RAM. The hider would therefore need to know the carrier wavelength of the seeker's SBR system when developing RAM.

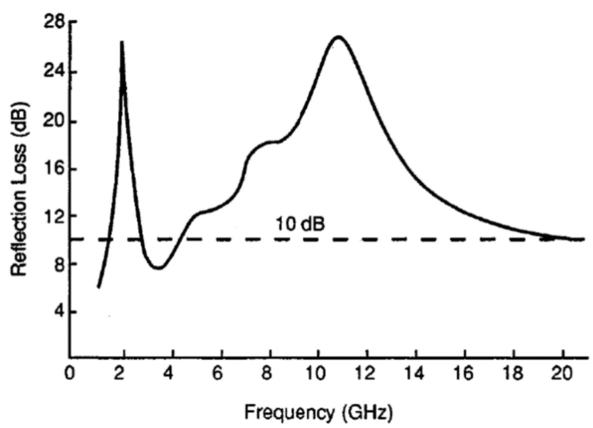


Figure 6-6: Spectral attenuation using a layered radar-absorbing material. Adapted from [117].

RAM are capable of decreasing the RCS by roughly two orders of magnitude in the X-band region [117]. While it is difficult to estimate the RCS of a TEL, assuming that it falls between the RCS of a truck ($\sim$ 20 dBsm) and a bomber ($\sim$ 10–30 dBsm) or a small ship (35 dBsm) would place it in the range of 20–30 dBsm [69]. A 30 dB RCS reduction in the X-band, if actualized, would place a TELs RCS in a similar range to that of a passenger vehicle [69,70]. This analysis has assumed a target RCS value of 10 dBsm, which is a conservative value, approximately that of a passenger vehicle or small truck. As such, successful application of RAM to reduce TEL RCS with RAM is already compensated for by the seeker in this model. It would however, increase the number of objects that could be confused with a TEL, increasing the burden on identification and making decoys more effective.

The challenges with RAM are cost and operational effectiveness. If the interface between the absorber and the air becomes dirty, the impedance can become mismatched, decreasing the efficiency of the absorber, and lessening the RCSR. TELs operating on remote roads would accumulate dust, water, leaves, *etc.*, all of which would degrade the effectiveness of the RAM. RAM coatings can also be fragile. The RAM utilized on F-22 Raptor airplanes must be reapplied before each flight [118]. RAM could be effective for increasing the burden of identification modalities, but would require significant operational demands in addition to any R&D efforts.

A second method of radar absorption is utilizing a geometric transition absorber [65]. A pattern of square pyramidal structures on the wall gradually changes the impedance of the surface, which allows for efficient coupling of the incident field into the structures. While lesser known because it is not useful for aircraft, this is utilized on the walls of indoor radar ranges to reduce the background signal of the room. Utilizing an absorptive material for the structures allows the field transmitted into the structure to be absorbed. While different in approach to RAM, the operational drawbacks of geometric transition absorbers are similar, any dirt or other debris that got caught within the textured surface would decrease the efficiency of absorption.

Shaping

A more robust method of RCSR is through shaping. Shaping allows the designer to control the RCS shape by removing shapes that intrinsically reflect large amounts of incident energy (like corner-cube cavities), and controlling the directions where the largest returns are directed. By considering the angles from which a TEL is expected to be viewed, shaping can be used to direct the largest returns out of these sectors, lowering the average measured RCS (even if the average RCS over the entire 4π steradian remains constant).

One of the primary modes where shaping is useful is in covering cavities and corners. Dihedral and trihedral corners have a large acceptance angle where the specular return is strong. The specular flash of a plate is oriented normal to the plate. Corners can produce specular returns over a wider array of acceptance angles. The approximate RCS values of features are shown in Table 6.1. For a feature of a fixed size, the trihedral corner makes the largest contribution to the RCS of the target, followed by the dihedral corner, and then a plate. For example, if a TEL has an internal trihedral corner that is visible, covering that cavity with a plate will decrease the overall RCS of the TEL.

Vehicles with sheer vertical sides form a dihedral corner with the ground which spans the length of the vehicle. This dihedral corner can be eliminated by inclining the sides of the vehicle. For the specular flash to be directed away from a SBR satellite, the main lobe of the wall must be directed off of the line-of-sight vector. For large rectangular panels, the half-power beam width of the specular flash is $\beta_w = \lambda/h$ where h is the height of the wall. For a TEL side wall, on the order of a meter, in the X-band, the beam width will be on the order of a degree. If the main lobe is completely directed away from the line-of-sight by inclining the sides of the vehicle by more than the beam width (*e.g.* a couple of degrees), the maximum RCS measured from the corner will drop by at least 13 dB.[11]

[11] While the main lobe can be directed off of the line-of-sight, side-lobes may still return incident power to a SBR satellite.

Feature	Average RCS
Plate	$4\pi\frac{A^2}{\lambda^2}$
Dihedral Corner	$8\pi\frac{A^2}{\lambda^2}$
Trihedral Corner	$12\pi\frac{A^4}{\lambda^2}$

Table 6.1: Approximate peak RCS values for different reflecting features where A is the physical area of a plate. Dihedral corners are formed by two plates meeting at a right angle. Trihderal corners are formed by three orthogonal plates meeting at a single point. Per [65].

The other consideration for TEL shaping is in managing the TEL's RCS in specific "threat sectors" – angle from which a TEL is likely to be viewed. There is a distinct set of angles from which a TEL can be observed by an overhead satellite, shown in figure 6-7. The first and most obvious boundary is that the TEL is only viewed from the 2π steradian above. Satellites are also limited by nadir hole effects and TELs cannot be viewed from below the horizon, so TEL can only be viewed when $20°< \theta \leq 90°$.

Limitations in azimuth are set by modality and sight-block. SMTI cannot observe a TEL when it's radial velocity is below the MDV of the system. This occurs when the TEL is traveling orthogonal to line-of-sight. Assuming $v_{\mathrm{TEL}} = 10\ \mathrm{m{\cdot}s^{-1}}$ and $v_{\mathrm{MDV}} = 2\ \mathrm{m{\cdot}s^{-1}}$, any observation made within $\pm$ 11°of TEL broadside will have a radial velocity below the MDV of the system when operating in SMTI mode.[12] If the TEL typically operates on wooded roads, the hider can assume that it will not be viewed from near-broadside at high incidence. These sectors, overhead and broadside at high incidence provide directions for stealthy TEL designers to direct the largest returns into without contributing to the average measured RCS of the TEL.

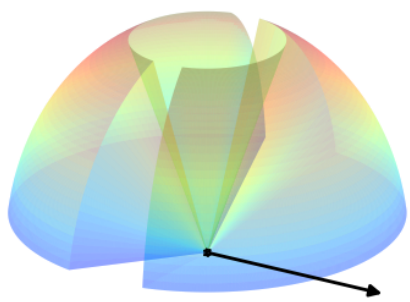


Figure 6-7: Threat sectors for a TEL for detection from SBR. Black dot represents the TEL location and the arrow represents its heading. The hole in at the top of the shape is caused by SBR nadir effects, the notches perpendicular to the TEL heading are from MDV effects. Color variation is only to aid visualization of the shape.

TELs could be designed to be stealthy, utilizing the shaping methods seen in stealthy ships or planes, such as the *USS Zumwalt* or the F-117. This would require a complete overhaul of the TEL fleet, which could be costly. The hider also has the option of retrofitting existing TELs to be stealthier via shaping. One method would be to place a radar-reflective cabinet around the TEL [119]. The cabinet consisting of flat panels, such as those used in the walls of semi-trailers, could be built with RCSR principles in mind, inclining the walls and directing returns out of threat sectors. This could contribute a significant amount of weight to the TEL, impairing speed and increasing fuel consumption. An alternative would be to drape a wire mesh, such as that suggested above for use in shelters, over an armature attached to the TEL. The armature, which would act as a skeleton under the mesh, could be made out of a lightweight material such as hollow aluminum tubing and shaped in order to create planes in the mesh which could be used to manage the directions of radar returns. In operation, the mesh would need to be made durable, or mended regularly, as any tears in the mesh would create cavities which would increase the average RCS.

The first side lobe is the largest in magnitude, and is 13 dB down from the main lobe peak.

[12]The TEL is still detectable with SAR from these angles.

6.1.5 Decoys

Decoys are objects that mimic some aspect of the appearance of a TEL in order to attempt to confuse the seeker. I group decoys into two kinds, those which are able to approximate the average RCS of a TEL, which can be used to deceive SMTI, and those which are able to approximate the appearance of a TEL to SAR and ISAR. If a decoy is capable of deceiving SAR, then it will also act as an SMTI decoy. SMTI-only decoys have less effect on the seeker, so the question becomes one of the complexity and cost of creating decoys. In this subsection, I will first introduce what comprise SMTI decoys, and how they could be used by the hider and then will introduce SAR decoys and their applications.

Average RCS Decoys

The simplest decoy is one that approximates the RCS distribution of a TEL. This means an average RCS on the order of ~ 25 dBsm that behaves as a Swerling target similar to the TEL the decoy is meant to represent.

Swerling I and II distributions approximate the RCS distribution of targets with multiple scattering centers of similar magnitude. Swerling III and IV approximate targets with a dominant scatterer.[13] Which the hider chooses to build will depend on the best model to represent the TEL. Regardless of Swerling target type, TELs are large objects with large RCS values, so decoys need either to be large themselves, or use radar reflective features.

Highly radar-reflective features such as corner reflectors can be used to boost the average RCS of a small vehicle to have it appear as a larger vehicle, such as a TEL. As mentioned in subsection 6.1.4, certain features, such as trihedral corners, are effective radar reflectors over a wide set of aspect angles. Utilizing sheet metal, corner reflectors can be fabricated to be relatively lightweight, and can be made collapsible so that the decoy can be activated and deactivated. Depending on the true RCS pattern of the TEL, a single large reflector or a number of smaller reflectors can be used to create the decoy.

Per Table 6.1, a trihedral corner reflector made from three 1m^2 plates would have an RCS of roughly 46 dBsm when viewed along the axis of symmetry of the corner in the X-band. To be viewable from all upward facing angles, the corner reflector would need to be 2 m in length and width (a corner reflector facing in each cardinal direction). While large, such a reflector could be carried on a small trailer or in the bed of a medium sized truck. If the TEL's RCS profile was best captured with a Swerling I/II distribution, a set of smaller reflectors totaling the same RCS could be used.[14] If used in conjunction with RCSR of the TELs, the size requirements of the reflector could be reduced. If the vertical plates were hinged, the reflector could be stowed when not in used, allowing the decoy to be activated and deactivated.

There are some spatial requirements for an average RCS decoy to be useful. The cross-range resolution of ISAR depends on the angle through which the target rotates, relative to the radar, over the course of the dwell. The inherent motion of a satellite in orbit creates some amount of apparent rotation, even if the TEL is traveling in a straight line. The cross-range resolution of ISAR is

[13] Whether a target behaves as Swerling I or II depends on how the radar is operated and the relative speed of the radar and target. Since Swerling II targets are less prone to fading, radar operators often frequency hop in order to have targets behave as Swerling II. The same is true of Swerling III versus Swerling IV.

[14] For RCS dominated by specular returns, the incoherent sum of the RCS values of scattering centers provides a roughly accurate estimation of the total average RCS of the target [65].

$$\delta_{xr} = \frac{c}{2f\Omega} \ , \tag{6.8}$$

where f is frequency and Ω is the angle through which the target appears to rotate. So long as Ω is small, for a satellite traveling at v_{sat}, the relative rotation *vis-a-vis* a stationary target is approximately

$$\Omega \approx \frac{v_{\text{sat}} T_{\text{dwell}}}{R} \ , \tag{6.9}$$

where R is the slant range. The minimum ISAR cross-range resolution is therefore

$$\delta_{xr} \approx \frac{cR}{2f v_{\text{sat}} T_{\text{Dwell}}} \ . \tag{6.10}$$

For range on the order of 1000 km, and a dwell time of 1 sec, δ_{xr} is roughly 2 m. For an average RCS decoy not to be immediately identifiable (why that is required is discussed below), it would need to approximate the RCS scatterer distribution of a TEL within roughly 2 m, meaning that it would need to span at least the length of a TEL (a few tens of meters). This could be accomplished by mounting reflectors on a trailer, or on two (or more) smaller vehicles traveling in series.[15] The cost of average RCS decoys would be driven by the cost of the vehicles on which they were mounted. The square meters of sheet metal used to make the reflectors would cost on the order of tens of dollars.

Per the Tracking Model in Chapter 4, the probability of a TEL moving without detection over a route is

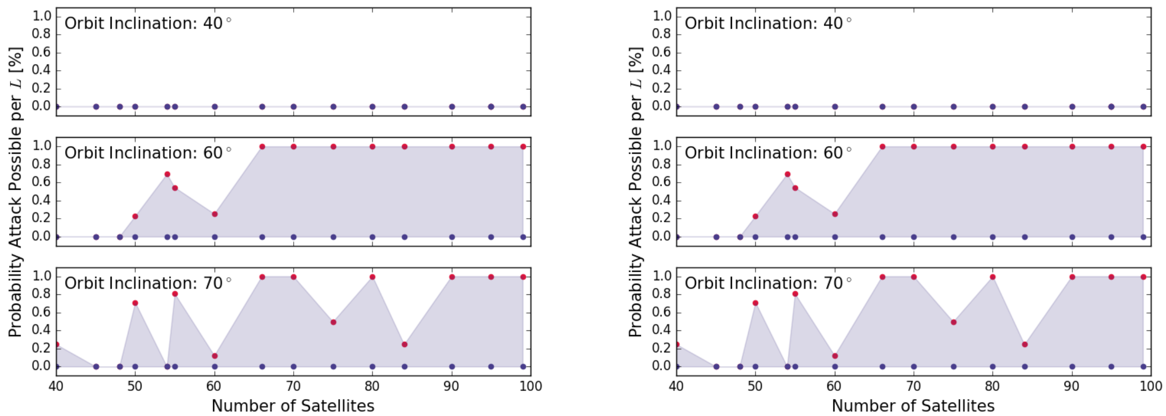
$$p_{\text{move}} = \left(1 - \frac{t_{\text{res}}}{\tau_{\text{rev}}}\right)^{n_{\text{obs}}} \ , \tag{6.11}$$

where p_{move} is the probability of moving without detection, t_{res} is a time interval, τ_{rev} is the revisit interval, and n_{obs} is the number of unobstructed views (*i.e.* no sight-block or MDV effects) of length t_{res} that the satellite can make over the move. For p_{move} that are close to unity, n_{obs} must be small unless $t_{\text{res}}/\tau_{\text{rev}}$ is very small. If p_{move} is high, which is when TELs are most likely to move, the seeker will only have a few opportunities over the course of the move to observe it. If the seeker does detect a TEL with SMTI, and if the hider utilizes average RCS decoys, the seeker should immediately attempt an ISAR measurement as a single successful ISAR measurement may be able to identify an average RCS decoy.

However, if average RCS decoys are operated like TELs, this can instill uncertainty in the seeker as to the identity of the sources of TEL detections that they register. The effect of this is that the seeker cannot reliably utilize detections in order to reduce the positional uncertainty over time. This will effectively remove the time-since last detection distribution from the Counterforce Model, and uncertainty will accrue at a rate of $R(p_{\text{min}})$ over the entire deployment lifetime of each TEL. This assumes that the hider is operating decoys in the vicinity of each deployed TEL. While the decoys themselves may be inexpensive, the hider would incur costs from the crews required to operate the decoys. The decoys would also need to operate behaviorally like a TEL, *e.g.* not exceeding the TEL's top speed, and moving with a convoy similar to that which moves with the TEL. Figure 6-8 shows the effect that operating average RCS decoys would have on the probability of successful attack.

This strategy would be most effective for p_{min} close to unity. As p_{move} decreases, the number of possible views over

[15] Relative motion between the vehicles would corrupt the ISAR image, which may be advantageous to the hider, making the ISAR image inconclusive.



(a) No Decoys

(b) Average RCS Decoys

Figure 6-8: The effect of average RCS decoys on the feasibility of counterforce attacks. Left shows the probability of a successful attack existing during a lifetime $L = 3$ days for an attack against a deployment zone in Irktusk utilizing warheads with $r_w = 5$ km. Right shows the same area, but with average RCS decoys such that the seeker cannot trust detections of moving TELs.

the route increases, and the probability of identification increases. Since average RCS decoys would be identifiable with relatively coarse ISAR resolution, they would not be amenable to being used at lower $p_{\min}$.

TEL-Shaped Decoy

An average RCS decoy is capable of deceiving SMTI, which only measures the RCS distribution. SAR and ISAR instead measure the spatial distribution of the scattering centers of the target, which can be used to form an image. An average RCS decoy can be constructed from a small number of corner reflectors while a TEL-shaped decoy mimics not just the sizes of the scattering centers, but also their spatial distribution.

One of the challenges of creating a TEL-shaped decoy is that TELs are large objects. Decoys are ideally easily and cheaply built, so that many can be used in conjunction (discussed in detail below). A shape decoy must be able to approximate the physical size of a real TEL, meaning that it will be on the order of tens of meters in length.

The challenges of creating a TEL-shaped decoy mirror those of shaping an actual TEL for RCSR. The same set of radar reflective materials, metal sheets or mesh can be used to control the RCS of a decoy. In the case of TEL RCSR, the TEL itself is large and can serve as a skeleton on which the outer radar reflective skin can be mounted. One possible decoy design would be to utilize an interior scaffolding made of inexpensive and lightweight material such as PVC tubing or aluminum struts to form the scaffold over which a wire mesh could be stretched. Integrating the mesh with a fabric or plastic liner, the same color as a TEL would make the decoy difficult to identify with optical modalities, in the case it was moved during the day.[16] In order to make such a simple decoy indistinguishable from an actual TEL, the same procedure could be used on actual TELs, *i.e.* decoys are made more convincing when used in conjunction with RCSR shaping.

Making the TEL decoy mobile would allow the hider additional modes for deceiving the seeker. Two simple methods for making a shape decoy mobile are mounting it on a smaller vehicle or on a trailer. In both cases, the

[16] A decoy would ideally move in the same manner as a real TEL, at night and under cloud cover, so deceiving optical modalities would not be a strict requirement, though could be accomplished with little additional effort.

decoy needs to be made lightweight enough that it can be carried by a vehicle such as one of the TEL convoy support vehicles. A TEL sized decoy could be skinned with 60–180 kg of material.[17] If scaffolding were constructed from aluminum tubing, it could weigh on the order of a few hundred kilograms.[18] Collectively, the decoy material would weight half a ton, approximately the payload weight capacity of a half-ton pickup truck. Such a decoy could be fabricated for roughly the cost of the pickup truck itself, on the order of tens of thousands of dollars. A less expensive option would be to mount the decoy on a trailer that could be towed by a security vehicle. This would have the advantage of being able to be disconnected and left in a shelter indefinitely.

Use of Decoys

To be convincing, decoys need to be operated in a similar manner to actual TELs, moving only when the probability of detection is low. Utilizing average RCS decoys in this manner, the hider can reduce the seeker's certainty in the source of detections made. Utilizing shape decoys, the hider can increase the number of targets that need to be tracked, and can undermine guided-weapon attacks (see Chapter 3).

Creating Ambiguous Detections. If an average RCS decoy is detected moving, the seeker can employ ISAR to attempt to identify it. If there are curves in the roads connecting shelters (which I assume to be the case generally), then the decoy can be imaged. Per the distribution in Chapter 2, a vehicle moving on a mixture of road types can be expected to be identifiable with ISAR at over up to one-fifth of its route. If there are multiple exposures to observation over a route, eventually the decoy will be expected to be identifiable. As such, average RCS decoys are most useful when traveling routes with low exposure to detection.

Increasing the number of targets. Shape decoys, while more sophisticated in their construction, are more straightforward in their effect. So long as they approximate the shape of a true TEL (and/or a TEL is shaped to resemble decoys), they will not be distinguishable from real TELs by SAR and ISAR alone. Identification would rest with behavioral identification and ELINT, which, as discussed in Chapter 4, are insufficient for identification as the signals can be modulated by the hider.

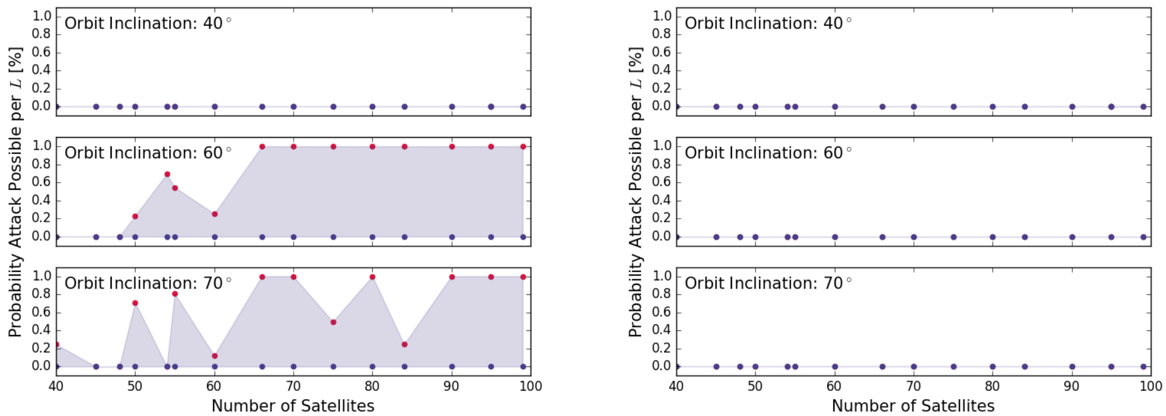
For each shape decoy that is operated like a true TEL (and so can't be distinguished by behavior or ELINT), the seeker is presented with another target to track and attack. Adding shape decoys in a deployment area has the same effect as increasing the number of TELs, from a deterrence standpoint.[19] Figure 6-9 shows the effect of increasing the number of targets in a deployment area on the probability of a successful attack existing.

Increasing the number of targets not only increases the number of targets to allocate warheads to, but also increases the burden of tracking, increasing $R(p_{\min})$ for all TELs and decoys by increasing the required τ_{rev}. In the case of the attack modeled in figure 6-9, the addition of 8 decoys is sufficient to render the attack infeasible. Shape decoys can also perform the same function as average RCS decoys if deployed near actual TELs. Deploying near actual TELs would have the added benefit of denying the attacker the ability to utilize guided-weapon attacks to improve attack efficiency.

[17] Galvanized hexagonal wire chicken netting with a 1.3 cm aperture (less than $\lambda/2$ for 10 GHz) can be purchased with a weight of 16 kg per 54 m^2. If a TEL is 20 meters long, 2 meters wide, and 3 meters tall, it has a (side and upward facing) surface area of approximately 200 m^2, which could be covered with 60 kg of wire. A heavy duty tarp has a weight of 18 oz per yd^2, or 0.6 kg m^{-2}. Covering 200 m^2 with a tarp for optical camouflage would add roughly 120 kg of weight.

[18] Using construction scaffolding to estimate the amount of aluminum tubing needed, construction scaffolding of dimension 2 m in height and 2.5 m in width, and cross-braced for stability weighs roughly 15 kg per unit [120]. These have a weight capacity of 150 kg each, roughly the weight of the entire skin. A scaffold 20 m in length and 2 m tall would require 8 units, if two are needed (one for each side of the decoy), then a total of 16 units, or 240 kg of scaffolding would be needed.

[19] A decoy and actual TEL differ in their ability to launch retaliation, though this is immaterial when considering deterrence.



(a) 16 TELs

(b) 16 TELs + 8 Decoys

Figure 6-9: The effect of TEL-shaped decoys on the feasibility of counterforce attacks. Left shows the probability of a successful attack existing during a lifetime $L = 3$ days for an attack against a deployment zone in Irktusk utilizing warheads with $r_w = 5$ km. Right shows the same area, but with 8 additional decoy targets, totalling 25 targets.

6.1.6 Jammers

Jammers are radio transmitters which transmit white noise across the operational band of a radar system in order to decrease the SNR such that detections are difficult or impossible to make. By increasing the noise floor, the probability of detection can be reduced. With sufficient noise injected into the receiver, the probability of detection can be reduced to an arbitrarily low level.

The efficacy of a jammer depends on the relative amount of noise that it can introduce into a radar receiver relative to the strength of the signal that it is attempting to jam. As noted in Chapter 2, radars are active imaging systems which experience a two-way transmission loss. One of the strengths of jammers in the context of a SBR system is that the jammer only experiences a one-way transmission loss, since it is located near the target. The jammer-to-noise ratio is

$$JNR = \frac{P_j G_j Q_0}{4\pi R^2} \cdot \frac{A}{N} \ , \tag{6.12}$$

where P_j and G_j are the jammer power and aperture gain, respectively, Q_0 is the noise quality, R is is the slant distance between the jammer and the radar receiver, A is the radar receiver aperture area, and N is the noise in the receiver. The radar quality Q_0 is a measure of how closely the transmitted noise mimics true white noise, I assume this value to be unity. Comparing the JNR to the SNR from the radar equation, the jammer-to-signal ratio (JSR)

$$JSR = \frac{1}{4\pi R^2} \frac{P_j G_j}{P_t G \sigma} \ , \tag{6.13}$$

can be determined. This ratio can be used to estimate the effect of a jammer relative to the signal from a given target.

For example, assume a SBR system with 10 kW peak power and a 40 m^2 aperture having approximately 60 dB of gain (operating in the X-band) at a distance of 1000 km from the target. If the jammer were isotropic ($G_j = 1$), for it to have an equivalent noise power to a 20 dBsm target, the jammer would need to have roughly 0.1 W of power.

This massive disparity between the SBR power and the jammer power is due to the jammer power only experiencing a one-way transmission loss. In order to prevent detection, the jammer operator would like to exceed the signal power significantly, by a factor of 10 or more. The JSR can be increased by increasing the power of the jammer, or equipping it with a directed antenna.

Stationary Jammer Arrays

A jammer array is a collection of jammers which are used to collectively deny radar operations in an area. Taking advantage of the huge disparity in the JSR, jammers can reach signal-equivalence at very modest powers and gains. If attempting to jam multiple satellites at once, jammers could be directed to cover the entire upward facing 2π steradian. More complicated arrays could use directed apertures, such as parabolic dish reflectors, to directly illuminate overflying satellites. These would be more complex and costly, but would avoid inadvertently jamming other systems. A jammer array could be conceived of on two scales, jamming a local area and overcoming array adaptation.

The more modest approach is to jam radar operations in a local area. This could be accomplished by distributing a number of low-power jammers in an area where the hider wishes to deny the seeker the ability to observe.[20] One possibility would be to mount these jammers on shelters. If the shelter spacing s is similar to or less than the beam swath width, then this would prevent the seeker from observing those shelters, or the avenues between them.

The jammers could be run continuously, or toggled. If run continuously, the area spanned by the local jammer array would be a permanent gap in coverage. In effect, this would create a single distributed shelter where the TEL could operate without risk of being detected, so long as the jammers were operating. If toggled, the jammers could be turned on when a TEL deployed to the local area, kept on for long enough for the TEL to be able to move anywhere within the local jammer array area.[21] The jammers could then be toggled off and the seeker would not know which of the shelters covered by the jammer array housed the TEL. The TEL could remain in that shelter throughout its patrol lifetime without needing to move.

This would change the dynamics of how the seeker's uncertainty would accrue over time. Rather than organically accruing over time, the seeker's uncertainty would grow linearly with time proportional to TEL velocity from when the jammer array was activated until the array was deactivated or the boundary of the jammer array was reached. The hider would have a strong incentive not to move after this point (so as not to risk detection) so the uncertainty would not necessarily grow thereafter. Utilizing a local jammer array, the hider has the ability to generate positional uncertainty effectively on demand, limited only by the costs of emplacing the jammers, and the provocativeness of utilizing them (discussed below). One such array would be needed for each TEL patrol to be protected.

The more expansive use of stationary jammer arrays would be to build a large network of low power jammers sufficient to overwhelm array adaptation. As mentioned in subsubsection 6.1.6, AESA can null out jamming from $n_{\text{element}} - 1$ directions simultaneously, where n_{element} is on the order of 1,000 elements. If the seeker installed an array of $\geq n_{\text{element}}$ low-power jammers in a deployment area, the interference from the $n^{\text{st}}_{\text{element}}$ and above jammers could

[20] Per the example above in subsection 6.1.6, a 10 W isotropic jammer (about the power consumption of an LED light bulb) would exceed the expected power of an echo from a TEL by a factor of approximately 100.

[21] The jammers would need to be kept on for $m_{\text{shelter}} s / v_{\text{TEL}}$ where m_{shelter} is the number of shelter-to-shelter moves needed to traverse the local jammer. If the array spans 10 shelters, with s on the order of a few km, the jammers would need to operate for roughly one hour.

not be canceled.[22] Such an array could jam any SBR satellite for which the jamming array fell within the satellite's field of view.

A wide-area jamming array could be operate in a similar manner to the local jammer array, continuously denying observation or being toggled to cover TEL movement. The difference would be in scale. A wide-area jamming array would require thousands of jammers, compared to the tens needed to main-lobe jam over a local area. Local jamming arrays could be built up into a wide-area jamming array over time. The blind spot created would similarly be much larger for the wide-area jammer, as it would not only jam main-lobe observations, but through the side-lobes of any receiver with line-of-sight to the jammer array. The effect on the seeker's uncertainty would be similar to that of using a local jamming array, but would effect all deployed TELs simultaneously (including ones that had been deployed for some amount of time). It could be considered more provocative than using a local jamming array, but the difference would be one of extent not kind.

Mobile Jammers

The other method for employing jammers to shield the movement of TELs is utilizing mobile jammers. Mobile jammers are already part of the electronic-warfare arsenal of Russia, includes the Krashuka-4 mobile jammer system. The U.S. Defense Intelligence Agency (DIA) has assessed that Russian mobile jammers have the capability to deny SAR observations as well interfere with satellite communications [121].

While mobile jammers could be utilized to transiently set up a jammer array in an area, this would be inefficient compared to a static jammer array (though more flexible). For example, to create an area of uncertainty 10 shelters in extent would require 10–61 mobile jammers for a linear or gridded road system, respectively. A more efficient method for utilizing jammers would be in exploiting their mobility.

A mobile jammer creates a moving blind spot for the seeker. By moving a TEL alongside the jammer, the hider can mask TEL movements. To cover a shelter-to-shelter movement, the jammer would need to approach the shelter containing a TEL, and move with it to another shelter where the TEL could encamp. The hider would be unable to determine if a TEL was present, or moved at all. So, if a mobile jammer passed through any TEL's area of uncertainty while operating its jammer, that TEL's possible positions would be spread over every shelter the jammer subsequently passed.

The hider could utilize this same method to mask the initial deployments of TELs from the base or the return of a TEL to the base, denying the seeker information on whether or not, or how far, any TELs moved during their deployment. By making a circuit around a deployment area, a mobile jammer could cause the seeker's uncertainty to extend along, and grow from, every shelter along the route of the jammer, creating a large number of virtual targets that need to be tracked and attacked. The only limitation to the number of virtual targets created is from how long the mobile jammer is able to operate its jamming system while in motion.[23]

Traveling across a deployment area at $v_{\text{TEL}} = 10\ \text{m}\cdot\text{s}^{-1}$ would take 7–15 hours for a 250 km by 500 km deployment area. This could be beyond the endurance of mobile jammers. However, given the low power requirements for jamming SBR systems, and the likelihood that jammers designed to also jam aircraft radar and communication satellites

[22] If a deployment area were partially in view of a SBR, fewer than n_{element} may be in view at a given time, so the hider would likely want to exceed the minimum number of jammers.

[23] If a particular model of mobile jammer is unable to operate while in motion, a pair of jammers could be used to leap-frog one another, though the process would be slower.

have directed antennae, the jammer may be able to operate at a reduced power, minimizing fuel consumption. Alternatively, the jammer route can be accomplished in segments, with the jammer stopping at shelters to refuel or otherwise replenish its power source.

There is also precedent for moving an electronic countermeasure vehicle with a TEL convoy. The Russian Peresvet system is an anti-satellite weapon designed to partially blind optical imaging satellites utilizing a high-powered laser on a mobile vehicle [122].[24] These systems have been deployed at TEL bases, such as Teykovo, near Moscow since 2019. Operating and training on Peresvet would provide Russia with experience directly applicable to operating mobile jammers to cover TEL movements.

Adaptive Mitigation of Jamming

If the hider jams one of the seeker's SBR satellites, there are avenues to mitigate the effects of the jamming [123]. The first is called burn-through. The seeker can compensate for increased noise by dwelling on an observation for longer. This is limited, however, by the constant motion of SBR satellites and their targets which places an upper limit on the coherent dwell times that a SBR satellite can achieve. As mentioned in Chapter 2, for SMTI, coherent dwells are limited by target acceleration and cannot be extended indefinitely. For SAR, dwell time is limited by the amount of time that the scene remains in the field of regard. If the jammer power is large enough that these dwell time limits are exceeded, then the jammer effectively creates a blind spot where the SBR system cannot make observations.

Some radar arrays can adapt to, and ignore, jamming from certain directions, a process called blanking. AESA, apertures made from a large array of individual TR modules can manipulate the weighting of the different elements to change the array's radiation pattern in order to null out jamming from specific directions. As noted in Chapter 2, the radiation pattern of arrays has a series of peaks, where the transmit and receive gains are high, and nulls, where little to no energy is transmitted or received. Changing the relative amplitude and phase of different elements in the array allows the radar operator to steer the beam. It also allows for the side lobe peaks and nulls to be moved. If a null is placed on the direction from which jamming power is originating, that power will not be measured in the receiver, canceling the effect of the jammer. For an AESA with n_{element} elements along a given axis, there are $n_{\text{element}} - 1$ nulls, which can be used to cancel $n_{\text{element}} - 1$ sources of jamming.[25]

Blanking can be used to counter side-lobe jamming, where noise is transmitted into any aperture for which there is a line-of-sight. Re-weighting elements cannot counter main lobe jamming however. Main-lobe jamming is the transmission of noise into the receiver from within the beam footprint. Placing a null on a source that is within the main lobe, is effectively equivalent looking away from that location. So while the radar can null out the jammer noise, the jammer creates a blind-spot that the SBR system cannot observe. The size of the blind spot depends on the gain of the beam. If the first null is used, the beam cannot be steered within a beam-width of the jammer without introducing jamming noise into the receiver.[26]

The limitation of main-lobe jamming is that it only creates a blind spot in the area surrounding the jammer. The swath of a SBR beam footprint varies with the gain and altitude of the SBR system but is typically on the order of kilometers in width and a few tens of kilometers in length. For a single jammer to prevent detection of a TEL,

[24] Anti-satellite weapons are discussed more in subsection 6.1.7.

[25] For a 40 m^2 aperture that is 16 m by 2.5 m, and operates in the X-band, there will be roughly 1000 elements along the long axis. Such an aperture could cancel the signal from 999 jammers.

[26] The null-to-null beam width is approximately $2\lambda/\ell$.

the TEL must be within a beam footprint of the jammer, an area that could be barraged with a modest number of nuclear warheads.[27] The hider would therefore have to use either an array of jammers, or mobile jammers.

Another method for reducing the effects of jamming is to use specialized radar waveforms, such as noise waveforms. Noise waveforms utilize pseudo-random white noise as the illuminating waveform rather than a repetitive waveform such as a linear frequency modulation chirp. Echoes are identified, and time of flight calculated utilizing the correlation between the measured echo and the transmitted waveform reference [124]. Noise waveforms typically operate in a ultra-wide band (UWB) mode[28] The unpredictable auto-correlation in frequency and the wide bandwidth make noise waveforms inherently resistant to certain forms of jamming in comparison to LFM chirps [124]. Noise waveforms have been shown to be up to 10 dB more resistant to than a standard LFM chirp against some forms of jamming [125].

The drawbacks of noise waveforms is that they require significantly more burdensome signal processing, have lower gains (due to the wide bandwidth), and require longer dwells. The hardware required to utilize noise waveforms is different from that which utilizes LFM, so noise waveforms would need to be part of the design process if the seeker expected an antagonistic jamming environment. Since the illuminating energy is spread out in frequency and time, it is also more difficult to detect, which could reduce the information imbalance between the hider and the seeker.[29] The longer dwell times also mean that the tracking performance will be worse than that modeled here.[30] Noise waveforms are an active area of research that presents the seeker with an opportunity to ameliorate the deleterious effects of jamming [126,127]. However, since $TL_{\text{one-way}} \sim 10^{13}$ for radar systems in LEO, jamming resistance will need to be significantly improved above the 10 dB effect demonstrated thus far. cite

Provocativeness

Of the countermeasures discussed thus far, jammers are the most provocative. Jammers are typically classified as methods of electronic warfare as they directly interfere with another country's military hardware. For example, in 2018, Russia jammed GPS signals during a NATO military exercise in Norway [128]. While this drew complaints, the non-destructive and temporary nature of electronic warfare places it on a lower level of belligerence than kinetic or nuclear conflict. However, utilizing jamming could create escalation pressures in two scenarios.

First is that the act of jamming satellites, especially if widespread, could be interpreted by the seeker as masking an offensive attack. For example, Russia could jam U.S. SBR systems to prevent detection of the preparations for a preemptive attack against the U.S. nuclear arsenal. The seeker is unable to directly distinguish between jamming to mask the preparations for an attack versus jamming to cover TEL movement. If the seeker believed that an attack was being prepared, they would face a pressure to "use it or lose it". As such, every instance in which jamming is used presents an opportunity for escalation, even inadvertently. Routine use of jamming during non-crisis periods may defray the novelty, and therefore potential for escalation, of jamming.

The second avenue through which jammers can prompt escalation is if they are used when a TEL fleet is already vulnerable. Jammers can be used to create positional uncertainty on relatively short time scales, essentially on de-

[27] Assuming variety in view angle from coverage with multiple satellites, a jammer could protect an circular area with radius $\beta_w R$ which is on the order of a few kilometers. This is of a similar scale to the blast effects of a single nuclear warhead.

[28] While there is no fixed definition of UWB, many definitions converge in the neighborhood of bandwidth in excess of 20% of the carrier frequency. In comparison, typical radar bandwidth is 10% of the carrier frequency.

[29] Note that in the foregoing analysis, the hider's planning of TEL movements does not require that the a TEL is able to detect when it has been detected. *i.e.* the forgoing analysis is consistent with the use of a low-probability of intercept (LPI) waveform, and any additional information available to the hider from detecting illumination could be used to improve TEL survivability above how it is modeled here.

[30] Increased dwell times increase burden of each modalities, increasing the interval between observations.

mand. If the TEL fleet somehow becomes vulnerable to preemption, through hider miscalculation or some temporary disablement of jammers, then the use of jammers to salvage the situation presents the seeker with a clear signal that their period of advantage is about to come to an end. If the seeker believes that the current crisis is only going to escalate further, they will feel a pressure to disarm their adversary while they have the opportunity.[31]

These escalation pressures can be mitigated somewhat. If jamming is selective in its application, such as through toggled local jamming arrays, or mobile jammers making circuits of a deployment zone, the purpose of the jamming can be signaled. Masking preparations for an attack requires utilizing jammers at the locations of all of the attacking weapons simultaneously. Utilizing jammers more sparsely can convey that they are being used to cover TEL movements, rather than masking an attack.

That these maneuvers are going to be undertaken can even be communicated in advance in order to prevent surprise.[32] While it may seem counter-intuitive to telegraph TEL movements to the seeker, so long as the other TELs in the fleet have sufficient positional uncertainty to deter attack, the seeker knowing the exact location of a given TEL is immaterial. Further, since jammers are capable of creating a large amount of uncertainty in a relatively short time frame, each of the hider's deployed TELs should have large positional uncertainties. The hider is also not beholden to move TELs when they engage their jamming systems, so these notifications could serve as bluffs without degrading their utility for diminishing surprise.

6.1.7 Anti-Satellite Weapons

Anti-satellite (ASAT) weapons are the most provocative countermeasure presented here. These weapons are used to destroy or disable satellites, such as the spacecraft which would carry an SBR system. Destroying SBR satellites could degrade or prevent tracking by removing sensors and creating gaps in coverage, albeit at the cost of a clearly aggressive action that could prompt significant escalation. In this subsection, I outline some ASAT technologies that can threaten satellites in LEO, and then discuss the timing limitations and escalation risks associated with utilizing ASATs to preserve TEL survivability. The three types of ASAT weapons most able to threaten SBR satellites in LEO are direct-ascent, co-orbital, and directed-energy weapons.

Direct-Ascent Weapons. Direct-ascent weapons follow a direct (usually ballistic) path to intercept a satellite while it is traveling through it's orbit. A typical direct-ascent ASAT would be a ballistic missile that delivers an exo-atmospheric kill vehicle (EKV) into the vicinity of an intercept point with the target satellite's orbit.[33] The EKV then homes in on the target, utilizing on-board sensors, and collides with it. The large orbital speeds of satellites in LEO ($\sim$ 7–8 km$\cdot$s^{-1}) provides sufficient energy that collisions are typically deleterious to the satellite. The U.S. and China have tested direct-ascent ASATs in the past, and Russia is suspected by the U.S. intelligence community, and independent experts, of developing the same capability [130,131,97].

Co-Orbital Weapons. Co-orbital ASATs are placed into the same orbit as the target satellite before closing the distance and intercepting it. This requires more preparation than direct-ascent ASATs as an object can be placed into an orbital plane only when it is directly over the ASAT launch site, which occurs twice a day. The Soviet Union tested a co-orbital ASAT during the Cold War which, after being placed into orbit, moved to the target over the

[31] As before, whether the pressures felt are sufficient to prompt the seeker to initiate a nuclear war is another question.

[32] This should not be done if the TEL fleet currently appears vulnerable, however, as it would provide more time for the seeker to prepare a preemptive attack.

[33] Mid-course ballistic missile defense interceptors, for example, are particularly well suited to act as direct-ascent ASATs, and have been tested as ASAT weapons in the past [129,130].

course of a few orbits (a process of a few hours) before detonating a conventional explosive in the vicinity of the target to damage it with shrapnel. Russia is believed to be testing a more sophisticated co-orbital ASAT capability utilizing its inspector satellites[131]. Inspector satellites could be maneuvered into close proximity to target satellites before damaging or destroying them with a short range weapon or through collision.[34] These have the advantage of being able to be placed into orbit well in advance of their use as ASATs.

Directed-Energy Weapons. Directed-energy weapons utilize high intensity electromagnetic radiation to cause damage to satellites. The energy source is typically an infrared laser, though more exotic sources of energy, such as nuclear-explosion-pumped X-ray lasers have been proposed in the past [132]. At low powers, lasers are capable of damaging sensitive optical imaging planes [133]. At higher powers, perhaps a factor of 10^6 more, lasers can cause structural damage to satellites through heat deposition [133]. Some ground-based megawatt-scale lasers, with the proper optics, are capable of causing structural damage to satellites in LEO on the time-scale of seconds [113]. The U.S. Army's Mid-Infrared Advanced Chemical Laser (MIRACL) is reported megawatt scale [113]. The current Russian mobile directed-energy system, Peresvet, is thought only to have sufficient power to blind or dazzle optical satellites, rather than cause structural damage [122]. The DIA has projected that China will likely develop directed-energy weapons capable of damaging satellites in LEO by the end of the 2020s, though DIA estimates have historically tended toward worst-case scenarios [121,134]. Independent assessments note that research on Chinese directed-energy weapons has received sustained support, there is very little public evidence of their deployment or use [97].

Disruption of Tracking With ASAT

ASAT systems could be used to degrade tracking by destructively removing SBR sensors from orbit. Satellites present tempting targets in a conflict as they base many valuable intelligence, surveillance, and reconnaissance (ISR) and command, communication, control, and intelligence (C3I) systems, such as SBR. Satellites are also very difficult to defend from attack [135]. TELs are sufficiently mobile that only a few tens of minutes without overhead remote-sensing may be sufficient to create enough positional uncertainty to make an attack prohibitively inefficient. Provided that the hider had sufficient ASAT capability to create tens of minutes of coverage loss, ASATs may prove attractive targets in a crisis if the hider feels insecure about their mobile nuclear arsenal. The timelines on which the different ASAT weapon types could attack SBR systems differ between modalities.

The least-prompt type of ASAT weapon is the co-orbital weapon that is launched directly into orbit, such as used in the early Soviet ASAT systems. The limitation that these weapons need to be launched when the target satellite's orbital plane passes over the launch site means that the average wait time to use such a weapon is 6 hours. Further hours would be needed for the ASAT to close in on the target. Co-orbital weapons which are pre-positioned, sometimes called space mines, remove the launch time component. If placed in crossing-orbits to the target, it may take hours for an intercept to be made [113]. If made to trail the target in the same orbit, the closing time could be significantly shorter.

Direct-ascent ASAT weapons are able to reach LEO in a matter of minutes once launched. If targeting specific satellites, it may take hours for the satellite's orbit to come into range of a launch site [113]. In the context of defeating TEL tracking, the most important SBR satellites to destroy are the ones that are approaching the TEL

[34] Potential short-range weapons include conventional explosives, high power microwave bursts, directed-energy weapons, or damage or dissection with a robotic arm, among others.

deployment areas. If direct-ascent ASATs were positioned in the same regions as the TEL deployment zones, they could be used to destroy overflying SBR satellites on the time-scale of minutes.

Directed-energy weapons can the most prompt as the energy is delivered at the speed of light. Though, if the goal is to cause structural damage to a satellite rather than dazzle a sensor, the effect may require dwelling on the target for minutes [133]. The weapon's power supply may also not be perfectly prompt. Similar to direct-ascent ASATs, their limitation that they are only able to attack satellites within their line-of-sight is less of a hindrance if they are located near the deployment areas they are used to protect. Directed-energy weapons can be hindered by poor weather, however, which can cause significant attenuation. As noted in Chapter 2, radar systems can operate in these conditions, so directed-energy weapons are not the best choice for countering SBR systems promptly.

ASATs degrade tracking capability by removing SBR sensors and creating gaps in coverage where TELs can move without detection. With a gap in coverage from a downed satellites (assuming coverage was not overlapping, or that all satellites in view were destroyed), positional uncertainty would grow with TEL speed until another satellite came into view. To increase the number of warheads required to be allocated to a given TEL by one, a gap in coverage would need to be r_w/v_{TEL} in duration.[35] For the weapons and TEL speeds considered here, this amounts to 300–500 seconds, about the duration of a satellite overflight. Creating a large enough gap in coverage to significantly increase the number of warheads required per TEL would therefore require either downing multiple satellites or utilizing ASATs long enough in advance of an attack that the gap in coverage would recur multiple times.

During an attack, one could envision attempting to use ASATs to reduce the effectiveness of guided weapons (see Chapter 3). To be effective during an attack, ASATs would need to disable satellites in a shorter time than the delivery of weapons. Flight times for depressed-trajectory SLBMs are roughly 7–10 minutes. In practice, the incoming attack would need to be detected and some time needed to decide to use ASAT weapons, and communicate that decision.[36] Due to the short timeline, it is difficult to envision ASATs as an effective response to an incoming attack. They would be more effective at degrading tracking if they were employed in advance of an attack, however this risks escalating the conflict.

The most obvious escalation risk with ASATs is if their use heightens the intensity of a conflict. If the hider's arsenal is sufficiently well tracked at the time the ASAT attack is launched, the ASAT attack itself may provoke the seeker into attacking. Depending on the type of ASAT attack used, conducting an ASAT may take anywhere from minutes to hours [113].[37] Including the few tens of minutes required for the TELs to be able to move following the destruction of the SBR satellites, there could be a window where the seeker may retain the opportunity to launch an attack in response to an ASAT attack. This could present the seeker with a "window of damage limitation" which can place pressure on the seeker to attack [7].

If the seeker were not in a position to undertake an attack when ASATs were employed, the "window of damage limitations" pressures would not be as acute. However, it could provoke escalation of the overall conflict as the first use of ASATs would be introducing a new domain into the conflict. Escalation and tit-for-tat retaliations are most easily understood when in-domain, *e.g.*, responding to an air raid with an air raid [2]. Extending the fight to include

[35]For a linear road model. For a gridded road model, the marginal increase depends on the initial number of allocated warheads.

[36]The decision could be automated to reduce the time required, but this would intensify escalation risks (see below).

[37]Some attacks may take days or more to prepare, such as co-orbital attacks where the attacking satellite needs to maneuver significantly to reach the target satellite. However, it may be ambiguous to the seeker whether these preparations are part of an attack.

space assets may be perceived as a significant escalation, owing to its novelty.

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Chapter 7

Conclusions

7.1 Conclusions

Currently, military SBR systems are deployed by the U.S., China, Germany, Russia, Japan, and India. None of these states deploy more than 5 SBR satellites. The analysis presented in Chapter 5 indicates that none of these constellations are dense enough to be able to track TELs to support a counterforce attack against either Russia or China. Simply put, the gaps in coverage are too large and too regular for TEL positional uncertainty to be meaningfully bounded with SBR alone under the assumption that TELs have many shelters. Building more dense constellations of SBR may enable tracking against permissive hiders, but SBR systems are inherently vulnerable to readily available countermeasures, such as jammers. As such, the widespread deployment of SBR systems would be unlikely to render TELs unsurvivable, however the countermeasures that would be required to safeguard TEL survivability could create escalation risks.

7.1.1 Critical Technologies

In the context of tracking TELs, SBR fits the mold of what I call a critical technology. I define a critical technology as a technology which is the sole existing means of completing some necessary task in a broader endeavor. As the sole means of completing some required task or tasks, critical technologies are necessary (though not necessarily sufficient) to the broader endeavor (tracking in this case). On the other side of that coin, critical technologies also represent single points of failure in the endeavor. If the critical technology fails, then so does the broader endeavor.

When tracking TELs, a major driver of uncertainty in TEL position is gaps in coverage with remote-sensing modalities. Gaps in coverage of length s/v_{TEL}, which can be on the scale of minutes, are long enough for TELs to move without risk detection. TEL operators can choose to move their TELs only during these periods, effectively never exposing their TELs to detection with remote-sensing systems. One of the major sources of gaps in coverage comes from some modalities, such as optical imaging, having limited capability to operate at night an during periods of bad weather, creating predictable gaps that can be hours in duration.

Without the capability to track TELs at night and through clouds, TELs cannot be meaningfully tracked. Of the existing technologies, SBR is the only one able to monitor large areas under these conditions. SBR can perform all of the tasks of tracking - detection, identification, and track association - whereas other remote-sensing technologies are insufficient to track TELs in all conditions, making SBR a critical technology.

I raise this definition as a means to draw attention to how central the consideration of countermeasures is to assessing the effect of an emerging technology. If some current capability does not exist (*e.g.*. the ability to track TELS), then the first technology that enables that capability, by accomplishing some previously unaccomplished task, will often be a critical technology.[1] This means that the fate of the larger endeavor hinges on not just the ability of the critical technology to perform its task(s), but also how durably it can do so in the face of accessible countermeasures. Therefore, when faced with some emerging technology, such as SBR, that has the potential to enable some capability, such as tracking TELs, then there are three questions which should be considered

1. How extensively does the technology need to be deployed to enable the capability?
2. Are there countermeasures to the technology and, if so, can the capability be retained if countermeasures are deployed?
3. What are the relative costs and availabilities of the technology and the countermeasures?

In this section, I summarize the findings from this, and the previous chapters which address these questions in the case of SBR and TEL tracking.

I find that the cost of a SBR system capable of tracking is high, on the order of hundreds of billions of dollars. I also find that there are countermeasures that could effectively increase the costs of an enabling system, or defeating tracking altogether. Importantly, the availability of those countermeasures means that it is unlikely that the widespread deployment of SBR would meaningfully undermine the survivability of TELs, though may increase the risks of escalation through entanglement by necessitating the use of provocative countermeasures. I contrast TEL tracking with SBR with the case of an underwater remote-sensing technology that undermined the survivability of Soviet submarines to show how the scenarios differ.

7.1.2 At What Point Does SBR Enable Tracking?

Since SBR is capable of completing all of the required tasks of tracking around the clock and in any weather, the question isn't whether it can be used to track TELs, but how many satellites would be required. The number of satellites drives both the effective revisit interval of each modality and by increasing the angular diversity of observations.[2] The increased rate of observations increases the likelihood that a TEL will be observed while within line-of-sight of a satellite, and angular diversity decreases the effect of sight-block.

As noted in Chapter 3, the requirements for enabling a counterforce attack depend on the parameters of the attack itself. One important factor is the shape of the local road network. Rather than try to assume some given road network, in this analysis I bounded the road shape between an isolated linear road, and a gridded road system. The majority of real road networks should fall between those bounds.[3] For attacks against either Russia or China, if the local TEL deployment area road networks are closer to the linear bound, then a SBR constellation of roughly 50 SBR satellites can be used to track TELs. If the road system is closer to the gridded road bound, the mobility of

[1] The exception would be if multiple technologies are fielded simultaneously or in close succession which redundantly accomplish those tasks.

[2] The revisit interval per satellite can be kept constant, but as more satellites are added, the effective revisit interval (the average time between observations from the perspective of a target) decreases.

[3] Isolated roads which curve back on themselves may require fewer warheads than the linear road estimates, and areas with more freedom of movement than a grid could require more warheads than the gridded model estimates.

TELs during the flight time (τ_f) of the attacking weapons, even from SLBMs on depressed trajectories, is sufficient to make any attack infeasible with lower yield warheads even if the TEL locations are perfectly known at the time the attack is launched.

In order to threaten TELs on a gridded road, the attacker needs to either use larger warheads, such as the W88, or decrease the TELs' ability to move during the attack. In the case of Russia, the U.S. does not maintain enough SLBM-deliverable W88s in its arsenal to attack all of the Russian TELs. Against China, the US could muster enough W88s if it uploaded a SSBN with those warheads in advance of the attack.[4]

A decreased ability for TELs to move during the attack would only occur if the U.S. were able to either disable the defender's early warning system or by successfully employing guided weapons. Disabling early warning would need to occur in advance of the attack itself. If done near the opening of the attack, it would mark the start of the attack itself, and the hider could start moving their TELs when the attack against the EWS was detected. This would provide no net result to the attacker. If done well in advance of the counterforce attack, it could provoke escalation and risks putting the defender in a use-it-or-lose-it situation. This would also provide the hider with time to bring slower, but effective, countermeasures such as direct-ascent ASATs to bear against the SBR system.

ICBMs from the continental U.S. could be used if the attacker were able to use a guided-weapon attack. The seeker would need to be confident that the hider was unable to utilize effective countermeasures during the attack. Countermeasures which are capable of defeating a guided-weapon attack are jamming, directed-energy ASAT, SAR-spoofing decoys, or extensive use of sight-blocking structures such as covered roads. If the hider has not expressed the ability or inclination to utilize any of these countermeasures, the seeker could successfully track TELs with a constellation of roughly 50 SBR satellites.

One key finding of Chapter 5 is that the latitudinal diversity in the deployment areas of China and Russia increase the number of satellites needed to track TELs. Constellation inclination impacts the coverage at different latitudes. The inclination is the maximum latitude to which the satellites in the constellation will travel. As the constellation inclination increases, coverage at higher latitude increases while coverage near the equator decreases. The latitudes of TEL deployment areas is therefore an important design parameter for SBR constellations. The minimum constellations that can track TELs at one latitude are not necessarily capable of tracking TELs at another, as was seen to be the case in Chapter 5, where the minimum constellation which could track Russian TELs could not track Chinese TELs and *vice versa*. A general-purpose constellation would therefore need to exceed the minimum constellations, requiring 70 satellites for the Baseline system.

The most obvious cost of deploying such a constellation is the monetary cost. As per the Cost Model in Chapter 5, the cost of an SBR system is measured in billions of dollars, when hardware, operational, and research costs of considered. While there are a number of factors which affect the total cost of a SBR system, the biggest driver is the number of satellites in the constellation. The general-purpose constealltion of 70 Baseline satellites would cost in the range of $220–300 billion.

[4] This would require significant pre-planning as the SSBN would need to be uploaded with warheads at the Strategic Weapons Facility Pacific in Bangor, Washington, and then sailed across the Pacific such that depressed trajectory SLBMs could be used.

7.1.3 Could SBR Track TELS Through Countermeasures?

In Chapter 6, I introduced a number of potential countermeasures to SBR tracking. As a critical technology, it is important to evaluate if and how SBR could operate in the presence of countermeasures. I define two ways in which countermeasures can negatively impact tracking, by increasing the cost of an enabling tracking system by degrading tracking performance, and by defeating tracking outright, preventing SBR from detecting TELs in an area.

Countermeasures to Increase Costs

The set of countermeasures which increase the performance requirements for an enabling constellation are increasing shelter density, increasing patrol lifetime, building sight-blocking structures, radar cross-section reduction, average-RCS decoys, small numbers of shape decoys, and jamming. Increasing shelter density and building sight-blocking structures both have the effect of increasing the probability that a TEL is able to move a certain distance without being observed, increasing $R(p_{\min})$. Both of these countermeasures must be undertaken far in advance of a crisis to allow time for construction, but increase the number of satellites that the seeker must deploy to achieve a manageable $R(p_{\min})$. RCSR can be used to increase τ_{rev} and therefore $R(p_{\min})$ in theory, though the performance requirements set out by the USG for SBR development may already take these effects into account. RCSR is more consequential when used in conjunction with decoys because it can make practical decoys easier to manufacture.

Increasing patrol lifetime increases the amount of uncertainty that a deployed TEL is able to accrue, increasing the average positional uncertainty across the TEL fleet. Owing to the non-linearity of $R(p_{\min})$ with focused tracking, increasing the average positional uncertainty increases the burden of tracking, increasing $R(p_{\min})$. It has the benefit of being easily implemented before or during a crisis without the seeker's knowledge, but is limited by the endurance of the TELs and their crews. The upper limit of the effectiveness of this countermeasure is in forcing the seeker into wide-area search where the entirety of the TEL deployment area must be monitored. An SBR constellation must be capable of tracking in a wide-area mode in order to be immune to the effects of increasing patrol lifetime.

Average RCS decoys, when operated similarly to TELs can create ambiguity in the source of TEL detections, limiting the seeker's ability to manage TEL uncertainty during a patrol. This indirectly increases the average $R(p_{\min})$ over a TEL's patrol by increasing the time-averaged positional uncertainty. Shape decoys can have the same effect as average RCS decoys if deployed near actual TELs. Small numbers of shape decoys also increase the tracking burden on the seeker's SBR system by increasing the number of individual targets that need to be tracked. This at least increases the number of warheads needed, and, if $R(p_{\min})$ is non-zero with the increased number of targets, increases the tracking burden and average $R(p_{\min})$.

The effects of jamming depend on the efficacy of counter-countermeasures such as jamming suppression with noise waveforms. If noise waveforms suppress jamming enough to allow SBR systems to operate in a jamming environment, albeit with longer dwell times, then jammers will have the effect of increasing the tracking burden in a similar manner to RSCR. If the JSR is sufficiently large that counter-countermeasures are insufficient, then jammers can be used to defeat tracking (discussed below).

All of the countermeasures that increase the performance requirements for the SBR system can be overcome by the seeker by deploying more SBR satellites. Each additional SBR satellite spreads the burden over more sensors and also lessens the effect of sight block by providing more diversity in view angle. Whether countering such countermeasures is effective therefore depends on the cost of the additional satellites (relative to the seeker's means) compared to the costs of the countermeasures (relative to the hider's means).

Each additional satellite added to a constellation would cost the seeker a few billion dollars over its lifetime. For a wealthy nation like the U.S., this may be a tolerable cost. Constructing new shelters and sight-blocking structures, when spread over a large area could present considerable costs to the hider. Russia, which has resource constraints owing to a weakened economy, may instead opt for more cost-effective countermeasures such as average RCS decoys and jamming. Whether the trade was worthwhile would depend on the exact scenario, but the comparatively lower costs of building and deploying things on the ground, relative to in space, would tend to favor the hider.

Countermeasures to Defeat Tracking

There are some countermeasures which are capable of defeating tracking outright. These are covered roads, large numbers of TEL-shaped decoys, and jamming (in the case where noise waveforms are insufficient), and ASAT.

Roads covered with radar-reflective material, such as wire mesh or chain-link, prevent SBR from being able to observe the protected road. If these structures are built over an extended length of road, they create, in effect, a distributed shelter where the TEL can hide. By building such structures to extend over kilometers of road, the hider can effectively set the minimum number of warheads needed to barrage a TEL in the protected area. This provides the hider with the ability to dictate the number of warheads required for a counterforce attack, and can make the exchange arbitrarily unfavorable to the attacker, limited by the amount of covered road they are capable of building. Building covered roads on a scale to protect an entire deployment zone would cost on the order of a few billion dollars, roughly the lifetime cost of a single SBR satellite.

Large numbers of TEL-shaped decoys have a similar effect. By increasing the number of targets, the hider is able to both increase the number of targets to which warheads must be allocated and increase the average number of warheads that need to be allocated to each target. Even if the seeker has enough SBR capability to track TELs perfectly (*i.e.*, $R(p_{\min})$ is zero for all relevant p_{min}), the hider can increase the number of targets and therefore warheads required, limited only by cost and operational capacity (*i.e.*, crews). Simple shape decoys (when used in conjunction with RSCR shaping of real TELs) should be constructible for a cost on the order of a few tens of thousands of dollars each. Crew availability and operational costs are likely to be more limiting than the hardware, though decoys can be deployed (even built) during, or in the lead-up to, a crisis, provided that the design, scaffolding, and skin material were stockpiled in advance.

Jammers, if not suppressible, can also defeat tracking outright. Arrays of static jammers can have the same effect as covered roads, creating areas where SBR systems simply cannot operate. Mobile jammers can also create transient paths through a deployment area where TELs can move without detection. Whether jammers are effective depends on the efficacy of countermeasure-resistant waveforms such as noise waveforms. These counter-countermeasures do not cancel jamming completely, but decrease its effect, so can be countered in turn by increasing jammer power or gain. Some jammer-resistant radar architectures may have different hardware requirements than traditional LFM pulsed radars, so which architecture was to be used would need to be decided in advance of radar deployment.

One countermeasure to any satellite-based remote-sensing system, such as SBR, are ASAT weapons. ASAT, if employed in advance of a loss in survivability, can defeat radar tracking entirely by simply removing sensors from the system. In the realm of ASAT systems, offensively destroying satellites is much more easily accomplished than defending them [135]. Since any repeating gap in coverage (of a few to tens of minutes) is sufficient to allow uncertainty to predictably grow over time, ASAT weapons could be decisive against SBR. The cost of relying on ASAT to defeat SBR is the risk of escalation inherent to attacking an adversary's military hardware.

Whether or not the seeker can depend on SBR to be able to track TELs depends on whether the hider is capable or willing to deploy these decisize countermeasures.

7.1.4 Availability of Countermeasures and Counter-Countermeasures

In this section, I discuss how the availability–both the monetary cost and the technological readiness–of SBR systems and countermeasures shape the competition between them. I employ a historical example, the tracking of Soviet SSBNs with passive sonar arrays, a competition for which the outcome is known, to contrast with the potential competition between SBR and TELs.

Availability and Accessibility of SBR Countermeasures

Cost is a large component of the availability of a technology. A technology is relevant only if a country has sufficient resources to deploy it and presumes a sufficient value to the technology to make the investment. SBR systems are very expensive. The 20-year lifetime cost for each satellite in a constellation is estimated on the order of a few billion dollars. The minimum enabling constellations found in Chapter 5 require at least 60 satellites, putting constellation cost on the order of a few hundred billion dollars. This required investment also has some caveats.

A SBR constellation achieves its capability to track TELs only when it exceeds the minimum requirements. Sparse constellations that are not near completion, which still have gaps in coverage, do not provide fractional tracking, *i.e.*, a constellation half the size of the requirement threshold does not track half of the TELs, it tracks no TELs. This is because gaps in coverage create opportunities for TELs to move freely without detection. Since the seeker's uncertainty grows whenever a TEL could have moved, as whether it has actually moved is not information directly available to the seeker, any coverage gaps will compound over time. Under these circumstances, the hider is under little pressure to move their TELs, except to signal to the seeker that they can.

Minimum constellations also do not provide other functionality in addition to TEL tracking during the portion spent overflying TEL deployment areas. SBR systems are used today for intelligence purposes. When meeting the minimum requirements for tracking, the whole of the SBR system's resources (what I have called the burden) need to be devoted to tracking. This does not provide time to acquire SAR images for intelligence collection. This is qualified in two ways. First, during tracking SAR images are produced of the TEL deployment area. Those images may be useful for intelligence purposes, however TELs may be deployed in remote areas with few points of interest. Second, when not overflying TELs, the SBR system would have the bandwidth to collect images for other purposes. The analysis in Chapter 5 indicates that solar power cells, especially if utilizing recent advances, should be sufficient to provide a near-unity capacity factor. An SBR system built to track TELs, and later defeated by countermeasures, would also retain value for other intelligence gathering endeavors.

So while SBR systems are expensive, in a competition between a hider and a seeker, the relevant metric of availability is the relative cost of the SBR system to the seeker and the relative cost of countermeasures to the hider normalized to the wealth of each nation. In the case of the U.S. in the role of the seeker, it can leverage its advantage in military budget, more than double that of China, and almost thirty times that of Russia[136]. This can place pressure on the hider to leverage low-cost countermeasures, or utilize technologies already at hand.

One dynamic of the deployment of countermeasures is that the hider can have an advantage where they are able to respond to the investments of the seeker. The requirements for countermeasures are more lax in the early phases of building an SBR constellation as the gaps in coverage will be longer. This creates additional flexibility for countermeasures which can be deployed or constructed quickly.

One significant up-front cost to the hider could be the construction of shelters. Individually, shelters are inexpensive, as they could be constructed from common materials such as corrugated sheet metal, chain-link, or wire mesh. Since shelters are needed on the order of one per few kilometers of road, the cost can become significant, on the order of hundreds of millions of dollars if every shelter is built *de novo*. Cost savings could be made by utilizing pre-existing structures as shelters. Such shelters would need to be large enough to to physically fit the TEL and its support vehicles for a relatively short period of time (on the order of days). Shelters are a baseline requirement for making TELs survivable while on patrol, so some investment in shelters must be made.[5] The simplicity of shelters means that the investment in shelters can be made after the seeker has committed their investment to SBR.

One inexpensive, technologically simple, and quick to deploy countermeasure is decoys. Average RCS decoys have some utility in degrading tracking and could be deployed simply by providing pre-existing vehicles will collapsible corner reflectors. While these decoys would not stand up to intense scrutiny with ISAR, they could be useful in bridging the gap to other, more effective countermeasures, at the cost of a few tens of dollars of sheet metal per reflector.

More convincing TEL decoys could be constructed utilizing commercially available products such as aluminum scaffolding, wire mesh, and small trucks. In order to make the decoys convincing, the same lightweight materials (with less required scaffolding) could be applied to real TELs to make them more difficult to distinguish. The up-front investment in these decoys would be in the raw materials. As the seeker built up their SBR constellation, decoys could be constructed and equipped on pre-existing vehicles and put into service. If the hider were primarily concerned about crises, rather than a bolt-from-the-blue attack, decoys could be put into service at the outset of a crisis. The worst case scenario for such a decoy is against an extremely capable SBR tracking system, such that uncertainty would not grow significantly over time. Even under these conditions, each decoy would need to be allocated one or more warheads during an attack, so the hider could out-build the seeker's nuclear arsenal with decoys. Against less capable systems, where uncertainty grows at a manageable rate for the seeker, each decoy not only reduces the number of warheads available to be allocated against real TELs, but also increases the tracking burden on the seeker. Given the simplicity and low cost of decoys, they could be employed in response to the seeker's investment in SBR. Decoys could potentially defeat tracking entirely by increasing the requirements of SBR, or by outnumbering the available warheads with targets.

Two other countermeasures could potentially decisively defeat SBR tracking, though with a much higher risk of escalation. Jamming, a form of electronic warfare, could be used to create blind spots for the seeker. By creating effective gaps in coverage, jamming would allow the seeker to spread the seeker's uncertainty in the location of a given TEL over an arbitrary number of shelters. The hider could effectively set the seeker's uncertainty to a level where a counterforce attack was unfeasible.

Jamming is likely to be available now. Russia, for example, is already known to have the capability to jam U.S. SAR imaging [131]. Future developments in electronic countermeasure (ECM) resistant radars, such as those

[5]The more heavily a seeker invests in jammers and/or decoys to degrade tracking, fewer shelters may be needed. Taken to the extreme, a continuously operating jammer network, or a large number of decoys could be used in place of shelters.

using noise waveforms may reduce the decisiveness of jammers. However, whether the gains in resilience required to overcome the 13 order of magnitude imbalance in JSR are achievable remains to be seen. One of the advantages of jammers, particularly in the near term, is that they have other roles in electronic warfare, such as jamming satellite communications, that could justify their cost to a hider. The other advantage to jammers is that, owing to these other roles, they are already deployed. This means that for an SBR system to be capable of tracking TELs, ECM-resistant radars would need to be developed, adapted for use in space, and deployed, before the already deployed jammers would have reduced utility in defeating tracking. Jammers therefore provide a near-term buffer for the hider in order to preserve TEL survivability.

If jammers are eventually invalidated by ECM-resistant systems one effective, if highly provocative, countermeasure would be ASAT weapons. For technologically advanced states, space has become an offense-advantaged domain, it is easier to destroy assets in space than it is to protect them [135]. China and Russia have both demonstrated ASAT capability against satellites in LEO, the orbits in which SBR would need to be deployed [97]. While ASAT would be highly provocative, and would degrade the usefulness of LEO to all parties, including the hider, if the hider were genuinely concerned about the degradation of the survivability of their TELs, ASAT would be an effective means of preventing a SBR tracking constellation from reaching effective numbers. ASAT would be of questionable use during an attack itself, given the short flight times of depressed-trajectory SLBMs, so would need to be used prior to a crisis, before an SBR system actually gained tracking capability. This presents a pressure for the hider to escalate to the use of ASAT in order to preserve their deterrent.

All of these countermeasures are either technologically simple, such as the construction of covered roads or decoys, or are already deployed or in development for other purposes, such as jammers and ASAT. Some countermeasures, such as RCSR with radar-absorbing material would require extensive research and development. Such countermeasures may play a future role in the competition between remote-sensing and TELs, RAM in particular since it is a sought after technology for use with aircraft, but currently the competition is overdetermined in the favor of the hider. Currently deployed jammers could defeat a SBR systems that could be deployed today, and may only be circumvented by the development of ECM-resistant radars which will take time. Inexpensive and simple decoys made of wire mesh, tarps, and scaffolding could serve to further bolster the survivability of TELs. Rather than attempt to perfectly replicate a real TEL with a decoy, the two can be made difficult to distinguish by making the TELs resemble the decoys.

Countermeasures that raise the cost of, and potentially defeat, tracking are accessible or already deployed, and have redundancy. This means that SBR, even if deployed in significant number, are unlikely to undermine the survivability of TELs. Some countermeasures, such as jamming and ASAT weapons, might present escalation risk, however.

A Comparative Historical Example

Currently, SBRs are deployed in relatively small numbers. A vigorous competition between SBR and TEL survivability would exist only in the future. Given the difficulty of making predictions, a historical example of competition between remote-sensing and nuclear weapons is useful. One such competition, for which there is some information publicly available, is the Cold War competition between Soviet SSBNs and the Sound Surveillance System (SOSUS), a passive sonar array that was deployed by the U.S [25]. This episode has also been referenced in the context of tracking TELs as an example of survivability being undermined by remote-sensing [9,10]. As with TELs, SSBNs provide survivability to the weapons they carry through a combination of mobility and concealment. The ability to track SSBNs therefore degrades their survivability, assuming the seeker has anti-submarine warfare (ASW) capabilities that they can bring to bear.

As noted in Chapter 1, starting in the 1950s, the U.S. developed a passive underwater sonar array that was capable of detecting Soviet SSBNs from long distances by relying on the sound-channeling effect of the Sound Fixing and Ranging channel (SOFAR), a deep-water sound channel [25].[6] By the 1970s, the U.S. developed a remote-sensing system that was capable of tracking SSBNs throughout their deployment. This era was referred to, in the U.S. ASW community, as "the Happy Time" [25]. SOSUS, like SBR, was a critical technology in a larger network of remote sensors.

SOSUS was capable of detecting Soviet SSBNs as they transited the GIUK gap, the choke point between Greenland, Iceland, and the United Kingdom that connects the Norwegian sea to the broader Atlantic. Russian SSBNs attempting to go on deterrence patrols in the Atlantic needed to pass through the GIUK gap. Soviet SSBNs of the era were sufficiently noisy that SOSUS (which was deployed near the North American coast) was capable of detecting them. This allowed the U.S. to cue attack submarines to follow the SSBNs and track them throughout their deployment.

Without SOSUS, U.S. attack submarines were not capable of tracking Soviet SSBNs indefinitely. If contact was lost between the trailing attack submarine, SOSUS was also used for re-acquisition [25]. This placed SOSUS in the category of a critical technology. Throughout the 1970s, the U.S. had the capacity to track Soviet SSBNs because SOSUS enabled it.

As with any critical technology, if not made redundant with other technologies, it is a single point of failure. Eventually, during the 1980s Soviet submarines were replaced with Delta-class submarines which were quieter than their predecessors, and carried SLBMs with sufficient range to threaten North American targets without having to transit through the GIUK gap [25]. Subsequently, the U.S. lost the ability to track Soviet (later Russian) SSBNs and SOSUS was dismantled.

This episode has value in the context of assessing whether SBR will undermine the survivability of TELs for a number of reasons. First, it provides an example of a critical technology enabling tracking. Second, it illustrates how countermeasures to a critical technology, once employed can defeat tracking. Importantly, it also highlights the importance of the availability and accessibility of countermeasures.

When SOSUS was first deployed, it was an effective remote-sensing system. The countermeasure to passive sonar arrays was, at the time, theoretically identifiable. For over a decade, however, countermeasures were either not attained, or not pursued by the Soviets. During this period of time, SOSUS was capable of tracking SSBNs, not because effective countermeasures were unknown, but because they were difficult to develop and deploy. Once deployed, however, tracking was defeated.

In contrast, in the competition between SBR and TELs, countermeasures that could prove decisive are either technologically simple (*e.g.*, decoys), or are currently in service for other purposes (*e.g.*, jammers and ASAT weapons). In other words, the countermeasures predate the capability. One would therefore not expect the deployment of SBR, even in number capable of tracking a permissive hider's TELs, to be able to undermine the survivability of TELs.

[6] Sound channeling can occur in the ocean as the speed of sound in water can vary with depth. If there is a horizontal layer of water where the speed of sound is at a minimum, sound waves above and below that layer will bend toward it. This prevents reflection and/or absorption off of the ocean floor or surface, allowing sound to be transmitted through the channel over great distances.

7.1.5 Conclusion and Recommendations

Based on this analysis, I offer some assessments and recommendations surrounding the competition between TELs and remote-sensing.

Tracking TELs requires the development of critical technologies. There are currently gaps where extant remote-sensing systems are not capable of tracking TELs, namely at night and during periods of bad weather. If a state wishes to develop the capability to track TELs, they should focus their efforts on technologies which address those gaps. Similarly, when evaluating emerging remote-sensing technologies, the focus of analysis should be on the technology's ability to close those gaps.

Analysis of critical technologies should focus on countermeasures. When evaluating the potential effects of an emerging technology, analysis should focus not only on the effectiveness of that technology, but also the existence, effectiveness, and availability of the countermeasures. Broad endeavors such as tracking are only as strong as their weakest link. Critical technologies are a naturally attractive target for countermeasures as they present a single point of failure.

SBR is not a promising means of tracking TELs. While SBR has the elements of a critical technology, and the ability to operate under conditions where other deployed modalities do not have tracking capability, SBR is vulnerable to simple countermeasures. While some countermeasures, such as RCSR with RAM, would require significant research and development effort, effective countermeasures such as decoys are technologically simple and should be deployable at scale for relatively low cost. Some countermeasures, such as jammers and ASAT weapons, are already in deployment as they serve other purposes, so SBR, if pursued, would be deployed into a hostile environment. Electronic warfare and ASAT weapons also present the risk of escalation through entanglement, so states should expect a decrease in their security if they choose to deploy an SBR system with capability to track (permissive) TELs.

Modest SBR capability should not undermine strategic stability. One key finding of this work is that a seeker's uncertainty in a TELs position grows during gaps in coverage of sufficient length. For the case of tracking TELs, if shelters are kilometers apart, regular gaps of length on the order of a few to ten minutes are sufficient to prevent tracking. Concern has been raised in the literature that the relatively modest collection of fewer than two dozen SBR satellites deployed by countries around the world would be sufficient to undermine the survivability of TELs [9]. Even if all deployed SBR satellites were utilized as a cohesive constellation (it is unclear if this is possible), there would be an average gap of more than twenty minutes between overflights [9].[7] The U.S. collection of six SBR satellites would present more than 90 minute gaps between overflights. These gaps are sufficiently large that a motivated hider could easily exploit them. The deployment of modest numbers of SBR satellites (*e.g.* twenty to thirty) should not undermine TEL survivability nor provoke escalatory countermeasures on the grounds of TEL survivability.

Fears that remote-sensing will inevitably undermine TEL survivability and damage strategic stability are overblown. Tracking with remote-sensing is potentially more challenging than is widely understood. In a game of hide-and-seek, the seeker does not contend with what that hider does, but what they could do. The hider does not need to be passive in the competition, as they can actively interfere with remote-sensing through countermeasures. The hider also has the advantage of being able to be reactive, they do not need to deploy countermeasures until after the seeker has committed to developing a particular remote-sensing system. That said, history has shown that countermeasures can sometimes be hard to leverage, and survivability can be undermined. Degradation of survivability is not inevitable,

[7] These numbers are for overflights of the Korean peninsula, approximately the same latitude as Central China in China.

however. Which way the balance will tip depends on the specific technologies in question, and in the case of SBR, the advantage lays with the hider.

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